COMPARISON PRINCIPLES FOR DEGENERATE SUB-ELLIPTIC EQUATIONS IN NON-DIVERGENCE FORM: BLUEPRINT OF THE LEAN FORMALIZATION

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Abstract. This document collects the natural-language statements and detailed proofs that accompany the Lean 4 formalization of Theorems 1.2 and 1.3 of J. J. Manfredi and S. Mukherjee, *Comparison principles for degenerate sub-elliptic equations in non-divergence form*. Five results that the paper cites from the literature (the strong maximum and minimum principles of Bardi and Goffi in the form of Corollary 2.4, Aleksandrov's theorem, Jensen's lemma, Pansu's differentiability theorem with [DMV13, Proposition 4.8], and parts (1)–(3) of Lemma 3.8, which are [CYW07, Proposition 3.3]) are not proved; they are stated precisely and carried as explicit hypotheses (`CitedResults`, `EuclidCitedResults`). Everything else is proved. The Carnot group enters through the structure `CarnotSetting`, whose fields are the structural facts of Section 2.4 of the paper. Each subsection corresponds to the Lean declaration of the same name in the directory `CompND/`; cross-references point to the results that a proof uses. Section 3 lists the formalized results of the paper. Within each section, items appear in dependency order.

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1. Definitions

1.1. Preamble.

Definition 1.1 (Ambient spaces, jets and matrices)**.** For $N \in \mathbb{N}$ we write $\mathbb{R}^N$ for the N-dimensional real inner product space `EuclideanSpace` $\mathbb{R}$ (`Fin` N), with the Euclidean scalar product $\langle \cdot, \cdot \rangle$ and the Euclidean norm $|x| = \langle x, x \rangle^{1/2}$, and $e_1, \dots, e_N$ for its standard basis. A 1-*jet* on $\mathbb{R}^N$ is a continuous linear functional $D : \mathbb{R}^N \to \mathbb{R}$, and a 2-*jet* is a continuous bilinear form $B : \mathbb{R}^N \times \mathbb{R}^N \to \mathbb{R}$; both are taken in the Mathlib sense of continuous (multi)linear maps. $\mathbb{R}^{k \times k}$ denotes the $k \times k$ real matrices, with Tr the trace and A^T the transpose. For a map F differentiable at x, $DF(x)$ is its Fréchet derivative (Jacobian) at x, and $\nabla u(x)$ is the differential of a scalar function u regarded as a 1-jet. Balls $B_\rho(x)$ and the measure $|E|$ are Euclidean and Lebesgue unless a homogeneous norm is named explicitly.

Definition 1.2 (Lie bracket of vector fields)**.** A *vector field* on $\mathbb{R}^n$ is a map $V : \mathbb{R}^n \to \mathbb{R}^n$. For two vector fields V, W the *Lie bracket* is the vector field

$$[V, W](x) \;=\; DW(x)\, V(x) \;-\; DV(x)\, W(x),$$

where $DV(x), DW(x)$ are Fréchet derivatives (and are read as 0 at points of non-differentiability, as in Mathlib's `fderiv`). Identifying V with the first-order operator $V \cdot \nabla$, this is the commutator of the two operators.

1.2. `CarnotGroup`.

Definition 1.3 (Carnot group in exponential coordinates: the group data)**.** A *Carnot group datum* consists of three integers n, m, r with $r \geq 1$ together with

- a map $* : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ (the group law),
- a continuous linear map $\pi_1 : \mathbb{R}^n \to \mathbb{R}^m$ (the projection onto the first layer),
- a map $\exp_{\mathrm{h}} : \mathbb{R}^m \to \mathbb{R}^n$ (the horizontal exponential),

subject to the following conditions.

(1) $(\mathbb{R}^n, *)$ is a group whose neutral element is the origin and whose inversion is $x \mapsto -x$; explicitly, $(x * y) * z = x * (y * z)$, $x * 0 = x$, $0 * x = x$, $x * (-x) = 0$ and $(-x) * x = 0$, for all $x, y, z \in \mathbb{R}^n$.
(2) The group law is C^∞ on $\mathbb{R}^n \times \mathbb{R}^n$.
(3) $\pi_1(x * y) = \pi_1(x) + \pi_1(y)$ for all $x, y \in \mathbb{R}^n$.
(4) $\exp_{\mathrm{h}}$ is continuous and $\pi_1(\exp_{\mathrm{h}}(\zeta)) = \zeta$ for every $\zeta \in \mathbb{R}^m$.
(5) If $r = 1$ then the group law is commutative: $x * y = y * x$ for all x, y.

These are the facts about a Carnot group $\mathbb{G} = (\mathbb{R}^n, *)$ of step r with m generators, read in the exponential coordinates $x = \exp\big(\sum_{j \leq r} \sum_{i \leq m_j} x_{i,j} X_{i,j}\big)$ of Section Section 2.4, that the rest of the development uses at the group level. Item (1) and item (2) are the statement recorded there that in these coordinates the group law is polynomial, hence smooth, the neutral element is 0 and $x^{-1} = -x$; writing the inverse as $-x$ rather than carrying it as separate data is a choice of coordinates, not an extra hypothesis. Item (3) is the identity $\pi_1(x * y) = \pi_1(x) + \pi_1(y)$: π_1 is a homomorphism onto the abelianization $\mathbb{G}/[\mathbb{G}, \mathbb{G}] \cong \mathbb{R}^m$, and it is linear because it is a coordinate projection. In item (4), $\exp_{\mathrm{h}}(\zeta) = \exp(\sum_i \zeta_i X_i)$ is the point whose first-layer coordinates are ζ and whose higher-layer coordinates vanish, so $\pi_1 \circ \exp_{\mathrm{h}} = \mathrm{id}$; in particular π_1 is surjective. Item (5) records that a Carnot group of step 1 has $\mathfrak{g} = \mathfrak{g}_1$ with $[\mathfrak{g}_1, \mathfrak{g}_1] = \{0\}$, so it is the Euclidean additive group.

The formalization note (item 3 of the note after Theorem Theorem 1.3) permits the structural facts of Section Section 2.4 to be built into the definition of the setting, provided every Carnot group in those coordinates satisfies the definition. Each condition above is one of those facts, so the proviso holds; in particular the Euclidean additive group on $\mathbb{R}^m$, with $n = m$, $r = 1$, $x * y = x + y$, $\pi_1 = \mathrm{id}$ and $\exp_{\mathrm{h}} = \mathrm{id}$, satisfies all five.

1.3. `CarnotNorm`.

Definition 1.4 (Carnot group with its homogeneous norm)**.** A *Carnot norm datum* is a Carnot group datum (`CarnotGroup`) together with a map $\|\cdot\| : \mathbb{R}^n \to \mathbb{R}$, the homogeneous norm, subject to the conditions below. Throughout, $q := 2\,r!$, the associated left-invariant distance is

$$d(x, y) := \|y^{-1} * x\| = \|(-y) * x\|,$$

and $B_\rho(x) := \{y \in \mathbb{R}^n : d(y,x) < \rho\}$.

(1) $\|x\| \geq 0$ for all x; $\|x\| = 0$ if and only if $x = 0$; $\|-x\| = \|x\|$; and $\|\cdot\|$ is continuous on $\mathbb{R}^n$.
(2) $\|\exp_{\mathrm{h}}(\zeta)\| = |\zeta|$ for every $\zeta \in \mathbb{R}^m$.
(3) $x \mapsto \|x\|^q$ is C^∞ on $\mathbb{R}^n$.
(4) (First inequality of (2.17).) For every $\nu > 0$ there is $c > 0$ such that $\|x*y\| \leq c\,(\|x\|+\|y\|)$ whenever $\|x\| < \nu$ and $\|y\| < \nu$.
(5) (Second inequality of (2.17).) For every $\nu > 0$ there is $c > 0$ such that $\|x^{-1} * y * x\| \leq c\,\|y\|^{1/r}$ whenever $\|x\| < \nu$ and $\|y\| < \nu$.
(6) (Hölder equivalence on bounded sets.) For every $\nu > 0$ there is $c > 0$ such that

$$c^{-1}\,|x-y| \;\leq\; d(x,y) \;\leq\; c\,|x-y|^{1/r} \qquad \text{whenever} \quad |x| \leq \nu \text{ and } |y| \leq \nu.$$

(7) For every $x \in \mathbb{R}^n$ and every $\rho > 0$ the ball $B_\rho(x)$ is non-empty and connected.
(8) $\|\cdot\|$ is proper: for every $C \in \mathbb{R}$ the sublevel set $\{x \in \mathbb{R}^n : \|x\| \leq C\}$ is bounded.

These are the properties of the homogeneous norm (2.15) recorded in Section Section 2.4. Item (1) is continuity, evenness and non-degeneracy of $\|x\| = \big(\sum_j (\sum_i |x_{i,j}|^2)^{r!/j}\big)^{1/2r!}$. Item (3) is the observation that $\|x\|^q = \|x\|^{2r!}$ is a polynomial, which is what makes the kernel $\|y * x^{-1}\|^q$ of the sup- and inf-convolutions (3.13) smooth; item (2) says that the horizontal exponential is an isometry from $(\mathbb{R}^m, |\cdot|)$ onto the first-layer directions, which is immediate from (2.15) since $\exp_{\mathrm{h}}(\zeta)$ has first-layer coordinates ζ and vanishing higher coordinates. Items (4) and (5) are the two inequalities (2.17), the second being the estimate $\|c_x(y)\| \leq c_{\mathbb{G}}\|y\|^{1/r}$ for the inner automorphism $c_x(y) = x^{-1} * y * x$, whose exponent $1/r$ is sharp. Item (6) is the Hölder equivalence of d with the Euclidean distance on bounded sets, which also says the two topologies agree. Item (7) is the statement that the balls $B_\rho(0)$ are connected, being star-shaped with respect to the dilations, and hence so are all balls by left invariance of d; it is non-empty because $d(x,x) = \|0\| = 0 < \rho$.

Left invariance $d(z*x, z*y) = d(x,y)$ and the right-translation identity (2.16) $d(x*h, x) = \|h\|$ are not listed: both follow from the group axioms of `CarnotGroup` alone, since $d(z*x, z*y) = \|(z*y)^{-1} * (z*x)\| = \|y^{-1} * x\|$ and $d(x*h, x) = \|x^{-1} * x * h\| = \|h\|$. The dilations themselves are not carried, because nothing below uses them other than through items (3), (5) and (7).

Item (8) is the properness of (2.15), and it too is immediate from that formula: if $\|x\| \leq C$ then $\big(\sum_i |x_{i,j}|^2\big)^{r!/j} \leq C^{2r!}$ for every layer j, hence $\sum_i |x_{i,j}|^2 \leq C^{2j}$, so all coordinates of x are bounded in terms of C alone. It is listed as a separate condition because it does not follow from the others: the quasi-triangle inequality (4) bounds $\|y\|$ in terms of $\|y * x^{-1}\|$ and $\|x\|$, which says nothing about the Euclidean size of y, and the lower Hölder bound of (6) is available only once both of its arguments are already known to be Euclidean-bounded, so neither of them yields boundedness of a sublevel set of $\|\cdot\|$. Properness is what makes a sublevel set of the kernel $y \mapsto \|y * x^{-1}\|^q$ of (3.15) relatively compact, hence the infimum in (3.15) locally an infimum over a compact set of y and therefore a continuous function of x; and that continuity is what makes the reduced domain $\Omega[\varepsilon]$ open, as the paper asserts without proof.

As with `CarnotGroup`, every Carnot group in the coordinates of Section Section 2.4 satisfies all eight conditions, so the proviso of item 3 of the formalization note holds. For the Euclidean additive group on $\mathbb{R}^m$ ($n = m$, $r = 1$, $x * y = x + y$, $\|x\| = |x|$) one has $q = 2$, $d(x,y) = |x-y|$, and each condition reduces to a standard fact with $c = 1$: (4) is the triangle inequality, (5) is $\|y\| \leq \|y\|$ since $x^{-1} * y * x = y$, (6) is an equality, (7) holds because Euclidean balls are convex, and (8) holds because a Euclidean ball is bounded.

1.4. `LieHull`.

Definition 1.5 (The Lie hull of a frame of vector fields)**.** Let $n, m \in \mathbb{N}$ and let $\sigma : \mathbb{R}^n \to \mathbb{R}^{n\times m}$, $\sigma(x) = [\sigma^1(x), \dots, \sigma^m(x)]$, be a frame of m vector fields on $\mathbb{R}^n$. Call a family S of vector fields on $\mathbb{R}^n$ *admissible for* σ if it contains each σ^i and is closed under the Lie bracket, i.e. $[V,W] \in S$ whenever $V, W \in S$. The *Lie hull* LieHull σ is the intersection of all admissible families:

$$\text{LieHull } \sigma \;=\; \bigcap \big\{\, S \;:\; S \text{ admissible for } \sigma \,\big\}.$$

The intersection of admissible families is admissible, so LieHull σ is the smallest family of vector fields containing $\sigma^1, \dots, \sigma^m$ and closed under brackets: it consists precisely of the iterated brackets of the frame fields. Its purpose is to express Hörmander's rank condition (1.6),

$$\dim\Big(\mathrm{Lie}\big(X_1(x), \dots, X_m(x)\big)\Big) = n \qquad \text{for all } x \in \mathbb{R}^n,$$

as the statement that the span of $\{V(x) : V \in \mathrm{LieHull}\ \sigma\}$ is all of $\mathbb{R}^n$ for every x. Taking the smallest such family, rather than merely asserting the existence of *some* bracket-closed family whose values span, is what makes the condition a restriction: the family of all vector fields is bracket-closed and its values span trivially.

1.5. `TwiceDiffAt`.

Definition 1.6 (Pointwise second-order (Peano) jet)**.** Let $N \in \mathbb{N}$, let $w : \mathbb{R}^N \to \mathbb{R}$, let $x \in \mathbb{R}^N$, let $D : \mathbb{R}^N \to \mathbb{R}$ be a 1-jet and let $B : \mathbb{R}^N \times \mathbb{R}^N \to \mathbb{R}$ be a 2-jet. We say that (D, B) is a *second-order jet of w at x*, written TwiceDiffAt $w\ x\ D\ B$, if

(1) B is symmetric, $B(u, v) = B(v, u)$ for all $u, v \in \mathbb{R}^N$; and
(2) for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$\Big|\, w(y) - w(x) - D(y - x) - \tfrac{1}{2}\, B(y - x,\, y - x) \,\Big| \;\le\; \varepsilon\, |y - x|^2 \qquad \text{whenever} \quad |y - x| < \delta.$$

Thus (D, B) is a second-order jet of w at x exactly when $w(y) = w(x) + D(y - x) + \frac{1}{2}B(y - x, y - x) + o(|y - x|^2)$ as $y \to x$, with B symmetric.

The existence of such a pair is the statement that w is *twice differentiable at x* in the pointwise sense used throughout the paper: it is the conclusion of Aleksandrov's theorem (Theorem Theorem 2.6), the property selected at the points produced by Jensen's lemma (Lemma Lemma 2.7), and the hypothesis under which a viscosity sub- or super-solution satisfies the equation classically (Remark Remark 2.2). When it exists the pair is unique, D being the Fréchet differential $\nabla w(x)$ and B the Hessian $D^2 w(x)$: if (D, B) and (D', B') both qualify then $C := B - B'$ is a symmetric bilinear form with $|C(h, h)| \le 4\varepsilon |h|^2$ for every $\varepsilon > 0$, so taking $h = tv$ and dividing by t^2 gives $C(v, v) = 0$ for every v, hence $C = 0$ by polarization, and then $D = D'$ by the first-order part. It is for this uniqueness that B is required to be bilinear rather than merely a function of two arguments: the quadratic part of the expansion controls $B(h, h)$ only as $h \to 0$, whereas the symmetrized horizontal second derivative built from a 2-jet evaluates B at the fixed vectors $\sigma^i(x)$, which are not small.

1.6. `HasPansuGradAt`.

Definition 1.7 (Pansu differentiability and the horizontal differential)**.** Let $n, m \in \mathbb{N}$, let $* : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ be a group law, $\pi_1 : \mathbb{R}^n \to \mathbb{R}^m$ a continuous linear map and $\|\cdot\| : \mathbb{R}^n \to \mathbb{R}$ a homogeneous norm. For $f : \mathbb{R}^n \to \mathbb{R}$, $x \in \mathbb{R}^n$ and $\xi \in \mathbb{R}^m$ we say that *f is Pansu differentiable at x with horizontal differential ξ*, written HasPansuGradAt $(*)\ \pi_1\ \|\cdot\|\ f\ x\ \xi$, if for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$\big|\, f(x * g) - f(x) - \langle \xi, \pi_1(g) \rangle \,\big| \;\le\; \varepsilon\, \|g\| \qquad \text{whenever} \quad \|g\| < \delta,$$

where $\langle \xi, \pi_1(g) \rangle = \sum_{i=1}^{m} \xi_i\, \pi_1(g)_i$. Equivalently

$$f(x * g) = f(x) + \langle \xi, \pi_1(g) \rangle + o\big(\|g\|\big) \qquad (g \to 0).$$

This is the expansion of Section Section 2.4: it is (2.18) when f is differentiable at x, in which case $\xi = \mathcal{X} f(x)$, and it is the conclusion of Pansu's theorem for a d-Lipschitz f at almost every point, where $\xi = \mathcal{X} f(x)$ is called the horizontal differential of f at x. Note that the increment is the *right* translation $g \mapsto x * g$, and that the error is measured against the homogeneous norm $\|g\|$, not against $|g|$; both are essential, and both are why the expansion does not transfer to left translations.

1.7. `hgrad`.

Definition 1.8 (Horizontal gradient of a 1-jet, (2.6))**.** Let $n, m \in \mathbb{N}$, let $\sigma : \mathbb{R}^n \to \mathbb{R}^{n\times m}$, $\sigma(x) = [\sigma^1(x), \dots, \sigma^m(x)]$, be a frame of m vector fields on $\mathbb{R}^n$, so that the associated first-order operators are $X_i = \sigma^i(x) \cdot \nabla$. For $x \in \mathbb{R}^n$ and a 1-jet $D : \mathbb{R}^n \to \mathbb{R}$, the *horizontal gradient of D at x* is the vector hgrad $\sigma\ x\ D \in \mathbb{R}^m$ with components

$$\big(\text{hgrad } \sigma\ x\ D\big)_i \ = \ D\big(\sigma^i(x)\big), \qquad i = 1, \dots, m.$$

If u is differentiable at x and $D = \nabla u(x)$, then $D(\sigma^i(x)) = \langle \nabla u(x), \sigma^i(x)\rangle = X_i u(x)$, so hgrad $\sigma\ x\ \nabla u(x) = \mathcal{X}u(x) = \sigma(x)^T \nabla u(x)$ is the horizontal gradient $(X_1 u(x), \dots, X_m u(x))$ of (2.6). Stating it on 1-jets rather than on functions is what (2.10) does: it makes the operator a function of a point and a jet, so that no notion of horizontal derivative of a non-smooth function is needed.

1.8. `Mfield`.

Definition 1.9 (The correction matrix $\mathcal{M}(x, \zeta)$, (2.7))**.** Let $n, m \in \mathbb{N}$ and let $\sigma : \mathbb{R}^n \to \mathbb{R}^{n\times m}$, $\sigma(x) = [\sigma^1(x), \dots, \sigma^m(x)]$, be a frame of m vector fields on $\mathbb{R}^n$. For $x \in \mathbb{R}^n$ and a 1-jet $D : \mathbb{R}^n \to \mathbb{R}$, playing the role of $\zeta = \nabla u(x)$, the matrix $\mathcal{M}(x, D) = \text{Mfield } \sigma\ x\ D \in \mathbb{R}^{m\times m}$ is defined by

$$\mathcal{M}(x, D)_{i,j} \ = \ \tfrac{1}{2}\Big(D\big(D\sigma^j(x)\,\sigma^i(x)\big) + D\big(D\sigma^i(x)\,\sigma^j(x)\big)\Big), \qquad i, j = 1, \dots, m,$$

where $D\sigma^k(x)$ is the Fréchet derivative (Jacobian) of σ^k at x and $D\sigma^k(x)\,\sigma^i(x)$ is its value on the vector $\sigma^i(x)$.

Writing ζ for the vector representing D, this is

$$\mathcal{M}(x, \zeta)_{i,j} = \tfrac{1}{2}\big(\langle D\sigma^j(x)\sigma^i(x), \zeta\rangle + \langle D\sigma^i(x)\sigma^j(x), \zeta\rangle\big),$$

which is exactly (2.7). It is symmetric in (i, j) by construction, linear in ζ because $D \mapsto D(v)$ is, and smooth in x when σ is smooth. It is the term by which the symmetrized horizontal second derivative differs from the conjugated Euclidean Hessian in (2.6); it is a first-order term, produced by differentiating the frame, and it vanishes identically when the frame is constant.

1.9. `hhessS`.

Definition 1.10 (Symmetrized horizontal second derivative of a 2-jet, (2.6))**.** Let $n, m \in \mathbb{N}$, let $\sigma : \mathbb{R}^n \to \mathbb{R}^{n\times m}$, $\sigma(x) = [\sigma^1(x), \dots, \sigma^m(x)]$, be a frame of m vector fields on $\mathbb{R}^n$, let $x \in \mathbb{R}^n$ and let (D, B) be a 1-jet and a 2-jet on $\mathbb{R}^n$. The *symmetrized horizontal second derivative of (D, B) at x* is the matrix hhessS $\sigma\ x\ D\ B \in \mathbb{R}^{m\times m}$ with entries

$$\big(\text{hhessS } \sigma\ x\ D\ B\big)_{i,j} \ = \ B\big(\sigma^i(x), \sigma^j(x)\big) \ + \ \mathcal{M}(x, D)_{i,j},$$

where $\mathcal{M}(x, D)$ is the correction matrix of `Mfield`.

If u is twice differentiable at x in the pointwise (Peano) sense, with second-order jet $(D, B) = (\nabla u(x), D^2 u(x))$, then $B(\sigma^i(x), \sigma^j(x)) = \langle D^2 u(x)\sigma^i(x), \sigma^j(x)\rangle = (\sigma(x)^T D^2 u(x)\,\sigma(x))_{i,j}$, so

$$\text{hhessS } \sigma\ x\ D\ B \ = \ \sigma(x)^T D^2 u(x)\,\sigma(x) + \mathcal{M}(x, \nabla u(x)) \ = \ \mathcal{X}\mathcal{X}^s u(x),$$

the symmetrized horizontal second derivative $\tfrac{1}{2}\big(X_i X_j u + X_j X_i u\big)_{i,j}$ of (2.6). The matrix is symmetric whenever B is, since $\mathcal{M}(x, D)$ is symmetric by construction.

1.10. `CarnotSetting`.

Definition 1.11 (The sub-elliptic setting)**.** A *Carnot setting* is a Carnot norm datum (`CarnotNorm`) together with a map $\sigma : \mathbb{R}^n \to \mathbb{R}^{n\times m}$, $\sigma(x) = [\sigma^1(x), \dots, \sigma^m(x)]$ — the horizontal frame, written $X_i = \sigma^i(x) \cdot \nabla$ — subject to the following conditions. Here $\mathcal{X}u(x) = \text{hgrad } \sigma\ x\ \nabla u(x)$ is the horizontal gradient of `hgrad`, $\mathcal{X}\mathcal{X}^s$ is the symmetrized horizontal second derivative of `hhessS`, a *second-order jet* is as in `TwiceDiffAt`, and $L_h(y) = h * y$ denotes left translation.

(1) Each $\sigma^i : \mathbb{R}^n \to \mathbb{R}^n$ is C^∞.
(2) For every $x \in \mathbb{R}^n$ the vectors $\sigma^1(x), \dots, \sigma^m(x)$ are linearly independent.

(3) (Hörmander's rank condition (1.6).) For every $x \in \mathbb{R}^n$,
$$\operatorname{span}\{\, V(x) \;:\; V \in \operatorname{LieHull}\,\sigma \,\} = \mathbb{R}^n,$$
with LieHull σ the Lie hull of the frame (`LieHull`).

(4) (The Taylor formula (2.18).) If $u : \mathbb{R}^n \to \mathbb{R}$ is differentiable at x with differential D, then u is Pansu differentiable at x (`HasPansuGradAt`) with horizontal differential hgrad $\sigma\ x\ D$; that is, $u(x * h) = u(x) + \langle \mathcal{X}u(x), \pi_1(h)\rangle + o(\|h\|)$ as $h \to 0$.

(5) (Left-invariance of the horizontal derivatives.) For all $h, x \in \mathbb{R}^n$ and every $\varphi : \mathbb{R}^n \to \mathbb{R}$ with a second-order jet (D, B) at $h*x$, the function $\varphi \circ L_h$ has a second-order jet (D', B') at x satisfying
$$\operatorname{hgrad}\,\sigma\ x\ D' = \operatorname{hgrad}\,\sigma\ (h * x)\ D, \qquad \operatorname{hessS}\,\sigma\ x\ D'\ B' = \operatorname{hessS}\,\sigma\ (h * x)\ D\ B.$$

(6) (The form of the X_i on functions of the first layer.) For every $\psi : \mathbb{R}^m \to \mathbb{R}$ with a second-order jet (E, C) at $\pi_1(x)$, the function $\psi \circ \pi_1$ has a second-order jet (D, B) at x satisfying
$$\operatorname{hgrad}\,\sigma\ x\ D = \big(E(e_i)\big)_{i=1}^m, \qquad \operatorname{hessS}\,\sigma\ x\ D\ B = \big(C(e_i, e_j)\big)_{i,j=1}^m,$$
with $e_1, \dots, e_m$ the standard basis of $\mathbb{R}^m$.

Items (1)–(3) are the hypotheses on the vector fields in Section Section 1.2: in exponential coordinates $X_i = \partial_{x_{i,1}} + \sum_{j\ge 2}\sum_k b^i_{k,j}(x)\partial_{x_{k,j}}$ with polynomial $b^i_{k,j}$, so the σ^i are smooth and, their first-layer components forming the identity matrix, pointwise linearly independent — in particular $\sigma(x) \neq 0$ — and $X_1, \dots, X_m$, being a basis of the first layer of a stratification, satisfy (1.6). Item (4) is (2.18), which the paper derives from the Euclidean expansion of u at x together with $x * h - x = \sum_i \pi_1(h)_i X_i(x) + O(\|h\|^2)$.

Items (5) and (6) are the two structural facts stated at the level of jets, which is where the development uses them. Item (5) is left-invariance of the X_i: since $X_i(\varphi \circ L_h) = (X_i\varphi) \circ L_h$ and $X_jX_i(\varphi \circ L_h) = (X_jX_i\varphi) \circ L_h$, and since L_h is a polynomial diffeomorphism, a second-order jet (D, B) of φ at $h * x$ transports to the jet $D' = D \circ DL_h(x)$, $B'(u, v) = B(DL_h(x)u, DL_h(x)v) + D(D^2L_h(x)(u, v))$ of $\varphi \circ L_h$ at x, whose horizontal gradient and symmetrized horizontal second derivative at x are those of φ at $h*x$. This is what makes Lemma Lemma 2.9 — left translates of sub-solutions are sub-solutions — a bookkeeping statement. Item (6) is the displayed computation in the proof of Theorem Theorem 1.3(iii): because $X_i = \partial_{x_{i,1}} + \sum_{j\ge 2}\sum_k b^i_{k,j}\partial_{x_{k,j}}$ kills the dependence on the higher layers, $X_i(\psi \circ \pi_1) = (\partial_i\psi) \circ \pi_1$, and iterating, $X_jX_i(\psi \circ \pi_1) = (\partial_j\partial_i\psi) \circ \pi_1$; hence $\mathcal{X}(\psi \circ \pi_1) = \nabla\psi \circ \pi_1$ and $\mathcal{X}\mathcal{X}^s(\psi \circ \pi_1) = D^2\psi \circ \pi_1$, which is the displayed identity $\mathcal{L}(\psi \circ \pi_1) = (\mathcal{L}^{\mathrm{eucl}}\psi) \circ \pi_1$ once the operator is applied. Both items are stated as the existence of a transported jet with the prescribed horizontal data, which is all the proofs use; the transported jet is in fact unique, as second-order jets always are.

The formalization note (item 3 of the note after Theorem Theorem 1.3) permits exactly these facts of Section Section 2.4 to be built into the definition of the setting, provided every Carnot group in those coordinates satisfies the definition, and each condition above is one of them. Deliberately absent is any statement about sub- or super-solutions of $\mathcal{L}$: the strong maximum principle, and a fortiori any comparison or strong comparison principle, is a cited result carried as a separate hypothesis, not a property of the setting. For the Euclidean additive group on $\mathbb{R}^m$ ($n = m$, $r = 1$, $x * y = x + y$, $\|x\| = |x|$, $\pi_1 = \mathrm{id}$, $\exp_{\mathrm{h}} = \mathrm{id}$) the frame is the constant one $\sigma^i(x) = e_i$: then (1) and (2) are clear, the Lie hull consists of the e_i and the zero field so (3) holds because $e_1, \dots, e_m$ span $\mathbb{R}^m$, $\mathcal{M}(x, D) = 0$ so $\mathcal{X}u = \nabla u$ and $\mathcal{X}\mathcal{X}^s u = D^2u$, (4) is the definition of the Fréchet derivative, and (5), (6) hold with $(D', B') = (D, B)$ and $(D, B) = (E, C)$ since $DL_h = I$, $D^2L_h = 0$ and $\pi_1 = \mathrm{id}$.

1.11. `TranslatedDomain`.

Definition 1.12. Let $\mathbb{G}$ be a sub-elliptic setting (`CarnotSetting`) with group law $*$ and homogeneous norm $\|\cdot\|$, and write $d(x, y) := \|y^{-1} * x\|$ for the left-invariant distance of Section 2.4. For $\Omega \subseteq \mathbb{R}^n$ and $\delta \in \mathbb{R}$ the *inner domain* Ω_δ of (3.8) is
$$\Omega_\delta := \big\{x \in \Omega :\ \delta < d(x, y) \text{ for every } y \in \partial\Omega\big\}.$$

The paper writes this as $\{x \in \Omega : \ \mathrm{dist}(x, \partial\Omega) > \delta\}$ with $\mathrm{dist}(x, \partial\Omega) = \inf\{d(x, y) : y \in \partial\Omega\}$, and the two agree: whenever Ω is bounded, $\partial\Omega$ is compact, so for $\partial\Omega \neq \emptyset$ the infimum is attained and "$\mathrm{dist}(x, \partial\Omega) > \delta$" is equivalent to "$\delta < d(x, y)$ for every $y \in \partial\Omega$"; and for $\partial\Omega = \emptyset$ the displayed condition is vacuous, giving $\Omega_\delta = \Omega$, which matches the paper's convention $\inf \emptyset = +\infty$. The quantified form is used here because it is the form in which the condition is consumed: every use below reads it as "every boundary point is d-far from x".

1.12. `BoundaryCore`.

Definition 1.13. Let $\mathbb{G}$ be a sub-elliptic setting (`CarnotSetting`) with group law $*$ and homogeneous norm $\|\cdot\|$, and write $d(x, y) := \|y^{-1} * x\|$ for the left-invariant distance of Section 2.4. For $\Omega \subseteq \mathbb{R}^n$ and $\rho \in \mathbb{R}$ the *compact core* K_ρ of (3.8) is

$$K_\rho := \big\{x \in \Omega : \ \rho \leq d(x, y) \text{ for every } y \in \partial\Omega\big\}.$$

The paper writes this as $\{x \in \Omega : \ \mathrm{dist}(x, \partial\Omega) \geq \rho\}$ with $\mathrm{dist}(x, \partial\Omega) = \inf\{d(x, y) : y \in \partial\Omega\}$; since an infimum is $\geq \rho$ exactly when every member of the set is $\geq \rho$, the two descriptions agree verbatim when $\partial\Omega \neq \emptyset$, and for $\partial\Omega = \emptyset$ the displayed condition is vacuous, giving $K_\rho = \Omega$, which matches the paper's convention $\inf \emptyset = +\infty$.

1.13. `MaxTranslate`.

Definition 1.14. Let $\mathbb{G}$ be a sub-elliptic setting (`CarnotSetting`) with group law $*$, let $\Omega \subseteq \mathbb{R}^n$, let $\delta \in \mathbb{R}$, let $u, v : \mathbb{R}^n \to \mathbb{R}$ and let $h \in \mathbb{R}^n$. Writing $u_h := u(\cdot * h)$ for the right translate of u by h and Ω_δ for the inner domain (`TranslatedDomain`), the first quantity of (3.9) is

$$M_\delta(h) := \sup_{x \in \Omega_\delta} \big\{u_h(x) - v(x)\big\} = \sup_{x \in \Omega_\delta} \big\{u(x * h) - v(x)\big\}.$$

The supremum is the least upper bound in $\mathbb{R}$; it is a genuine supremum of a non-empty set of reals bounded above exactly when $\Omega_\delta \neq \emptyset$ and $u_h - v$ is bounded above on Ω_δ, which is the situation in every use below.

1.14. `Lop`.

Definition 1.15 (The operator $\mathcal{L}$ on 2-jets, (1.8) through (2.10))**.** Let $n, m \in \mathbb{N}$, let $\sigma : \mathbb{R}^n \to \mathbb{R}^{n \times m}$, $\sigma(x) = [\sigma^1(x), \dots, \sigma^m(x)]$, be a frame of m vector fields on $\mathbb{R}^n$, and let $A : \mathbb{R}^m \to \mathbb{R}^{m \times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$. For $x \in \mathbb{R}^n$, a 1-jet D and a 2-jet B on $\mathbb{R}^n$ we set

$$\mathrm{Lop}\ \sigma\ A\ H\ x\ D\ B \ := \ -\mathrm{Tr}\Big(A\big(\mathrm{hgrad}\ \sigma\ x\ D\big) \cdot \big(\mathrm{hhessS}\ \sigma\ x\ D\ B\big)\Big) \ + \ H\big(\mathrm{hgrad}\ \sigma\ x\ D\big),$$

with hgrad as in `hgrad` and hhessS as in `hhessS`.

If u is twice differentiable at x with second-order jet $(D, B) = (\nabla u(x), D^2 u(x))$ then $\mathrm{hgrad}\ \sigma\ x\ D = \mathcal{X}u(x)$ and $\mathrm{hhessS}\ \sigma\ x\ D\ B = \mathcal{X}\mathcal{X}^s u(x)$, so this is

$$\mathcal{L}u(x) = -\mathrm{Tr}\big(A(\mathcal{X}u(x))\, \mathcal{X}\mathcal{X}^s u(x)\big) + H(\mathcal{X}u(x)),$$

the operator (1.8). Putting it on jets is the paper's own reduction (2.10) of $G(x, u, \mathcal{X}u, \mathcal{X}\mathcal{X}^s u) = 0$ to $F(x, u, \nabla u, D^2 u) = 0$ via $F(x, r, \zeta, X) = G(x, r, \sigma(x)^T \zeta, \sigma(x)^T X \sigma(x) + \mathcal{M}(x, \zeta))$. The operator has no zeroth-order term, so the value $u(x)$ does not appear; and A and H are only ever evaluated at hgrad $\sigma\ x\ D$, so only their values away from the origin matter at the jets admitted by Definition Definition 2.1.

1.15. `IsViscSubsol`.

Definition 1.16 (Viscosity sub-solution, Definition 2.1)**.** Let $\mathbb{G}$ be a Carnot setting (`CarnotSetting`) with horizontal frame σ, let $A : \mathbb{R}^m \to \mathbb{R}^{m \times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$, let $\Omega \subset \mathbb{R}^n$ and let $u : \mathbb{R}^n \to \mathbb{R}$. We say that u is a *viscosity sub-solution of* $\mathcal{L}u = 0$ *in* Ω, written IsViscSubsol $\mathbb{G}\ A\ H\ \Omega\ u$, if

(1) u is upper semi-continuous on Ω; and
(2) for every $x \in \Omega$ and every $\varphi \in C^2(\Omega)$ with $x \in \mathrm{argmax}_\Omega(u - \varphi)$, every second-order jet (D, B) of φ at x (`TwiceDiffAt`) with hgrad $\sigma\ x\ D \neq 0$ satisfies

$$\mathrm{Lop}\ \sigma\ A\ H\ x\ D\ B \ \leq \ 0,$$

with hgrad as in `hgrad` and Lop as in `Lop`.

Here $x \in \operatorname{argmax}_\Omega(u-\varphi)$ means, as in (2.8), that $x \in \Omega$ and $(u-\varphi)(y) \le (u-\varphi)(x)$ for every $y \in \Omega$.

This is Definition Definition 2.1 for the operator (1.8). The admissible test functions are those of the class (2.11), $\mathcal{A}_x^+(u,\Omega) = \{\varphi \in C^2(\Omega) : x \in \operatorname{argmax}_\Omega(u-\varphi),\ \mathcal{X}\varphi(x) \neq 0\}$, and the required inequality is $\mathcal{L}\varphi(x) \le 0$. The restriction $\mathcal{X}\varphi(x) \neq 0$ — not merely $\nabla\varphi(x) \neq 0$, which is weaker when $m < n$ — is what allows A and H to be undefined at the origin, and it makes the class of sub-solutions larger than the one used by Bardi and Goffi.

Two points of translation. First, $\mathcal{L}\varphi(x)$ is written through the jet of φ at x: for Ω open, $\varphi \in C^2(\Omega)$ and $x \in \Omega$ a second-order jet of φ at x exists and is unique, namely $(D,B) = (\nabla\varphi(x), D^2\varphi(x))$, so quantifying over all jets of φ at x names that one pair, and then hgrad $\sigma\ x\ D = \mathcal{X}\varphi(x)$ and hhessS $\sigma\ x\ D\ B = \mathcal{X}\mathcal{X}^s\varphi(x)$, so the displayed inequality is $\mathcal{L}\varphi(x) \le 0$. Second, the operator (1.8) has no zeroth-order term, so the value $\varphi(x)$ does not enter and no monotonicity in r has to be recorded. The paper also notes that, since φ may be modified away from x, requiring x to be a global maximum point of $u-\varphi$ on Ω is equivalent to requiring it to be a local one; the global form is used here, following (2.11). The notion is only ever used with Ω an open set, as in Definition Definition 2.1, which is what makes the quantification over second-order jets of φ at x non-vacuous. Writing $\mathcal{L}u \le 0$ in Ω in the viscosity sense means that u is a viscosity sub-solution in this sense.

1.16. `IsViscSupersol`.

Definition 1.17 (Viscosity super-solution, Definition 2.1)**.** Let $\mathbb{G}$ be a Carnot setting (`CarnotSetting`) with horizontal frame σ, let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$, let $\Omega \subset \mathbb{R}^n$ and let $v : \mathbb{R}^n \to \mathbb{R}$. We say that v is a *viscosity super-solution of* $\mathcal{L}v = 0$ *in* Ω, written IsViscSupersol $\mathbb{G}\ A\ H\ \Omega\ v$, if

(1) v is lower semi-continuous on Ω; and
(2) for every $x \in \Omega$ and every $\varphi \in C^2(\Omega)$ with $x \in \operatorname{argmin}_\Omega(v-\varphi)$, every second-order jet (D,B) of φ at x (`TwiceDiffAt`) with hgrad $\sigma\ x\ D \neq 0$ satisfies

$$\text{Lop}\ \sigma\ A\ H\ x\ D\ B\ \ge\ 0,$$

with hgrad as in `hgrad` and Lop as in `Lop`.

Here $x \in \operatorname{argmin}_\Omega(v-\varphi)$ means, as in (2.8), that $x \in \Omega$ and $(v-\varphi)(x) \le (v-\varphi)(y)$ for every $y \in \Omega$.

This is the super-solution half of Definition Definition 2.1 for the operator (1.8): the admissible test functions are those of the class $\mathcal{A}_x^-(v,\Omega)$, defined as in (2.11) with argmin in place of argmax, and the required inequality is $\mathcal{L}\varphi(x) \ge 0$. As for sub-solutions, $\mathcal{L}\varphi(x)$ is written through the second-order jet of φ at x, which for Ω open, $\varphi \in C^2(\Omega)$ and $x \in \Omega$ exists, is unique, and equals $(\nabla\varphi(x), D^2\varphi(x))$; the condition $\mathcal{X}\varphi(x) \neq 0$ keeps A and H away from the origin; and the operator carries no zeroth-order term. Writing $\mathcal{L}v \ge 0$ in Ω in the viscosity sense means that v is a viscosity super-solution in this sense. The notion is only ever used with Ω an open set, as in Definition Definition 2.1, which is what makes the quantification over second-order jets of φ at x non-vacuous. A viscosity solution is a function that is both a sub-solution and a super-solution.

1.17. `LocallySemiConvexOn`.

Definition 1.18 (Local semi-convexity and semi-concavity)**.** Let $N \in \mathbb{N}$, let $\Omega \subset \mathbb{R}^N$ and let $w : \mathbb{R}^N \to \mathbb{R}$. We say that w is *locally semi-convex on* Ω, written LocallySemiConvexOn $\Omega\ w$, if for every $x \in \Omega$ there are $\rho > 0$ and $\Lambda \ge 0$ such that the Euclidean ball $B_\rho(x)$ is contained in Ω and

$$y\ \longmapsto\ w(y) + \tfrac{\Lambda}{2}\,|y|^2$$

is convex on $B_\rho(x)$. A function w is *locally semi-concave on* Ω when $-w$ is locally semi-convex on Ω; no separate name is introduced for that notion, and the phrase is always used in that sense below.

This is Definition Definition 2.5 of the paper with the semi-convexity constant allowed to depend on the point. The localization is forced by the paper's own use of the notion and is not a weakening that loses content where the paper uses a global constant. Property (4) of Section Section 2.3 concludes only that $\psi \circ w$ and $w \circ \Theta$ are semi-convex *on compact subsets*, with

constants depending on the local Lipschitz constant of w, so no single Λ serves on all of Ω for the composed functions; and those composed functions are exactly the ones fed to Proposition Proposition 3.3 in Theorem Theorem 1.3(iii) ($u \circ R_h$) and inside Proposition 3.3 itself ($\psi_\lambda \circ u$). Conversely Proposition 3.3 never uses more than a constant valid on a ball around one point (Step 4, near x_1), so the local notion carries the whole argument. A function that is semi-convex on Ω in the sense of Definition 2.5, with a global Λ, is locally semi-convex on Ω when Ω is open.

1.18. `EuclidCitedResults`.

Definition 1.19 (The two Euclidean cited results, carried as a hypothesis package)**.** A *Euclidean cited-results package* is a proof of both statements below. Throughout, $|E|$ denotes Lebesgue measure on $\mathbb{R}^N$.

(1) (ALEKSANDROV; Theorem Theorem 2.6.) For every $N \in \mathbb{N}$, every open $\Omega \subseteq \mathbb{R}^N$ and every $w : \mathbb{R}^N \to \mathbb{R}$ that is locally semi-convex on Ω (`LocallySemiConvexOn`): for almost every $x \in \mathbb{R}^N$ with respect to Lebesgue measure, if $x \in \Omega$ then w has a second-order jet at x (`TwiceDiffAt`), i.e. there are a 1-jet D and a 2-jet B with TwiceDiffAt $w\ x\ D\ B$.

(2) (JENSEN; Jensen's Lemma Lemma 2.7, [Cr-Ish-Lions, Lemma A.3].) For every $N \in \mathbb{N}$, every $w : \mathbb{R}^N \to \mathbb{R}$ and every $\hat{x} \in \mathbb{R}^N$ such that
- w is locally semi-convex on some ball $B_\nu(\hat{x})$, $\nu > 0$, and
- $\hat{x}$ is a strict local maximum point of w: there is $\nu > 0$ with $w(y) < w(\hat{x})$ for every $y \in B_\nu(\hat{x})$ with $y \neq \hat{x}$,

there are $\rho_0 > 0$ and $\delta_0 > 0$ such that for all $\rho \in (0, \rho_0)$ and all $\delta \in (0, \delta_0)$ the set of (2.12),

$$J_{\rho,\delta}(\hat{x}) = \bigcup_{|p|<\delta} \Big\{ x \in B_\rho(\hat{x}) :\ w_p(y) \le w_p(x) \text{ for all } y \in B_\rho(\hat{x}) \Big\}, \qquad w_p(y) := w(y) + \langle p, y \rangle,$$

has positive Lebesgue measure. (Here $\{x \in B_\rho(\hat{x}) : w_p(y) \le w_p(x)\ \forall y \in B_\rho(\hat{x})\}$ is $B_\rho(\hat{x}) \cap \operatorname{argmax}_{B_\rho(\hat{x})}(w_p)$ in the notation (2.8).)

These are items (2) and (3) of the list in item 2 of the paper's note for the formalization, which names five results as cited from the literature and *not* to be proved. They are separated from the other three — the Bardi–Goffi principles, Pansu's theorem together with [DMV13, Prop. 4.8], and parts (1)–(3) of Lemma 3.8, which make up the cited-results package at a sub-elliptic setting — on the following principle, which is also what fixes the strength of this package: *a cited result is indexed by the sub-elliptic setting exactly when its content mentions one.* Aleksandrov's theorem and Jensen's lemma mention no group law, no homogeneous norm and no horizontal frame; they are theorems about an arbitrary $\mathbb{R}^N$ with its Euclidean structure, quantified here over all N at once, in the form the paper uses. The other three results are theorems about Carnot groups whose proofs use the dilations and the stratification, so they may be assumed only at a setting that is known to be a Carnot group, and the package carrying them carries that setting as a parameter.

Two consequences of the split are worth recording, because they are the reason for it. First, a node whose own statement is Euclidean — for instance the one combining item 2 here with two applications of item 1 at a strict maximum — can take this package as its hypothesis without having to name a sub-elliptic setting it never uses. Second, a node holding a Carnot-side package at a setting $\mathbb{G}$ still has both results here at every N, because such a package is by definition a package of the present kind together with the three setting-dependent results.

Item 1 is kept at the strength of Theorem 2.6: a second-order jet exists at almost every point of Ω, and nothing is asserted about the jet off that full-measure set. Item 2 is kept at exactly the strength of Lemma 2.7: it asserts only that $J_{\rho,\delta}(\hat{x})$ has positive measure. It does *not* assert the existence of a point at which a given pair of functions is twice differentiable; that step combines item 2 with two applications of item 1 — one exceptional null set for each of the two functions — and is to be proved separately.

Finally, both items are satisfiable, which is the property the package exists to preserve: Theorem 2.6 and Lemma 2.7 are theorems of classical real analysis on $\mathbb{R}^N$, proved in the references

the paper cites, so a proof of the conjunction of items 1 and 2 exists. Assuming it therefore makes no result vacuous.

1.19. `StructureConditionsES`.

Definition 1.20 (The structure conditions (E) and (S))**.** Let $m \in \mathbb{N}$, let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$. We say that (A, H) *satisfies the structure conditions*, written StructureConditionsES $m\ A\ H$, if

(1) for every $\xi \neq 0$ the matrix $A(\xi)$ is symmetric, $A(\xi)^T = A(\xi)$;
(2) for every $\xi \neq 0$ the matrix $A(\xi)$ is positive semidefinite: $\langle A(\xi)\eta, \eta\rangle = \sum_i \sum_j A_{i,j}(\xi)\eta_j\eta_i \geq 0$ for all $\eta \in \mathbb{R}^m$;
(3) A is continuous on $\mathbb{R}^m \setminus \{0\}$;
(4) H is continuous on $\mathbb{R}^m \setminus \{0\}$;
(5) (E), (1.9): $\mathcal{E}(\xi) := \langle A(\xi)\xi, \xi\rangle > 0$ for every $\xi \in \mathbb{R}^m \setminus \{0\}$;
(6) (S), (1.10): there is a function $\kappa : [1, \infty) \times (\mathbb{R}^m \setminus \{0\}) \to (0, \infty)$ such that

$$A(t\xi) = \kappa(t, \xi)\, A(\xi) \quad \text{and} \quad H(t\xi) = t\,\kappa(t, \xi)\, H(\xi) \qquad \text{for all } t \geq 1 \text{ and } \xi \in \mathbb{R}^m \setminus \{0\}.$$

These are exactly the hypotheses imposed on A and H in Section Section 1.2 for the equation (1.8), $\mathcal{L}u = -\mathrm{Tr}(A(\mathcal{X}u)\mathcal{X}\mathcal{X}^s u) + H(\mathcal{X}u) = 0$: items (1)–(4) are the standing requirements $A_{i,j}, H \in C(\mathbb{R}^m \setminus \{0\})$ with $A(\xi)$ symmetric and positive semidefinite; (5) is the strict ellipticity in the gradient direction (E); and (6) is the homogeneity along rays (S), which is the form taken, for the operators (1.8), by the scaling hypothesis of the strong maximum principle of Bardi and Goffi. Nothing is required at $\xi = 0$, and no local Lipschitz continuity is required: the viscosity sub- and super-solutions of (1.8) are defined with test functions whose horizontal gradient does not vanish at the point of contact (Definition Definition 2.1), so A and H are never evaluated at the origin, and the exactness of the relabelling identity under (S) is what replaces the Lipschitz hypothesis (BB1) of Barles–Busca. In (6), H is required to scale exactly, with the factor $t\,\kappa(t, \xi)$ and not merely an inequality, because both the sub-solution and the super-solution side of the scaling hypothesis are used (Remark Remark 1.1(b)). The positivity of κ is part of (S); note that (E) determines it, $\kappa(t, \xi) = \mathcal{E}(t\xi)/(t^2\mathcal{E}(\xi))$, so no generality is gained or lost by quantifying over κ existentially.

1.20. `SupConv`.

Definition 1.21 (Sup-convolution, (3.13) and (3.14))**.** Let $\mathbb{G}$ be a Carnot setting (`CarnotSetting`) with group law $*$ and homogeneous norm $\|\cdot\|$, put $q := 2\,r!$, let $\Omega \subset \mathbb{R}^n$, let $w : \mathbb{R}^n \to \mathbb{R}$, let $\varepsilon > 0$ and let $x \in \mathbb{R}^n$. The *sup-convolution* of w is

$$w^\varepsilon(x) \;:=\; \sup\Big\{\, w(z * x) - \frac{\|z\|^q}{2\varepsilon} \;:\; z \in \mathbb{R}^n,\ z * x \in \overline{\Omega} \,\Big\}.$$

This is (3.14), which is (3.13), $w^\varepsilon(x) = \sup_{y\in\overline{\Omega}}\{w(y) - \|y * x^{-1}\|^q/2\varepsilon\}$, after the substitution $y = z * x$: the sup-convolution is a supremum of *left* translates of w, which is what makes the kernel compatible with the equation, since the horizontal vector fields are left-invariant. The exponent $q = 2\,r!$ is chosen so that $\|\cdot\|^q$ is a polynomial, hence smooth; this is the kernel used by Wang for Carnot groups.

The supremum is taken over the set of admissible z, which is non-empty exactly when $x \in \overline{\Omega}$, since $z = 0$ gives $0 * x = x$ and $\|0\| = 0$ — in which case $w^\varepsilon(x) \geq w(x)$. It is finite whenever Ω is bounded and w is continuous on $\overline{\Omega}$, because then w is bounded on the compact set $\overline{\Omega}$ and the penalty $-\|z\|^q/2\varepsilon$ is non-positive, so the family is bounded above by $\sup_{\overline{\Omega}} w$. In the formal statement the supremum is the supremum of the indexed family of real numbers, which by convention returns 0 when the family is empty or unbounded above; the nodes that use w^ε therefore carry the hypotheses Ω bounded, $w \in C(\overline{\Omega})$ and $x \in \overline{\Omega}$ under which neither degenerate case occurs, and under which the above is the supremum in the ordinary sense.

1.21. `InfConv`.

Definition 1.22 (Inf-convolution, (3.13))**.** Let $\mathbb{G}$ be a Carnot setting (`CarnotSetting`) with group law $*$ and homogeneous norm $\|\cdot\|$, put $q := 2\,r!$, let $\Omega \subset \mathbb{R}^n$, let $w : \mathbb{R}^n \to \mathbb{R}$, let $\varepsilon > 0$ and let $x \in \mathbb{R}^n$. The *inf-convolution* of w is

$$w_\varepsilon(x) \;:=\; \inf\Big\{\, w(z * x) + \frac{\|z\|^q}{2\varepsilon} \;:\; z \in \mathbb{R}^n,\; z * x \in \overline{\Omega} \,\Big\}.$$

This is the inf-convolution of (3.13), $w_\varepsilon(x) = \inf_{y \in \overline{\Omega}}\{w(y) + \|y * x^{-1}\|^q/2\varepsilon\}$, after the substitution $y = z * x$, so that w_ε is an infimum of *left* translates of w; it is the form in which the paper uses it in the proof of Lemma Lemma 3.8(4). One has $w_\varepsilon = -\big((-w)^\varepsilon\big)$, where $(-w)^\varepsilon$ is the sup-convolution of $-w$, which is why the inf-convolution of a super-solution is semi-concave where the sup-convolution of a sub-solution is semi-convex.

The infimum is taken over the set of admissible z, which is non-empty exactly when $x \in \overline{\Omega}$, since $z = 0$ gives $0 * x = x$ and $\|0\| = 0$ — in which case $w_\varepsilon(x) \le w(x)$. It is finite whenever Ω is bounded and w is continuous on $\overline{\Omega}$, because the family is then bounded below by $\inf_{\overline{\Omega}} w$. In the formal statement the infimum is the infimum of the indexed family of real numbers, which by convention returns 0 when the family is empty or unbounded below; the nodes that use w_ε carry the hypotheses Ω bounded, $w \in C(\overline{\Omega})$ and $x \in \overline{\Omega}$ under which neither degenerate case occurs.

1.22. `ReducedDomain`.

Definition 1.23 (The reduced domains (3.15))**.** Let $\mathbb{G}$ be a Carnot setting (`CarnotSetting`) with group law $*$ and homogeneous norm $\|\cdot\|$, put $q := 2\,r!$, let $\Omega \subset \mathbb{R}^n$ and let $R, \varepsilon \in \mathbb{R}$. The *reduced domain* is

$$\Omega[\varepsilon] \;:=\; \Big\{\, x \in \Omega \;:\; \inf_{y \in \mathbb{R}^n \setminus \Omega} \|y * x^{-1}\|^q \;>\; (1 + 4R)\,\varepsilon \,\Big\},$$

where $x^{-1} = -x$, so that $y * x^{-1} = y * (-x)$.

This is (3.15). In the paper the parameter is fixed to $R = 2\max\{\|u\|_{L^\infty}, \|v\|_{L^\infty}\}$ for the two functions u, v under consideration; here it is carried as a parameter, so that the nodes which use $\Omega[\varepsilon]$ state their own bound on R and the definition does not have to mention u and v. The quantity being compared with $(1+4R)\varepsilon$ is the q-th power of the right-invariant distance from x to the complement of Ω, and the condition says that x is deep inside Ω in that sense; the point of the threshold $(1 + 4R)\varepsilon$ is that the displacement z_x realizing the sup- or inf-convolution at a point of $\Omega[\varepsilon]$ satisfies $\|z_x\|^q \le 4\varepsilon\|u\|_{L^\infty} \le 2R\varepsilon < (1 + 4R)\varepsilon$, so that $z_x * x'$ still lies in Ω for the relevant x'.

The infimum is over $\{y : y \notin \Omega\}$, which is non-empty whenever Ω is bounded — as it is wherever the reduced domains are used — and the family is bounded below by 0, so the infimum is the infimum in the ordinary sense. (In the formal statement the infimum of an empty family returns 0 by convention, in which case the defining condition would be $(1+4R)\varepsilon < 0$.) For $\varepsilon > 0$ and $R \ge 0$ the sets increase as ε decreases, and $\Omega[\varepsilon] \uparrow \Omega$ as $\varepsilon \to 0^+$.

1.23. `CitedResults`.

Definition 1.24 (The cited results at a setting, carried as a hypothesis package)**.** Let $\mathbb{G}$ be a Carnot setting (`CarnotSetting`) with dimension n, number of generators m, step r, group law $*$, first-layer projection π_1, homogeneous norm $\|\cdot\|$ and horizontal frame σ; let $d(x, y) := \|y^{-1} * x\| = \|(-y) * x\|$ be the associated left-invariant distance; let $|E|$ denote Lebesgue measure; and let $\mathcal{L}$ be the operator (1.8) determined by (A, H). A *cited-results package at* $\mathbb{G}$ is a Euclidean cited-results package (`EuclidCitedResults`) together with a proof of all seven statements below. The setting $\mathbb{G}$ is fixed throughout: each of the seven is a statement about $\mathbb{G}$ itself, and none of them is asserted for any other setting.

(1) (strongMax; Bardi–Goffi, Corollary Corollary 2.4, sub-solution side.) For every $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$ satisfying the structure conditions (`StructureConditionsES`), every *connected* open $U \subseteq \mathbb{R}^n$ and every $u : \mathbb{R}^n \to \mathbb{R}$ that is a viscosity sub-solution of $\mathcal{L}u = 0$ in U (`IsViscSubsol`): if $x_0 \in U$ satisfies $u(y) \le u(x_0)$ for all $y \in U$ and $u(x_0) \ge 0$, then $u(y) = u(x_0)$ for all $y \in U$.

(2) (strongMin; Corollary Corollary 2.4, super-solution side.) For every $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$ satisfying the structure conditions (`StructureConditionsES`), every *connected* open $U \subseteq \mathbb{R}^n$ and every $v : \mathbb{R}^n \to \mathbb{R}$ that is a viscosity super-solution of $\mathcal{L}v = 0$ in U (`IsViscSupersol`): if $x_0 \in U$ satisfies $v(x_0) \le v(y)$ for all $y \in U$ and $v(x_0) \le 0$, then $v(y) = v(x_0)$ for all $y \in U$.

(3) (pansuDiff; Pansu's differentiability theorem [P89], as stated in Section Section 2.4.) For every open $\Omega \subseteq \mathbb{R}^n$, every $f : \mathbb{R}^n \to \mathbb{R}$ and every $L \in \mathbb{R}$ such that $|f(x) - f(y)| \le L\, d(x, y)$ for all $x, y \in \Omega$: for almost every $x \in \mathbb{R}^n$ with respect to Lebesgue measure, if $x \in \Omega$ then there is $\xi \in \mathbb{R}^m$ such that f is Pansu differentiable at x with horizontal differential ξ (`HasPansuGradAt`).

(4) (pansuConst; [DMV13, Proposition 4.8].) For every *connected* open $\Omega \subseteq \mathbb{R}^n$, every $f : \mathbb{R}^n \to \mathbb{R}$ and every $L \in \mathbb{R}$ such that $|f(x) - f(y)| \le L\, d(x, y)$ for all $x, y \in \Omega$: if for almost every $x \in \mathbb{R}^n$ it holds that $x \in \Omega$ implies that f is Pansu differentiable at x with horizontal differential 0, then $f(x) = f(y)$ for all $x, y \in \Omega$.

(5) (wang1; part (1) of Lemma Lemma 3.8, [CYW07, Proposition 3.3].) For every bounded open $\Omega \subseteq \mathbb{R}^n$, every $w : \mathbb{R}^n \to \mathbb{R}$ continuous on $\overline{\Omega}$ and every $\varepsilon > 0$: the sup-convolution w^ε (`SupConv`) is locally semi-convex on Ω, and the inf-convolution w_ε (`InfConv`) is locally semi-concave on Ω, i.e. $-w_\varepsilon$ is locally semi-convex on Ω.

(6) (wang2; part (2) of Lemma Lemma 3.8.) For every bounded open $\Omega \subseteq \mathbb{R}^n$ and every $w : \mathbb{R}^n \to \mathbb{R}$ continuous on $\overline{\Omega}$, the convergences $w^\varepsilon \downarrow w$ and $w_\varepsilon \uparrow w$ hold uniformly on $\overline{\Omega}$ as $\varepsilon \to 0^+$, in the following three explicit senses:
   (a) for every $\varepsilon > 0$ and every $x \in \overline{\Omega}$, $w(x) \le w^\varepsilon(x)$ and $w_\varepsilon(x) \le w(x)$;
   (b) for all $\varepsilon, \varepsilon'$ with $0 < \varepsilon' \le \varepsilon$ and every $x \in \overline{\Omega}$, $w^{\varepsilon'}(x) \le w^\varepsilon(x)$ and $w_\varepsilon(x) \le w_{\varepsilon'}(x)$;
   (c) for every $\eta > 0$ there is $\varepsilon_0 > 0$ such that for every $\varepsilon \in (0, \varepsilon_0)$ and every $x \in \overline{\Omega}$, $|w^\varepsilon(x) - w(x)| \le \eta$ and $|w_\varepsilon(x) - w(x)| \le \eta$.

(7) (wang3; part (3) of Lemma Lemma 3.8.) For every (A, H) satisfying the structure conditions (`StructureConditionsES`), every bounded open $\Omega \subseteq \mathbb{R}^n$, all $u, v : \mathbb{R}^n \to \mathbb{R}$ continuous on $\overline{\Omega}$, every $R \in \mathbb{R}$ with $|u| \le R/2$ and $|v| \le R/2$ on $\overline{\Omega}$, and every $\varepsilon > 0$: if u is a viscosity sub-solution and v a viscosity super-solution of $\mathcal{L}\cdot = 0$ in Ω, then u^ε is a viscosity sub-solution and v_ε a viscosity super-solution of $\mathcal{L}\cdot = 0$ in the reduced domain $\Omega[\varepsilon]$ (`ReducedDomain`).

Item 2 of the paper's note for the formalization lists five results as cited from the literature and *not* to be proved, and prescribes that each be stated once, in the form the paper uses, and carried as an explicit hypothesis, so that the formalized theorems read "cited results ⇒ Theorem Theorem 1.2 / Theorem Theorem 1.3". The five are: (a) the Bardi–Goffi principles, here items 1 and 2; (b) Aleksandrov's theorem and (c) Jensen's lemma, which are the two items of `EuclidCitedResults` and are inherited by the present package; (d) Pansu's theorem together with [DMV13, Prop. 4.8], here items 3 and 4; and (e) parts (1)–(3) of Lemma 3.8, here items 5–7. No axiom is introduced: the package is a hypothesis, and every result that uses one of the nine statements takes a package as an argument, at the setting where the statement is used.

Why the setting is a parameter. This is the point at which it would be easiest to assume too much, and the error is not conservative. Items 1–7 are theorems about genuine Carnot groups, and their proofs use the dilations and the stratification of the Lie algebra. Neither is a field of `CarnotSetting`: that structure records the exponential coordinates, the polynomial group law, the frame, the properties of the homogeneous quasi-norm, Hörmander's rank condition (1.6), (2.18), left-invariance of the horizontal derivatives and the form of the X_i on the first layer — and nothing about dilations or a graded decomposition. So for an *arbitrary* instance of `CarnotSetting` items 1–7 are not known, and may be false. Asserting them for every instance at once would therefore risk a package that no proof can ever exhibit, and a hypothesis package that is unsatisfiable makes every theorem depending on it vacuously true while every verification lane still passes. Carrying $\mathbb{G}$ as a parameter removes that risk at the point where it arises: a consumer assumes the package only at the setting it is actually working in, and a consumer

that works in two settings — as case (iii) of Theorem 1.3 does, descending to $\mathbb{R}^m$ — assumes it separately at each.

Item 3 of the paper's note for the formalization is one-directional in exactly the same way, and it is worth stating why it does not license more. It permits building the structural facts of Section Section 2.4 into the definition of the setting "provided every Carnot group in the coordinates of Section Section 2.4 satisfies that definition". The proviso guarantees that every real Carnot group *is* an instance of `CarnotSetting`; it says nothing about every instance being a real Carnot group. It therefore licenses stating the cited results *for* a given group, which is all items 1–7 do, and does not license quantifying them over all instances.

Four points that fix the strength of the package.

First, items 1 and 2 are Corollary Corollary 2.4, not Theorem Theorem 2.3: the note says the special case used suffices, and Corollary 2.4 is that case. Its domain hypothesis is retained in full — U must be *connected* — because that is where Corollary 2.4 is stated for a connected Ω, and the proofs below apply it on balls and on reduced domains. Nothing comparison-flavoured appears here: Bardi and Goffi also prove a *strong comparison* principle, which the paper explicitly distinguishes from the comparison principle being proved (end of Section Section 1.2), and assuming that would make both targets vacuous. Note also what items 1 and 2 need from the setting even at a fixed $\mathbb{G}$: Theorem Theorem 2.3, from which Corollary 2.4 is derived, requires Hörmander's rank condition (1.6) on $X_1, \dots, X_m$, and (1.6) is a field of `CarnotSetting`. Without that field the package would be unsatisfiable at some instances for a second and independent reason: one can exhibit a structure meeting all the other conditions in which a single horizontal field admits no admissible test function at an interior maximum, so that a non-constant function is vacuously a sub-solution attaining a non-negative interior maximum and items 1–2 fail. The field must therefore not be removed from the setting.

Second, items 3 and 4 are the two halves of (d), stated for $\mathbb{G}$'s own quasi-distance d and horizontal differential; the d-Lipschitz hypothesis is the one the consuming nodes verify, and nothing is asserted about the Pansu differential off the full-measure set.

Third, item 4 retains the connectedness of Ω, as in [DMV13, Proposition 4.8]: a d-Lipschitz function with vanishing horizontal differential is constant on a connected open set, and on a disconnected one it is only locally constant.

Fourth, item 7 is stated for every R with $|u|, |v| \le R/2$ on $\overline{\Omega}$ rather than for the single value $R = 2\max\{\|u\|_{L^\infty}, \|v\|_{L^\infty}\}$ used in (3.15). Since $\Omega[\varepsilon]$ shrinks as R grows, quantifying over all admissible R is weaker than fixing the paper's value, and it is the paper's value that the consuming nodes will supply. Part (4) of Lemma 3.8 is deliberately absent: the note states it is to be proved, and it is a separate proof node.

Satisfiability. Because the note forbids proving the cited results, this is the one thing the node cannot establish internally, and the parameter is what keeps it reachable. For any $\mathbb{G}$ that is a Carnot group in the coordinates of Section Section 2.4, items 1–7 are the cited theorems themselves and a package at $\mathbb{G}$ exists. Two instances where this is used are worth naming. A theorem assuming a package at the $\mathbb{G}$ it is about is satisfiable whenever $\mathbb{G}$ is a real Carnot group, which is the only case the paper is about. And whenever $\mathbb{G}$ is a setting whose group law is addition on $\mathbb{R}^n$ with $n = m$ and whose homogeneous norm is the Euclidean one — the Euclidean setting, which is the second place a package is assumed — a package exists for an elementary reason: there $d(x, y) = |x - y|$, Pansu differentiability is ordinary differentiability and item 3 is Rademacher's theorem, items 1 and 2 are the classical strong maximum and minimum principles, and items 5–7 are the classical sup- and inf-convolution facts.

1.24. `Ellip`.

Definition 1.25 (The ellipticity function $\mathcal{E}$, (1.9))**.** Let $m \in \mathbb{N}$ and let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ be a matrix-valued function. The *ellipticity function of* A is

$$\mathcal{E}_A(\xi) \;:=\; \langle A(\xi)\,\xi,\ \xi\rangle \;=\; \sum_{i=1}^{m}\sum_{j=1}^{m} A_{i,j}(\xi)\,\xi_j\,\xi_i, \qquad \xi \in \mathbb{R}^m,$$

written Ellip $A\ \xi$ when the argument A needs to be displayed.

This is the quantity named $\mathcal{E}$ in (1.9). The paper's condition (E) of strict ellipticity in the gradient direction is exactly the statement that $\mathcal{E}_A(\xi) > 0$ for every $\xi \in \mathbb{R}^m \setminus \{0\}$, which is how it occurs among the structure conditions; and $\mathcal{E}_A$ is the quantity that appears in the lower bound (2.4), in the identification $\kappa(t,\xi) = \mathcal{E}_A(t\xi)/(t^2\mathcal{E}_A(\xi))$ of the scaling factor in Remark 1.1(a), and in the strict sub-solution estimate of Corollary Corollary 3.2, where the ellipticity of $\mathcal{L}$ in the direction of the horizontal gradient is what produces a quantitative gain.

Nothing is assumed about A: the definition makes sense for every matrix-valued A and every ξ, and in particular no symmetry or positivity is built in. Note $\mathcal{E}_A(0) = 0$ for every A, so the restriction to $\xi \neq 0$ in (E) is not a convention but a necessity. Only the value $A(\xi)$ of A at the point ξ enters $\mathcal{E}_A(\xi)$, so $\mathcal{E}_A$ inherits the homogeneity of A along rays: if $A(t\xi) = A(\xi)$ for $t > 0$ then $\mathcal{E}_A(t\xi) = t^2\mathcal{E}_A(\xi)$, because $(t\xi)_k = t\,\xi_k$ contributes one factor t from each of the two slots.

1.25. `NormalizedAH`.

Definition 1.26 (The normalization (2.3))**.** Let $m \in \mathbb{N}$, let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$. We say that (A, H) is *normalized*, written NormalizedAH $m\ A\ H$, if

$$A(t\xi) = A(\xi) \qquad \text{and} \qquad H(t\xi) = t\,H(\xi) \qquad \text{for all } t > 0 \text{ and all } \xi \in \mathbb{R}^m \setminus \{0\},$$

that is, A is 0-homogeneous and H is 1-homogeneous along every ray through the origin.

This is (2.3). It is the form to which the pair (A, H) is reduced in Section Section 2.1: under (E) and (S) one has $A(\xi) = a(\xi)A_0(\xi/|\xi|)$ and $H(\xi) = |\xi|a(\xi)H_0(\xi/|\xi|)$ with $a > 0$ continuous, and replacing (A, H) by $(A_0(\xi/|\xi|),\ |\xi|H_0(\xi/|\xi|))$ changes neither the viscosity sub-solutions nor the viscosity super-solutions of (1.8), while making (S) hold with $\kappa \equiv 1$. Being normalized is strictly stronger than (S) — it is (S) for all $t > 0$ with $\kappa \equiv 1$ — and it is the standing assumption of the paper from Section Section 2 on. Two consequences used repeatedly: $\mathcal{E}(\xi) = \langle A(\xi)\xi, \xi\rangle$ is 2-homogeneous, so under (E) it is bounded below by $e_0|\xi|^2$ with $e_0 = \min_{|\omega|=1} \mathcal{E}(\omega) > 0$ ((2.4)); and the exact relabelling identity (2.5), $-t\,\mathrm{Tr}(A(t\xi)X) + H(t\xi) = t\,[-\mathrm{Tr}(A(\xi)X) + H(\xi)]$, holds for every $t > 0$, every $\xi \neq 0$ and every symmetric X. Note also that A and H are then determined by their values on the unit sphere, hence bounded on bounded sets.

1.26. `MatLe`.

Definition 1.27 (The order on symmetric matrices, (2.1))**.** Let $m \in \mathbb{N}$ and $A, B \in \mathbb{R}^{m\times m}$. We write MatLe $A\ B$, read $A \le B$, if

$$\langle A\xi, \xi\rangle \ \le\ \langle B\xi, \xi\rangle \qquad \text{for all } \xi \in \mathbb{R}^m, \qquad \text{where} \quad \langle A\xi, \xi\rangle = \sum_{i=1}^{m}\sum_{j=1}^{m} A_{i,j}\,\xi_j\,\xi_i.$$

This is the order (2.1) of the paper's notation section. No symmetry is required of A or B in the definition; the relation only sees their symmetric parts, and it is applied in the paper exclusively to symmetric matrices, where it is the usual partial order. Two uses fix the strength needed: the trace monotonicity (2.2) ($\mathrm{Tr}(AY) \le \mathrm{Tr}(AX)$ whenever $A \ge 0$ and $Y \le X$), and the comparison $\mathcal{X}\mathcal{X}^s u(x_0) \le \mathcal{X}\mathcal{X}^s v(x_0)$ of the symmetrized horizontal second derivatives of a semi-convex u and a semi-concave v at a local maximum point x_0 of $u - v$, which opens the proof of Proposition Proposition 3.3.

1.27. `EuclidSetting`.

Definition 1.28 (The Euclidean setting on $\mathbb{R}^N$, named)**.** For each $N \in \mathbb{N}$, let $\mathbb{G}_E(N)$ denote one sub-elliptic setting (`CarnotSetting`) witnessing `EuclideanCarnotSettingExists` at N, chosen once and for all. Writing $n_E, m_E, r_E, *_E, \|\cdot\|_E, \pi_1^E, \exp_E, \sigma_E$ for the data of $\mathbb{G}_E(N)$, the choice therefore satisfies the seven identities that `EuclideanCarnotSettingExists` supplies:

$$n_E = N, \quad m_E = N, \quad r_E = 1, \quad x *_E y = x + y, \quad \|x\|_E = |x|, \quad \exp_E\left(\pi_1^E(x)\right) = x, \quad \sigma_E(x)_i = \exp_E(e_i)$$

for all $x, y \in \mathbb{R}^{n_E}$ and every i, where e_i is the i-th standard basis vector of $\mathbb{R}^N$.

Those seven identities, together with the fields that every sub-elliptic setting carries (`CarnotSetting`), are all that this definition makes available about $\mathbb{G}_E(N)$: it fixes a witness and asserts nothing

further about it. In particular π_1^E is *not* the identity map here, and no consumer may treat it as one; the displayed identities relate π_1^E and $\exp_E$ only as mutually inverse maps between $\mathbb{R}^{n_E}$ and $\mathbb{R}^N$.

The purpose of fixing a witness, rather than invoking `EuclideanCarnotSettingExists` where it is needed, is that a setting obtained inside a proof cannot be named in a statement. Case (iii) of Theorem 1.3 descends to $\mathbb{R}^m$ and applies the cited results of the literature *there*, at the Euclidean setting; those results are asserted only at the setting carried as a parameter of the cited-results package, so a hypothesis asserting them at the Euclidean setting has to be able to refer to that setting by name. With $\mathbb{G}_E(m)$ named, the hypothesis can be written; without it, it cannot.

2. Lemmas

2.1. `InnerDomainProperties`.

Theorem 2.1 (The inner domains and compact cores of (3.8))**.** *Let $\mathbb{G}$ be a sub-elliptic setting (`CarnotSetting`) with group law $*$ and homogeneous norm $\|\cdot\|$, write $d(x,y) := \|y^{-1} * x\|$, and let $\Omega \subseteq \mathbb{R}^n$ be open and bounded. Write Ω_δ for the inner domain (`TranslatedDomain`) and K_ρ for the compact core (`BoundaryCore`). Then*

1. *$\Omega_\delta \subseteq \Omega$ for every $\delta \in \mathbb{R}$;*
2. *$K_\rho \subseteq \Omega$ for every $\rho \in \mathbb{R}$;*
3. *Ω_δ is open for every $\delta \in \mathbb{R}$;*
4. *K_ρ is compact for every $\rho > 0$;*
5. *$K_\rho \subseteq \Omega_\delta$ whenever $\delta < \rho$;*
6. *for every $\delta > 0$ and every $x \in \Omega_\delta$, the d-ball $B_\delta(x) = \{z : \ d(z,x) < \delta\}$ is contained in Ω;*
7. *for every $\rho > 0$ and every $x \in K_\rho$, the d-ball $B_\rho(x)$ is contained in Ω.*

Proof. These are the properties of (3.8) that the paper records in the sentence following it ("so that K_ρ is compact and $\Omega_\delta \supset K_\rho$ for $\rho > \delta$") together with the two facts it uses without proof: that Ω_δ is open, and that a d-ball about an interior point whose boundary distance is at least its radius stays in Ω. The latter is the inclusion behind "$x * h \in \Omega$ whenever $x \in \Omega_\delta$ and $\|h\| < \delta$" and behind "$B_\eta(y) = y * B_\eta(0)$, a ball contained in Ω" in Lemma 3.7. Note that neither uses a triangle inequality for d: pm-0013 records that the homogeneous norm is only a quasi-norm, and (2.17) carries a constant $c_{\mathbb{G}}$, so an exact triangle inequality is not available. Item (7) instead comes from the connectedness of d-balls.

Throughout we use from `CarnotSetting`: associativity of $*$, the identities $0 * x = x$, $x * 0 = x$, $x * (-x) = 0$, $(-x) * x = 0$, smoothness of $*$, continuity of $\|\cdot\|$, $\|x\| \geq 0$, $\|x\| = 0 \iff x = 0$, $\|-x\| = \|x\|$, and the connectedness of the balls $\{z : \ d(z,x) < \rho\}$ for $\rho > 0$. Since Ω is open, $\partial\Omega = \bar{\Omega} \setminus \Omega$.

Step 0: the group inverse is $-$, and d is symmetric. First, if $a * b = 0$ then $b = -a$: multiplying on the left by $-a$ and using associativity, $b = 0 * b = ((-a) * a) * b = (-a) * (a * b) = (-a) * 0 = -a$. Next,

$$(x*y)*\big((-y)*(-x)\big) = x*\Big(y*\big((-y)*(-x)\big)\Big) = x*\Big(\big(y*(-y)\big)*(-x)\Big) = x*\big(0*(-x)\big) = x*(-x) = 0,$$

so by the previous sentence $(-y) * (-x) = -(x * y)$. Applying this with x replaced by $-y$ and y by x gives $(-x) * y = -\big((-y) * x\big)$, whence, by $\|-z\| = \|z\|$,

$$d(x,y) = \big\|(-y) * x\big\| = \big\|(-x) * y\big\| = d(y,x). \tag{1}$$

Step 1: continuity of the distance kernel. The map $(x,y) \mapsto (-y) * x$ is continuous, being the composition of the continuous $(x,y) \mapsto (-y,x)$ with the group law, which is C^∞ and in particular continuous. Composing with the continuous $\|\cdot\|$, the function $(x,y) \mapsto d(x,y)$ is jointly continuous; in particular $x \mapsto d(x,y)$ is continuous for fixed y, and $y \mapsto d(x,y)$ is continuous for fixed x.

Step 2: $\partial\Omega$ is compact. $\partial\Omega$ is closed, and it is bounded because $\partial\Omega \subseteq \bar{\Omega}$ and $\bar{\Omega}$ is bounded (Ω is bounded). A closed bounded subset of $\mathbb{R}^n$ is compact.

Step 3: items (1), (2) and (5). Items (1) and (2) are immediate from `TranslatedDomain` and `BoundaryCore`, whose defining conditions both contain "$x \in \Omega$". For (5), let $\delta < \rho$ and $x \in K_\rho$: then $x \in \Omega$ and $\rho \le d(x,y)$ for every $y \in \partial\Omega$, so $\delta < \rho \le d(x,y)$, i.e. $x \in \Omega_\delta$.

Step 4: item (3), Ω_δ is open. If $\partial\Omega = \emptyset$ then the condition "$\delta < d(x,y)$ for every $y \in \partial\Omega$" is vacuous, so $\Omega_\delta = \Omega$, which is open.

Assume $\partial\Omega \neq \emptyset$ and set $f(x) := \inf\{d(x,y) : \ y \in \partial\Omega\}$, the paper's $\operatorname{dist}(x, \partial\Omega)$. The set $\{d(x,y) : y \in \partial\Omega\}$ is non-empty and bounded below by 0, so $f(x)$ is a well-defined real number and

$$f(x) \le d(x,y) \qquad \text{for every } y \in \partial\Omega. \tag{2}$$

By Step 1 the function $y \mapsto d(x,y)$ is continuous and by Step 2 $\partial\Omega$ is compact and non-empty, so its image is compact and non-empty and the infimum is attained:

$$\text{there is } y_x \in \partial\Omega \text{ with } f(x) = d(x, y_x). \tag{3}$$

By Step 1 again d is *jointly* continuous, and $\partial\Omega$ is compact, so f is continuous. (This is the point at which compactness is needed and the paper's one-line assertion requires care: an infimum of continuous functions over an arbitrary index set is only upper semi-continuous; over a compact index set, with jointly continuous integrand, it is continuous.)

Now

$$\Omega_\delta = \Omega \cap \{x : \ \delta < f(x)\}. \tag{4}$$

Indeed, if $x \in \Omega_\delta$ then $\delta < d(x, y_x) = f(x)$ by (3); conversely if $x \in \Omega$ and $\delta < f(x)$ then $\delta < f(x) \le d(x,y)$ for every $y \in \partial\Omega$ by (2). The right-hand side of (4) is the intersection of the open set Ω with the preimage of an open half-line under the continuous f, hence open.

Step 5: item (4), K_ρ is compact for $\rho > 0$. Put

$$T := \{x : \ \rho \le d(x,y) \text{ for every } y \in \partial\Omega\} = \bigcap_{y \in \partial\Omega} \{x : \ \rho \le d(x,y)\},$$

a closed set, being an intersection of sets closed by the continuity of $x \mapsto d(x,y)$ (Step 1). We claim

$$K_\rho = \bar{\Omega} \cap T. \tag{5}$$

"$\subseteq$": if $x \in K_\rho$ then $x \in \Omega \subseteq \bar{\Omega}$ and $x \in T$. "$\supseteq$": let $x \in \bar{\Omega} \cap T$; we must show $x \in \Omega$. If not, then $x \in \bar{\Omega} \setminus \Omega = \partial\Omega$, so taking $y := x$ in the definition of T gives $\rho \le d(x,x) = \|(-x) * x\| = \|0\| = 0$, contradicting $\rho > 0$. Hence $x \in \Omega$, and $x \in K_\rho$.

By (5), K_ρ is closed (an intersection of two closed sets) and bounded (it is contained in the bounded set $\bar{\Omega}$), hence compact.

Step 6: the ball inclusion. We prove the following statement, which gives both (6) and (7): *let $x \in \Omega$, let $\delta > 0$ and assume $\delta \le d(x,y)$ for every $y \in \partial\Omega$; then $S := \{z : \ d(z,x) < \delta\} \subseteq \Omega$.*

By `CarnotSetting` the set S is connected (and non-empty). First, S misses $\partial\Omega$: if $z \in S$ and $z \in \partial\Omega$ then the hypothesis at $y := z$ gives $\delta \le d(x,z)$, while (1) gives $d(x,z) = d(z,x) < \delta$ — a contradiction.

Second, S is covered by the two *open* sets Ω and $(\bar{\Omega})^c$: let $z \in S$. If $z \notin \bar{\Omega}$ then $z \in (\bar{\Omega})^c$. If $z \in \bar{\Omega}$ and $z \notin \Omega$ then $z \in \bar{\Omega} \setminus \Omega = \partial\Omega$, contradicting the previous paragraph; so $z \in \Omega$.

Third, $x \in S$, because $d(x,x) = \|(-x) * x\| = \|0\| = 0 < \delta$; and $x \in \Omega$ by hypothesis, so $S \cap \Omega \neq \emptyset$. The two covering sets are disjoint from each other *on* S in the strong sense that $\Omega \cap (\bar{\Omega})^c = \emptyset$ (a point of Ω lies in $\bar{\Omega}$). If some $w \in S$ had $w \notin \Omega$, then $w \in (\bar{\Omega})^c$ by the cover, so S would meet both Ω and $(\bar{\Omega})^c$; since S is connected and $\Omega \cup (\bar{\Omega})^c \supseteq S$ with Ω, $(\bar{\Omega})^c$ open and $\Omega \cap (\bar{\Omega})^c \cap S = \emptyset$, this is impossible. Hence $S \subseteq \Omega$.

For (6), let $\delta > 0$ and $x \in \Omega_\delta$: then $x \in \Omega$ and $\delta < d(x,y)$, so $\delta \le d(x,y)$, for every $y \in \partial\Omega$, and the statement applies. For (7), let $\rho > 0$ and $x \in K_\rho$: then $x \in \Omega$ and $\rho \le d(x,y)$ for every $y \in \partial\Omega$, and the statement applies with $\delta := \rho$. □

2.2. `BoundaryLocalization`.

Lemma 2.2 (Lemma 3.4, with the quasi-triangle constant carried)**.** *Let $\mathbb{G}$ be a sub-elliptic setting (`CarnotSetting`) with group law $*$, homogeneous norm $\|\cdot\|$ and $d(x,y) := \|y^{-1} * x\|$, and let $\Omega \subset \mathbb{R}^n$ be a bounded open set. Let $u, v : \mathbb{R}^n \to \mathbb{R}$ be continuous on $\bar{\Omega}$, let $\omega_u, \omega_v : \mathbb{R} \to \mathbb{R}$ be non-decreasing with*

$$|u(a) - u(b)| \le \omega_u\big(d(a,b)\big), \qquad |v(a) - v(b)| \le \omega_v\big(d(a,b)\big) \qquad (a, b \in \bar{\Omega}),$$

let $\tau > 0$, assume $u - v \le -\tau$ on $\partial\Omega$, let $x_M \in \bar{\Omega}$ be a point at which $u - v$ attains its maximum over $\bar{\Omega}$ and assume $\tau \le M_0 := u(x_M) - v(x_M)$. Let $\rho_0 > 0$ satisfy $\omega_u(\rho_0) + \omega_v(\rho_0) \le \tau/4$. Write Ω_δ for the inner domain (`TranslatedDomain`), K_ρ for the compact core (`BoundaryCore`), $M_\delta(h)$ for the translated maximum value (`MaxTranslate`),

$$E := \big\{ y \in \Omega : \ u - v \text{ attains its maximum over } \bar{\Omega} \text{ at } y \big\} \quad \big(= \operatorname*{argmax}_{\Omega}(u - v) \big),$$

and $u_h := u(\cdot * h)$. Then there is $c' \ge 1$ such that, setting

$$\rho_1 := \frac{\rho_0}{2c'}, \qquad \rho_2 := \frac{\rho_0}{4c'^2}, \qquad \delta_1 := \frac{\rho_0}{4c'^2},$$

the following hold.

- (a) *$E \subseteq K_{\rho_0}$; and for every δ with $0 < \delta \le \delta_1$, $M_\delta(0) = M_0$ and $\{x \in \Omega_\delta : \ u_0 - v$ is maximal over Ω_δ at $x\} = E$.*
- (a′) *For every δ with $0 < \delta \le \delta_1$, every $x \in K_{\rho_1}$ and every g with $\|g\| < \delta$, one has $x * g \in K_{\rho_2}$.*
- (b) *For every δ with $0 < \delta \le \delta_1$ and every h with $\|h\| < \delta$, $u_h - v \le -\tau/2$ on $\Omega_\delta \setminus K_{\rho_1}$.*
- (c) *For every δ with $0 < \delta \le \delta_1$ and every h with $\|h\| < \delta$: $M_\delta(h) \ge M_0 - \tau/4$; the supremum defining $M_\delta(h)$ is attained, i.e. there is $x \in \Omega_\delta$ at which $u_h - v$ is maximal over Ω_δ and $M_\delta(h) = u(x * h) - v(x)$; and every such x satisfies $x \in K_{\rho_1}$ and $x * h \in K_{\rho_2}$.*

Proof. This is Lemma 3.4 together with the sentence of (3.11) that fixes δ_1.

Why the constants are explicit, and why that is not a deviation. The paper sets $\delta_1 := \rho_0/4$ in (3.11) and then argues in (b) "$\operatorname{dist}(x, \partial\Omega) < \rho_0/2$, so $\operatorname{dist}(x*h, \partial\Omega) < \rho_0/2 + \delta < \rho_0$" and in (c) "$\operatorname{dist}(y * h^{-1}, \partial\Omega) \ge \rho_0 - \delta > \delta$". Both are steps with triangle constant 1. But the homogeneous norm (2.15) is a *quasi*-norm: (2.17) itself is stated as $\|x * y\| \le c_{\mathbb{G}}(\|x\| + \|y\|)$ with $c_{\mathbb{G}} = c(\mathbb{G}, \nu)$ for $\|x\|, \|y\| < \nu$, and that is exactly what `CarnotSetting` provides (pm-0013 is the governing constraint). So the radii have to carry $c_{\mathbb{G}}$. We therefore produce $c' \ge 1$ dominating $c_{\mathbb{G}}$ on a scale large enough for every application below, and replace the paper's $\rho_0/2$, $\rho_0/4$ by $\rho_1 = \rho_0/(2c')$ and $\rho_2 = \delta_1 = \rho_0/(4c'^2)$. Nothing in any *statement* changes: ρ_0 still satisfies (3.11) verbatim, and (a), (b), (c) are the paper's assertions with the paper's $\rho_0/2$, $\rho_0/4$ replaced by the two constants it would have produced had (2.17) been applied with its own constant. This is the one place in the formalization where the paper's constants are made explicit; the paper is read precisely, not corrected. Part (a) uses no triangle inequality at all, and we keep it that way so that the reader can see where the constants enter and where they do not.

Step 0: preliminaries. Since $x_M \in \bar{\Omega}$, $\bar{\Omega} \ne \emptyset$. Taking $a = b = x_M$ in the modulus hypothesis for u and using $d(x_M, x_M) = \|(-x_M) * x_M\| = \|0\| = 0$ (`CarnotSetting`) gives $0 \le \omega_u(0)$, hence $\omega_u(s) \ge 0$ for every $s \ge 0$ by monotonicity, and likewise for ω_v. Combined with $\omega_u(\rho_0) + \omega_v(\rho_0) \le \tau/4$ this gives

$$0 \le \omega_u(\rho_0) \le \tau/4, \qquad 0 \le \omega_v(\rho_0) \le \tau/4. \tag{1}$$

Also $M_0 \ge \tau > 0$, and $u - v \le M_0$ on $\bar{\Omega}$ by the maximality at x_M. Since Ω is open, $\partial\Omega = \bar{\Omega} \setminus \Omega$.

$E \ne \emptyset$, indeed $x_M \in E$. If $x_M \in \partial\Omega$ then $M_0 = u(x_M) - v(x_M) \le -\tau < 0$, contradicting $M_0 \ge \tau > 0$. So $x_M \in \bar{\Omega} \setminus \partial\Omega = \Omega$, and $u - v$ is maximal over $\bar{\Omega}$ at x_M by hypothesis; thus $x_M \in E$. Note also

$$E = \{ y \in \Omega : \ u(y) - v(y) = M_0 \}, \tag{2}$$

because $u - v \le M_0$ on $\bar{\Omega}$, so "maximal over $\bar{\Omega}$ at y" is the same as "$(u - v)(y) = M_0$" for $y \in \Omega \subseteq \bar{\Omega}$.

Step 1: the scale ν and the constant c'. The group law $*$ is C^∞ and the homogeneous norm $\|\cdot\|$ is continuous (`CarnotSetting`), and $x \mapsto -x$ is continuous, so $(z,w) \mapsto d(z,w) = \|(-w)*z\|$ is a composition of continuous maps and is continuous on $\mathbb{R}^n \times \mathbb{R}^n$. The set $\bar\Omega$ is compact (Ω is bounded) and non-empty, so $\bar\Omega \times \bar\Omega$ is compact and non-empty and

$$\Delta := \max\{d(z,w) : \ z,w \in \bar\Omega\}$$

is a finite real number, with $\Delta \geq 0$. Put $\nu := \Delta + \rho_0 + 1 > 0$. By the quasi-triangle inequality (2.17) of `CarnotSetting`, applied at this ν, there is $c_{\mathbb{G}} > 0$ with

$$\|a*b\| \leq c_{\mathbb{G}}\big(\|a\| + \|b\|\big) \qquad \text{whenever } \|a\| < \nu,\ \|b\| < \nu.$$

Set $c' := \max\{c_{\mathbb{G}}, 1\}$, so $c' \geq 1$, $c' \geq c_{\mathbb{G}}$ and, since $\|a\| + \|b\| \geq 0$,

$$\|a*b\| \leq c'\big(\|a\| + \|b\|\big) \qquad \text{whenever } \|a\| < \nu,\ \|b\| < \nu. \tag{3}$$

With this c' define ρ_1, ρ_2, δ_1 as in the statement. Then

$$0 < \delta_1 = \rho_2 = \frac{\rho_0}{4c'^2} \leq \frac{\rho_0}{4}, \qquad \rho_1 = \frac{\rho_0}{2c'} \leq \frac{\rho_0}{2}, \qquad \delta_1 = \frac{\rho_1}{2c'} \leq \frac{\rho_1}{2} < \rho_1 < \rho_0. \tag{4}$$

In particular $\delta \leq \delta_1 < \rho_1 < \rho_0 < \nu$ for every admissible δ, and $\|h\| < \delta < \nu$ for every admissible h.

We also record once and for all the following, used repeatedly below. *For every $\delta > 0$, every $x \in \Omega_\delta$ and every g with $\|g\| < \delta$ one has $x * g \in \Omega \subseteq \bar\Omega$.* Indeed, by associativity and $0 * g = g$ (`CarnotSetting`), $d(x*g, x) = \|(-x)*(x*g)\| = \|((-x)*x)*g\| = \|0*g\| = \|g\| < \delta$ — this is the exact identity (2.16), kept inline by pm-0012 — so $x*g$ lies in the d-ball of radius δ about x, which is contained in Ω by item (6) of `InnerDomainProperties`. (4′)

Step 2: the boundary estimate. *If $z \in \bar\Omega$ and $y \in \partial\Omega$ satisfy $d(z,y) \leq \rho_0$, then $(u-v)(z) \leq -3\tau/4$.* Indeed $y \in \partial\Omega \subseteq \bar\Omega$, so the moduli apply to the pair (z,y), and using $u - v \leq -\tau$ at y, monotonicity of ω_u, ω_v and (1),

$$(u-v)(z) = (u-v)(y) + \big(u(z)-u(y)\big) - \big(v(z)-v(y)\big) \leq -\tau + \omega_u\big(d(z,y)\big) + \omega_v\big(d(z,y)\big) \leq -\tau + \frac{\tau}{4} = -\frac{3\tau}{4}. \tag{5}$$

Step 3: part (a). $E \subseteq K_{\rho_0}$. Let $y \in E$ and $y' \in \partial\Omega$. We must show $\rho_0 \leq d(y,y')$ (membership $y \in \Omega$ is part of the definition of E). Suppose $d(y,y') < \rho_0$. Then $y \in \Omega \subseteq \bar\Omega$ and $d(y,y') \leq \rho_0$, so (5) gives $(u-v)(y) \leq -3\tau/4 < 0 < \tau \leq M_0$, contradicting $(u-v)(y) = M_0$ from (2). Hence $\rho_0 \leq d(y,y')$, and $y \in K_{\rho_0}$ by `BoundaryCore`. No triangle inequality was used.

$M_\delta(0) = M_0$ and $A_\delta(0) = E$, for $0 < \delta \leq \delta_1$. Since $\delta \leq \delta_1 < \rho_0$ by (4), item (5) of `InnerDomainProperties` gives $K_{\rho_0} \subseteq \Omega_\delta$, so by the previous paragraph

$$E \subseteq K_{\rho_0} \subseteq \Omega_\delta \subseteq \Omega \subseteq \bar\Omega, \tag{6}$$

the third inclusion being item (1) of `InnerDomainProperties`. Now $u_0 = u$, because $x * 0 = x$ (`CarnotSetting`). For $x \in \Omega_\delta$ we have $x \in \bar\Omega$ by (6), hence $u_0(x) - v(x) = (u-v)(x) \leq M_0$; and $x_M \in E \subseteq \Omega_\delta$ with $u_0(x_M) - v(x_M) = M_0$. So $u_0 - v$ is bounded above on the non-empty set Ω_δ by M_0 and attains the value M_0 there; therefore its supremum over Ω_δ, which is $M_\delta(0)$ by `MaxTranslate`, equals M_0, and the points of Ω_δ at which $u_0 - v$ is maximal over Ω_δ are exactly the points $x \in \Omega_\delta$ with $(u-v)(x) = M_0$. By (2) and (6) that set is E: if $x \in \Omega_\delta$ and $(u-v)(x) = M_0$ then $x \in \Omega$ by (6), so $x \in E$ by (2); conversely every $y \in E$ lies in Ω_δ by (6) and has $(u-v)(y) = M_0$. This is part (a).

Step 4: part (b). Let $0 < \delta \leq \delta_1$, $\|h\| < \delta$, and let $x \in \Omega_\delta \setminus K_{\rho_1}$. By `BoundaryCore` and $x \in \Omega$, failing $x \in K_{\rho_1}$ means there is $y \in \partial\Omega$ with

$$d(x,y) < \rho_1. \tag{7}$$

By (4′),

$$x * h \in \Omega \subseteq \bar\Omega. \tag{8}$$

Now estimate $d(x*h,y)$. Using associativity, $(-y)*(x*h)=\big((-y)*x\big)*h$, and $\|(-y)*x\|=d(x,y)<\rho_1<\nu$ by (7) and (4), while $\|h\|<\delta<\nu$; so (3) applies and, with (7), (4) and $c'\ge 1$, (9)

$$d(x*h,y)=\big\|\big((-y)*x\big)*h\big\|\le c'\big(d(x,y)+\|h\|\big)<c'\Big(\frac{\rho_0}{2c'}+\frac{\rho_0}{4c'^2}\Big)=\frac{\rho_0}{2}+\frac{\rho_0}{4c'}\le\frac{\rho_0}{2}+\frac{\rho_0}{4}=\frac{3\rho_0}{4}<\rho_0.$$

This is the step where the paper writes $\operatorname{dist}(x*h,\partial\Omega)<\rho_0/2+\delta<\rho_0$; the constants ρ_1 and δ_1 are chosen precisely so that the conclusion $d(x*h,y)<\rho_0$ survives the factor c'.

By (8), (9) and Step 2 applied to $z:=x*h$,

$$(u-v)(x*h)\le-\frac{3\tau}{4}. \tag{10}$$

Finally, $x\in\bar\Omega$ and $x*h\in\bar\Omega$ by (6) and (8), so the modulus of v applies to the pair $(x*h,x)$ and, with $d(x*h,x)=\|h\|<\delta\le\delta_1\le\rho_0$, monotonicity and (1),

$$\big|v(x*h)-v(x)\big|\le\omega_v\big(\|h\|\big)\le\omega_v(\rho_0)\le\frac{\tau}{4}. \tag{11}$$

Therefore, by (10) and (11),

$$(u_h-v)(x)=u(x*h)-v(x)=\big(u(x*h)-v(x*h)\big)+\big(v(x*h)-v(x)\big)\le-\frac{3\tau}{4}+\frac{\tau}{4}=-\frac{\tau}{2},$$

which is part (b).

Step 5: part (c), the lower bound for $M_\delta(h)$. Let $0<\delta\le\delta_1$ and $\|h\|<\delta$, and fix $y\in E$ (possible, $x_M\in E$ by Step 0). We claim $y*(-h)\in\Omega_\delta$.

First, $y\in K_{\rho_0}$ by Step 3, so by item (7) of `InnerDomainProperties` the d-ball of radius ρ_0 about y lies in Ω; and $d\big(y*(-h),y\big)=\|(-y)*\big(y*(-h)\big)\|=\|\big((-y)*y\big)*(-h)\|=\|-h\|=\|h\|<\delta<\rho_0$, using associativity, $0*z=z$ and $\|-z\|=\|z\|$. Hence

$$y*(-h)\in\Omega\subseteq\bar\Omega. \tag{12}$$

Second, let $y'\in\partial\Omega$. Since $h*(-h)=0$ and $z*0=z$ we have $\big(y*(-h)\big)*h=y*\big((-h)*h\big)=y*0=y$, so by associativity

$$(-y')*y=(-y')*\Big(\big(y*(-h)\big)*h\Big)=\Big((-y')*\big(y*(-h)\big)\Big)*h.$$

Here $\|(-y')*\big(y*(-h)\big)\|=d\big(y*(-h),y'\big)\le\Delta<\nu$ because $y*(-h)\in\bar\Omega$ by (12) and $y'\in\partial\Omega\subseteq\bar\Omega$, and $\|h\|<\delta<\nu$; so (3) applies, and with $\rho_0\le d(y,y')$ from Step 3,

$$\rho_0\le d(y,y')\le c'\Big(d\big(y*(-h),y'\big)+\|h\|\Big),$$

whence, using $\|h\|<\delta\le\delta_1=\rho_0/(4c'^2)$ and $c'\ge1$,

$$d\big(y*(-h),y'\big)\ \ge\ \frac{\rho_0}{c'}-\|h\|\ >\ \frac{\rho_0}{c'}-\frac{\rho_0}{4c'^2}\ \ge\ \frac{\rho_0}{c'}-\frac{\rho_0}{4c'}\ =\ \frac{3\rho_0}{4c'}\ \ge\ \frac{3\rho_0}{4c'^2}\ >\ \frac{\rho_0}{4c'^2}=\delta_1\ \ge\ \delta.$$

So $\delta<d\big(y*(-h),y'\big)$ for every $y'\in\partial\Omega$, and with (12) this gives $y*(-h)\in\Omega_\delta$ by `TranslatedDomain`. This is the paper's "$\operatorname{dist}(y*h^{-1},\partial\Omega)\ge\rho_0-\delta>\delta$, so $y*h^{-1}\in\Omega_\delta$", with the constant carried.

Next, $\big(y*(-h)\big)*h=y$ as computed above, so

$$(u_h-v)\big(y*(-h)\big)=u(y)-v\big(y*(-h)\big).$$

By (12), $y\in\bar\Omega$ and $y*(-h)\in\bar\Omega$, and $d\big(y*(-h),y\big)=\|h\|\le\rho_0$, so the modulus of v and (1) give $v\big(y*(-h)\big)\le v(y)+\omega_v(\|h\|)\le v(y)+\tau/4$. Hence

$$(u_h-v)\big(y*(-h)\big)\ \ge\ u(y)-v(y)-\frac{\tau}{4}\ =\ M_0-\frac{\tau}{4}\ \ge\ \tau-\frac{\tau}{4}\ =\ \frac{3\tau}{4}\ >\ 0, \tag{13}$$

using (2) and $M_0\ge\tau$.

Step 6: part (c), attainment. Keep δ and h as in Step 5. By (4), $\delta\le\delta_1<\rho_1$, so item (5) of `InnerDomainProperties` gives

$$K_{\rho_1}\subseteq\Omega_\delta, \tag{14}$$

and item (4) of `InnerDomainProperties` gives that K_{ρ_1} is compact (note $\rho_1>0$ by (4)).

$u_h - v$ is continuous on K_{ρ_1}. The map $z \mapsto z * h$ is continuous (`CarnotSetting`) and, by (14) and (8), sends K_{ρ_1} into $\Omega \subseteq \bar{\Omega}$; u is continuous on $\bar{\Omega}$; so $u_h = u(\cdot * h)$ is continuous on K_{ρ_1}. Also $K_{\rho_1} \subseteq \Omega \subseteq \bar{\Omega}$ by item (2) of `InnerDomainProperties`, so v is continuous on K_{ρ_1}. Hence so is $u_h - v$.

$K_{\rho_1} \neq \emptyset$. The point $y*(-h)$ of Step 5 lies in Ω_δ and, by (13), $(u_h - v)\big(y*(-h)\big) \geq 3\tau/4 > -\tau/2$; so by part (b) it cannot lie in $\Omega_\delta \setminus K_{\rho_1}$, i.e. $y * (-h) \in K_{\rho_1}$.

Therefore $u_h - v$ attains its maximum over the non-empty compact set K_{ρ_1}, say at $x_0 \in K_{\rho_1} \subseteq \Omega_\delta$. We claim $u_h - v$ is maximal over all of Ω_δ at x_0. Let $x \in \Omega_\delta$. If $x \in K_{\rho_1}$ then $(u_h - v)(x) \leq (u_h - v)(x_0)$ by the choice of x_0. If $x \notin K_{\rho_1}$ then part (b) gives $(u_h - v)(x) \leq -\tau/2$, while $(u_h - v)(x_0) \geq (u_h - v)\big(y * (-h)\big) \geq 3\tau/4 > -\tau/2$ by the choice of x_0, $y * (-h) \in K_{\rho_1}$ and (13); so again $(u_h - v)(x) \leq (u_h - v)(x_0)$.

Consequently $u_h - v$ is bounded above on the non-empty set Ω_δ by $(u_h - v)(x_0)$, and attains that value there, so by `MaxTranslate`

$$M_\delta(h) = u(x_0 * h) - v(x_0), \tag{15}$$

which is the asserted attainment, and by (13) and the maximality of x_0,

$$M_\delta(h) = (u_h - v)(x_0) \;\geq\; (u_h - v)\big(y * (-h)\big) \;\geq\; M_0 - \frac{\tau}{4},$$

which is the asserted lower bound.

Step 7: part (a′). Let $0 < \delta \leq \delta_1$, let $x \in K_{\rho_1}$ and let $\|g\| < \delta$; write $h := g$ for uniformity with the computation below. By (14), $x \in K_{\rho_1} \subseteq \Omega_\delta$, so $x * h \in \Omega$ by (8).

Let $y' \in \partial\Omega$. Since $h * (-h) = 0$ and $z * 0 = z$, $(x * h) * (-h) = x$, so by associativity

$$(-y') * x = (-y') * \big((x * h) * (-h)\big) = \Big((-y') * (x * h)\Big) * (-h).$$

Here $\|(-y') * (x * h)\| = d(x * h, y') \leq \Delta < \nu$ because $x * h \in \bar{\Omega}$ by (8) and $y' \in \bar{\Omega}$, and $\| - h\| = \|h\| < \delta < \nu$; so (3) applies, and with $\rho_1 \leq d(x, y')$ (from $x \in K_{\rho_1}$, `BoundaryCore`),

$$\rho_1 \leq d(x, y') \leq c'\Big(d(x * h, y') + \|h\|\Big),$$

whence, using $\|h\| < \delta \leq \delta_1 = \rho_0/(4c'^2)$ and $\rho_1 = \rho_0/(2c')$,

$$d(x * h, y') \;\geq\; \frac{\rho_1}{c'} - \|h\| \;>\; \frac{\rho_0}{2c'^2} - \frac{\rho_0}{4c'^2} \;=\; \frac{\rho_0}{4c'^2} \;=\; \rho_2.$$

In particular $\rho_2 \leq d(x * h, y')$ for every $y' \in \partial\Omega$, so $x * h \in K_{\rho_2}$ by `BoundaryCore`. This is part (a′); note that it used only $x \in K_{\rho_1}$ and $\|h\| < \delta$, not any maximality of x, which is why it is stated separately. It is the paper's "$\mathcal{A}_\delta(h) * h \subseteq K_{\rho_0/4}$", with the constant carried.

Step 8: part (c), $A_\delta(h) \subseteq K_{\rho_1}$ and $A_\delta(h) * h \subseteq K_{\rho_2}$. Let $0 < \delta \leq \delta_1$, $\|h\| < \delta$, and let $x \in \Omega_\delta$ be any point at which $u_h - v$ is maximal over Ω_δ. Then $(u_h - v)(x) = M_\delta(h) \geq M_0 - \tau/4 \geq 3\tau/4 > -\tau/2$ by Step 6 and (13); so by part (b), $x \in K_{\rho_1}$. Part (a′) applied with $g := h$ then gives $x * h \in K_{\rho_2}$, which completes part (c). □

2.3. `AddConstPreservesSubsol`.

Lemma 2.3 (Adding a constant preserves viscosity sub- and super-solutions)**.** *Let $\mathbb{G}$ be a Carnot setting (`CarnotSetting`) with horizontal frame σ, let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$, let $\Omega \subseteq \mathbb{R}^n$, let $u : \mathbb{R}^n \to \mathbb{R}$ and let $c \in \mathbb{R}$. If u is a viscosity sub-solution of $\mathcal{L}u = 0$ in Ω (`IsViscSubsol`), then so is $u + c$; and if u is a viscosity super-solution of $\mathcal{L}u = 0$ in Ω (`IsViscSupersol`), then so is $u + c$.*

Proof. Write $u_c := u + c$, i.e. $u_c(y) = u(y) + c$ for every y.

Step 1: semi-continuity is preserved. Suppose u is upper semi-continuous on Ω, i.e. for every $x \in \Omega$ and every $y \in \mathbb{R}$ with $u(x) < y$ the inequality $u(x') < y$ holds for all x' in some neighbourhood of x relative to Ω. Let $x \in \Omega$ and $y \in \mathbb{R}$ with $u_c(x) < y$, i.e. $u(x) + c < y$, i.e. $u(x) < y - c$. Applying upper semi-continuity of u at x with the value $y - c$ gives a neighbourhood of x relative to Ω on which $u(x') < y - c$, hence on which $u_c(x') = u(x') + c < y$. So u_c is upper semi-continuous on Ω.

Symmetrically, if u is lower semi-continuous on Ω and $y < u_c(x)$, then $y - c < u(x)$, so $y - c < u(x')$ and hence $y < u_c(x')$ on a neighbourhood of x relative to Ω; so u_c is lower semi-continuous on Ω.

Step 2: transport of test functions. Let $x \in \Omega$, let $\varphi \in C^2(\Omega)$, let (D, B) be a second-order jet of φ at x (`TwiceDiffAt`) with hgrad $\sigma\ x\ D \neq 0$ (`hgrad`), and suppose $x \in \operatorname{argmax}_\Omega(u_c - \varphi)$. Put

$$\psi := \varphi - c, \qquad \text{i.e.} \qquad \psi(y) = \varphi(y) - c.$$

We check that (x, ψ, D, B) is admissible data for u.

(i) $\psi \in C^2(\Omega)$. ψ is the difference of $\varphi \in C^2(\Omega)$ and the constant function c, which is C^2 on Ω; a difference of C^2 functions is C^2.

(ii) $x \in \operatorname{argmax}_\Omega(u - \psi)$. For every y,

$$u(y) - \psi(y) = u(y) - \varphi(y) + c = \big(u(y) + c\big) - \varphi(y) = u_c(y) - \varphi(y),$$

so the functions $u - \psi$ and $u_c - \varphi$ are equal. Since $x \in \Omega$ and $(u_c - \varphi)(y) \leq (u_c - \varphi)(x)$ for every $y \in \Omega$, the same inequalities hold with $u - \psi$ in place of $u_c - \varphi$, i.e. $x \in \operatorname{argmax}_\Omega(u - \psi)$.

(iii) (D, B) *is a second-order jet of* ψ *at* x. By `TwiceDiffAt` this means that B is symmetric and that for every $\varepsilon > 0$ there is $\delta > 0$ with

$$\Big|\psi(y) - \psi(x) - D(y - x) - \tfrac{1}{2}B(y - x, y - x)\Big| \leq \varepsilon |y - x|^2 \qquad \text{whenever } |y - x| < \delta.$$

Symmetry of B is part of the hypothesis on (D, B). For the estimate, note $\psi(y) - \psi(x) = \big(\varphi(y) - c\big) - \big(\varphi(x) - c\big) = \varphi(y) - \varphi(x)$, so the quantity inside the absolute value is identical to the corresponding quantity for φ; the estimate for φ therefore holds verbatim for ψ, with the same δ for each ε.

(iv) hgrad $\sigma\ x\ D \neq 0$. This condition mentions only σ, x and D, which are unchanged.

Step 3: conclusion for sub-solutions. Assume u is a viscosity sub-solution of $\mathcal{L}u = 0$ in Ω. By Step 1 the function u_c is upper semi-continuous on Ω, which is the first clause of `IsViscSubsol` for u_c. For the second clause, let x, φ, D, B be as in Step 2. By Step 2 the data (x, ψ, D, B) with $\psi = \varphi - c$ is admissible for u, so the second clause of `IsViscSubsol` for u gives

$$\text{Lop } \sigma\ A\ H\ x\ D\ B\ \leq\ 0.$$

But this inequality is exactly what has to be shown for u_c at the data (x, φ, D, B): by `Lop` the value Lop $\sigma\ A\ H\ x\ D\ B = -\operatorname{Tr}\big(A(\text{hgrad } \sigma\ x\ D) \cdot \text{hhessS } \sigma\ x\ D\ B\big) + H(\text{hgrad } \sigma\ x\ D)$ is computed from σ, x, D, B, A and H alone; neither the function u nor the test function φ nor its value $\varphi(x)$ enters it. In particular replacing ψ by $\varphi = \psi + c$ does not change it, since ψ and φ have the same jet (D, B) at x by Step 2(iii). Hence u_c satisfies the second clause, and u_c is a viscosity sub-solution of $\mathcal{L}u = 0$ in Ω.

Step 4: conclusion for super-solutions. Assume u is a viscosity super-solution. By Step 1, u_c is lower semi-continuous on Ω. For the second clause of `IsViscSupersol`, let $x \in \Omega$, $\varphi \in C^2(\Omega)$ and (D, B) be a second-order jet of φ at x with hgrad $\sigma\ x\ D \neq 0$, and suppose $x \in \operatorname{argmin}_\Omega(u_c - \varphi)$. With $\psi := \varphi - c$, items (i), (iii) and (iv) of Step 2 are unchanged, and item (ii) becomes: $u - \psi$ and $u_c - \varphi$ are the same function, so $x \in \operatorname{argmin}_\Omega(u - \psi)$. The second clause of `IsViscSupersol` for u then gives $0 \leq$ Lop $\sigma\ A\ H\ x\ D\ B$, which again is literally the inequality required of u_c at (x, φ, D, B).

Remark. The only structural feature of $\mathcal{L}$ used is that it carries no zeroth-order term: by `Lop` the operator (1.8) is $-\operatorname{Tr}\big(A(\mathcal{X}u)\mathcal{X}\mathcal{X}^s u\big) + H(\mathcal{X}u)$, in which the value of u does not appear. That is what allows the additive constant to be moved from the solution into the test function without changing the inequality to be verified. □

2.4. `ViscTestLocalization`.

Theorem 2.4 (Localization of test functions, the sentence after Definition 2.1)**.** *Let* $\mathbb{G}$ *be a sub-elliptic setting (`CarnotSetting`) with frame* $\sigma = [\sigma^1, \dots, \sigma^m]$*, let* $A : \mathbb{R}^m \to \mathbb{R}^{m \times m}$ *and* $H : \mathbb{R}^m \to \mathbb{R}$*, let* $\Omega \subset \mathbb{R}^n$ *and let* $u : \mathbb{R}^n \to \mathbb{R}$*.*

(1) *Assume u is bounded above on Ω, i.e. there is $M_u \in \mathbb{R}$ with $u \le M_u$ on Ω, and that u is a viscosity sub-solution of $\mathcal{L}u = 0$ in Ω (`IsViscSubsol`). Then:*
   (a) *for every $x \in \Omega$, every $\varphi : \mathbb{R}^n \to \mathbb{R}$, every $r > 0$ with $B(x,r) \subset \Omega$ and $\varphi \in C^2(B(x,r))$, if $u - \varphi \le (u-\varphi)(x)$ on $B(x,r)$ and (D,B) is a second-order jet of φ at x (`TwiceDiffAt`) with* hgrad σ x $D \neq 0$ *(`hgrad`), then* Lop σ A H x D $B \le 0$ *(`Lop`);*
   (b) *u is a viscosity sub-solution of $\mathcal{L}u = 0$ in every open $V \subset \Omega$.*

(2) *Dually, assume u is bounded below on Ω, i.e. there is $m_u \in \mathbb{R}$ with $m_u \le u$ on Ω, and that u is a viscosity super-solution in Ω (`IsViscSupersol`). Then:*
   (a) *for every $x \in \Omega$, every φ, every $r > 0$ with $B(x,r) \subset \Omega$ and $\varphi \in C^2(B(x,r))$, if $(u-\varphi)(x) \le u - \varphi$ on $B(x,r)$ and (D,B) is a second-order jet of φ at x with* hgrad σ x $D \neq 0$*, then $0 \le$* Lop σ A H x D B*;*
   (b) *u is a viscosity super-solution in every open $V \subset \Omega$.*

Here $B(x,r)$ is the Euclidean ball of $\mathbb{R}^n$.

Proof. The paper's warrant is the single sentence following Definition 2.1: "since φ may be modified away from x, it is equivalent to require in (2.11) that x be a local maximum of $w - \varphi$". We write the modification out. The point of doing so is that our rendering of (2.11) is literal: `IsViscSubsol` requires x to maximize $u - \varphi$ over *all* of Ω and requires $\varphi \in C^2(\Omega)$, so neither the passage from a local to a global touching nor the restriction to a smaller open set is available for free. Both halves of the statement are instances of one construction.

Where the boundedness hypothesis comes from. The modification replaces φ, away from x, by a constant K so large that the competitor $u - K$ can never beat the value $(u-\varphi)(x)$ on Ω. Such a K exists precisely when u is bounded above on Ω (for part (2), bounded below). It is not a free hypothesis, and it is the reason both this node and its consumers carry it; in the use sites of Section 3 it is supplied by Ω bounded and $u \in C(\bar{\Omega})$.

Step 1: the cutoff. Fix $x \in \Omega$, $r > 0$ with $B(x,r) \subset \Omega$, and $\varphi \in C^2(B(x,r))$. Let $\chi : \mathbb{R}^n \to \mathbb{R}$ be a smooth bump function centred at x with inner radius $r/4$ and outer radius $r/2$, so that

$$\chi \in C^\infty(\mathbb{R}^n), \qquad 0 \le \chi \le 1, \qquad \chi \equiv 1 \text{ on } \bar{B}(x, r/4), \qquad \{\chi \neq 0\} = B(x, r/2). \tag{1}$$

Such a χ exists on any inner product space, in particular on $\mathbb{R}^n$: take the standard radial bump.

Step 2: the modified test function. Put

$$K := M_u - (u-\varphi)(x), \qquad \tilde{\varphi}(y) := \chi(y)\varphi(y) + K\big(1 - \chi(y)\big) \qquad (y \in \mathbb{R}^n). \tag{2}$$

We check the three properties needed to feed $\tilde{\varphi}$ to `IsViscSubsol`.

(2a) $\tilde{\varphi} \in C^2(\Omega)$. Let $y \in \Omega$. If $|y - x| > r/2$, then on the open set $\{z : |z - x| > r/2\}$, which contains y, we have $\chi = 0$ by (1), hence $\tilde{\varphi} \equiv K$ there, so $\tilde{\varphi}$ is C^2 near y. If $|y - x| \le r/2$, then $y \in B(x,r)$, which is open, so φ is C^2 near y; since χ is smooth, $\tilde{\varphi} = \chi\varphi + K(1-\chi)$ is C^2 near y. In both cases $\tilde{\varphi}$ is C^2 on a neighbourhood of y, and $y \in \Omega$ was arbitrary. This is the step for which the two radii of (1) are needed: χ is supported strictly inside the ball on which φ is known to be C^2, so the product is never formed where φ might fail to be regular.

(2b) $x \in \operatorname{argmax}_\Omega(u - \tilde{\varphi})$. First, $\chi(x) = 1$ by (1), so $\tilde{\varphi}(x) = \varphi(x)$ and the value to beat is $(u-\varphi)(x)$. Next, for every y,

$$u(y) - \tilde{\varphi}(y) = \chi(y)\big(u(y) - \varphi(y)\big) + \big(1 - \chi(y)\big)\big(u(y) - K\big), \tag{3}$$

by (2) and a rearrangement. By (1) the two coefficients $\chi(y)$ and $1 - \chi(y)$ are non-negative and sum to 1, so (3) exhibits $u(y) - \tilde{\varphi}(y)$ as a convex combination of the two entries, and it therefore suffices to bound each entry by $(u-\varphi)(x)$ whenever its coefficient is non-zero.

For the second entry, let $y \in \Omega$. By the choice of K in (2) and $u(y) \le M_u$,

$$u(y) - K = u(y) - M_u + (u-\varphi)(x) \;\le\; (u-\varphi)(x). \tag{4}$$

For the first entry we may assume $\chi(y) \neq 0$, and then $y \in B(x, r/2) \subset B(x,r)$ by (1), so the local hypothesis $u - \varphi \le (u-\varphi)(x)$ on $B(x,r)$ applies and gives

$$u(y) - \varphi(y) \;\le\; (u-\varphi)(x). \tag{5}$$

If $\chi(y) = 0$, (3) reduces to $u(y) - K$ and (4) finishes; otherwise multiplying (5) by $\chi(y) \geq 0$ and (4) by $1 - \chi(y) \geq 0$ and adding gives $u(y) - \tilde{\varphi}(y) \leq (u - \varphi)(x) = u(x) - \tilde{\varphi}(x)$. Since $y \in \Omega$ was arbitrary, x maximizes $u - \tilde{\varphi}$ over Ω. Note that the hypothesis was used only where $\chi \neq 0$, i.e. only on $B(x, r/2)$: the local touching is exactly enough.

(2c) (D, B) is a second-order jet of $\tilde{\varphi}$ at x. By (1), $\chi \equiv 1$ on $\bar{B}(x, r/4)$, so $\tilde{\varphi} = \varphi$ there; in particular $\tilde{\varphi}(x) = \varphi(x)$. The symmetry of B is part of the hypothesis and is unchanged. For the Peano estimate, let $\varepsilon > 0$ and let $\delta > 0$ be given by `TwiceDiffAt` for φ, (D, B) and ε. Then $\min\{\delta, r/4\} > 0$, and for $|y - x| < \min\{\delta, r/4\}$ we have $\tilde{\varphi}(y) = \varphi(y)$ and $\tilde{\varphi}(x) = \varphi(x)$, so

$$\left|\tilde{\varphi}(y) - \tilde{\varphi}(x) - D(y-x) - \tfrac{1}{2}B(y-x, y-x)\right| = \left|\varphi(y) - \varphi(x) - D(y-x) - \tfrac{1}{2}B(y-x, y-x)\right| \leq \varepsilon |y-x|^2.$$

Thus (D, B) is a second-order jet of $\tilde{\varphi}$ at x. This is where the inner radius is used: χ is identically 1, not merely non-zero, near x, so the jet at x is literally unchanged and the conclusion is a statement about φ's jet.

Step 3: conclusion of (1a). By (2a), (2b), (2c) and hgrad $\sigma\ x\ D \neq 0$, the function $\tilde{\varphi}$ is admissible for `IsViscSubsol` at x with the jet (D, B), whence Lop $\sigma\ A\ H\ x\ D\ B \leq 0$. This is (1a).

Step 4: proof of (1b). Let $V \subset \Omega$ be open. Upper semi-continuity of u on V follows from upper semi-continuity on Ω by restriction. Let $x \in V$, let $\varphi \in C^2(V)$ with $x \in \operatorname{argmax}_V(u - \varphi)$, and let (D, B) be a second-order jet of φ at x with hgrad $\sigma\ x\ D \neq 0$. Since V is open there is $r > 0$ with $B(x, r) \subset V$. Then $B(x, r) \subset \Omega$, $\varphi \in C^2(B(x, r))$ by restriction, and $u - \varphi \leq (u - \varphi)(x)$ on $B(x, r)$ because $B(x, r) \subset V$. So (1a) applies and yields Lop $\sigma\ A\ H\ x\ D\ B \leq 0$. As x, φ and (D, B) were arbitrary, u is a viscosity sub-solution in V in the sense of `IsViscSubsol`.

Step 5: the super-solution half. Part (2) is proved by the same construction with the inequalities reversed, and it is not a formal consequence of part (1) (our `IsViscSupersol` is a separate predicate, and $-u$ is not in play). Take $K := m_u - (u - \varphi)(x)$, so that $u(y) - K = u(y) - m_u + (u - \varphi)(x) \geq (u - \varphi)(x)$ for $y \in \Omega$, which is (4) reversed; (5) is reversed by hypothesis; (3) is an identity and unchanged, and the same convex-combination argument now bounds $u - \tilde{\varphi}$ from below by $(u - \varphi)(x)$ on Ω, i.e. $x \in \operatorname{argmin}_\Omega(u - \tilde{\varphi})$. Steps (2a), (2c) do not involve u and are verbatim. Feeding $\tilde{\varphi}$ to `IsViscSupersol` gives $0 \leq$ Lop $\sigma\ A\ H\ x\ D\ B$, which is (2a) of the statement, and Step 4 with argmin in place of argmax and lower semi-continuity in place of upper gives (2b). □

2.5. `ClassicalAtTwiceDiff`.

Theorem 2.5 (Remark 2.2)**.** *Let $\mathbb{G}$ be a sub-elliptic setting (`CarnotSetting`) with frame $\sigma = [\sigma^1, \dots, \sigma^m]$, let $A : \mathbb{R}^m \to \mathbb{R}^{m \times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$, let $\Omega \subset \mathbb{R}^n$ be a bounded open set, and let $u : \mathbb{R}^n \to \mathbb{R}$ be continuous on $\bar{\Omega}$. Let $x \in \Omega$ and let (D, B) be a second-order jet of u at x (`TwiceDiffAt`) with* hgrad $\sigma\ x\ D \neq 0$ *(`hgrad`). Then*

1. *if u is a viscosity sub-solution of $\mathcal{L}u = 0$ in Ω (`IsViscSubsol`), then* Lop $\sigma\ A\ H\ x\ D\ B \leq 0$ *(`Lop`);*
2. *if u is a viscosity super-solution of $\mathcal{L}u = 0$ in Ω (`IsViscSupersol`), then* $0 \leq$ Lop $\sigma\ A\ H\ x\ D\ B$.

Proof. This is Remark 2.2. We prove (1); (2) is the same argument with the three inequalities reversed, and is recorded separately because `IsViscSupersol` is a separate predicate (the proof of Proposition 3.3, Step 3, uses (1) for u and (2) for v).

Step 0: u is bounded on Ω. Since Ω is bounded, $\bar{\Omega}$ is closed and bounded in $\mathbb{R}^n$, hence compact; u is continuous on $\bar{\Omega}$, so $|u| \leq C_u$ on $\bar{\Omega}$ for some C_u, and in particular $-C_u \leq u \leq C_u$ on $\Omega \subset \bar{\Omega}$. This is exactly the hypothesis that `ViscTestLocalization` requires, and it is the reason this node carries Ω bounded and $u \in C(\bar{\Omega})$: without a bound on u the modification of the test function away from x cannot be built, and the remark is then not available as stated.

Step 1: the test functions. Let $\langle \cdot, \cdot \rangle$ be the Euclidean scalar product of $\mathbb{R}^n$, viewed as a continuous symmetric bilinear form J, so that $J(h, h) = |h|^2$. For $c \in \mathbb{R}$ put

(1)

$$B_c := B + cJ, \qquad \varphi_c(y) := u(x) + D(y-x) + \tfrac{1}{2}B_c(y-x, y-x) = u(x) + D(y-x) + \tfrac{1}{2}B(y-x, y-x) + \tfrac{c}{2}|y-x|^2.$$

Thus φ_c is the second-order Taylor polynomial of u at x — which is meaningful precisely because (D, B) is a second-order jet of u at x — plus $\frac{c}{2}|y-x|^2$; the paper's test function is $\varphi_{2\eta}$ for the sub-solution case and $\varphi_{-2\eta}$ for the super-solution case, with $\eta > 0$. Three facts about φ_c:

(a) $\varphi_c \in C^\infty(\mathbb{R}^n)$, being a polynomial of degree 2: it is the sum of a constant, a continuous linear functional applied to $y-x$, and a continuous bilinear form applied to $(y-x, y-x)$.
(b) $\varphi_c(x) = u(x)$, since $D(0) = 0$ and $B_c(0,0) = 0$.
(c) (D, B_c) is a second-order jet of φ_c at x, with no error term at all: by (1) and (b), $\varphi_c(y) - \varphi_c(x) - D(y-x) - \frac{1}{2}B_c(y-x, y-x) = 0$ for every y, so the Peano estimate of `TwiceDiffAt` holds with any δ for any $\varepsilon > 0$. The required symmetry of B_c follows from that of B (part of the hypothesis) and the symmetry of the scalar product.

Step 2: $\mathcal{L}$ is affine in the second-order jet, at fixed first jet. Set $\xi_0 := \mathrm{hgrad}\ \sigma\ x\ D$ and let

$$\mathcal{G} \in \mathbb{R}^{m\times m}, \qquad \mathcal{G}_{i,j} := \langle \sigma^i(x), \sigma^j(x)\rangle, \qquad T := \mathrm{Tr}\left(A(\xi_0)\,\mathcal{G}\right). \tag{2}$$

The point of the perturbation is that it does not touch the first jet: φ_c and u have the *same* 1-jet D at x, so by `hgrad` their horizontal gradients at x are the same vector ξ_0, and the numbers $A(\xi_0)$ and $H(\xi_0)$ occurring in $\mathcal{L}$ are literally unchanged. By `hhessS` and $B_c = B + cJ$,

$$\mathrm{hhessS}\ \sigma\ x\ D\ B_c = \Big(B_c\big(\sigma^i(x), \sigma^j(x)\big)\Big)_{i,j} + \mathcal{M}(x, D) = \mathrm{hhessS}\ \sigma\ x\ D\ B + c\,\mathcal{G},$$

the correction matrix $\mathcal{M}(x, D)$ of `Mfield` being unchanged because it depends only on the 1-jet D. Hence, by `Lop` and the linearity of $X \mapsto \mathrm{Tr}(A(\xi_0)X)$,

$$\mathrm{Lop}\ \sigma\ A\ H\ x\ D\ B_c = \mathrm{Lop}\ \sigma\ A\ H\ x\ D\ B - cT \qquad \text{for every } c \in \mathbb{R}. \tag{3}$$

This identity is our form of the paper's "$\mathcal{L}$ is continuous in $(\mathcal{X}\varphi, \mathcal{X}\mathcal{X}^s\varphi)$ at points with $\mathcal{X}\varphi \neq 0$". In the jet encoding the statement is stronger and elementary: because the perturbation leaves the horizontal gradient alone, no continuity of A or H is needed — the dependence on the second-order jet is affine, with the single constant T of (2). That is why this node does not carry the structure conditions of Section 1.2; A and H are arbitrary here. The non-vanishing $\xi_0 \neq 0$ is still essential, but as the admissibility condition on the test function, not as a continuity point.

Step 3: the perturbed test function touches u locally. Let $\eta > 0$. Since Ω is open there is $r_0 > 0$ with $B(x, r_0) \subset \Omega$. Apply `TwiceDiffAt` to u, (D, B) with accuracy $\varepsilon := \eta$: there is $\delta > 0$ such that

$$\left|u(y) - u(x) - D(y-x) - \tfrac{1}{2}B(y-x, y-x)\right| \;\le\; \eta\,|y-x|^2 \qquad \text{for } |y-x| < \delta. \tag{4}$$

Put $r := \min\{r_0, \delta\} > 0$, so $B(x, r) \subset \Omega$. For $y \in B(x, r)$, by (1), (b) and (4),

$$u(y) - \varphi_{2\eta}(y) = \Big(u(y) - u(x) - D(y-x) - \tfrac{1}{2}B(y-x, y-x)\Big) - \eta|y-x|^2 \;\le\; \eta|y-x|^2 - \eta|y-x|^2 = 0 = u(x) - \varphi_{2\eta}(x).$$

So $u - \varphi_{2\eta} \le (u - \varphi_{2\eta})(x)$ on $B(x, r)$: the touching is local, which is all the paper claims, and the accuracy $\varepsilon = \eta$ of the Peano expansion is exactly matched to the size of the added quadratic. This is the "$\eta > 0$" and the "δ from the Peano expansion" of Remark 2.2.

Step 4: the localization step. We now have, for the function $\varphi_{2\eta}$: $\varphi_{2\eta} \in C^2(B(x, r))$ by (a), $B(x, r) \subset \Omega$, $u - \varphi_{2\eta} \le (u - \varphi_{2\eta})(x)$ on $B(x, r)$ by Step 3, $(D, B_{2\eta})$ a second-order jet of $\varphi_{2\eta}$ at x by (c), and $\mathrm{hgrad}\ \sigma\ x\ D = \xi_0 \neq 0$ by hypothesis. Together with the bound on u of Step 0 and the sub-solution property, part (1a) of `ViscTestLocalization` applies and gives

$$\mathrm{Lop}\ \sigma\ A\ H\ x\ D\ B_{2\eta} \;\le\; 0.$$

This is the paper's clause "belongs to $\mathcal{A}_x^+(u, \Omega)$ after a modification away from x": the modification is carried out inside `ViscTestLocalization`, and invoking that node is precisely what writing the clause out amounts to. By (3) with $c = 2\eta$,

$$\mathrm{Lop}\ \sigma\ A\ H\ x\ D\ B \;\le\; 2\eta\,T \qquad \text{for every } \eta > 0. \tag{5}$$

Step 5: $\eta \to 0^+$. Let $\varepsilon > 0$ and take $\eta := \varepsilon/\big(2(|T|+1)\big) > 0$. Then

$$2\eta T \;\le\; 2\eta\,|T| \;=\; \frac{\varepsilon\,|T|}{|T|+1} \;\le\; \varepsilon,$$

so (5) gives Lop σ A H x D $B \le \varepsilon$. As $\varepsilon > 0$ was arbitrary, Lop σ A H x D $B \le 0$, which is (1). Note that the constant T of (2) is fixed once and for all by x, σ, A and D; it does not depend on η, which is what makes the limit trivial.

Step 6: the super-solution half. For (2), run Steps 3–5 with $\varphi_{-2\eta}$, i.e. the second-order Taylor polynomial *minus* $\eta|y-x|^2$, as Remark 2.2 prescribes. The left inequality of (4) gives, for $y \in B(x,r)$,

$$u(y)-\varphi_{-2\eta}(y) = \Big(u(y)-u(x)-D(y-x)-\tfrac{1}{2}B(y-x,y-x)\Big)+\eta|y-x|^2 \;\ge\; -\eta|y-x|^2+\eta|y-x|^2 = 0 = u(x)-\varphi_{-2\eta}(x)$$

so $(u-\varphi_{-2\eta})(x) \le u-\varphi_{-2\eta}$ on $B(x,r)$; with the lower bound $-C_u \le u$ of Step 0, part (2a) of `ViscTestLocalization` gives $0 \le$ Lop σ A H x D $B_{-2\eta}$, and (3) with $c = -2\eta$ turns this into $-2\eta T \le$ Lop σ A H x D B for every $\eta > 0$. The same choice of η as in Step 5 gives $-\varepsilon \le$ Lop σ A H x D B for every $\varepsilon > 0$, i.e. $0 \le$ Lop σ A H x D B. □

2.6. `ExpRelabelling`.

Lemma 2.6 (The relabelling of Proposition 3.3, Step 1)**.** *Let $\lambda > 0$ and $M \in \mathbb{R}$, and let*

$$\psi_\lambda(s) := s + \lambda e^s \qquad (s \in \mathbb{R}).$$

Then

(1) $\psi_\lambda \in C^2(\mathbb{R})$*;*
(2) $\psi_\lambda'(s) = 1 + \lambda e^s$ *for every s, and $\psi_\lambda' \ge 1$ everywhere;*
(3) $\psi_\lambda''(s) = \lambda e^s$ *for every s, and $\psi_\lambda'' \ge 0$ everywhere;*
(4) $\lambda e^{-M} > 0$*, and $\psi_\lambda''(s) \ge \lambda e^{-M}$ whenever $|s| \le M$;*
(5) $|\psi_\lambda(s) - s| \le \lambda e^M$ *whenever $|s| \le M$.*

Proof. **Why this ψ_λ and not the paper's.** Proposition 3.3, Step 1, uses the relabelling $\psi_\lambda(s) = s + \lambda(s-u_0)^2$ with $u_0 = \inf_\Omega u$, and asserts for it "$\psi_\lambda' \ge 1$ *on* $[u_0,\infty)$", $\psi_\lambda'' = 2\lambda$ and $\|u_\lambda - u\|_{L^\infty} \le 4\lambda\|u\|_{L^\infty}^2$. The paper is right, and the bound $\psi_\lambda' \ge 1$ it claims is a bound on $[u_0,\infty)$ only — which is all the paper needs, because ψ_λ is only ever composed with $u \ge u_0$. Our rendering of the composition property Section 2.3(4), however, requires $\psi' \ge 1$ and $\psi'' \ge 0$ *globally* on $\mathbb{R}$, and the quadratic $s+\lambda(s-u_0)^2$ has $\psi_\lambda'(s) = 1+2\lambda(s-u_0) < 1$ for $s < u_0$. So the paper's ψ_λ does not instantiate that statement. Rather than weaken it, we choose a relabelling that satisfies its hypotheses globally; $\psi_\lambda(s) = s + \lambda e^s$ does. The cost is that $\psi_\lambda'' = \lambda e^s$ is no longer the constant 2λ and has no positive global lower bound, which is why items (4) and (5) are stated on the range $|s| \le M$ of a bounded u: on $\bar\Omega$ compact and $u \in C(\bar\Omega)$ we have $|u| \le M$ for some M, so (4) supplies the one uniform constant $\lambda' := \lambda e^{-M}$ that Corollary 3.2 needs at every point, and (5) replaces the paper's $\|u_\lambda - u\|_{L^\infty} \le 4\lambda\|u\|_{L^\infty}^2$ by $\|\psi_\lambda \circ u - u\|_{L^\infty(\bar\Omega)} \le \lambda e^M$, which likewise tends to 0 as $\lambda \to 0^+$. No conclusion of Proposition 3.3 changes; only the explicit constants do.

(1) and (2). The identity function and $s \mapsto e^s$ are C^∞ on $\mathbb{R}$, hence so is $\psi_\lambda = \mathrm{id} + \lambda\exp$; in particular $\psi_\lambda \in C^2$. Differentiating, $\psi_\lambda'(s) = 1 + \lambda e^s$, by the sum rule, $\frac{d}{ds}s = 1$ and $\frac{d}{ds}e^s = e^s$. Since $\lambda > 0$ and $e^s > 0$ for every s, we get $\lambda e^s > 0$ and therefore $\psi_\lambda'(s) = 1 + \lambda e^s \ge 1$ for every $s \in \mathbb{R}$.

(3). Differentiating $\psi_\lambda'(s) = 1 + \lambda e^s$ once more gives $\psi_\lambda''(s) = \lambda e^s$, and $\lambda > 0$, $e^s > 0$ give $\psi_\lambda''(s) > 0 \ge 0$ for every s.

(4). $\lambda e^{-M} > 0$ because $\lambda > 0$ and $e^{-M} > 0$. Let $|s| \le M$. Then $-M \le s$, and exp is monotone increasing, so $e^{-M} \le e^s$; multiplying by $\lambda > 0$ and using (3),

$$\lambda e^{-M} \;\le\; \lambda e^s \;=\; \psi_\lambda''(s).$$

(5). Let $|s| \le M$. Then $s \le M$, so $e^s \le e^M$ by monotonicity of exp. Moreover $\psi_\lambda(s) - s = \lambda e^s$, which is ≥ 0 since $\lambda > 0$, so

$$\big|\psi_\lambda(s) - s\big| = \lambda e^s \;\le\; \lambda e^M.$$

□

2.7. `JensenTwiceDiffPoint`.

Theorem 2.7 (Maximal points of linear perturbations at which both functions are twice differentiable)**.** *Let $N \in \mathbb{N}$, let the two Euclidean cited results hold (`EuclidCitedResults`), let $\Omega \subseteq \mathbb{R}^N$ and let $u, v : \mathbb{R}^N \to \mathbb{R}$ be such that u is locally semi-convex on Ω and v is locally semi-concave on Ω, i.e. $-v$ is locally semi-convex on Ω (`LocallySemiConvexOn`). Let $x_1 \in \mathbb{R}^N$ and assume that x_1 is a strict local maximum point of $u - v$ inside Ω: there is $\nu > 0$ with $B_\nu(x_1) \subseteq \Omega$ and*

$$u(y) - v(y) \;<\; u(x_1) - v(x_1) \qquad \text{for every } y \in B_\nu(x_1) \text{ with } y \neq x_1.$$

Then there are sequences $(p_k)_{k\in\mathbb{N}}$ and $(z_k)_{k\in\mathbb{N}}$ in $\mathbb{R}^N$ with $p_k \to 0$ and $z_k \to x_1$ such that for every k:

1. *$z_k \in \Omega$;*
2. *z_k is a local maximum point of $y \mapsto u(y) - v(y) + \langle p_k, y\rangle$, where $\langle p, y\rangle = \sum_{i=1}^N p_i\, y_i$;*
3. *u has a second-order jet at z_k (`TwiceDiffAt`); and*
4. *v has a second-order jet at z_k.*

Proof. Write $w := u - v$ and fix $\nu > 0$ as in the hypothesis; put $W := B_\nu(x_1)$, an open set with $W \subseteq \Omega$.

Step 0: local semi-convexity restricts to open subsets. If $f : \mathbb{R}^N \to \mathbb{R}$ is locally semi-convex on Ω and $V \subseteq \Omega$ is open, then f is locally semi-convex on V. Indeed, let $x \in V$. By `LocallySemiConvexOn` there are $\rho > 0$ and $\Lambda \ge 0$ with $B_\rho(x) \subseteq \Omega$ and $f + \frac{\Lambda}{2}|\cdot|^2$ convex on $B_\rho(x)$. Since V is open there is $\sigma > 0$ with $B_\sigma(x) \subseteq V$. Put $\rho' := \min\{\rho, \sigma\} > 0$; then $B_{\rho'}(x) \subseteq V$, and $f + \frac{\Lambda}{2}|\cdot|^2$ is convex on the convex subset $B_{\rho'}(x)$ of $B_\rho(x)$. So the requirement of `LocallySemiConvexOn` holds at x for V. We use this three times below, with $V := W$.

Step 1: $w = u - v$ is locally semi-convex on Ω, hence on W. Let $x \in \Omega$. By `LocallySemiConvexOn` applied to u there are $\rho_1 > 0$ and $\Lambda_1 \ge 0$ with $B_{\rho_1}(x) \subseteq \Omega$ and $u + \frac{\Lambda_1}{2}|\cdot|^2$ convex on $B_{\rho_1}(x)$; applied to $-v$ it gives $\rho_2 > 0$, $\Lambda_2 \ge 0$ with $B_{\rho_2}(x) \subseteq \Omega$ and $-v + \frac{\Lambda_2}{2}|\cdot|^2$ convex on $B_{\rho_2}(x)$. Put $\rho := \min\{\rho_1, \rho_2\} > 0$ and $\Lambda := \Lambda_1 + \Lambda_2 \ge 0$. Both functions are convex on the convex subset $B_\rho(x)$ of their respective balls, and

$$w + \frac{\Lambda}{2}|\cdot|^2 = \Big(u + \frac{\Lambda_1}{2}|\cdot|^2\Big) + \Big(-v + \frac{\Lambda_2}{2}|\cdot|^2\Big)$$

pointwise, a sum of two convex functions on $B_\rho(x)$, hence convex there. Since $B_\rho(x) \subseteq \Omega$, the requirement of `LocallySemiConvexOn` holds at x. Thus w is locally semi-convex on Ω, and by Step 0 also on W.

Step 2: Jensen's lemma applies at x_1. The clause JENSEN of `EuclidCitedResults` requires two hypotheses for the function w at the point x_1. The first is that w be locally semi-convex on some ball $B_{\nu'}(x_1)$ with $\nu' > 0$: Step 1 gives it with $\nu' := \nu$, since $W = B_\nu(x_1)$. The second is that x_1 be a strict local maximum point of w: there is $\nu' > 0$ with $w(y) < w(x_1)$ for every $y \in B_{\nu'}(x_1)$ with $y \neq x_1$; this is the hypothesis of the theorem, again with $\nu' := \nu$.

Hence JENSEN furnishes $\rho_0 > 0$ and $\delta_0 > 0$ such that for all $\rho \in (0, \rho_0)$ and all $\delta \in (0, \delta_0)$ the set

$$J(\rho, \delta) := \bigcup_{|p|<\delta} \Big\{ z \in B_\rho(x_1) \;:\; w(y) + \langle p, y\rangle \le w(z) + \langle p, z\rangle \text{ for all } y \in B_\rho(x_1) \Big\}$$

has positive Lebesgue measure: $|J(\rho, \delta)| > 0$. (This is the set $J_{\rho,\delta}(x_1)$ of (2.12), written out: the inner set is $B_\rho(x_1) \cap \operatorname{argmax}_{B_\rho(x_1)}(w_p)$ for $w_p = w + \langle p, \cdot\rangle$.)

Step 3: the two Aleksandrov null sets. By Step 0, u is locally semi-convex on the open set W, and so is $-v$. The clause ALEKSANDROV of `EuclidCitedResults`, applied with the open set

W to the locally semi-convex function u, says that for Lebesgue-almost every $z \in \mathbb{R}^N$, if $z \in W$ then u has a second-order jet at z. Equivalently, the set

$$\mathcal{N}_u := \{z \in W \ : \ u \text{ has no second-order jet at } z\}$$

is Lebesgue-null, $|\mathcal{N}_u| = 0$. Applying the same clause with the open set W to the locally semi-convex function $-v$, the set

$$\mathcal{N}_v := \{z \in W \ : \ -v \text{ has no second-order jet at } z\}$$

is Lebesgue-null, $|\mathcal{N}_v| = 0$. Consequently

$$|\mathcal{N}_u \cup \mathcal{N}_v| \ \le \ |\mathcal{N}_u| + |\mathcal{N}_v| \ = \ 0. \tag{1}$$

This is the point the paper compresses: its closing sentence applies Aleksandrov's theorem to one function, whereas the conclusion needed here concerns the pair (u, v), so two exceptional sets must be removed.

We also record the elementary transfer from $-v$ to v: *if (D, B) is a second-order jet of $-v$ at z, then $(-D, -B)$ is a second-order jet of v at z.* Indeed $-B$ is symmetric because B is, and for every y

$$v(y)-v(z)-(-D)(y-z)-\tfrac{1}{2}(-B)(y-z,y-z) = -\Big[(-v)(y)-(-v)(z)-D(y-z)-\tfrac{1}{2}B(y-z,y-z)\Big],$$

so the two quantities have the same absolute value and the Peano estimate of `TwiceDiffAt` for $-v$ with (D, B) is literally the estimate for v with $(-D, -B)$, with the same δ for each ε. Hence

$$z \in W \setminus \mathcal{N}_v \implies v \text{ has a second-order jet at } z. \tag{2}$$

Step 4: construction of the two sequences. For $k \in \mathbb{N}$ put

$$\rho_k := \tfrac{1}{2}\min\Big\{\rho_0,\ \nu,\ \tfrac{1}{k+1}\Big\} > 0, \qquad \delta_k := \tfrac{1}{2}\min\Big\{\delta_0,\ \tfrac{1}{k+1}\Big\} > 0.$$

Then $0 < \rho_k < \rho_0$, $0 < \delta_k < \delta_0$, $\rho_k \le \frac{1}{2(k+1)}$, $\delta_k \le \frac{1}{2(k+1)}$, and $\rho_k < \nu$, so that $B_{\rho_k}(x_1) \subseteq B_\nu(x_1) = W$.

By Step 2, $|J(\rho_k, \delta_k)| > 0$. If we had $J(\rho_k, \delta_k) \subseteq \mathcal{N}_u \cup \mathcal{N}_v$, then monotonicity of Lebesgue measure together with (1) would give $|J(\rho_k, \delta_k)| \le 0$, contradicting positivity. Hence

$$J(\rho_k, \delta_k) \setminus \big(\mathcal{N}_u \cup \mathcal{N}_v\big) \ \neq \ \emptyset,$$

and we may choose z_k in it. By the definition of $J(\rho_k, \delta_k)$ there is a vector p_k with $|p_k| < \delta_k$ such that

$$z_k \in B_{\rho_k}(x_1) \qquad \text{and} \qquad w(y) + \langle p_k, y\rangle \le w(z_k) + \langle p_k, z_k\rangle \ \text{ for all } y \in B_{\rho_k}(x_1). \tag{3}$$

We verify the four assertions for this choice.

(i) $z_k \in \Omega$. $z_k \in B_{\rho_k}(x_1) \subseteq W \subseteq \Omega$.

(ii) z_k is a local maximum point of $w + \langle p_k, \cdot\rangle$. The set $B_{\rho_k}(x_1)$ is open and contains z_k, so it contains a ball $B_\sigma(z_k)$ with $\sigma > 0$; by (3), $w(y) + \langle p_k, y\rangle \le w(z_k) + \langle p_k, z_k\rangle$ for every $y \in B_\sigma(z_k)$. Since $w = u - v$, this says precisely that $y \mapsto u(y) - v(y) + \langle p_k, y\rangle$ has a local maximum at z_k.

(iii) u has a second-order jet at z_k. We have $z_k \in W$ and $z_k \notin \mathcal{N}_u$, so by the definition of $\mathcal{N}_u$ the function u has a second-order jet at z_k.

(iv) v has a second-order jet at z_k. We have $z_k \in W$ and $z_k \notin \mathcal{N}_v$, so (2) applies and v has a second-order jet at z_k.

Convergence. From $z_k \in B_{\rho_k}(x_1)$ we get $|z_k - x_1| < \rho_k \le \frac{1}{2(k+1)} \to 0$, so $z_k \to x_1$; and $|p_k| < \delta_k \le \frac{1}{2(k+1)} \to 0$, so $p_k \to 0$.

This produces the required sequences and completes the proof.

Remark. The hypothesis that the maximum be *strict* is used only through JENSEN, where it is indispensable: if w were constant near x_1 then w_p would have no interior maximum in $B_\rho(x_1)$ for $p \neq 0$ and $J(\rho, \delta)$ could be a single point, of measure zero. In Proposition Proposition 3.3 the strictness is arranged by a quartic penalization before this node is applied. The clause JENSEN is used at exactly the strength at which it is stated — positivity of the measure of $J(\rho, \delta)$ — and the twice-differentiability of the pair is obtained here, not assumed there. □

2.8. `SemiConvexLocLipschitz`.

Theorem 2.8 (Section 2.3(3): the local Lipschitz bound)**.** *Let $N \in \mathbb{N}$, let $\Omega \subseteq \mathbb{R}^N$ and let $w : \mathbb{R}^N \to \mathbb{R}$ be locally semi-convex on Ω (`LocallySemiConvexOn`). Then*

(1) *w is continuous on Ω;*
(2) *w is locally Lipschitz on Ω: for every $x \in \Omega$ there are $r > 0$ and $C \geq 0$ such that $B_r(x) \subseteq \Omega$ and*
$$|w(y) - w(z)| \leq C\,|y - z| \qquad \text{for all } y, z \in B_r(x);$$
(3) *w is Lipschitz on every compact subset of Ω: for every compact $K \subseteq \Omega$ there is $C \geq 0$ with $|w(y) - w(z)| \leq C\,|y - z|$ for all $y, z \in K$;*
(4) *the derivative of w is bounded on compact subsets of Ω wherever it exists: for every compact $K \subseteq \Omega$ there is $C \geq 0$ such that $\|D\| \leq C$ whenever $x \in K$ and D is the Fréchet derivative of w at x.*

Here $B_r(x)$ is the Euclidean open ball and $|\cdot|$ the Euclidean norm.

Proof. Throughout we use one standard fact of convex analysis, which is not a tablet node: a convex function that is finite on an open convex subset of a finite-dimensional normed space is continuous there. We prove (2) first, then deduce (1), (3) and (4) from it.

Step 0: Ω is open. By `LocallySemiConvexOn`, every $x \in \Omega$ admits $\rho > 0$ with $B_\rho(x) \subseteq \Omega$. Hence Ω is open. (This is used only to know that the balls produced below are neighbourhoods inside Ω.)

Step 1: proof of (2). Fix $x \in \Omega$. By `LocallySemiConvexOn` there are $\rho > 0$ and $\Lambda \geq 0$ with $B_\rho(x) \subseteq \Omega$ such that
$$g(y) := w(y) + \frac{\Lambda}{2}|y|^2$$
is convex on $B_\rho(x)$. Put $s := \rho/2$, so that the closed ball $\overline{B}_s(x)$ satisfies $\overline{B}_s(x) \subseteq B_\rho(x)$ (if $|y - x| \leq s = \rho/2$ then $|y - x| < \rho$).

(1a) g is bounded on $\overline{B}_s(x)$. $B_\rho(x)$ is open and convex and g is convex and real-valued on it, so by the standard fact quoted above g is continuous on $B_\rho(x)$. The set $\overline{B}_s(x)$ is closed and bounded in the finite-dimensional space $\mathbb{R}^N$, hence compact, and it is contained in $B_\rho(x)$. A continuous real function on a non-empty compact set is bounded, so
$$M := \sup_{y \in \overline{B}_s(x)} |g(y)| < \infty,$$
and $M \geq 0$ since $\overline{B}_s(x) \ni x$ is non-empty.

(1b) g is Lipschitz with constant $4M/s$ on $B_{s/2}(x)$. Let $y, z \in B_{s/2}(x)$. If $y = z$ the inequality $|g(y) - g(z)| \leq (4M/s)|y - z|$ is trivial, so assume $y \neq z$ and set
$$\sigma := |y - z| > 0, \qquad e := \frac{y - z}{\sigma}, \qquad q := y + \frac{s}{2}\,e.$$
Then $|e| = 1$ and
$$|q - x| \leq |y - x| + \frac{s}{2} < \frac{s}{2} + \frac{s}{2} = s,$$
so $q \in \overline{B}_s(x) \subseteq B_\rho(x)$; also $y, z \in B_{s/2}(x) \subseteq B_\rho(x)$. Put
$$t := \frac{\sigma}{\sigma + s/2} \in (0, 1).$$
Then $y = (1-t)z + t\,q$: indeed $(1-t)z + tq = z + t(q - z)$ and $q - z = (y - z) + \frac{s}{2}e = \sigma e + \frac{s}{2}e = (\sigma + \frac{s}{2})e$, so $t(q - z) = \frac{\sigma}{\sigma + s/2}(\sigma + \frac{s}{2})e = \sigma e = y - z$. Since $B_\rho(x)$ is convex and g is convex on it,
$$g(y) \leq (1 - t)g(z) + t\,g(q), \qquad \text{i.e.} \qquad g(y) - g(z) \leq t\big(g(q) - g(z)\big) \leq t \cdot 2M,$$
using $|g(q)| \leq M$ and $|g(z)| \leq M$ (both q and z lie in $\overline{B}_s(x)$, as $|z - x| < s/2 \leq s$). Now
$$t \cdot 2M = \frac{2M\sigma}{\sigma + s/2} \leq \frac{2M\sigma}{s/2} = \frac{4M}{s}\,\sigma,$$

because $\sigma + s/2 \geq s/2 > 0$ and $2M\sigma \geq 0$. Hence $g(y) - g(z) \leq (4M/s)|y - z|$. Exchanging the roles of y and z (which exchanges e for $-e$ and leaves σ unchanged) gives $g(z) - g(y) \leq (4M/s)|y - z|$, so

$$|g(y) - g(z)| \leq \frac{4M}{s}\,|y - z| \qquad \text{for all } y, z \in B_{s/2}(x). \tag{1}$$

(1c) subtracting the quadratic term. For $y, z \in B_{s/2}(x)$ we have $|y| \leq |x| + s/2$ and $|z| \leq |x| + s/2$, so

$$\frac{\Lambda}{2}\big|\,|y|^2 - |z|^2\big| = \frac{\Lambda}{2}\big|\,|y| - |z|\,\big| \cdot \big(|y| + |z|\big) \leq \frac{\Lambda}{2}\,|y - z| \cdot 2\Big(|x| + \frac{s}{2}\Big) = \Lambda\Big(|x| + \frac{s}{2}\Big)|y - z|,$$

where $\big|\,|y| - |z|\,\big| \leq |y - z|$ is the triangle inequality. Since $w = g - \frac{\Lambda}{2}|\cdot|^2$, combining with (1),

$$|w(y) - w(z)| \leq |g(y) - g(z)| + \frac{\Lambda}{2}\big|\,|y|^2 - |z|^2\big| \leq \Big[\frac{4M}{s} + \Lambda\Big(|x| + \frac{s}{2}\Big)\Big]\,|y - z|.$$

So (2) holds with

$$r := \frac{s}{2} = \frac{\rho}{4} > 0, \qquad C := \frac{4M}{s} + \Lambda\Big(|x| + \frac{s}{2}\Big) \;\geq\; 0,$$

and $B_r(x) \subseteq B_\rho(x) \subseteq \Omega$.

Step 2: proof of (1). Let $x \in \Omega$ and take r, C from Step 1. Then $|w(y) - w(x)| \leq C|y - x|$ for $y \in B_r(x)$, so $w(y) \to w(x)$ as $y \to x$; thus w is continuous at x (as a function on $\mathbb{R}^N$, hence a fortiori its restriction to Ω is continuous at x within Ω). As $x \in \Omega$ was arbitrary, w is continuous on Ω.

Step 3: proof of (3). Let $K \subseteq \Omega$ be compact. If $K = \emptyset$ take $C := 0$; the assertion is vacuous. So assume $K \neq \emptyset$.

By Step 1, for each $x \in K \subseteq \Omega$ choose $r_x > 0$ and $C_x \geq 0$ with $B_{r_x}(x) \subseteq \Omega$ and w C_x-Lipschitz on $B_{r_x}(x)$. The balls $\{B_{r_x/3}(x)\}_{x \in K}$ form an open cover of K, so by compactness there are finitely many points $x_1, \ldots, x_n \in K$ (with $n \geq 1$, since $K \neq \emptyset$) such that $K \subseteq \bigcup_{i=1}^n B_{r_i/3}(x_i)$, where $r_i := r_{x_i}$ and $C_i := C_{x_i}$. Put

$$r := \min_{1 \leq i \leq n} \frac{r_i}{3} > 0, \qquad C_0 := \max_{1 \leq i \leq n} C_i \;\geq 0.$$

(3a) the short-range estimate. Let $y, z \in K$ with $|y - z| < r$. Choose i with $y \in B_{r_i/3}(x_i)$. Then

$$|z - x_i| \leq |z - y| + |y - x_i| < r + \frac{r_i}{3} \leq \frac{r_i}{3} + \frac{r_i}{3} < r_i,$$

so both y and z lie in $B_{r_i}(x_i)$, on which w is C_i-Lipschitz. Hence $|w(y) - w(z)| \leq C_i|y - z| \leq C_0|y - z|$.

(3b) the long-range estimate. By Step 2, w is continuous on Ω, hence its restriction to the non-empty compact set K is continuous and bounded: put $M_K := \sup_K |w| < \infty$, $M_K \geq 0$. If $y, z \in K$ with $|y - z| \geq r$, then

$$|w(y) - w(z)| \leq |w(y)| + |w(z)| \leq 2M_K = \frac{2M_K}{r} \cdot r \leq \frac{2M_K}{r}\,|y - z|.$$

Taking $C := \max\{C_0,\, 2M_K/r\} \geq 0$, both cases give $|w(y) - w(z)| \leq C|y - z|$ for all $y, z \in K$, which is (3).

Step 4: proof of (4). Let $K \subseteq \Omega$ be compact. If $K = \emptyset$ take $C := 0$. Otherwise, with $x_1, \ldots, x_n$, r_i, C_i and $C_0 = \max_i C_i$ exactly as in Step 3, put $C := C_0$.

Let $x \in K$ and let D be the Fréchet derivative of w at x. Choose i with $x \in B_{r_i/3}(x_i)$, and let $\eta := r_i/3 - |x - x_i| > 0$, so that $B_\eta(x) \subseteq B_{r_i}(x_i)$: indeed if $|y - x| < \eta$ then $|y - x_i| \leq |y - x| + |x - x_i| < r_i/3 \leq r_i$. Since w is C_i-Lipschitz on $B_{r_i}(x_i)$, it is C_i-Lipschitz on $B_\eta(x)$.

Fix a unit vector $e \in \mathbb{R}^N$ and let $0 < \tau < \eta$. Then $x + \tau e \in B_\eta(x)$ and

$$\big|w(x + \tau e) - w(x)\big| \leq C_i\,|\tau e| = C_i\,\tau.$$

By differentiability, $w(x+\tau e)-w(x)=\tau\,D(e)+o(\tau)$ as $\tau\to 0^+$; dividing the displayed inequality by $\tau>0$ gives

$$\Big|D(e)+\frac{o(\tau)}{\tau}\Big|\le C_i,$$

and letting $\tau\to 0^+$ yields $|D(e)|\le C_i$. As e ranges over the unit sphere, this gives $\|D\|=\sup_{|e|=1}|D(e)|\le C_i\le C_0=C$, which is (4).

Remark on strength. The paper states (3) with the explicit constant $c(N)\big(\operatorname{osc}_\Omega w+\Lambda\sup_\Omega|x|\operatorname{diam}\Omega\big)\operatorname{dist}(K,\partial\Omega$ Only the finiteness of a Lipschitz constant on compacts, and the resulting bound on ∇w, are ever used downstream, so the statement above asserts existence of a finite constant rather than that particular formula; Step 1 nevertheless exhibits a concrete admissible constant on each ball. The hypothesis is the local notion of semi-convexity, as fixed for this formalization, which is also why the conclusion of (2) is stated on a ball contained in Ω rather than with one constant for all of Ω. □

2.9. `SemiConvexGradPartialCont`.

Theorem 2.9 (Section 2.3(2): partial continuity of the gradient)**.** *Let $N\in\mathbb{N}$, let $\Omega\subseteq\mathbb{R}^N$ and let $w:\mathbb{R}^N\to\mathbb{R}$ be locally semi-convex on Ω* or *locally semi-concave on Ω — that is, either w or $-w$ is locally semi-convex on Ω (`LocallySemiConvexOn`). Let $x\in\Omega$ and let $(x_k)_{k\in\mathbb{N}}$ be a sequence in Ω with $x_k\to x$. Assume w is differentiable at x, with Fréchet derivative D, and differentiable at each x_k, with Fréchet derivative D_k. Then $D_k\to D$ in the operator norm.*

Proof. **Step 0: reduction to the semi-convex case.** Suppose the statement is proved when w is locally semi-convex on Ω, and suppose instead that $-w$ is locally semi-convex on Ω. The function $-w$ is differentiable at x with derivative $-D$ and at each x_k with derivative $-D_k$. The proved case, applied to $-w$, gives $-D_k\to -D$, hence $D_k\to D$ because $\|D_k-D\|=\|(-D_k)-(-D)\|$. So we may and do assume from now on that w is locally semi-convex on Ω.

Step 1: the convex function g and its data. By `LocallySemiConvexOn` applied at x there are $\rho>0$ and $\Lambda\ge 0$ with $B_\rho(x)\subseteq\Omega$ such that

$$g(y):=w(y)+\frac{\Lambda}{2}|y|^2$$

is convex on $B_\rho(x)$. By part (2) of `SemiConvexLocLipschitz` applied at x there are $r_0>0$ and $C_0\ge 0$ with $B_{r_0}(x)\subseteq\Omega$ and $|w(y)-w(z)|\le C_0|y-z|$ for all $y,z\in B_{r_0}(x)$. Put

$$r:=\min\{\rho,r_0\}>0,\qquad L:=C_0+\Lambda\big(|x|+r\big)\ \ge 0.$$

Then g is L-Lipschitz on $B_r(x)$: for $y,z\in B_r(x)$ we have $|y|,|z|\le|x|+r$, so

$$(1)\quad |g(y)-g(z)|\le|w(y)-w(z)|+\frac{\Lambda}{2}\big|\,|y|^2-|z|^2\big|\le C_0|y-z|+\frac{\Lambda}{2}\big(|y|+|z|\big)\big|\,|y|-|z|\,\big|\le L\,|y-z|,$$

using $\big|\,|y|-|z|\,\big|\le|y-z|$. In particular g is continuous on $B_r(x)$.

Since $y\mapsto\frac{\Lambda}{2}|y|^2$ is differentiable everywhere with derivative $h\mapsto\Lambda\langle y,h\rangle$ at y, the differentiability hypotheses on w transfer to g: g is differentiable at x with derivative

$$q:=D+\Lambda\langle x,\cdot\rangle,$$

and, for each k, differentiable at x_k with derivative

$$(2)\qquad q_k:=D_k+\Lambda\langle x_k,\cdot\rangle.$$

Finally, since $x_k\to x$, there is k_0 with $|x_k-x|<r/3$ for all $k\ge k_0$; all statements below about x_k are for $k\ge k_0$, which suffices for a statement about limits.

Step 2: the derivative of a convex function is a subgradient. We record the following standard fact of convex analysis, with its proof, since it is used twice. *Let U be open and convex, let g be convex on U, let $p\in U$ and suppose g is differentiable at p with derivative ℓ. Then*

$$(3)\qquad g(y)\ \ge\ g(p)+\ell(y-p)\qquad\text{for all } y\in U.$$

Indeed, fix $y \in U$ and consider $\varphi(t) := g\big(p+t(y-p)\big)$ for $t \in [0,1]$, which is defined because U is convex, and is a convex function of t. For $0 < t \le 1$ convexity of φ gives $\varphi(t) \le (1-t)\varphi(0)+t\varphi(1)$, i.e.

$$\frac{\varphi(t)-\varphi(0)}{t} \;\le\; \varphi(1)-\varphi(0) = g(y)-g(p).$$

On the other hand, differentiability of g at p gives $\varphi(t)-\varphi(0) = t\,\ell(y-p)+o(t)$ as $t \to 0^+$, so the left-hand side tends to $\ell(y-p)$. Passing to the limit in the displayed inequality yields (3).

Step 3: the sequence (q_k) is bounded, by L. Let $k \ge k_0$ and let $e \in \mathbb{R}^N$ with $|e| = 1$. Put $\tau := r/3$ and $y := x_k + \tau e$. Then

$$|y-x| \le |x_k - x| + \tau < \frac{r}{3} + \frac{r}{3} < r,$$

so $y \in B_r(x)$, and also $x_k \in B_r(x)$. Applying (3) on the open convex set $B_r(x)$ (on which g is convex, since $B_r(x) \subseteq B_\rho(x)$) at the differentiability point $p = x_k$, with derivative q_k,

$$g(y) \;\ge\; g(x_k) + q_k(y-x_k) = g(x_k) + \tau\, q_k(e).$$

Hence, using (1), $\tau\, q_k(e) \le g(y)-g(x_k) \le L|y-x_k| = L\tau$, and dividing by $\tau > 0$ gives $q_k(e) \le L$. Applying this to $-e$ gives $q_k(e) \ge -L$, so $|q_k(e)| \le L$ for every unit vector e, i.e.

$$\|q_k\| \;\le\; L \qquad \text{for all } k \ge k_0. \tag{4}$$

Step 4: q is the only possible subsequential limit. Suppose $(q_{k_j})_j$ is a subsequence with $q_{k_j} \to \bar q$ in the operator norm. Fix $y \in B_r(x)$. By (3) applied at $p = x_{k_j}$,

$$g(y) \;\ge\; g(x_{k_j}) + q_{k_j}\big(y - x_{k_j}\big) \qquad \text{for all } j.$$

As $j \to \infty$ we have $x_{k_j} \to x$, hence $g(x_{k_j}) \to g(x)$ because g is continuous on $B_r(x)$ by (1), and $q_{k_j}(y-x_{k_j}) \to \bar q(y-x)$ because

$$\big|q_{k_j}(y-x_{k_j}) - \bar q(y-x)\big| \le \big|(q_{k_j}-\bar q)(y-x_{k_j})\big| + \big|\bar q(x-x_{k_j})\big| \le \|q_{k_j}-\bar q\|\,\big|y-x_{k_j}\big| + \|\bar q\|\,\big|x-x_{k_j}\big| \;\to\; 0,$$

the first factor being bounded since $|y-x_{k_j}| \le |y-x| + |x-x_{k_j}|$ is bounded. Passing to the limit,

$$g(y) \;\ge\; g(x) + \bar q(y-x) \qquad \text{for all } y \in B_r(x). \tag{5}$$

Now use that g is differentiable at x with derivative q. Let e be a unit vector and $0 < \tau < r$; taking $y := x+\tau e$ in (5) gives $g(x+\tau e) - g(x) \ge \tau\,\bar q(e)$, while differentiability gives $g(x+\tau e) - g(x) = \tau\, q(e) + o(\tau)$. Hence $\tau q(e) + o(\tau) \ge \tau \bar q(e)$; dividing by $\tau > 0$ and letting $\tau \to 0^+$ gives $q(e) \ge \bar q(e)$. Applying this with $-e$ gives $q(e) \le \bar q(e)$, so $q(e) = \bar q(e)$ for every unit vector e and therefore $\bar q = q$.

Step 5: $q_k \to q$. Suppose not. Then there are $\eta > 0$ and a subsequence (q_{k_j}) with $\|q_{k_j} - q\| \ge \eta$ for all j. By (4) this subsequence is bounded in the space of continuous linear functionals on $\mathbb{R}^N$, which is finite-dimensional, so by the Bolzano–Weierstrass theorem it has a further subsequence converging to some $\bar q$. By Step 4, $\bar q = q$; but $\|q_{k_j} - q\| \ge \eta$ for all terms of that further subsequence gives $\|\bar q - q\| \ge \eta > 0$, a contradiction. Hence $q_k \to q$.

Step 6: transfer back to w. By (2), $D_k = q_k - \Lambda\langle x_k, \cdot\rangle$ and $D = q - \Lambda\langle x, \cdot\rangle$. For every unit vector e, $\big|\Lambda\langle x_k, e\rangle - \Lambda\langle x, e\rangle\big| = \Lambda\big|\langle x_k - x, e\rangle\big| \le \Lambda|x_k - x|$, so $\big\|\Lambda\langle x_k,\cdot\rangle - \Lambda\langle x,\cdot\rangle\big\| \le \Lambda|x_k - x| \to 0$. Therefore

$$\|D_k - D\| \le \|q_k - q\| + \Lambda\,|x_k - x| \;\longrightarrow\; 0,$$

i.e. $D_k \to D$, which is the assertion.

Remark. The hypothesis is the disjunction "locally semi-convex or locally semi-concave", exactly as the paper states this property, and Step 0 is the reduction of the second case to the first; the conclusion is a statement about w alone, so no separate semi-concave conclusion is needed. The proof is the one the paper indicates — local boundedness and closedness of the subdifferential graph of $w + \frac{\Lambda}{2}|\cdot|^2$, which is a singleton at a point of differentiability — written out: Step 3 is the local boundedness, Step 4 the closedness together with the singleton property, and Step 5 the compactness argument that upgrades "every subsequential limit is q" to convergence. □

2.10. `RelabellingGradientBound`.

Theorem 2.10 ((3.4)). *Let $n, m \in \mathbb{N}$, let $\sigma = [\sigma^1, \dots, \sigma^m]$ be a frame of m vector fields on $\mathbb{R}^n$ with each $\sigma^i : \mathbb{R}^n \to \mathbb{R}^n$ continuous, let $\Omega \subset \mathbb{R}^n$ and let $u : \mathbb{R}^n \to \mathbb{R}$ be locally semi-convex on Ω (`LocallySemiConvexOn`). Let $E \subset \Omega$ be compact, let $\theta > 0$ and assume*

(1) *u is differentiable at every point of E;*
(2) *$\big|\mathrm{hgrad}\ \sigma\ x\ D\big| \geq \theta$ whenever $x \in E$ and D is the derivative of u at x (`hgrad`).*

Let $\big(E_\lambda\big)_{\lambda>0}$ be a family of subsets of $\mathbb{R}^n$ such that

(3) *$E_\lambda \subset \Omega$ for every $\lambda > 0$;*
(4) *$\sup_{E_\lambda} \mathrm{dist}(\cdot, E) \to 0$ as $\lambda \to 0^+$, i.e. for every $\eta > 0$ there is $\lambda_0 > 0$ such that every $x \in E_\lambda$ with $0 < \lambda < \lambda_0$ admits $z \in E$ with $|x - z| \leq \eta$.*

Then there is $\lambda_0 > 0$ such that for every $\lambda \in (0, \lambda_0)$, every $x \in E_\lambda$ and every derivative D of u at x,

$$\big|\mathrm{hgrad}\ \sigma\ x\ D\big| \ \geq\ \frac{\theta}{2}.$$

Proof. This is (3.4) together with the contradiction argument that follows it. We first record where the data come from in Proposition 3.3, since that fixes the reading of the hypotheses.

The intended instantiation. In Proposition 3.3, Step 1: Ω is the bounded open set of Section 3, u is the semi-convex sub-solution, v the semi-concave super-solution, $E = \mathrm{argmax}_\Omega(u - v)$ and $E_\lambda = \mathrm{argmax}_\Omega(u_\lambda - v)$ with $u_\lambda = \psi_\lambda \circ u$ and $\psi_\lambda(s) = s + \lambda e^s$. Hypothesis (1) is Section 2.3(1) (differentiability at maximal points of $u - v$) and hypothesis (2) is the definition $\theta := \min_E |\mathcal{X}u|$, which is positive because E is compact, u is differentiable on E and $\mathcal{X}u$ is continuous on E by Section 2.3(2). Hypothesis (4) is the paper's "$E_\lambda \to E$ in the sense $\sup_{E_\lambda} \mathrm{dist}(\cdot, E) \to 0$ as $\lambda \to 0^+$, because $u_\lambda - v \to u - v$ uniformly", which Proposition 3.3 establishes from $\|u_\lambda - u\|_{L^\infty} \leq \lambda e^M$ before invoking the present statement. Note also that the conclusion is conditional on D being a derivative of u at x, so we do not need differentiability on E_λ as a hypothesis; in Proposition 3.3 it holds, every point of E_λ being a differentiability point of u_λ and hence of u, since $\psi'_\lambda \geq 1 > 0$.

Step 1: setting up the contradiction. Suppose the conclusion fails. Then for every $\lambda_0 > 0$ there exist $\lambda \in (0, \lambda_0)$, a point $x \in E_\lambda$ and a derivative D of u at x with $|\mathrm{hgrad}\ \sigma\ x\ D| < \theta/2$. Applying this with $\lambda_0 := \frac{1}{k+1}$ for $k \in \mathbb{N}$ produces sequences

$$(1)\quad \lambda_k \in \Big(0, \tfrac{1}{k+1}\Big), \qquad x_k \in E_{\lambda_k}, \qquad D_k \text{ a derivative of } u \text{ at } x_k, \qquad \big|\mathrm{hgrad}\ \sigma\ x_k\ D_k\big| < \frac{\theta}{2}.$$

In particular $\lambda_k \to 0^+$, and $x_k \in \Omega$ for every k by hypothesis (3). This is the paper's "otherwise there are $\lambda_k \to 0$, $x_k \in E_{\lambda_k}$ with $|\mathcal{X}u(x_k)| < \theta/2$".

Step 2: $\mathrm{dist}(x_k, E) \to 0$, and a convergent subsequence. Let $j \in \mathbb{N}$ and apply hypothesis (4) with $\eta := \frac{1}{j+1} > 0$: there is $\lambda_0^{(j)} > 0$ such that every $x \in E_\lambda$ with $0 < \lambda < \lambda_0^{(j)}$ admits $z \in E$ with $|x - z| \leq \frac{1}{j+1}$. Since $\lambda_k \to 0^+$ there is K_j with $\lambda_k < \lambda_0^{(j)}$ for all $k \geq K_j$; hence for $k \geq K_j$ there is $z \in E$ with $|x_k - z| \leq \frac{1}{j+1}$, i.e. $\mathrm{dist}(x_k, E) \leq \frac{1}{j+1}$. As j was arbitrary, $\mathrm{dist}(x_k, E) \to 0$. Choose $z_k \in E$ with

$$(2)\qquad |x_k - z_k| \ \leq\ \mathrm{dist}(x_k, E) + \tfrac{1}{k+1} \ \longrightarrow\ 0.$$

(Such a z_k exists for each k by the definition of the infimum; $E \neq \emptyset$, because $E = \emptyset$ would make hypothesis (4) force $E_\lambda = \emptyset$ for small λ, contradicting $x_k \in E_{\lambda_k}$ for all k.)

Since E is compact, the sequence (z_k) has a subsequence (z_{k_j}) converging to some $\hat{x} \in E$. By (2), $|x_{k_j} - \hat{x}| \leq |x_{k_j} - z_{k_j}| + |z_{k_j} - \hat{x}| \to 0$, so

$$(3)\qquad x_{k_j} \ \longrightarrow\ \hat{x}\ \in\ E\ \subset\ \Omega.$$

Step 3: partial continuity of the gradient. By hypothesis (1) there is a derivative $\hat{D}$ of u at $\hat{x}$. We now have: u locally semi-convex on Ω; $\hat{x} \in \Omega$ with derivative $\hat{D}$; the points $x_{k_j} \in \Omega$

with derivatives D_{k_j}; and $x_{k_j} \to \hat{x}$ by (3). Therefore `SemiConvexGradPartialCont` applies and gives

$$D_{k_j} \longrightarrow \hat{D} \qquad \text{in the operator norm.} \tag{4}$$

This is the paper's "partial continuity of the gradient at $\hat{x}$ gives $\mathcal{X}u(x_k) \to \mathcal{X}u(\hat{x})$" — except that partial continuity is a statement about the Euclidean derivatives, and the passage to the horizontal gradients is the next step.

Step 4: the horizontal gradients converge. By `hgrad`, the i-th component of hgrad σ x D is $D(\sigma^i(x))$. Fix $i \in \{1, \dots, m\}$. Then

$$\Big|D_{k_j}\big(\sigma^i(x_{k_j})\big) - \hat{D}\big(\sigma^i(\hat{x})\big)\Big| \le \Big|D_{k_j}\big(\sigma^i(x_{k_j}) - \sigma^i(\hat{x})\big)\Big| + \Big|\big(D_{k_j} - \hat{D}\big)\big(\sigma^i(\hat{x})\big)\Big| \le \|D_{k_j}\|\,\big|\sigma^i(x_{k_j}) - \sigma^i(\hat{x})\big| + \|D_{k_j} - \hat{D}\|\,|\sigma^i$$

The sequence $\|D_{k_j}\|$ is bounded, being convergent by (4); σ^i is continuous and $x_{k_j} \to \hat{x}$, so $|\sigma^i(x_{k_j}) - \sigma^i(\hat{x})| \to 0$; and $\|D_{k_j} - \hat{D}\| \to 0$ by (4). Hence each component converges: $D_{k_j}(\sigma^i(x_{k_j})) \to \hat{D}(\sigma^i(\hat{x}))$. Since $\mathbb{R}^m$ carries the Euclidean norm and m is finite, componentwise convergence is convergence, so

$$\text{hgrad } \sigma\ x_{k_j}\ D_{k_j} \longrightarrow \text{hgrad } \sigma\ \hat{x}\ \hat{D}, \qquad \text{hence} \qquad \big|\text{hgrad } \sigma\ x_{k_j}\ D_{k_j}\big| \longrightarrow \big|\text{hgrad } \sigma\ \hat{x}\ \hat{D}\big| \tag{5}$$

by continuity of the norm.

Step 5: the contradiction. By (1), $|\text{hgrad } \sigma\ x_{k_j}\ D_{k_j}| < \theta/2$ for every j, so passing to the limit in (5),

$$\big|\text{hgrad } \sigma\ \hat{x}\ \hat{D}\big| \le \frac{\theta}{2}.$$

On the other hand $\hat{x} \in E$ and $\hat{D}$ is a derivative of u at $\hat{x}$, so hypothesis (2) gives $|\text{hgrad } \sigma\ \hat{x}\ \hat{D}| \ge \theta$. Combining, $\theta \le \theta/2$, i.e. $\theta \le 0$, contradicting $\theta > 0$. Hence the conclusion holds. This is the paper's "$|\mathcal{X}u(\hat{x})| \ge \theta$, a contradiction". □

2.11. `SemiConvexCompose`.

Theorem 2.11 (Section 2.3(4): composition)**.** *Let $N \in \mathbb{N}$, let $\Omega \subseteq \mathbb{R}^N$ and let $w : \mathbb{R}^N \to \mathbb{R}$ be locally semi-convex on Ω (`LocallySemiConvexOn`). Then:*

1. *for every $\psi \in C^2(\mathbb{R})$ with $\psi' \ge 1$ and $\psi'' \ge 0$ everywhere, the function $\psi \circ w$ is locally semi-convex on Ω;*
2. *for every $\Theta : \mathbb{R}^N \to \mathbb{R}^N$ of class C^2, the function $w \circ \Theta$ is locally semi-convex on $\Theta^{-1}(\Omega)$.*

Proof. The proof runs through the *chord form* of local semi-convexity, established in Step 1 in both directions: it is consumed for w and produced for $\psi \circ w$ and $w \circ \Theta$.

Step 0: the parallelogram identity. For all $x, y \in \mathbb{R}^N$ and $t \in \mathbb{R}$,

$$\big|t\,x + (1-t)\,y\big|^2 = t\,|x|^2 + (1-t)\,|y|^2 - t(1-t)\,|x-y|^2. \tag{0}$$

Indeed, expanding the left side, $|tx + (1-t)y|^2 = t^2|x|^2 + 2t(1-t)\langle x, y\rangle + (1-t)^2|y|^2$, while expanding the right side,

$$t|x|^2 + (1-t)|y|^2 - t(1-t)\big(|x|^2 - 2\langle x, y\rangle + |y|^2\big) = t\big(1-(1-t)\big)|x|^2 + (1-t)(1-t)|y|^2 + 2t(1-t)\langle x, y\rangle,$$

and $t(1-(1-t)) = t^2$, so the two agree.

Step 1: the chord form. Let $S \subseteq \mathbb{R}^N$, let $f : \mathbb{R}^N \to \mathbb{R}$ and let $L \in \mathbb{R}$. Consider the two statements

$$\text{(C)}\quad f\big(tx + (1-t)y\big) \le t\,f(x) + (1-t)\,f(y) + \frac{L}{2}\,t(1-t)\,|x-y|^2 \ \text{ for all } x, y \in S,\ t \in [0,1];$$

$$\text{(V)}\quad y \mapsto f(y) + \frac{L}{2}|y|^2 \text{ is convex on } S.$$

Then (V) implies (C); and if S is convex, (C) implies (V).

(V)⇒(C). Let $x, y \in S$ and $t \in [0,1]$. Put $a := t$, $b := 1-t$, so $a, b \geq 0$ and $a+b = 1$. Convexity of $f + \frac{L}{2}|\cdot|^2$ on S gives

$$f\big(ax+by\big) + \frac{L}{2}\big|ax+by\big|^2 \leq a\Big(f(x) + \frac{L}{2}|x|^2\Big) + b\Big(f(y) + \frac{L}{2}|y|^2\Big).$$

Substituting (0) for $|ax+by|^2 = |tx+(1-t)y|^2$ on the left and cancelling the common terms $\frac{L}{2}\big(t|x|^2 + (1-t)|y|^2\big)$ from both sides leaves exactly

$$f\big(tx+(1-t)y\big) - \frac{L}{2}\,t(1-t)|x-y|^2 \;\leq\; t\,f(x) + (1-t)\,f(y),$$

which is (C).

(C)⇒(V) when S is convex. Let $x, y \in S$ and $a, b \geq 0$ with $a+b = 1$; then $b = 1-a$ and $a \in [0,1]$. By (C) with $t := a$,

$$f\big(ax+(1-a)y\big) \;\leq\; af(x) + (1-a)f(y) + \frac{L}{2}\,a(1-a)|x-y|^2.$$

Adding $\frac{L}{2}|ax+(1-a)y|^2$ to both sides and rewriting it by (0) as $\frac{L}{2}\big(a|x|^2 + (1-a)|y|^2\big) - \frac{L}{2}a(1-a)|x-y|^2$, the two terms $\frac{L}{2}a(1-a)|x-y|^2$ cancel and we get

$$f\big(ax+(1-a)y\big) + \frac{L}{2}\big|ax+(1-a)y\big|^2 \leq a\Big(f(x) + \frac{L}{2}|x|^2\Big) + (1-a)\Big(f(y) + \frac{L}{2}|y|^2\Big),$$

which, together with the convexity of S, is (V).

Step 2: proof of (1), the outer composition. Fix $x_0 \in \Omega$. By `LocallySemiConvexOn` there are $\rho > 0$ and $\Lambda \geq 0$ with $B_\rho(x_0) \subseteq \Omega$ such that $w + \frac{\Lambda}{2}|\cdot|^2$ is convex on $B_\rho(x_0)$. Put $r := \rho/2$, so $\overline{B}_r(x_0) \subseteq B_\rho(x_0)$ and $B_r(x_0)$ is an open convex subset of $B_\rho(x_0)$. By Step 1 ((V)⇒(C), with $S := B_r(x_0)$, noting that a convex function on $B_\rho(x_0)$ is convex on the convex subset $B_r(x_0)$),

$$(2)\quad w\big(ty+(1-t)z\big) \;\leq\; t\,w(y) + (1-t)\,w(z) + \frac{\Lambda}{2}\,t(1-t)\,|y-z|^2 \qquad \text{for } y, z \in B_r(x_0),\ t \in [0,1].$$

(2a) a compact interval containing all relevant values. By part (1) of `SemiConvexLocLipschitz`, w is continuous on Ω, hence on the compact set $\overline{B}_r(x_0) \subseteq \Omega$, so $M := \sup_{\overline{B}_r(x_0)} |w| < \infty$ and $M \geq 0$. For $y, z \in B_r(x_0)$ and $t \in [0,1]$ put

$$a := t\,w(y) + (1-t)\,w(z), \qquad h := \frac{\Lambda}{2}\,t(1-t)\,|y-z|^2 \;\geq 0.$$

Then $|a| \leq M$, being a convex combination of two numbers of absolute value $\leq M$; and since $|y-z| \leq |y-x_0| + |x_0-z| < 2r$ and $t(1-t) \leq \frac{1}{4}$,

$$0 \leq h \leq \frac{\Lambda}{2} \cdot \frac{1}{4} \cdot (2r)^2 = \frac{\Lambda r^2}{2}.$$

Also $ty + (1-t)z \in B_r(x_0)$ because balls are convex, so $|w(ty+(1-t)z)| \leq M$. Therefore all of $w(ty+(1-t)z)$, a and $a+h$ lie in the compact interval

$$I := \Big[-M-1,\ M + \frac{\Lambda r^2}{2} + 1\Big].$$

(2b) properties of ψ. Since $\psi \in C^2(\mathbb{R})$, ψ' is continuous, so $L_\psi := \max_{s \in I} |\psi'(s)| < \infty$ and $L_\psi \geq 1$ because $\psi' \geq 1$ on I and I is non-empty. By the mean value theorem, for $s_1 \leq s_2$ in I there is θ between them (hence in I, as I is an interval) with $\psi(s_2) - \psi(s_1) = \psi'(\theta)(s_2 - s_1)$; consequently

$$(3)\qquad \psi(s_2) - \psi(s_1) \leq L_\psi\,(s_2 - s_1) \qquad \text{for } s_1 \leq s_2 \text{ in } I,$$

and also $\psi(s_1) \leq \psi(s_2)$, i.e. ψ is nondecreasing — in fact ψ is nondecreasing on all of $\mathbb{R}$, since $\psi' \geq 1 > 0$ everywhere. Finally ψ is convex on $\mathbb{R}$, because $\psi'' \geq 0$ everywhere.

(2c) the chord inequality for $\psi \circ w$. Let $y, z \in B_r(x_0)$ and $t \in [0,1]$. Applying the nondecreasing ψ to (2),

$$\psi\Big(w\big(ty+(1-t)z\big)\Big) \;\leq\; \psi(a+h).$$

Since $a \le a+h$ and both lie in I, (3) gives $\psi(a+h) \le \psi(a) + L_\psi h$. Since ψ is convex and $a = t\,w(y) + (1-t)\,w(z)$,

$$\psi(a) \;\le\; t\,\psi\big(w(y)\big) + (1-t)\,\psi\big(w(z)\big).$$

Combining the three displays and substituting the value of h,

$$(\psi\circ w)\big(ty+(1-t)z\big) \;\le\; t\,(\psi\circ w)(y) + (1-t)\,(\psi\circ w)(z) + \frac{L_\psi\Lambda}{2}\,t(1-t)\,|y-z|^2.$$

This is statement (C) of Step 1 for $f := \psi\circ w$ on $S := B_r(x_0)$ with $L := L_\psi\Lambda \ge 0$. As $B_r(x_0)$ is convex, Step 1 ((C)⇒(V)) gives that $\psi\circ w + \frac{L_\psi\Lambda}{2}|\cdot|^2$ is convex on $B_r(x_0)$. Since $r>0$, $L_\psi\Lambda \ge 0$ and $B_r(x_0) \subseteq B_\rho(x_0) \subseteq \Omega$, the point x_0 satisfies the requirement of `LocallySemiConvexOn` for $\psi\circ w$. As $x_0 \in \Omega$ was arbitrary, $\psi\circ w$ is locally semi-convex on Ω.

Step 3: a second-order bound for C^2 maps. Let $\Theta \in C^2$, let $K \subseteq \mathbb{R}^N$ be convex with $C_1 := \sup_K \|D\Theta\| < \infty$ and $C_2 := \sup_K \|D^2\Theta\| < \infty$. Then for all $y, z \in K$ and $t \in [0,1]$,

$$\big|\Theta y - \Theta z\big| \le C_1|y-z|, \tag{4}$$

$$\Big|\Theta\big(ty+(1-t)z\big) - t\,\Theta y - (1-t)\,\Theta z\Big| \;\le\; \frac{C_2}{2}\,t(1-t)\,|y-z|^2. \tag{5}$$

(4) is the mean value inequality along the segment $[z,y] \subseteq K$. For (5), set $\gamma(\tau) := \tau y + (1-\tau)z$ for $\tau \in [0,1]$, which lies in K by convexity, and

$$\phi(\tau) := \Theta\big(\gamma(\tau)\big) - \tau\,\Theta y - (1-\tau)\,\Theta z.$$

Then ϕ is C^2 on $[0,1]$ with $\phi(0) = \phi(1) = 0$, and since $\gamma' = y - z$, $\gamma'' = 0$,

$$\phi''(\tau) = D^2\Theta\big(\gamma(\tau)\big)(y-z,\,y-z), \qquad \text{so} \qquad |\phi''(\tau)| \le C_2\,|y-z|^2 =: S.$$

Fix $t \in [0,1]$ and choose a linear functional ℓ on $\mathbb{R}^N$ with $\|\ell\| = 1$ and $\ell(\phi(t)) = |\phi(t)|$ (take $\ell = \langle \phi(t)/|\phi(t)|, \cdot\rangle$ if $\phi(t) \ne 0$, and any unit functional otherwise). Put $\varphi := \ell\circ\phi$, a real C^2 function on $[0,1]$ with $\varphi(0) = \varphi(1) = 0$ and $|\varphi''| \le S$. Let $\pi(\tau) := \frac{S}{2}\tau(1-\tau)$, so $\pi(0) = \pi(1) = 0$ and $\pi'' \equiv -S$. The function $\eta := \varphi - \pi$ satisfies $\eta'' = \varphi'' + S \ge 0$, so η is convex on $[0,1]$, and $\eta(0) = \eta(1) = 0$; a convex function on $[0,1]$ vanishing at both endpoints is ≤ 0 throughout, since $\eta(\tau) \le (1-\tau)\eta(0) + \tau\eta(1) = 0$. Hence $\varphi \le \pi$ on $[0,1]$. The same argument applied to $-\varphi$ gives $-\varphi \le \pi$. Therefore

$$|\phi(t)| = \varphi(t) \le \pi(t) = \frac{S}{2}\,t(1-t) = \frac{C_2}{2}\,t(1-t)\,|y-z|^2,$$

which is (5).

Step 4: proof of (2), the inner composition. Let $y_0 \in \Theta^{-1}(\Omega)$ and put $x_0 := \Theta(y_0) \in \Omega$.

(4a) data at x_0. By `LocallySemiConvexOn` there are $\rho > 0$ and $\Lambda \ge 0$ with $B_\rho(x_0) \subseteq \Omega$ and $w + \frac{\Lambda}{2}|\cdot|^2$ convex on $B_\rho(x_0)$. By part (2) of `SemiConvexLocLipschitz` at x_0 there are $r_0 > 0$ and $C_0 \ge 0$ with $B_{r_0}(x_0) \subseteq \Omega$ and $|w(P) - w(Q)| \le C_0|P-Q|$ for all $P, Q \in B_{r_0}(x_0)$. Put $\rho' := \min\{\rho, r_0\} > 0$. On the open convex set $B_{\rho'}(x_0)$ both properties hold: $w + \frac{\Lambda}{2}|\cdot|^2$ is convex there (it is convex on the larger convex set $B_\rho(x_0)$), and w is C_0-Lipschitz there. By Step 1 ((V)⇒(C)),

$$w\big(tP+(1-t)Q\big) \le t\,w(P) + (1-t)\,w(Q) + \frac{\Lambda}{2}\,t(1-t)|P-Q|^2 \quad \text{for } P, Q \in B_{\rho'}(x_0),\ t \in [0,1]. \tag{6}$$

(4b) data at y_0. Fix any $s_0 > 0$ and let $K := \overline{B}_{s_0}(y_0)$, a compact convex set. Since Θ is C^2, $D\Theta$ and $D^2\Theta$ are continuous, hence bounded on K:

$$C_1 := \sup_K \|D\Theta\| < \infty, \qquad C_2 := \sup_K \|D^2\Theta\| < \infty,$$

both ≥ 0. Θ is continuous, so there is $s_1 \in (0, s_0]$ with $\Theta\big(B_{s_1}(y_0)\big) \subseteq B_{\rho'/2}(x_0)$. Choose

$$s := \min\left\{s_1,\ \sqrt{\tfrac{\rho'}{8(C_2+1)}}\right\} > 0,$$

so that in particular $B_s(y_0) \subseteq B_{s_1}(y_0) \subseteq K$, $\Theta(B_s(y_0)) \subseteq B_{\rho'/2}(x_0)$ and

$$\frac{C_2}{2} \cdot \frac{1}{4} \cdot (2s)^2 = \frac{C_2 s^2}{2} \le \frac{(C_2+1)}{2} \cdot \frac{\rho'}{8(C_2+1)} = \frac{\rho'}{16} < \frac{\rho'}{2}. \tag{7}$$

Note also $B_s(y_0) \subseteq \Theta^{-1}(\Omega)$, since $\Theta(B_s(y_0)) \subseteq B_{\rho'/2}(x_0) \subseteq B_\rho(x_0) \subseteq \Omega$.

(4c) the chord inequality for $w \circ \Theta$. Let $y, z \in B_s(y_0)$ and $t \in [0,1]$, and write

$$P := \Theta y, \qquad Q := \Theta z, \qquad R := \Theta\big(ty + (1-t)z\big) - \big(tP + (1-t)Q\big).$$

Since balls are convex, $ty+(1-t)z \in B_s(y_0)$, so $\Theta(ty+(1-t)z) \in B_{\rho'/2}(x_0)$, and $P, Q \in B_{\rho'/2}(x_0)$. Hence $tP + (1-t)Q \in B_{\rho'/2}(x_0)$, being a convex combination of two of its points. By Step 3 applied on the convex set $K \supseteq B_s(y_0)$,

$$|R| \le \frac{C_2}{2} t(1-t)|y-z|^2 \quad \text{and} \quad |P-Q| \le C_1|y-z|. \tag{8}$$

Using $|y-z| < 2s$ and $t(1-t) \le \frac{1}{4}$, the first bound in (8) together with (7) gives $|R| < \rho'/2$, so

$$\big|\big(tP + (1-t)Q\big) + R - x_0\big| \le \big|tP + (1-t)Q - x_0\big| + |R| < \frac{\rho'}{2} + \frac{\rho'}{2} = \rho',$$

i.e. both $tP + (1-t)Q$ and $\big(tP + (1-t)Q\big) + R = \Theta(ty + (1-t)z)$ lie in $B_{\rho'}(x_0)$. Therefore, w being C_0-Lipschitz on $B_{\rho'}(x_0)$,

$$w\Big(\Theta\big(ty+(1-t)z\big)\Big) \;\le\; w\big(tP+(1-t)Q\big) + C_0\,|R| \;\le\; w\big(tP+(1-t)Q\big) + \frac{C_0 C_2}{2}\,t(1-t)\,|y-z|^2.$$

Applying (6) to $P, Q \in B_{\rho'}(x_0)$ and then the second bound in (8),

$$w\big(tP+(1-t)Q\big) \le t\,w(P)+(1-t)\,w(Q)+\frac{\Lambda}{2}t(1-t)|P-Q|^2 \le t\,w(P)+(1-t)\,w(Q)+\frac{\Lambda C_1^2}{2}t(1-t)|y-z|^2.$$

Adding the two displays and recalling $P = \Theta y$, $Q = \Theta z$,

$$(w\circ\Theta)\big(ty+(1-t)z\big) \le t\,(w\circ\Theta)(y) + (1-t)\,(w\circ\Theta)(z) + \frac{\Lambda C_1^2 + C_0 C_2}{2}\,t(1-t)\,|y-z|^2.$$

This is statement (C) of Step 1 for $f := w \circ \Theta$ on the convex set $S := B_s(y_0)$ with $L := \Lambda C_1^2 + C_0 C_2 \ge 0$, so Step 1 ((C)⇒(V)) gives that $w\circ\Theta + \frac{L}{2}|\cdot|^2$ is convex on $B_s(y_0)$. Since $s > 0$, $L \ge 0$ and $B_s(y_0) \subseteq \Theta^{-1}(\Omega)$, the point y_0 satisfies the requirement of `LocallySemiConvexOn` for $w \circ \Theta$ on $\Theta^{-1}(\Omega)$. As $y_0 \in \Theta^{-1}(\Omega)$ was arbitrary, $w \circ \Theta$ is locally semi-convex on $\Theta^{-1}(\Omega)$.

Remark on the hypotheses and the route. The paper justifies this property by the identities $D^2(\psi \circ w) = \psi'' \nabla w \otimes \nabla w + \psi' D^2 w$ and $D^2(w \circ \Theta) = D\Theta^T D^2 w\, D\Theta + \nabla w \cdot D^2\Theta$ "at points of twice differentiability, and by approximation elsewhere". The approximation step is not supplied there, and it is avoided entirely above: the chord form of Step 1 is a statement about values of w, so it needs no differentiability of w at all, and the two constants $L_\psi \Lambda$ and $\Lambda C_1^2 + C_0 C_2$ are the chord-form counterparts of the two displayed Hessian identities.

The hypotheses on ψ are the paper's. The hypothesis on Θ is that it is C^2; the paper asks in addition that Θ be a diffeomorphism with bounded first and second derivatives. Neither addition is needed for the conclusion as stated, and both would be unusable in the intended application: the conclusion is the *local* notion of semi-convexity, so only the bounds C_1, C_2 on the compact set K enter, and those come from C^2 alone; while the map the development feeds in is the right translation $R_h(x) = x * h$ of a Carnot group, whose first and second derivatives are polynomial in x and hence bounded on bounded sets but not globally. Reading the paper's hypothesis on the bounded set where it is used — Ω is bounded throughout — is exactly the C^2 hypothesis above, and it is what the smoothness of the group law supplies. Injectivity of Θ is never used, because local semi-convexity of $w \circ \Theta$ is checked at each point of $\Theta^{-1}(\Omega)$ separately. □

2.12. `SemiConvexDiffAtMax`.

Theorem 2.12 (Section 2.3(1): differentiability at maximal points)**.** *Let $N \in \mathbb{N}$, let $\Omega \subseteq \mathbb{R}^N$ and let $u, v : \mathbb{R}^N \to \mathbb{R}$ be such that u is locally semi-convex on Ω and v is locally semi-concave on Ω, i.e. $-v$ is locally semi-convex on Ω (`LocallySemiConvexOn`). Then:*

1. *for every $x_0 \in \Omega$ at which $u - v$ has a local maximum there is a continuous linear functional D on $\mathbb{R}^N$ which is the Fréchet derivative of u at x_0 and also the Fréchet derivative of v at x_0 — so u and v are both differentiable at x_0 and $\nabla u(x_0) = \nabla v(x_0)$;*
2. *the same conclusion holds for every $x_0 \in \operatorname{argmax}_\Omega(u - v)$, i.e. for every $x_0 \in \Omega$ with $(u - v)(y) \le (u - v)(x_0)$ for all $y \in \Omega$.*

Proof. We use one standard fact of convex analysis, which is not a tablet node: *if g is convex and real-valued on an open convex set $U \subseteq \mathbb{R}^N$ and $p_0 \in U$, then $\partial g(p_0) \neq \emptyset$*, i.e. there is $q \in \mathbb{R}^N$ with $g(y) \ge g(p_0) + \langle q, y - p_0\rangle$ for all $y \in U$. (A convex function finite on an open convex subset of a finite-dimensional space is continuous there, so its epigraph has non-empty interior and a supporting hyperplane at the boundary point $(p_0, g(p_0))$ is non-vertical; its slope is such a q.)

Step 1: a one-sided quadratic minorant for u at x_0. Let $x_0 \in \Omega$. By `LocallySemiConvexOn` applied to u there are $\rho_1 > 0$ and $\Lambda_1 \ge 0$ with $B_{\rho_1}(x_0) \subseteq \Omega$ such that $g_1(y) := u(y) + \frac{\Lambda_1}{2}|y|^2$ is convex on $B_{\rho_1}(x_0)$. By the standard fact there is q_1 with

$$g_1(y) \;\ge\; g_1(x_0) + \langle q_1,\, y - x_0\rangle \qquad \text{for all } y \in B_{\rho_1}(x_0).$$

Rewriting with $u = g_1 - \frac{\Lambda_1}{2}|\cdot|^2$ and using the identity

$$\frac{\Lambda_1}{2}\big(|x_0|^2 - |y|^2\big) = -\frac{\Lambda_1}{2}\langle y - x_0,\, y + x_0\rangle = -\Lambda_1\langle x_0,\, y - x_0\rangle - \frac{\Lambda_1}{2}|y - x_0|^2$$

(which is $|y|^2 - |x_0|^2 = \langle y - x_0, y + x_0\rangle$ and $y + x_0 = (y - x_0) + 2x_0$), we obtain, with $p := q_1 - \Lambda_1 x_0$,

$$u(y) \;\ge\; u(x_0) + \langle p,\, y - x_0\rangle - \frac{\Lambda_1}{2}|y - x_0|^2 \qquad \text{for all } y \in B_{\rho_1}(x_0). \tag{1}$$

Step 2: a one-sided quadratic majorant for v at x_0. Applying Step 1 verbatim to the locally semi-convex function $-v$ gives $\rho_2 > 0$, $\Lambda_2 \ge 0$ with $B_{\rho_2}(x_0) \subseteq \Omega$ and a vector $\tilde{p}$ with

$$-v(y) \;\ge\; -v(x_0) + \langle \tilde{p},\, y - x_0\rangle - \frac{\Lambda_2}{2}|y - x_0|^2 \qquad \text{on } B_{\rho_2}(x_0),$$

that is, with $p' := -\tilde{p}$,

$$v(y) \;\le\; v(x_0) + \langle p',\, y - x_0\rangle + \frac{\Lambda_2}{2}|y - x_0|^2 \qquad \text{for all } y \in B_{\rho_2}(x_0). \tag{2}$$

Step 3: proof of (1), the local-maximum case. Assume now that $u - v$ has a local maximum at x_0: there is $\rho_3 > 0$ with

$$u(y) - v(y) \;\le\; u(x_0) - v(x_0) \qquad \text{for all } y \in B_{\rho_3}(x_0). \tag{3}$$

Put $\rho := \min\{\rho_1, \rho_2, \rho_3\} > 0$ and $\Lambda := \Lambda_1 + \Lambda_2 \ge 0$. For $y \in B_\rho(x_0)$, chaining (1), (3) and (2),

$$u(x_0) + \langle p, y - x_0\rangle - \frac{\Lambda_1}{2}|y - x_0|^2 \overset{(1)}{\le} u(y) \overset{(3)}{\le} v(y) + u(x_0) - v(x_0) \overset{(2)}{\le} u(x_0) + \langle p', y - x_0\rangle + \frac{\Lambda_2}{2}|y - x_0|^2,$$

whence, cancelling $u(x_0)$,

$$\langle p - p',\, y - x_0\rangle \;\le\; \frac{\Lambda}{2}\,|y - x_0|^2 \qquad \text{for all } y \in B_\rho(x_0). \tag{4}$$

(3a) $p = p'$. Let $e \in \mathbb{R}^N$ with $|e| = 1$ and let $0 < \tau < \rho$. Taking $y := x_0 + \tau e$ in (4), which is legitimate since $|y - x_0| = \tau < \rho$, gives $\tau\langle p - p', e\rangle \le \frac{\Lambda}{2}\tau^2$; dividing by $\tau > 0$,

$$\langle p - p',\, e\rangle \;\le\; \frac{\Lambda}{2}\,\tau \qquad \text{for every } \tau \in (0, \rho).$$

Letting $\tau \to 0^+$ gives $\langle p - p', e\rangle \le 0$. Applying this with $-e$ in place of e gives $\langle p - p', e\rangle \ge 0$, so $\langle p - p', e\rangle = 0$ for every unit vector e, hence $p - p' = 0$, i.e. $p = p'$.

(3b) u is differentiable at x_0 with derivative $\langle p, \cdot\rangle$. For $y \in B_\rho(x_0)$, (1) gives the lower bound and the chain (3), (2) together with $p = p'$ gives the upper bound:

$$u(x_0) + \langle p, y - x_0\rangle - \frac{\Lambda_1}{2}|y - x_0|^2 \;\le\; u(y) \;\le\; u(x_0) + \langle p, y - x_0\rangle + \frac{\Lambda_2}{2}|y - x_0|^2,$$

that is,

$$\big|\, u(y) - u(x_0) - \langle p, y - x_0\rangle \,\big| \;\le\; \frac{\Lambda}{2}\,|y - x_0|^2 \qquad \text{for all } y \in B_\rho(x_0).$$

Since $\frac{\Lambda}{2}|y - x_0|^2 = o(|y - x_0|)$ as $y \to x_0$, the continuous linear functional $D := \langle p, \cdot\rangle$ is the Fréchet derivative of u at x_0.

(3c) v is differentiable at x_0 with the same derivative. For $y \in B_\rho(x_0)$, (2) with $p' = p$ gives the upper bound, and (3) together with (1) gives the lower bound:

$$v(y) \;\ge\; u(y) - \big(u(x_0) - v(x_0)\big) \;\ge\; v(x_0) + \langle p, y - x_0\rangle - \frac{\Lambda_1}{2}|y - x_0|^2,$$

$$v(y) \;\le\; v(x_0) + \langle p, y - x_0\rangle + \frac{\Lambda_2}{2}|y - x_0|^2,$$

so $\big|v(y) - v(x_0) - \langle p, y - x_0\rangle\big| \le \frac{\Lambda}{2}|y - x_0|^2$ on $B_\rho(x_0)$, and $D = \langle p, \cdot\rangle$ is also the Fréchet derivative of v at x_0. This proves (1) with this D.

Step 4: proof of (2), the argmax **case.** Let $x_0 \in \Omega$ with $(u - v)(y) \le (u - v)(x_0)$ for every $y \in \Omega$. By `LocallySemiConvexOn` applied to u at x_0 there is $\rho_0 > 0$ with $B_{\rho_0}(x_0) \subseteq \Omega$. Hence $(u - v)(y) \le (u - v)(x_0)$ for all $y \in B_{\rho_0}(x_0)$, i.e. $u - v$ has a local maximum at x_0. Part (1), already proved, gives the conclusion.

Remark. Both forms are stated because both are used: the paper's line stating this property lists the points of $\operatorname{argmax}_\Omega(u - v)$ and "every local maximum point of $u - v$", and the local form is the one Proposition Proposition 3.3 applies, after the maximum has been localized by a penalization. The hypotheses are the local notion of semi-convexity, so the constants Λ_1, Λ_2 depend on x_0; the proof uses them only on the single ball $B_\rho(x_0)$, which is why the local notion suffices. □

2.13. `EllipticityLowerBound`.

Theorem 2.13 ((2.4): the ellipticity lower bound)**.** *Let $m \in \mathbb{N}$ and let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$ satisfy the structure conditions (E) and (S) (`StructureConditionsES`) and be normalized in the sense of (2.3) (`NormalizedAH`). Write $\mathcal{E}_A$ for the ellipticity function of A (`Ellip`). Then there is a constant $e_0 > 0$ such that*

$$e_0\,|\xi|^2 \;\le\; \mathcal{E}_A(\xi) \qquad \textit{for every } \xi \in \mathbb{R}^m.$$

Proof. **Step 1: $\mathcal{E}_A$ is 2-homogeneous.** Let $t > 0$ and $\eta \neq 0$. `NormalizedAH` gives $A(t\eta) = A(\eta)$, that being the first of its two identities, valid for every $t > 0$ and every $\eta \in \mathbb{R}^m \setminus \{0\}$. Hence, using $(t\eta)_k = t\,\eta_k$ for every k, which contributes one factor t in each of the two vector slots of the quadratic form of `Ellip`,

$$\mathcal{E}_A(t\eta) = \sum_i \sum_j A_{i,j}(t\eta)\,(t\eta)_j\,(t\eta)_i = t^2 \sum_i \sum_j A_{i,j}(\eta)\,\eta_j\,\eta_i = t^2\,\mathcal{E}_A(\eta). \tag{1}$$

Step 2: $\mathcal{E}_A(0) = 0$. By `Ellip`, $\mathcal{E}_A(0) = \sum_i \sum_j A_{i,j}(0)\cdot 0 \cdot 0 = 0$.

Step 3: the degenerate case $m = 0$. If $m = 0$ then $\mathbb{R}^0 = \{0\}$, so the only ξ to consider is $\xi = 0$. Take $e_0 := 1 > 0$. By Step 2 and $|0| = 0$ the required inequality reads $1 \cdot 0 \le 0$, which holds. For the rest of the proof assume $m \ge 1$.

Step 4: the minimum of $\mathcal{E}_A$ over the unit sphere. Let

$$S := \{\omega \in \mathbb{R}^m : \; |\omega| = 1\}.$$

S is bounded, and it is closed as the preimage of the closed set $\{1\}$ under the continuous map $\eta \mapsto |\eta|$; being a closed bounded subset of the finite-dimensional space $\mathbb{R}^m$, it is compact. It is

non-empty: since $m \geq 1$ the first standard basis vector e_1 exists and $|e_1| = 1$, so $e_1 \in S$. Finally $S \subseteq \mathbb{R}^m \setminus \{0\}$, because $|\omega| = 1 \neq 0$ forces $\omega \neq 0$.

By clause (3) of `StructureConditionsES`, A is continuous on $\mathbb{R}^m \setminus \{0\}$, hence so is each entry $\eta \mapsto A_{i,j}(\eta)$, and therefore so is

$$\eta \;\longmapsto\; \mathcal{E}_A(\eta) = \sum_i \sum_j A_{i,j}(\eta)\, \eta_j\, \eta_i,$$

being a finite sum of products of functions continuous on $\mathbb{R}^m \setminus \{0\}$ (`Ellip`). In particular $\mathcal{E}_A$ is continuous on S. A real-valued continuous function on a non-empty compact set attains its infimum, so there is $\omega_0 \in S$ with

$$\mathcal{E}_A(\omega_0) \;\leq\; \mathcal{E}_A(\omega) \qquad \text{for every } \omega \in S. \tag{2}$$

Set $e_0 := \mathcal{E}_A(\omega_0)$. Since $\omega_0 \in S$ we have $\omega_0 \neq 0$, so clause (5) of `StructureConditionsES` — condition (E), (1.9) — applies at ω_0 and gives

$$e_0 = \mathcal{E}_A(\omega_0) > 0. \tag{3}$$

Step 5: the bound, for $\xi = 0$. By Step 2 and $|0| = 0$, both sides of the asserted inequality vanish: $e_0\,|0|^2 = 0 = \mathcal{E}_A(0)$. So the inequality holds at $\xi = 0$, with equality.

Step 6: the bound, for $\xi \neq 0$. Put

$$t := |\xi| > 0, \qquad \omega := t^{-1}\xi.$$

Then $|\omega| = |\xi|^{-1}|\xi| = 1$, so $\omega \in S$ and in particular $\omega \neq 0$; and $\xi = t\omega$. Step 1 applied to the pair (t, ω) — legitimate because $t > 0$ and $\omega \neq 0$ — gives

$$\mathcal{E}_A(\xi) = \mathcal{E}_A(t\omega) = t^2\, \mathcal{E}_A(\omega) = |\xi|^2\, \mathcal{E}_A(\omega).$$

By (2), $\mathcal{E}_A(\omega) \geq e_0$, and $|\xi|^2 > 0$, so multiplying that inequality by the non-negative number $|\xi|^2$ preserves it:

$$\mathcal{E}_A(\xi) = |\xi|^2\, \mathcal{E}_A(\omega) \;\geq\; |\xi|^2\, e_0 = e_0\, |\xi|^2.$$

Step 7: conclusion. With e_0 as in Step 4 we have $e_0 > 0$ by (3), and the inequality $e_0|\xi|^2 \leq \mathcal{E}_A(\xi)$ holds for $\xi = 0$ by Step 5 and for $\xi \neq 0$ by Step 6, hence for every $\xi \in \mathbb{R}^m$. This is (2.4), with $e_0 = \min_{|\omega|=1} \mathcal{E}_A(\omega)$ as in the paper. □

2.14. `LocalMaxSecondOrder`.

Theorem 2.14 (The elementary observation of Section 3.1)**.** *Let $n, m \in \mathbb{N}$, let $\sigma : \mathbb{R}^n \to \mathbb{R}^{n\times m}$, $\sigma(x) = [\sigma^1(x), \dots, \sigma^m(x)]$, be a frame of m vector fields on $\mathbb{R}^n$, let $u, v : \mathbb{R}^n \to \mathbb{R}$ and let $x_0 \in \mathbb{R}^n$. Assume that (D_u, B_u) is a second-order jet of u at x_0 and (D_v, B_v) is a second-order jet of v at x_0 (`TwiceDiffAt`), and that $u - v$ has a local maximum at x_0. Then*

1. *$D_u = D_v$;*
2. *$B_u(h, h) \leq B_v(h, h)$ for every $h \in \mathbb{R}^n$;*
3. *hgrad σ x_0 D_u = hgrad σ x_0 D_v (`hgrad`); and*
4. *hhessS σ x_0 D_u B_u $\leq$ hhessS σ x_0 D_v B_v in the order (2.1) (`MatLe`, `hhessS`).*

Proof. Items (1) and (2) are the paper's "$\nabla u(x_0) = \nabla v(x_0)$ and $D^2u(x_0) \leq D^2v(x_0)$"; items (3) and (4) are their transport through (2.6) to the horizontal jets. The order matters: (1) must be established before (4), because it is exactly what makes the correction terms of `hhessS` cancel.

Step 1: the basic inequality. Since $u - v$ has a local maximum at x_0 there is $r > 0$ with

$$u(y) - v(y) \;\leq\; u(x_0) - v(x_0) \qquad \text{whenever } |y - x_0| < r. \tag{1}$$

We claim: for every $\varepsilon > 0$ there is $\delta > 0$ with $\delta \leq r$ such that

$$D_u(y - x_0) - D_v(y - x_0) + \tfrac{1}{2}\Big(B_u(y - x_0, y - x_0) - B_v(y - x_0, y - x_0)\Big) \;\leq\; 2\varepsilon\, |y - x_0|^2 \tag{2}$$

whenever $|y - x_0| < \delta$.

Indeed, let $\varepsilon > 0$. By `TwiceDiffAt` applied to u there is $\delta_1 > 0$ with

$$\Big|u(y) - u(x_0) - D_u(y - x_0) - \tfrac{1}{2}B_u(y - x_0, y - x_0)\Big| \le \varepsilon |y - x_0|^2 \qquad \text{for } |y - x_0| < \delta_1,$$

and by `TwiceDiffAt` applied to v there is $\delta_2 > 0$ with the corresponding estimate for v with (D_v, B_v). Put $\delta := \min\{r, \delta_1, \delta_2\} > 0$ and let $|y - x_0| < \delta$. Dropping the absolute values in the direction we need,

$$\begin{aligned} u(y) &\ge u(x_0) + D_u(y - x_0) + \tfrac{1}{2}B_u(y - x_0, y - x_0) - \varepsilon|y - x_0|^2, \\ v(y) &\le v(x_0) + D_v(y - x_0) + \tfrac{1}{2}B_v(y - x_0, y - x_0) + \varepsilon|y - x_0|^2. \end{aligned}$$

Combining these with (1) in the form $u(y) \le v(y) + u(x_0) - v(x_0)$,

$$u(x_0)+D_u(y-x_0)+\tfrac{1}{2}B_u(y-x_0,y-x_0)-\varepsilon|y-x_0|^2 \le u(x_0)+D_v(y-x_0)+\tfrac{1}{2}B_v(y-x_0,y-x_0)+\varepsilon|y-x_0|^2,$$

and cancelling $u(x_0)$ and rearranging gives (2).

Step 2: proof of (1). Fix $h \in \mathbb{R}^n$. We show $D_u(h) \le D_v(h)$. If $h = 0$ this is the equality $0 \le 0$, both sides vanishing by linearity, so assume $h \ne 0$ and put

$$\kappa := 2\,|h|^2 - \tfrac{1}{2}\Big(B_u(h,h) - B_v(h,h)\Big).$$

Apply Step 1 with $\varepsilon := 1$ and let $\delta > 0$ be the resulting radius. For $0 < \tau < \delta/|h|$ put $y := x_0 + \tau h$; then $y - x_0 = \tau h$ and $|y - x_0| = \tau|h| < \delta$, so (2) applies. By linearity of D_u, D_v and bilinearity of B_u, B_v,

$$D_u(\tau h) - D_v(\tau h) = \tau\big(D_u(h) - D_v(h)\big), \qquad B_u(\tau h, \tau h) - B_v(\tau h, \tau h) = \tau^2\big(B_u(h,h) - B_v(h,h)\big),$$

and $|y - x_0|^2 = \tau^2|h|^2$, so (2) becomes

$$\tau\big(D_u(h) - D_v(h)\big) + \tfrac{\tau^2}{2}\big(B_u(h,h) - B_v(h,h)\big) \le 2\tau^2|h|^2,$$

i.e., after dividing by $\tau > 0$,

$$D_u(h) - D_v(h) \le \tau\,\kappa \qquad \text{for every } \tau \in \big(0, \delta/|h|\big). \tag{3}$$

Now let $\eta > 0$ be arbitrary and choose $\tau := \min\{\eta/(|\kappa| + 1),\ \delta/(2|h|)\}$, which is positive and lies in $(0, \delta/|h|)$. Then $\tau\kappa \le \tau|\kappa| \le \eta|\kappa|/(|\kappa| + 1) \le \eta$, so (3) gives $D_u(h) - D_v(h) \le \eta$. As $\eta > 0$ was arbitrary, $D_u(h) - D_v(h) \le 0$, i.e. $D_u(h) \le D_v(h)$.

Applying this to $-h$ in place of h gives $D_u(-h) \le D_v(-h)$, i.e. $-D_u(h) \le -D_v(h)$, i.e. $D_v(h) \le D_u(h)$. Hence $D_u(h) = D_v(h)$ for every $h \in \mathbb{R}^n$, i.e. $D_u = D_v$. This is (1).

Step 3: proof of (2). Fix $h \in \mathbb{R}^n$; we show $B_u(h,h) - B_v(h,h) \le 0$. If $h = 0$ both terms vanish by bilinearity, so assume $h \ne 0$. Let $\eta > 0$ and set

$$\varepsilon := \frac{\eta}{4\,|h|^2 + 1} > 0.$$

Let $\delta > 0$ be the radius furnished by Step 1 for this ε, pick any $\tau \in (0, \delta/|h|)$ and put $y := x_0 + \tau h$, so that $|y - x_0| = \tau|h| < \delta$ and (2) applies. By Step 1 and the computation of Step 2, (2) reads

$$\tau\big(D_u(h) - D_v(h)\big) + \tfrac{\tau^2}{2}\big(B_u(h,h) - B_v(h,h)\big) \le 2\varepsilon\,\tau^2|h|^2.$$

By Step 1 (already proved) $D_u = D_v$, so the first term vanishes. Dividing by $\tau^2/2 > 0$,

$$B_u(h,h) - B_v(h,h) \le 4\varepsilon\,|h|^2 = \frac{4\eta\,|h|^2}{4|h|^2 + 1} \le \eta.$$

As $\eta > 0$ was arbitrary, $B_u(h,h) - B_v(h,h) \le 0$, which is (2).

Step 4: proof of (3). By `hgrad`, the i-th component of hgrad $\sigma\ x_0\ D$ is $D(\sigma^i(x_0))$, for any 1-jet D. Since $D_u = D_v$ by Step 2, the two vectors hgrad $\sigma\ x_0\ D_u$ and hgrad $\sigma\ x_0\ D_v$ have the same components $D_u(\sigma^i(x_0)) = D_v(\sigma^i(x_0))$ for every i, hence are equal. (In the paper's notation this is $\mathcal{X}u(x_0) = \mathcal{X}v(x_0) =: \xi_0$.)

Step 5: proof of (4). By `hhessS`,

$$\big(\text{hhessS } \sigma\ x_0\ D\ B\big)_{i,j} = B\big(\sigma^i(x_0), \sigma^j(x_0)\big) + \mathcal{M}(x_0, D)_{i,j},$$

where $\mathcal{M}(x_0, D)$ is the correction matrix of `Mfield`. The crucial point is that $\mathcal{M}(x_0, D)$ depends only on the 1-jet D (and on x_0 and σ): by `Mfield`, $\mathcal{M}(x_0, D)_{i,j} = \frac{1}{2}\big(D(D\sigma^j(x_0)\sigma^i(x_0)) + D(D\sigma^i(x_0)\sigma^j(x_0))\big)$, in which the 2-jet does not occur. Hence, by Step 2,

$$\mathcal{M}(x_0, D_u) = \mathcal{M}(x_0, D_v) =: \mathcal{M}. \tag{4}$$

This is precisely why the correction terms cancel below, and it is available only because the first jets have already been shown to agree.

Now let $\xi \in \mathbb{R}^m$ and put

$$h_\xi := \sum_{i=1}^{m} \xi_i \, \sigma^i(x_0) \ \in \ \mathbb{R}^n.$$

For any 2-jet B, bilinearity gives

$$B\big(h_\xi, h_\xi\big) = B\Big(\sum_i \xi_i \sigma^i(x_0),\, \sum_j \xi_j \sigma^j(x_0)\Big) = \sum_i \sum_j \xi_i \, \xi_j \, B\big(\sigma^i(x_0), \sigma^j(x_0)\big) = \sum_i \sum_j B\big(\sigma^i(x_0), \sigma^j(x_0)\big)\, \xi_j \, \xi_i. \tag{5}$$

Therefore, using the entrywise formula for hhessS, splitting the double sum, and then (4) and (5),

$$\sum_i \sum_j \big(\text{hhessS } \sigma \ x_0 \ D_u \ B_u\big)_{i,j} \, \xi_j \, \xi_i = B_u\big(h_\xi, h_\xi\big) + \sum_i \sum_j \mathcal{M}_{i,j} \, \xi_j \, \xi_i,$$

$$\sum_i \sum_j \big(\text{hhessS } \sigma \ x_0 \ D_v \ B_v\big)_{i,j} \, \xi_j \, \xi_i = B_v\big(h_\xi, h_\xi\big) + \sum_i \sum_j \mathcal{M}_{i,j} \, \xi_j \, \xi_i,$$

with the *same* second summand in both lines by (4). By Step 3 applied to the vector h_ξ, $B_u(h_\xi, h_\xi) \le B_v(h_\xi, h_\xi)$. Adding the common term preserves the inequality, so

$$\sum_i \sum_j \big(\text{hhessS } \sigma \ x_0 \ D_u \ B_u\big)_{i,j} \, \xi_j \, \xi_i \ \le \ \sum_i \sum_j \big(\text{hhessS } \sigma \ x_0 \ D_v \ B_v\big)_{i,j} \, \xi_j \, \xi_i.$$

Since $\xi \in \mathbb{R}^m$ was arbitrary, this is exactly the relation hhessS $\sigma \ x_0 \ D_u \ B_u \le$ hhessS $\sigma \ x_0 \ D_v \ B_v$ of `MatLe`, i.e. (4) of the statement. In the paper's notation, $\mathcal{X}\mathcal{X}^s u(x_0) \le \mathcal{X}\mathcal{X}^s v(x_0)$.

Remark. No semi-convexity is assumed: the hypothesis is only the existence of the two second-order jets at the single point x_0 together with the local maximum, which is all the paper's sentence uses. The role of semi-convexity in Section 3.1 is to produce such points, which is the business of Aleksandrov's theorem and Jensen's lemma, not of this observation. Combined with the trace monotonicity (2.2), items (3) and (4) give $0 \le \operatorname{Tr}\big(A(\xi_0)(\mathcal{X}\mathcal{X}^s v(x_0) - \mathcal{X}\mathcal{X}^s u(x_0))\big) = \mathcal{L}u(x_0) - \mathcal{L}v(x_0)$, the terms $H(\xi_0)$ cancelling because the horizontal gradients coincide by (3); that is how the observation is used. □

2.15. `PerturbationIdentity`.

Lemma 2.15 (Lemma 3.1)**.** *Let $n, m \in \mathbb{N}$, let $\sigma : \mathbb{R}^n \to \mathbb{R}^{n\times m}$, $\sigma = [\sigma^1, \dots, \sigma^m]$, be a frame of m vector fields on $\mathbb{R}^n$, and let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$, $H : \mathbb{R}^m \to \mathbb{R}$ satisfy the normalization (2.3) (`NormalizedAH`). Let $\psi \in C^2(\mathbb{R})$, let $w : \mathbb{R}^n \to \mathbb{R}$, let $x_0 \in \mathbb{R}^n$ and let (D, B) be a second-order jet of w at x_0 (`TwiceDiffAt`). Put*

$$\xi_0 := \text{hgrad } \sigma \ x_0 \ D \quad (\texttt{hgrad}), \qquad t := \psi'\big(w(x_0)\big),$$

and assume $\xi_0 \neq 0$ and $t > 0$. Let $D \otimes D$ denote the continuous bilinear form $(h, k) \mapsto D(h)\, D(k)$. Then:

1. *$\big(tD, \ tB + \psi''(w(x_0))\, D \otimes D\big)$ is a second-order jet of $\psi \circ w$ at x_0;*
2. *hgrad $\sigma \ x_0 \ (tD) = t\, \xi_0$;*
3. *with that jet, (3.3) holds (`Lop`, `Ellip`):*

$$\text{Lop } \sigma \ A \ H \ x_0 \ (tD) \ \big(tB + \psi''(w(x_0))\, D \otimes D\big) \ = \ t\Big[\text{Lop } \sigma \ A \ H \ x_0 \ D \ B - \frac{\psi''(w(x_0))}{\psi'(w(x_0))}\, \mathcal{E}\big(A, \xi_0\big)\Big].$$

Proof. Write $a := w(x_0)$, $t = \psi'(a)$, $c := \psi''(a)$, and $h := y - x_0$ for the increment. Items (1) and (2) are the paper's two displayed identities $\mathcal{X}(\psi \circ w) = \psi'(w)\mathcal{X}w = t\xi_0$ and $\mathcal{X}\mathcal{X}^s(\psi \circ w) = \psi'(w)\mathcal{X}\mathcal{X}^s w + \psi''(w)\mathcal{X}w \otimes \mathcal{X}w$ at x_0, read on jets; item (3) is the computation of Lemma 3.1's proof, whose last step is (2.5). Since (2.5) is a two-line consequence of (2.3) (pm-0012 keeps it inline rather than as a node), we derive it in Step 4 below.

Step 1: a Peano expansion for ψ at a. We claim: for every $\varepsilon > 0$ there is $\rho > 0$ such that

$$\left|\psi(a+s) - \psi(a) - t\,s - \tfrac{1}{2}c\,s^2\right| \;\le\; \varepsilon\, s^2 \qquad \text{whenever } |s| < \rho. \tag{1}$$

Indeed, ψ'' is continuous because $\psi \in C^2$, so there is $\rho > 0$ with $|\psi''(\theta) - \psi''(a)| \le 2\varepsilon$ whenever $|\theta - a| \le \rho$. Let $|s| < \rho$. If $s = 0$ the claim is trivial. Otherwise, by Taylor's theorem with Lagrange remainder applied to ψ on the segment between a and $a+s$, there is θ strictly between a and $a + s$ with

$$\psi(a+s) = \psi(a) + \psi'(a)s + \tfrac{1}{2}\psi''(\theta)s^2,$$

and $|\theta - a| < |s| < \rho$, so $|\psi(a+s) - \psi(a) - ts - \frac{1}{2}cs^2| = \frac{1}{2}|\psi''(\theta) - \psi''(a)|\,s^2 \le \varepsilon s^2$. This is (1). (Only $\psi \in C^2$ near a is used; the hypothesis $t > 0$ plays no role here.)

Step 2: proof of (1). Set

$$D_\psi := tD, \qquad B_\psi := tB + c\,D \otimes D.$$

B_ψ is symmetric: B is symmetric by `TwiceDiffAt` and $(D \otimes D)(h,k) = D(h)D(k) = (D \otimes D)(k,h)$. It remains to prove the Peano estimate.

Write, for $y \in \mathbb{R}^n$ and $h = y - x_0$,

$$S(h) := w(y) - w(x_0), \qquad R(h) := S(h) - D(h) - \tfrac{1}{2}B(h,h),$$

so that by `TwiceDiffAt` applied to w: for every $\varepsilon_1 > 0$ there is $\delta_1 > 0$ with

$$|R(h)| \le \varepsilon_1 |h|^2 \qquad \text{for } |h| < \delta_1. \tag{2}$$

Fix once and for all $\delta_1^0 > 0$ obtained from (2) with $\varepsilon_1 = 1$, and put

$$C_0 := \|D\| + \tfrac{1}{2}\|B\| + 1.$$

For $|h| < \min\{\delta_1^0, 1\}$ we get from (2) with $\varepsilon_1 = 1$ that

$$|S(h)| \le |D(h)| + \tfrac{1}{2}|B(h,h)| + |h|^2 \le \big(\|D\| + \tfrac{1}{2}\|B\|\,|h| + |h|\big)|h| \le C_0|h|, \tag{3}$$

using $|h| \le 1$. Likewise, for such h,

$$|S(h) - D(h)| \le \tfrac{1}{2}\|B\|\,|h|^2 + |h|^2 = C_1|h|^2, \qquad C_1 := \tfrac{1}{2}\|B\| + 1, \tag{4}$$

and $|S(h) + D(h)| \le (C_0 + \|D\|)|h|$ by (3).

Now let $\varepsilon > 0$ be given. Choose, in this order:

- $\varepsilon_2 := \dfrac{\varepsilon}{3(C_0^2 + 1)} > 0$ and, by Step 1 with ε_2, a radius $\rho > 0$ for which (1) holds;
- $\varepsilon_1 := \dfrac{\varepsilon}{3(|t| + 1)} > 0$ and, by (2), a radius $\delta_1 > 0$ for which (2) holds with this ε_1;
- $\delta := \min\left\{\delta_1^0,\, 1,\, \delta_1,\, \dfrac{\rho}{C_0 + 1},\, \dfrac{2\varepsilon}{3(|c| + 1)C_1(C_0 + \|D\| + 1)}\right\} > 0.$

Let $|h| < \delta$. By (3) and $|h| < \rho/(C_0 + 1)$ we have $|S(h)| \le C_0|h| < \rho$, so (1) applies with $s := S(h)$. Decompose, using $\psi(w(y)) = \psi(a + S(h))$ and $\psi(w(x_0)) = \psi(a)$:

$$\psi(w(y)) - \psi(w(x_0)) - D_\psi(h) - \tfrac{1}{2}B_\psi(h,h) = \underbrace{\Big[\psi(a+S) - \psi(a) - tS - \tfrac{1}{2}cS^2\Big]}_{=:T_1} + \underbrace{t\Big[S - D(h) - \tfrac{1}{2}B(h,h)\Big]}_{=:T_2} + \underbrace{\tfrac{1}{2}c\Big[S^2 -}_{=:}$$

where $S = S(h)$; this is an identity, obtained by expanding $D_\psi(h) = tD(h)$ and $\frac{1}{2}B_\psi(h,h) = \frac{t}{2}B(h,h) + \frac{c}{2}D(h)^2$ and cancelling. We estimate the three terms.

$|T_1| \le \varepsilon_2 S^2 \le \varepsilon_2 C_0^2 |h|^2 \le \frac{\varepsilon}{3}|h|^2$ by (1), (3) and the choice of ε_2.

$|T_2| = |t|\,|R(h)| \le |t|\varepsilon_1|h|^2 \le \frac{\varepsilon}{3}|h|^2$ by (2) and the choice of ε_1.

$|T_3| = \frac{1}{2}|c|\,|S - D(h)|\,|S + D(h)| \le \frac{1}{2}|c|\,C_1|h|^2(C_0 + \|D\|)|h| \le \frac{\varepsilon}{3}|h|^2$ by (4), (3) and the last entry in the definition of δ, which gives $|h| \le \frac{2\varepsilon}{3(|c|+1)C_1(C_0+\|D\|+1)}$.

Adding, $|\psi(w(y)) - \psi(w(x_0)) - D_\psi(h) - \frac{1}{2}B_\psi(h,h)| \le \varepsilon|h|^2$ for $|h| < \delta$. Since $\varepsilon > 0$ was arbitrary, (D_ψ, B_ψ) is a second-order jet of $\psi \circ w$ at x_0. This is (1). (The cubic term T_3 is where the paper's $\mathcal{X}w \otimes \mathcal{X}w$ term comes from: it is $\frac{1}{2}c\,D(h)^2$ that survives, the discrepancy $S^2 - D(h)^2$ being of order $|h|^3$.)

Step 3: proof of (2), and the splitting of the second jet. By `hgrad`, $(\text{hgrad } \sigma\ x_0\ D)_i = D(\sigma^i(x_0))$, which is linear in the 1-jet; hence $\text{hgrad } \sigma\ x_0\ (tD) = \big(tD(\sigma^i(x_0))\big)_i = t\,\xi_0$, which is (2). In particular $t\xi_0 \neq 0$, since $t > 0$ and $\xi_0 \neq 0$.

Next, by `hhessS` and `Mfield`,

$$\big(\text{hhessS } \sigma\ x_0\ D'\ B'\big)_{i,j} = B'\big(\sigma^i(x_0), \sigma^j(x_0)\big) + \mathcal{M}(x_0, D')_{i,j}, \qquad \mathcal{M}(x_0, D')_{i,j} = \tfrac{1}{2}\Big(D'\big(D\sigma^j\sigma^i\big) + D'\big(D\sigma^i\sigma^j\big)\Big),$$

so $\mathcal{M}(x_0, \cdot)$ is linear in the 1-jet and $\mathcal{M}(x_0, tD) = t\,\mathcal{M}(x_0, D)$. Moreover $(D\otimes D)(\sigma^i(x_0), \sigma^j(x_0)) = D(\sigma^i(x_0))D(\sigma^j(x_0)) = (\xi_0)_i(\xi_0)_j$ by `hgrad`. Hence, writing $\Xi := \big((\xi_0)_i(\xi_0)_j\big)_{i,j} = \xi_0 \otimes \xi_0$ and $M := \text{hhessS } \sigma\ x_0\ D\ B$,

$$\text{(5)}\quad \text{hhessS } \sigma\ x_0\ (tD)\ \big(tB + c\,D\otimes D\big) = \big(tB(\sigma^i, \sigma^j) + c(\xi_0)_i(\xi_0)_j\big)_{i,j} + t\,\mathcal{M}(x_0, D) = t\,M + c\,\Xi.$$

This is the paper's $\mathcal{X}\mathcal{X}^s(\psi \circ w) = \psi'(w)\mathcal{X}\mathcal{X}^s w + \psi''(w)\mathcal{X}w \otimes \mathcal{X}w$.

Step 4: (2.5), and the proof of (3). First, the trace against Ξ is the ellipticity function: by `Ellip`,

$$\text{(6)}\qquad \operatorname{Tr}\big(A(\xi_0)\,\Xi\big) = \sum_i \sum_j A(\xi_0)_{i,j}\,\Xi_{j,i} = \sum_i \sum_j A(\xi_0)_{i,j}(\xi_0)_j(\xi_0)_i = \mathcal{E}(A, \xi_0).$$

Second, (2.5). By (2.3) (`NormalizedAH`) and $t > 0$, $\xi_0 \neq 0$,

$$\text{(7)}\qquad A(t\xi_0) = A(\xi_0), \qquad H(t\xi_0) = t\,H(\xi_0),$$

whence, for any symmetric $X \in \mathbb{R}^{m\times m}$,

$$-t\operatorname{Tr}\big(A(t\xi_0)X\big) + H(t\xi_0) = -t\operatorname{Tr}\big(A(\xi_0)X\big) + t\,H(\xi_0) = t\Big[-\operatorname{Tr}\big(A(\xi_0)X\big) + H(\xi_0)\Big],$$

which is (2.5); those are the two lines promised.

Now compute, using `Lop`, then (2) and (5), then (7), then linearity of the trace and (6):

$$\begin{aligned}
\text{Lop } \sigma\ A\ H\ x_0\ (tD)\ \big(tB + c\,D\otimes D\big) &= -\operatorname{Tr}\Big(A\big(\text{hgrad } \sigma\ x_0\ (tD)\big)\,\text{hhessS } \sigma\ x_0\ (tD)\ \big(tB + c\,D\otimes D\big)\Big) \\
&\qquad + H\big(\text{hgrad } \sigma\ x_0\ (tD)\big) \\
&= -\operatorname{Tr}\Big(A(t\xi_0)\big(tM + c\,\Xi\big)\Big) + H(t\xi_0) \\
&= -\operatorname{Tr}\Big(A(\xi_0)\big(tM + c\,\Xi\big)\Big) + t\,H(\xi_0) \\
&= -t\,\operatorname{Tr}\big(A(\xi_0)M\big) - c\,\operatorname{Tr}\big(A(\xi_0)\Xi\big) + t\,H(\xi_0) \\
&= t\Big[-\operatorname{Tr}\big(A(\xi_0)M\big) + H(\xi_0)\Big] - c\,\mathcal{E}(A, \xi_0) \\
&= t\,\text{Lop } \sigma\ A\ H\ x_0\ D\ B - c\,\mathcal{E}(A, \xi_0).
\end{aligned}$$

Since $t > 0$ we may factor t out of the last line:

$$\text{Lop } \sigma\ A\ H\ x_0\ (tD)\ \big(tB + c\,D\otimes D\big) = t\Big[\text{Lop } \sigma\ A\ H\ x_0\ D\ B - \frac{c}{t}\,\mathcal{E}(A, \xi_0)\Big],$$

which is (3), i.e. (3.3) with $c = \psi''(w(x_0))$ and $t = \psi'(w(x_0))$.

Remark on the hypothesis on ψ. Only $\psi'(w(x_0)) > 0$ is assumed, not the paper's global $\psi' > 0$. That is all the proof uses: $t > 0$ enters exactly twice, in (7) — where (2.3) needs a positive factor — and in the division by t at the end. Stating it pointwise is necessary for the use made of this lemma later: the relabelling $\psi_\lambda(s) = s + \lambda e^s$ has $\psi_\lambda' > 0$ everywhere but its companion corollary needs a pointwise lower bound on ψ_λ'', and keeping both hypotheses pointwise keeps

the two statements uniform. Nothing is lost, since a ψ with $\psi' > 0$ everywhere satisfies the hypothesis at every point. □

2.16. `StrictSubsolutionPerturbation`.

Corollary 2.16 (Corollary 3.2)**.** *Let $n, m \in \mathbb{N}$, let $\sigma = [\sigma^1, \dots, \sigma^m]$ be a frame of m vector fields on $\mathbb{R}^n$, and let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$, $H : \mathbb{R}^m \to \mathbb{R}$ satisfy the normalization (2.3) (`NormalizedAH`). Let $e_0 > 0$ be a constant with*

$$e_0\,|\xi|^2 \;\le\; \mathcal{E}(A,\xi) \qquad \text{for all } \xi \in \mathbb{R}^m \quad (\texttt{Ellip}),$$

let $\psi \in C^2(\mathbb{R})$, let $w : \mathbb{R}^n \to \mathbb{R}$, let $x_0 \in \mathbb{R}^n$ and let (D, B) be a second-order jet of w at x_0 (`TwiceDiffAt`). Put $\xi_0 :=$ hgrad σ x_0 D (`hgrad`), $t := \psi'(w(x_0))$ and $c := \psi''(w(x_0))$, and assume

$$\theta > 0, \qquad \theta \le |\xi_0|, \qquad \lambda > 0, \qquad t > 0, \qquad \lambda \le c, \qquad \mathrm{Lop}\ \sigma\ A\ H\ x_0\ D\ B \le 0 \quad (\texttt{Lop}).$$

Then $\big(tD,\ tB + c\,D \otimes D\big)$ is a second-order jet of $\psi \circ w$ at x_0, where $(D \otimes D)(h,k) = D(h)D(k)$, and

$$\mathrm{Lop}\ \sigma\ A\ H\ x_0\ (tD)\ \big(tB + c\,D\otimes D\big) \;\le\; -\gamma, \qquad \gamma := \lambda\, e_0\, \theta^2 > 0.$$

Proof. This is Corollary 3.2. The constant e_0 is the one of (2.4): it exists, is positive and depends only on A (not on x_0, w or ψ) by `EllipticityLowerBound`, applied under the structure conditions (E) and (2.3). We carry it as a parameter rather than re-deriving it here, precisely so that the resulting γ is one constant valid at every point at which this corollary is applied — which is what Proposition 3.3 needs in its Steps 3 and 4.

Step 1: $\xi_0 \neq 0$. From $0 < \theta \le |\xi_0|$ we get $|\xi_0| > 0$, so $\xi_0 \neq 0$. This is the admissibility condition needed to invoke (2.3).

Step 2: the perturbation identity. All the hypotheses of `PerturbationIdentity` are in place: (2.3) for (A, H), $\psi \in C^2$, (D, B) a second-order jet of w at x_0, $\xi_0 \neq 0$ by Step 1, and $t > 0$. It gives that $(tD,\ tB + c\,D \otimes D)$ is a second-order jet of $\psi \circ w$ at x_0 — the first assertion — and the identity (3.3)

(1)

$$\mathrm{Lop}\,\sigma\,A\,H\,x_0\,(tD)\,\big(tB{+}c\,D{\otimes}D\big) = t\Big[\mathrm{Lop}\,\sigma\,A\,H\,x_0\,D\,B{-}\frac{c}{t}\,\mathcal{E}(A,\xi_0)\Big] = t\,\mathrm{Lop}\,\sigma\,A\,H\,x_0\,D\,B{-}c\,\mathcal{E}(A,\xi_0),$$

the second equality because $t > 0$, so $t \cdot (c/t) = c$.

Step 3: the three inequalities of line 477. First, since $t > 0$ and $\mathrm{Lop}\ \sigma\ A\ H\ x_0\ D\ B \le 0$,

$$t\,\mathrm{Lop}\ \sigma\ A\ H\ x_0\ D\ B \;\le\; 0,$$

so (1) gives

$$\mathrm{Lop}\ \sigma\ A\ H\ x_0\ (tD)\ \big(tB + c\,D\otimes D\big) \;\le\; -c\,\mathcal{E}(A,\xi_0). \tag{2}$$

This is the paper's first inequality $\mathcal{L}(\psi\circ w)(x_0) \le -\psi''(w(x_0))\mathcal{E}(\xi_0)$.

Second, by the hypothesis on e_0 and $\theta \le |\xi_0|$ with $\theta > 0$,

$$\mathcal{E}(A,\xi_0) \;\ge\; e_0\,|\xi_0|^2 \;\ge\; e_0\,\theta^2 \;>\; 0. \tag{3}$$

(The middle step uses $0 < \theta \le |\xi_0|$, hence $\theta^2 \le |\xi_0|^2$, and $e_0 > 0$.)

Third, $c \ge \lambda > 0$, so multiplying (3) by $c > 0$ and then using $\lambda \le c$ with $e_0\theta^2 > 0$,

$$c\,\mathcal{E}(A,\xi_0) \;\ge\; c\,e_0\,\theta^2 \;\ge\; \lambda\,e_0\,\theta^2 = \gamma > 0.$$

Combining with (2),

$$\mathrm{Lop}\ \sigma\ A\ H\ x_0\ (tD)\ \big(tB + c\,D\otimes D\big) \;\le\; -\gamma,$$

which is the assertion. The chain of the three displays is exactly the paper's $\mathcal{L}(\psi\circ w)(x_0) \le -\psi''\mathcal{E}(\xi_0) \le -\lambda e_0|\xi_0|^2 \le -\lambda e_0\theta^2$.

Remark 1 (pointwise hypotheses). The paper assumes $\psi' > 0$ and $\psi'' \ge \lambda > 0$ globally; we assume only $\psi'(w(x_0)) > 0$ and $\lambda \le \psi''(w(x_0))$. This is a strictly weaker hypothesis, so the corollary as stated is stronger, and it is the form that is needed: the relabelling used in Proposition 3.3 is $\psi_\lambda(s) = s + \lambda e^s$, whose second derivative λe^s has no positive global lower bound, while on

the range of a bounded u — say $|u| \le M$ on $\bar\Omega$ — it satisfies $\psi_\lambda''(u(x_0)) \ge \lambda e^{-M} > 0$. So Proposition 3.3 will apply this corollary with $\lambda' := \lambda e^{-M}$ in place of λ, giving one uniform γ. Every ψ satisfying the paper's global hypotheses satisfies ours at every point, so nothing is lost.

Remark 2 (no super-solution version). The paper notes at line 479 that Lemma 3.1 and this corollary have obvious super-solution versions, with $\psi'' \le -\lambda < 0$, "which we shall not need: the hypothesis of Proposition Proposition 3.3 of the paper below is symmetric in u and v, because $\mathcal{X}u = \mathcal{X}v$ at the maximal points of $u - v$". We record this so that the super-solution version is not added later: it is genuinely unnecessary, since Proposition 3.3 perturbs only the sub-solution u, and the non-degeneracy hypothesis there — that $\mathcal{X}u$ does not vanish on $\operatorname{argmax}_\Omega(u-v)$ — is equivalent to the same statement for $\mathcal{X}v$ by the first conclusion of `LocalMaxSecondOrder`. □

2.17. `TraceMonotone`.

Theorem 2.17 ((2.2): monotonicity of $Z \mapsto \operatorname{Tr}(AZ)$)**.** *Let $m \in \mathbb{N}$ and let $A, Y, X \in \mathbb{R}^{m\times m}$ satisfy*

1. *A is symmetric, $A^T = A$, and non-negative definite: $\langle A\eta, \eta\rangle = \sum_i \sum_j A_{i,j}\eta_j\eta_i \ \ge\ 0$ for every $\eta \in \mathbb{R}^m$;*
2. *Y and X are symmetric;*
3. *$Y \le X$ in the order (2.1) (`MatLe`), i.e. $\langle Y\eta, \eta\rangle \le \langle X\eta, \eta\rangle$ for every $\eta \in \mathbb{R}^m$.*

Then $\operatorname{Tr}(AY) \le \operatorname{Tr}(AX)$.

Proof. **Step 1: reduction to a single trace.** Put $Z := X - Y$. Matrix multiplication distributes over subtraction, $A(X - Y) = AX - AY$, and the trace is additive, $\operatorname{Tr}(P - Q) = \operatorname{Tr}(P) - \operatorname{Tr}(Q)$. Hence

$$\operatorname{Tr}(AX) - \operatorname{Tr}(AY) = \operatorname{Tr}(AX - AY) = \operatorname{Tr}\big(A(X - Y)\big) = \operatorname{Tr}(AZ),$$

so the assertion $\operatorname{Tr}(AY) \le \operatorname{Tr}(AX)$ is equivalent to

$$\operatorname{Tr}(AZ) \ \ge\ 0. \tag{1}$$

Step 2: Z is symmetric and non-negative definite. $Z = X - Y$ is symmetric, being a difference of the symmetric matrices X and Y of hypothesis (2): $Z^T = X^T - Y^T = X - Y = Z$. For non-negativity, let $\eta \in \mathbb{R}^m$; since $Z_{i,j} = X_{i,j} - Y_{i,j}$ and finite sums split over subtraction,

$$\langle Z\eta, \eta\rangle = \sum_i \sum_j \big(X_{i,j} - Y_{i,j}\big)\eta_j\eta_i = \sum_i \sum_j X_{i,j}\eta_j\eta_i - \sum_i \sum_j Y_{i,j}\eta_j\eta_i = \langle X\eta, \eta\rangle - \langle Y\eta, \eta\rangle \ \ge\ 0,$$

the last inequality being hypothesis (3), i.e. `MatLe` applied to the vector η.

Step 3: symmetric non-negative definite square roots. We use the following standard fact of linear algebra: *every symmetric non-negative definite $M \in \mathbb{R}^{m\times m}$ has a symmetric square root*, i.e. a symmetric $\sqrt{M} \in \mathbb{R}^{m\times m}$ with $\sqrt{M}\sqrt{M} = M$. Indeed, by the spectral theorem for real symmetric matrices there are an orthogonal Q and reals $\lambda_1, \dots, \lambda_m$ with $M = Q\operatorname{diag}(\lambda_1, \dots, \lambda_m)\, Q^T$; each λ_k is non-negative, because if v_k denotes the k-th column of Q then $|v_k| = 1$ and $\langle Mv_k, v_k\rangle = \lambda_k \ge 0$ by non-negative definiteness of M; so one may set $\sqrt{M} := Q\operatorname{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_m})\, Q^T$, which is symmetric and satisfies $\sqrt{M}\sqrt{M} = Q\operatorname{diag}(\lambda_1, \dots, \lambda_m)\, Q^T = M$ because $Q^TQ = I$.

Hypothesis (1) says A is symmetric and non-negative definite, and Step 2 says Z is; so both $\sqrt{A}$ and $\sqrt{Z}$ exist, are symmetric, and satisfy $\sqrt{A}\sqrt{A} = A$, $\sqrt{Z}\sqrt{Z} = Z$.

Step 4: the identity $\operatorname{Tr}(AZ) = |\sqrt{A}\sqrt{Z}|^2$. Put $M := \sqrt{A}\,\sqrt{Z}$. Using $(PQ)^T = Q^TP^T$, the symmetry of $\sqrt{A}$ and $\sqrt{Z}$, the commutation $\operatorname{Tr}(PQ) = \operatorname{Tr}(QP)$ of the trace, and the two square-root identities,

$$\operatorname{Tr}\big(M^TM\big) = \operatorname{Tr}\big(\sqrt{Z}^T\sqrt{A}^T\sqrt{A}\sqrt{Z}\big) = \operatorname{Tr}\big(\sqrt{Z}\,A\,\sqrt{Z}\big) = \operatorname{Tr}\big(A\,\sqrt{Z}\,\sqrt{Z}\big) = \operatorname{Tr}(AZ),$$

where the third equality is $\mathrm{Tr}(PQ) = \mathrm{Tr}(QP)$ with $P = \sqrt{Z}$ and $Q = A\sqrt{Z}$. On the other hand, writing out the trace of M^TM entrywise,

$$\mathrm{Tr}\big(M^TM\big) = \sum_i \big(M^TM\big)_{i,i} = \sum_i \sum_k \big(M^T\big)_{i,k} M_{k,i} = \sum_i \sum_k M_{k,i}^2 \;\geq\; 0,$$

every summand being a square of a real number. (This is the Frobenius norm identity $\mathrm{Tr}(M^TM) = |M|^2$ of the paper's notation section.) Combining the two displays gives (1):

$$\mathrm{Tr}(AZ) = \mathrm{Tr}\big(M^TM\big) = \big|\sqrt{A}\sqrt{Z}\big|^2 \;\geq\; 0.$$

Step 5: conclusion. By Step 1 and (1), $\mathrm{Tr}(AX) - \mathrm{Tr}(AY) = \mathrm{Tr}(AZ) \geq 0$, i.e. $\mathrm{Tr}(AY) \leq \mathrm{Tr}(AX)$. □

2.18. `SecondOrderJetCalculus`.

Lemma 2.18 (The second-order jet calculus)**.** *Let $N \in \mathbb{N}$. Throughout, "(D, B) is a second-order jet of w at z" means* TwiceDiffAt *w z D B in the sense of* `TwiceDiffAt`*: the continuous bilinear form B on $\mathbb{R}^N$ is symmetric, and for every $\varepsilon > 0$ there is $\delta > 0$ such that*

$$\Big|w(y) - w(z) - D(y-z) - \tfrac{1}{2}B(y-z, y-z)\Big| \;\leq\; \varepsilon|y-z|^2 \qquad \text{whenever } |y-z| < \delta.$$

Then the following hold.

1. *(Cubic remainder criterion.) Let $g : \mathbb{R}^N \to \mathbb{R}$, let $z \in \mathbb{R}^N$, let D be a continuous linear functional and B a continuous bilinear form on $\mathbb{R}^N$, and let $C \in \mathbb{R}$. If B is symmetric, $C \geq 0$, and*

$$\Big|g(y) - g(z) - D(y-z) - \tfrac{1}{2}B(y-z, y-z)\Big| \;\leq\; C|y-z|^3 \qquad \text{for every } y \text{ with } |y-z| \leq 1,$$

   then (D, B) is a second-order jet of g at z.
2. *(Additivity.) If (D, B) is a second-order jet of w at z and (D', B') one of w' at z, then $(D + D', B + B')$ is a second-order jet of $w + w'$ at z.*
3. *(Negation.) If (D, B) is a second-order jet of w at z, then $(-D, -B)$ is a second-order jet of $-w$ at z.*
4. *(Subtractivity.) If (D, B) is a second-order jet of w at z and (D', B') one of w' at z, then $(D - D', B - B')$ is a second-order jet of $w - w'$ at z.*
5. *(Linear functionals.) For all $p, z \in \mathbb{R}^N$, the function $y \mapsto \sum_i p_i y_i$ has at z the second-order jet $\big(\langle p, \cdot\rangle, 0\big)$, with vanishing second-order part.*

Proof. **Part (1).** Symmetry of B is one of the hypotheses, so only the Peano estimate of `TwiceDiffAt` has to be produced. Let $\varepsilon > 0$ and put

$$\delta := \min\Big\{1,\; \frac{\varepsilon}{C+1}\Big\}.$$

Then $\delta > 0$: the first entry is positive, and the second is positive because $\varepsilon > 0$ and $C + 1 > 0$ (here $C \geq 0$). Let y satisfy $|y - z| < \delta$. Then $|y - z| < 1$, in particular $|y - z| \leq 1$, so the assumed cubic bound applies at y; and $|y - z| < \varepsilon/(C + 1)$. Multiplying the latter by $C \geq 0$ gives

$$C|y-z| \;\leq\; C \cdot \frac{\varepsilon}{C+1} \;\leq\; \varepsilon,$$

the second inequality because $C\varepsilon \leq (C+1)\varepsilon$, i.e. because $0 \leq \varepsilon$, and $C + 1 > 0$. Hence, using $|y - z|^2 \geq 0$,

$$\Big|g(y) - g(z) - D(y-z) - \tfrac{1}{2}B(y-z, y-z)\Big| \;\leq\; C|y-z|^3 \;=\; \big(C|y-z|\big)\,|y-z|^2 \;\leq\; \varepsilon|y-z|^2.$$

This is the required estimate, with this δ for this ε.

Part (2). Symmetry: for all $a, b \in \mathbb{R}^N$, $(B + B')(a, b) = B(a, b) + B'(a, b) = B(b, a) + B'(b, a) = (B + B')(b, a)$, using symmetry of B and of B'. For the estimate, let $\varepsilon > 0$. Since $\varepsilon/2 > 0$, the Peano estimates for w and w' supply $\delta_1 > 0$ and $\delta_2 > 0$ with

$$\Big|w(y) - w(z) - D(y-z) - \tfrac{1}{2}B(y-z, y-z)\Big| \leq \tfrac{\varepsilon}{2}|y-z|^2 \quad \text{for } |y-z| < \delta_1,$$

and the analogous bound for w', D', B' for $|y-z|<\delta_2$. Put $\delta:=\min\{\delta_1,\delta_2\}>0$ and let $|y-z|<\delta$; then both bounds apply. Since $(D+D')(y-z)=D(y-z)+D'(y-z)$ and $(B+B')(y-z,y-z)=B(y-z,y-z)+B'(y-z,y-z)$, rearranging gives the identity

$$\big(w+w'\big)(y)-\big(w+w'\big)(z)-(D+D')(y-z)-\tfrac12(B+B')(y-z,y-z)$$

$$=\Big(w(y)-w(z)-D(y-z)-\tfrac12 B(y-z,y-z)\Big)+\Big(w'(y)-w'(z)-D'(y-z)-\tfrac12 B'(y-z,y-z)\Big).$$

By the triangle inequality the absolute value of the left-hand side is at most the sum of the absolute values of the two brackets, hence at most $\frac{\varepsilon}{2}|y-z|^2+\frac{\varepsilon}{2}|y-z|^2=\varepsilon|y-z|^2$.

Part (3). Symmetry: $(-B)(a,b)=-B(a,b)=-B(b,a)=(-B)(b,a)$. For the estimate, let $\varepsilon>0$ and take the same $\delta>0$ that the Peano estimate for w supplies for this ε. For $|y-z|<\delta$, the identities $(-D)(y-z)=-D(y-z)$ and $(-B)(y-z,y-z)=-B(y-z,y-z)$ give

$$-w(y)-\big(-w(z)\big)-(-D)(y-z)-\tfrac12(-B)(y-z,y-z)=-\Big(w(y)-w(z)-D(y-z)-\tfrac12 B(y-z,y-z)\Big),$$

and $|-t|=|t|$, so the left-hand side has the same absolute value as the remainder for w, which is at most $\varepsilon|y-z|^2$.

Part (4). Symmetry of $B-B'$ follows as in (2): $(B-B')(a,b)=B(a,b)-B'(a,b)=B(b,a)-B'(b,a)=(B-B')(b,a)$. For the estimate, apply (3) to w' to get that $(-D',-B')$ is a second-order jet of $-w'$ at z, and then (2) to the pair w, $-w'$: $(D+(-D'),B+(-B'))$ is a second-order jet of $y\mapsto w(y)+(-w'(y))$ at z. Let $\varepsilon>0$ and let $\delta>0$ be the radius that this jet's Peano estimate supplies. For $|y-z|<\delta$, the remainder of $w-w'$ against $(D-D',B-B')$ coincides with the remainder of $w+(-w')$ against $(D+(-D'),B+(-B'))$: indeed $w(y)-w'(y)=w(y)+(-w'(y))$, $(D-D')(y-z)=(D+(-D'))(y-z)$ and $(B-B')(y-z,y-z)=(B+(-B'))(y-z,y-z)$, since subtraction of linear maps and of bilinear forms is addition of the negative. So the bound $\varepsilon|y-z|^2$ transfers verbatim.

Part (5). Fix $p,z\in\mathbb{R}^N$ and put $g(y):=\sum_i p_i y_i$. We apply (1) with $D:=\langle p,\cdot\rangle$, $B:=0$ and $C:=0$. The zero form is symmetric and $0\geq 0$, so it remains to check the cubic bound, and we show the remainder is identically zero. On $\mathbb{R}^N$ the inner product is $\langle p,w\rangle=\sum_i p_i w_i$, so $D(y-z)=\sum_i p_i(y-z)_i$. Since subtraction is coordinatewise, $(y-z)_i=y_i-z_i$, and a finite sum splits over subtraction, so

$$D(y-z)=\sum_i p_i(y_i-z_i)=\sum_i\big(p_iy_i-p_iz_i\big)=\sum_i p_iy_i-\sum_i p_iz_i=g(y)-g(z).$$

As $B=0$ gives $\frac12 B(y-z,y-z)=0$, the remainder is $g(y)-g(z)-D(y-z)-0=0$, whose absolute value is $0\leq 0\cdot|y-z|^3$ for every y. Part (1) therefore applies and yields that $\big(\langle p,\cdot\rangle,0\big)$ is a second-order jet of g at z. □

2.19. `ExpRelabellingJetDescent`.

Lemma 2.19 (Descent of a second-order jet through the relabelling ψ_λ)**.** *Let $N\in\mathbb{N}$, let $\lambda>0$, let $u:\mathbb{R}^N\to\mathbb{R}$, let $z\in\mathbb{R}^N$ and let $t,c\in\mathbb{R}$. Let D_j,D_u be continuous linear functionals and B_j,B_u continuous bilinear forms on $\mathbb{R}^N$, and assume*

1. *$t=1+\lambda e^{u(z)}$ and $c=\lambda e^{u(z)}$;*
2. *$D_uh=t^{-1}D_jh$ for every $h\in\mathbb{R}^N$;*
3. *$B_u(h,h')=t^{-1}\big(B_j(h,h')-c\,D_u(h)\,D_u(h')\big)$ for all $h,h'\in\mathbb{R}^N$;*
4. *B_u is symmetric, $B_u(a,b)=B_u(b,a)$ for all a,b;*
5. *(D_j,B_j) is a second-order jet of $y\mapsto u(y)+\lambda e^{u(y)}$ at z, in the sense of `TwiceDiffAt`.*

Then (D_u,B_u) is a second-order jet of u at z, again in the sense of `TwiceDiffAt`.

Proof. Write $\psi(s):=s+\lambda e^s$, so that hypothesis (5) says that (D_j,B_j) is a second-order jet of $\psi\circ u$ at z. Since $\psi'(s)=1+\lambda e^s$ and $\psi''(s)=\lambda e^s$, hypothesis (1) reads $t=\psi'(u(z))$ and $c=\psi''(u(z))$. No inverse function ψ^{-1} is used anywhere below, and no separate first-order chain rule: the first-order expansion of u is produced from the second-order estimate itself in Step 9.

Step 1: the constants t and c. Since $\lambda > 0$ and $e^{u(z)} > 0$ we get $c = \lambda e^{u(z)} > 0$, and then $t = 1 + c \geq 1$ by (1). In particular $t > 0$, so $t \neq 0$, $t^{-1} > 0$, and multiplying $1 \leq t$ by $t^{-1} > 0$ gives $t^{-1} \leq t^{-1}t = 1$, i.e.

$$t^{-1} \leq 1. \tag{1}$$

Step 2: ψ is expanding. For all $s, s' \in \mathbb{R}$,

$$|s - s'| \;\leq\; |\psi(s) - \psi(s')|. \tag{2}$$

Indeed $\psi(s) - \psi(s') = (s - s') + \lambda\big(e^s - e^{s'}\big)$. If $s' \leq s$ then $s - s' \geq 0$ and, since $t \mapsto e^t$ is non-decreasing and $\lambda > 0$, also $\lambda(e^s - e^{s'}) \geq 0$; hence $\psi(s) - \psi(s') \geq 0$ and

$$|\psi(s) - \psi(s')| = (s - s') + \lambda\big(e^s - e^{s'}\big) \;\geq\; s - s' = |s - s'|.$$

If $s \leq s'$ the two summands are both ≤ 0 and the same computation with signs reversed gives $|\psi(s) - \psi(s')| = -(s - s') - \lambda(e^s - e^{s'}) \geq -(s - s') = |s - s'|$.

Step 3: a cubic Taylor bound for ψ. For all $a, s \in \mathbb{R}$ with $|s| \leq 1$,

$$\Big|\psi(a + s) - \psi(a) - \big(1 + \lambda e^a\big)s - \tfrac{1}{2}\big(\lambda e^a\big)s^2\Big| \;\leq\; \lambda e^a |s|^3. \tag{3}$$

Indeed, $e^{a+s} = e^a e^s$ gives the exact identity

$$\psi(a + s) - \psi(a) - \big(1 + \lambda e^a\big)s - \tfrac{1}{2}\big(\lambda e^a\big)s^2 = \lambda e^a\Big(e^s - \big(1 + s + \tfrac{s^2}{2}\big)\Big),$$

since the terms $a + s - a - s$ cancel. Now $\big|e^s - (1 + s + \frac{s^2}{2})\big| \leq |s|^3$ for $|s| \leq 1$: this is the standard remainder estimate for the exponential series truncated after three terms, which bounds the remainder by $|s|^3 \cdot \frac{n+1}{n!\,n}$ with $n = 3$, and $\frac{4}{6 \cdot 3} = \frac{2}{9} \leq 1$. Taking absolute values in the identity and using $\lambda e^a > 0$ gives (3).

Step 4: reduction, and the choice of δ. By (4) the form B_u is symmetric, which is the first clause of `TwiceDiffAt` for (D_u, B_u). So let $\varepsilon > 0$; we must produce $\delta > 0$ with

$$\Big|u(y) - u(z) - D_u(y - z) - \tfrac{1}{2}B_u(y - z, y - z)\Big| \leq \varepsilon |y - z|^2 \qquad \text{for } |y - z| < \delta. \tag{4}$$

Apply the Peano estimate of `TwiceDiffAt` for the jet (D_j, B_j) of $\psi \circ u$ twice, with the accuracies $1 > 0$ and $\varepsilon/4 > 0$, obtaining $\delta_1 > 0$ and $\delta_2 > 0$ such that for $|y - z| < \delta_1$, resp. $|y - z| < \delta_2$,

$$\Big|\psi(u(y)) - \psi(u(z)) - D_j(y - z) - \tfrac{1}{2}B_j(y - z, y - z)\Big| \leq |y - z|^2, \tag{5a}$$

$$\Big|\psi(u(y)) - \psi(u(z)) - D_j(y - z) - \tfrac{1}{2}B_j(y - z, y - z)\Big| \leq \tfrac{\varepsilon}{4}|y - z|^2. \tag{5b}$$

Set

$$L := 1 + \|D_j\| + \|B_j\|, \qquad Q := 1 + cL^3 + \tfrac{cL^2}{2} + \tfrac{\|B_j\|}{2}, \qquad S := cL^3 + cQL,$$

so that $L \geq 1 > 0$ (operator norms being non-negative) and $Q \geq 0$, $S \geq 0$ (as $c > 0$ by Step 1), and put

$$\delta := \min\Big\{1,\ \delta_1,\ \delta_2,\ \tfrac{1}{L},\ \tfrac{\varepsilon}{2(S+1)}\Big\} \;>\; 0,$$

every entry being positive. Fix y with $|y - z| < \delta$ and abbreviate

$$h := y - z, \qquad s := u(y) - u(z).$$

Then $|h| \leq 1$, $|h| < \delta_1$, $|h| < \delta_2$, $|h| < 1/L$ and $|h| < \varepsilon/(2(S + 1))$; and from $0 \leq |h| \leq 1$ we get

$$|h|^3 \leq |h|^2 \leq |h|. \tag{6}$$

We shall also use the operator-norm bounds $|D_j h| \leq \|D_j\|\,|h|$ and $|B_j(h, h)| \leq \|B_j\|\,|h|\,|h|$.

Step 5: the crude bound $|s| \leq L|h|$. By (2) applied to $u(y)$ and $u(z)$, $|s| \leq |\psi(u(y)) - \psi(u(z))|$. Writing $\psi(u(y)) - \psi(u(z))$ as the sum of the remainder in (5a), of $D_j h$ and of $\frac{1}{2}B_j(h, h)$, the triangle inequality and (5a) give

$$|s| \;\leq\; |h|^2 + |D_j h| + \tfrac{1}{2}|B_j(h, h)|.$$

Now $|h|^2 \le |h|$ by (6); $|D_j h| \le \|D_j\|\,|h|$; and $\frac{1}{2}|B_j(h,h)| \le \frac{1}{2}\|B_j\|\,|h|\,|h| \le \|B_j\|\,|h|$ because $|h| \le 1$. Summing and using $1 + \|D_j\| + \|B_j\| = L$,

$$|s| \;\le\; \big(1 + \|D_j\| + \|B_j\|\big)|h| \;=\; L|h|. \tag{7}$$

Since $|h| < 1/L$ and $L > 0$, this gives $|s| \le L|h| < L \cdot \frac{1}{L} = 1$, so in particular

$$|s| \le 1. \tag{8}$$

Step 6: the Taylor expansion of ψ at $u(z)$. By (8) we may apply (3) with $a := u(z)$ and this s; since $u(z) + s = u(y)$ and, by hypothesis (1), $1 + \lambda e^{u(z)} = t$ and $\lambda e^{u(z)} = c$, it reads

$$\Big|\psi(u(y)) - \psi(u(z)) - t\,s - \tfrac{1}{2}c\,s^2\Big| \;\le\; c\,|s|^3.$$

Cubing (7) gives $|s|^3 \le L^3|h|^3$, and multiplying by $c > 0$,

$$\Big|\psi(u(y)) - \psi(u(z)) - t\,s - \tfrac{1}{2}c\,s^2\Big| \;\le\; cL^3|h|^3. \tag{9}$$

Step 7: the central combination. The algebraic identity

$$t\,s+\tfrac{1}{2}c\,s^2-D_jh-\tfrac{1}{2}B_j(h,h) = -\Big(\psi(u(y))-\psi(u(z))-t\,s-\tfrac{1}{2}c\,s^2\Big)+\Big(\psi(u(y))-\psi(u(z))-D_jh-\tfrac{1}{2}B_j(h,h)\Big)$$

holds because the two copies of $\psi(u(y)) - \psi(u(z))$ cancel. Applying the triangle inequality together with (9) and (5a), and using $|h|^3 \le |h|^2$ from (6),

$$\Big|t\,s + \tfrac{1}{2}c\,s^2 - D_jh - \tfrac{1}{2}B_j(h,h)\Big| \;\le\; cL^3|h|^3 + |h|^2 \;\le\; \big(1 + cL^3\big)|h|^2; \tag{10a}$$

applying it instead with (9) and (5b),

$$\Big|t\,s + \tfrac{1}{2}c\,s^2 - D_jh - \tfrac{1}{2}B_j(h,h)\Big| \;\le\; \tfrac{\varepsilon}{4}|h|^2 + cL^3|h|^3. \tag{10b}$$

Step 8: the bootstrap $|s - t^{-1}D_jh| \le Q|h|^2$. From

$$t\,s - D_jh = \Big(t\,s + \tfrac{1}{2}c\,s^2 - D_jh - \tfrac{1}{2}B_j(h,h)\Big) + \Big(-\tfrac{1}{2}c\,s^2\Big) + \tfrac{1}{2}B_j(h,h)$$

and the triangle inequality for three summands we get, using (10a), the bound $\big|-\frac{1}{2}c\,s^2\big| = \frac{1}{2}c\,s^2 \le \frac{1}{2}cL^2|h|^2$ (which is (7) squared, multiplied by $c/2 > 0$), and $\frac{1}{2}|B_j(h,h)| \le \frac{1}{2}\|B_j\|\,|h|^2$,

$$\big|t\,s - D_jh\big| \;\le\; \Big(1 + cL^3 + \tfrac{cL^2}{2} + \tfrac{\|B_j\|}{2}\Big)|h|^2 \;=\; Q|h|^2.$$

Since $t \neq 0$ we have $s - t^{-1}D_jh = t^{-1}\big(t\,s - D_jh\big)$, so taking absolute values and using $t^{-1} > 0$ together with (1),

$$\big|s - t^{-1}D_jh\big| = t^{-1}\big|t\,s - D_jh\big| \le \big|t\,s - D_jh\big| \le Q|h|^2. \tag{11}$$

Step 9: $\big|(t^{-1}D_jh)^2 - s^2\big| \le 2QL|h|^3$. Factor

$$\big(t^{-1}D_jh\big)^2 - s^2 = -\Big(\big(s - t^{-1}D_jh\big)\big(s + t^{-1}D_jh\big)\Big).$$

For the second factor, $|t^{-1}D_jh| = t^{-1}|D_jh| \le |D_jh| \le \|D_j\|\,|h| \le L|h|$ by (1) and $\|D_j\| \le L$, so with (7) the triangle inequality gives $\big|s + t^{-1}D_jh\big| \le |s| + |t^{-1}D_jh| \le 2L|h|$. Multiplying this by (11), both bounds having non-negative sides,

$$\big|(t^{-1}D_jh)^2 - s^2\big| \le Q|h|^2 \cdot 2L|h| = 2QL|h|^3. \tag{12}$$

Step 10: assembly. Substituting hypotheses (2) and (3) for D_u and B_u and using $t \neq 0$, one has the field identity

$$\begin{aligned} u(y) - u(z) - D_uh - \tfrac{1}{2}B_u(h,h) &= s - t^{-1}D_jh - \tfrac{1}{2}t^{-1}\Big(B_j(h,h) - c\big(t^{-1}D_jh\big)^2\Big) \\ &= t^{-1}\Big[\Big(t\,s + \tfrac{1}{2}c\,s^2 - D_jh - \tfrac{1}{2}B_j(h,h)\Big) + \tfrac{1}{2}c\Big(\big(t^{-1}D_jh\big)^2 - s^2\Big)\Big]. \end{aligned}$$

Bound the bracket. By the triangle inequality, (10b), and (12) multiplied by $\frac{1}{2}c > 0$,

$$\Big|\cdots\Big| \;\le\; \tfrac{\varepsilon}{4}|h|^2 + cL^3|h|^3 + \tfrac{1}{2}c\cdot 2QL|h|^3 \;=\; \tfrac{\varepsilon}{4}|h|^2 + \big(cL^3 + cQL\big)|h|^3.$$

For the cubic term, $cL^3 + cQL = S$ and, since $|h| < \varepsilon/(2(S+1))$ and $S \ge 0$,

$$S|h| \;\le\; S\cdot\frac{\varepsilon}{2(S+1)} \;\le\; \frac{\varepsilon}{2},$$

the second inequality because $S\varepsilon \le (S+1)\varepsilon$ (as $\varepsilon > 0$) and $2(S+1) > 0$. Hence

$$\big(cL^3 + cQL\big)|h|^3 = S|h|^3 = \big(S|h|\big)|h|^2 \le \tfrac{\varepsilon}{2}|h|^2,$$

and the bracket is bounded in absolute value by $\frac{\varepsilon}{4}|h|^2 + \frac{\varepsilon}{2}|h|^2 \le \varepsilon|h|^2$, using $\varepsilon|h|^2 \ge 0$. Finally, taking absolute values in the identity and using $t^{-1} > 0$ with (1),

$$\Big|u(y) - u(z) - D_u h - \tfrac{1}{2}B_u(h,h)\Big| = t^{-1}\Big|\cdots\Big| \;\le\; \Big|\cdots\Big| \;\le\; \varepsilon|h|^2.$$

This is exactly (4) with the δ of Step 4, so both clauses of `TwiceDiffAt` hold for (D_u, B_u) and u at z. □

2.20. `QuarticPenaltyCalculus`.

Lemma 2.20 (The calculus of the quartic penalty $P_{\mu,c}(x) = \mu\|x-c\|^4$)**.** *Let $N \in \mathbb{N}$. Let*

$$P : \mathbb{R}\times\mathbb{R}^N\times\mathbb{R}^N \to \mathbb{R}, \qquad D : \mathbb{R}\times\mathbb{R}^N\times\mathbb{R}^N \to (\mathbb{R}^N)^*, \qquad Q : \mathbb{R}\times\mathbb{R}^N\times\mathbb{R}^N \to \mathrm{Bil}(\mathbb{R}^N),$$

write $P_{\mu,c}(x) := P(\mu,c,x)$, $D_{\mu,c}(x) := D(\mu,c,x)$ and $Q_{\mu,c}(x) := Q(\mu,c,x)$ (so $D_{\mu,c}(x)$ is a continuous linear functional and $Q_{\mu,c}(x)$ a continuous bilinear form on $\mathbb{R}^N$), and let B be a continuous bilinear form on $\mathbb{R}^N$. Assume that for all $\mu \in \mathbb{R}$ and all $c, x \in \mathbb{R}^N$, and all $a, b \in \mathbb{R}^N$,

(H1) $$P_{\mu,c}(x) = \mu\|x-c\|^4,$$

(H2) $$D_{\mu,c}(x) = \big(4\mu\|x-c\|^2\big)\,\langle x-c, \cdot\rangle,$$

(H3) $$B(a,b) = \langle a, b\rangle,$$

(H4) $$Q_{\mu,c}(x) = \big(4\mu\|x-c\|^2\big)\,B + \big(8\mu\big)\,\big(\langle x-c,\cdot\rangle \otimes \langle x-c,\cdot\rangle\big).$$

Then the following hold.

1. *$Q_{\mu,c}(x)(a,b) = Q_{\mu,c}(x)(b,a)$ for all μ and all c, x, a, b.*
2. *If $\mu \ge 0$ then $\|D_{\mu,c}(x)\| \le 4\mu\|x-c\|^3$ for all c, x.*
3. *If $\mu \ge 0$ then $\|Q_{\mu,c}(x)\| \le 12\mu\|x-c\|^2$ for all c, x.*
4. *If $\mu \ge 0$ then, for all c, z, the pair $\big(D_{\mu,c}(z), Q_{\mu,c}(z)\big)$ is a second-order jet of $P_{\mu,c}$ at z in the sense of `TwiceDiffAt`.*
5. *$P_{\mu,c}(c) = 0$ for all μ, c.*
6. *If $\mu > 0$ then $P_{\mu,c}(x) > 0$ for all c and all $x \ne c$.*
7. *For all $a, b \in \mathbb{R}^N$ and $t \in \mathbb{R}$,*
$$\|ta + (1-t)b\|^2 = t\|a\|^2 + (1-t)\|b\|^2 - t(1-t)\|a-b\|^2.$$
8. *If $\mu \ge 0$, $R \ge 0$, and $x, y \in \mathbb{R}^N$ satisfy $\|x-c\| \le R$ and $\|y-c\| \le R$, then for every $t \in [0,1]$*
$$t\,P_{\mu,c}(x) + (1-t)P_{\mu,c}(y) - P_{\mu,c}\big(tx + (1-t)y\big) \;\le\; 6\mu R^2\big(t(1-t)\|x-y\|^2\big).$$

Proof. **Step 0: the pointwise formulas.** Evaluating (H2) at h, and (H4) at (h,k) using (H3), gives for all μ and all c, x, h, k

(1) $$D_{\mu,c}(x)h = 4\mu\|x-c\|^2\,\langle x-c, h\rangle,$$

(2) $$Q_{\mu,c}(x)(h,k) = 4\mu\|x-c\|^2\,\langle h,k\rangle + 8\mu\,\langle x-c,h\rangle\langle x-c,k\rangle.$$

We use the Cauchy–Schwarz inequality $|\langle u,v\rangle| \le \|u\|\,\|v\|$ throughout.

Step 1: part (1). By (2) applied to (a,b) and to (b,a), symmetry of the real inner product ($\langle a,b\rangle = \langle b,a\rangle$) and commutativity of multiplication in $\mathbb{R}$ give $Q_{\mu,c}(x)(a,b) = Q_{\mu,c}(x)(b,a)$.

Step 2: part (2). Let $\mu \geq 0$. The quantity $4\mu\|x-c\|^3$ is non-negative, so by the characterisation of the operator norm it suffices to bound $|D_{\mu,c}(x)h|$ by $4\mu\|x-c\|^3\|h\|$ for every h. Since $4\mu\|x-c\|^2 \geq 0$, (1) and Cauchy–Schwarz give

$$|D_{\mu,c}(x)h| = 4\mu\|x-c\|^2\,\big|\langle x-c, h\rangle\big| \leq 4\mu\|x-c\|^2\big(\|x-c\|\,\|h\|\big) = 4\mu\|x-c\|^3\|h\|.$$

Step 3: part (3). Let $\mu \geq 0$; then $12\mu\|x-c\|^2 \geq 0$. Applying the operator-norm characterisation twice, it suffices to bound $|Q_{\mu,c}(x)(h,k)|$ by $12\mu\|x-c\|^2\|h\|\,\|k\|$. By Cauchy–Schwarz, $|\langle h,k\rangle| \leq \|h\|\|k\|$, $|\langle x-c,h\rangle| \leq \|x-c\|\|h\|$ and $|\langle x-c,k\rangle| \leq \|x-c\|\|k\|$; so, using $4\mu\|x-c\|^2 \geq 0$ and $8\mu \geq 0$, the two summands of (2) are bounded in absolute value by $4\mu\|x-c\|^2\|h\|\|k\|$ and by $8\mu\big(\|x-c\|\|h\|\big)\big(\|x-c\|\|k\|\big) = 8\mu\|x-c\|^2\|h\|\|k\|$. The triangle inequality and $4+8=12$ finish the step.

Step 4: part (4). Let $\mu \geq 0$ and fix c, z. We apply part (1) of `SecondOrderJetCalculus` — the cubic-remainder criterion — with the function $P_{\mu,c}$, the base point z, the jet data $\big(D_{\mu,c}(z), Q_{\mu,c}(z)\big)$ and the constant

$$C := 4\mu\|z-c\| + \mu \;\geq\; 0,$$

which is non-negative because $\mu \geq 0$ and $\|z-c\| \geq 0$. Symmetry of $Q_{\mu,c}(z)$ is part (1). So it remains to prove the cubic bound: for every y with $\|y-z\| \leq 1$,

$$\Big|P_{\mu,c}(y) - P_{\mu,c}(z) - D_{\mu,c}(z)(y-z) - \tfrac{1}{2}Q_{\mu,c}(z)(y-z,y-z)\Big| \;\leq\; C\|y-z\|^3. \tag{3}$$

Abbreviate $A := \|z-c\|^2$, $\beta := \langle z-c, y-z\rangle$ and $E := \|y-z\|^2$. Since $y-c = (z-c)+(y-z)$, expanding the squared norm of a sum gives

$$\|y-c\|^2 = A + 2\beta + E. \tag{4}$$

Now compute the remainder exactly. By (H1) and $\|w\|^4 = \big(\|w\|^2\big)^2$ together with (4), $P_{\mu,c}(y) = \mu(A+2\beta+E)^2$ and $P_{\mu,c}(z) = \mu A^2$; by (1), $D_{\mu,c}(z)(y-z) = 4\mu A\beta$; and by (2) with $h = k = y-z$, using $\langle y-z, y-z\rangle = E$,

$$\tfrac{1}{2}Q_{\mu,c}(z)(y-z,y-z) = \tfrac{1}{2}\big(4\mu AE + 8\mu\beta^2\big) = 2\mu AE + 4\mu\beta^2.$$

Since $(A+2\beta+E)^2 = A^2+4\beta^2+E^2+4A\beta+2AE+4\beta E$, the four displayed quantities combine to

$$P_{\mu,c}(y) - P_{\mu,c}(z) - D_{\mu,c}(z)(y-z) - \tfrac{1}{2}Q_{\mu,c}(z)(y-z,y-z) = \mu\big(4\beta E + E^2\big). \tag{5}$$

It remains to bound $\big|4\beta E + E^2\big|$. First, by Cauchy–Schwarz and $E \geq 0$,

$$|4\beta E| = 4|\beta|E \leq 4\big(\|z-c\|\,\|y-z\|\big)\|y-z\|^2 = 4\|z-c\|\,\|y-z\|^3.$$

Second, $E^2 = \|y-z\|^4 \leq \|y-z\|^3$, because $\|y-z\| \leq 1$ and $\|y-z\|^3 \geq 0$. Hence, by the triangle inequality and $\mu \geq 0$, taking absolute values in (5),

$$\Big|\cdots\Big| = \mu\big|4\beta E + E^2\big| \leq \mu\Big(4\|z-c\|\,\|y-z\|^3 + \|y-z\|^3\Big) = \big(4\mu\|z-c\| + \mu\big)\|y-z\|^3,$$

which is (3). Part (1) of `SecondOrderJetCalculus` therefore applies and yields the jet property, i.e. part (4).

Step 5: parts (5) and (6). By (H1), $P_{\mu,c}(c) = \mu\|c-c\|^4$ and $c-c=0$, so $\|c-c\| = 0$ and $P_{\mu,c}(c) = 0$. If $\mu > 0$ and $x \neq c$ then $x-c \neq 0$, so $\|x-c\| > 0$ and hence $\|x-c\|^4 > 0$; multiplying by $\mu > 0$ gives $P_{\mu,c}(x) = \mu\|x-c\|^4 > 0$.

Step 6: part (7). Expanding the squared norms of a sum and of a difference, $\|u+v\|^2 = \|u\|^2 + 2\langle u,v\rangle + \|v\|^2$ and $\|u-v\|^2 = \|u\|^2 - 2\langle u,v\rangle + \|v\|^2$, and using $\|\lambda w\|^2 = \lambda^2\|w\|^2$ together with the homogeneity $\langle \lambda u, \mu v\rangle = \lambda\mu\langle u,v\rangle$ of the real inner product,

$$\|ta + (1-t)b\|^2 = t^2\|a\|^2 + 2t(1-t)\langle a,b\rangle + (1-t)^2\|b\|^2,$$

while

$$t\|a\|^2 + (1-t)\|b\|^2 - t(1-t)\|a-b\|^2 = t\|a\|^2 + (1-t)\|b\|^2 - t(1-t)\Big(\|a\|^2 - 2\langle a,b\rangle + \|b\|^2\Big).$$

The coefficient of $\|a\|^2$ on the right is $t-t(1-t)=t^2$, that of $\|b\|^2$ is $(1-t)-t(1-t)=(1-t)^2$, and the cross term is $2t(1-t)\langle a,b\rangle$. The two expressions agree, which is (7).

Step 7: part (8). Let $\mu\ge 0$, $R\ge 0$, $\|x-c\|\le R$, $\|y-c\|\le R$ and $0\le t\le 1$. Put

$$a:=\|x-c\|^2,\qquad b:=\|y-c\|^2,\qquad q:=t(1-t)\|x-y\|^2,\qquad S:=ta+(1-t)b.$$

Since $0\le t$ and $0\le 1-t$, we have $t(1-t)\ge 0$ and hence $q\ge 0$.

(7a) The chord value. As $tx+(1-t)y-c=t(x-c)+(1-t)(y-c)$ and $(x-c)-(y-c)=x-y$, part (7) applied with the vectors $x-c$ and $y-c$ in place of its a and b gives

$$\big\|tx+(1-t)y-c\big\|^2=t\|x-c\|^2+(1-t)\|y-c\|^2-t(1-t)\|x-y\|^2=S-q.$$

Therefore, by (H1) and $\|w\|^4=(\|w\|^2)^2$,

$$P_{\mu,c}(x)=\mu a^2,\qquad P_{\mu,c}(y)=\mu b^2,\qquad P_{\mu,c}\big(tx+(1-t)y\big)=\mu(S-q)^2,$$

so that

$$t\,P_{\mu,c}(x)+(1-t)P_{\mu,c}(y)-P_{\mu,c}\big(tx+(1-t)y\big)=\mu\Big(ta^2+(1-t)b^2-(S-q)^2\Big). \tag{6}$$

(7b) The algebraic identity. Since $S=ta+(1-t)b$ and $t+(1-t)=1$, one checks by expanding that $ta^2+(1-t)b^2-S^2=t(1-t)(a-b)^2$. Adding $S^2-(S-q)^2=2Sq-q^2$,

$$ta^2+(1-t)b^2-(S-q)^2=t(1-t)(a-b)^2+2Sq-q^2. \tag{7}$$

(7c) The three bounds. First, $|a-b|\le 2R\|x-y\|$: indeed

$$|a-b|=\Big|\|x-c\|^2-\|y-c\|^2\Big|=\Big|\|x-c\|-\|y-c\|\Big|\cdot\Big(\|x-c\|+\|y-c\|\Big),$$

the factorisation being legitimate because $\|x-c\|+\|y-c\|\ge 0$; now $\big|\|x-c\|-\|y-c\|\big|\le\|(x-c)-(y-c)\|=\|x-y\|$ by the reverse triangle inequality, and $\|x-c\|+\|y-c\|\le 2R$ by hypothesis. Squaring, $(a-b)^2\le 4R^2\|x-y\|^2$, and multiplying by $t(1-t)\ge 0$,

$$t(1-t)(a-b)^2\le 4R^2q.$$

Second, $a\le R^2$ and $b\le R^2$: from $0\le\|x-c\|\le R$ and $0\le\|y-c\|\le R$, squaring preserves the inequalities. Hence $S=ta+(1-t)b\le tR^2+(1-t)R^2=R^2$, and multiplying by $q\ge 0$,

$$2Sq\le 2R^2q.$$

Third, $-q^2\le 0$. Substituting the three bounds into (7),

$$ta^2+(1-t)b^2-(S-q)^2\ \le\ 4R^2q+2R^2q\ =\ 6R^2q.$$

(7d) Conclusion. Multiplying the last inequality by $\mu\ge 0$ and using (6),

$$t\,P_{\mu,c}(x)+(1-t)P_{\mu,c}(y)-P_{\mu,c}\big(tx+(1-t)y\big)\ \le\ 6\mu R^2q\ =\ 6\mu R^2\big(t(1-t)\|x-y\|^2\big),$$

which is part (8). □

2.21. `JetHessianLowerBoundOfConvex`.

Lemma 2.21 (The jet Hessian sign from convexity)**.** *Let $N\in\mathbb{N}$, let $w:\mathbb{R}^N\to\mathbb{R}$, let $\Lambda,r\in\mathbb{R}$ and let $x_c\in\mathbb{R}^N$. Assume that*

$$y\ \longmapsto\ w(y)+\tfrac{\Lambda}{2}|y|^2$$

is convex on the ball $B(x_c,r)$, let $z\in B(x_c,r)$, and let (D,B) be a second-order jet of w at z in the sense of `TwiceDiffAt`. Then

$$B(h,h)\ \ge\ -\Lambda|h|^2\qquad \textit{for every } h\in\mathbb{R}^N.$$

Proof. Put $g(y):=w(y)+\frac{\Lambda}{2}|y|^2$, so that g is convex on $B(x_c,r)$ by hypothesis, and fix $h\in\mathbb{R}^N$. The conclusion is a statement about a *pointwise jet* and not about a classical Hessian, which is the only form available in this development: the hypothesis actually supplied by the surrounding argument is convexity of $w+\frac{\Lambda}{2}|\cdot|^2$ on a ball and nothing more, so no second derivative of w is assumed to exist anywhere.

Step 1: the jet of g at z. Write $q(y) := \frac{\Lambda}{2}|y|^2$. We claim q has at z the second-order jet $\big(\Lambda\langle z,\cdot\rangle,\ \Lambda\langle\cdot,\cdot\rangle\big)$. The form $\Lambda\langle\cdot,\cdot\rangle$ is symmetric, by symmetry of the real inner product. For the remainder, apply $|a+b|^2 = |a|^2 + 2\langle a,b\rangle + |b|^2$ with $a := z$ and $b := y - z$, whose sum is y:

$$|y|^2 = |z|^2 + 2\langle z, y-z\rangle + |y-z|^2.$$

Substituting this,

$$q(y) - q(z) - \Lambda\langle z, y-z\rangle - \tfrac{1}{2}\Lambda\langle y-z, y-z\rangle$$
$$= \tfrac{\Lambda}{2}\Big(|z|^2 + 2\langle z,y-z\rangle + |y-z|^2\Big) - \tfrac{\Lambda}{2}|z|^2 - \Lambda\langle z,y-z\rangle - \tfrac{\Lambda}{2}|y-z|^2 = 0$$

for *every* y, using $\langle y-z, y-z\rangle = |y-z|^2$. In particular the remainder is bounded in absolute value by $0\cdot|y-z|^3$, so part (1) of `SecondOrderJetCalculus` applies with $C := 0$ and gives the claim.

By part (2) of `SecondOrderJetCalculus` (additivity of jets) applied to w and q, the function $g = w + q$ has at z the second-order jet (D_g, B_g) with

$$D_g := D + \Lambda\langle z,\cdot\rangle, \qquad B_g := B + \Lambda\langle\cdot,\cdot\rangle,$$

so that in particular

$$B_g(a,b) = B(a,b) + \Lambda\langle a,b\rangle \qquad (a,b\in\mathbb{R}^N). \tag{1}$$

Step 2: homogeneity in the direction h. For every $s\in\mathbb{R}$,

$$D_g(sh) = s\,D_g(h), \qquad B_g(sh,sh) = s^2 B_g(h,h), \qquad |sh|^2 = s^2|h|^2,$$

the first by linearity of D_g, the second by bilinearity of B_g, and the third from $|sh| = |s|\,|h|$ together with $|s|^2 = s^2$.

Step 3: a good scale t. Let $\varepsilon > 0$. The Peano estimate of `TwiceDiffAt` for the jet (D_g, B_g) of g at z supplies $\delta > 0$ with

$$\Big|g(y) - g(z) - D_g(y-z) - \tfrac{1}{2}B_g(y-z,y-z)\Big| \ \le\ \varepsilon|y-z|^2 \qquad \text{whenever } |y-z|<\delta. \tag{2}$$

Since $z\in B(x_c,r)$ we have $d(z,x_c) < r$, and $|h|+1 > 0$. Put

$$t := \min\Big\{\frac{\delta}{2(|h|+1)},\ \frac{r-d(z,x_c)}{2(|h|+1)}\Big\}\ >\ 0,$$

both entries being positive. Multiplying the two bounds $t \le \delta/(2(|h|+1))$ and $t \le (r - d(z,x_c))/(2(|h|+1))$ by $|h|+1>0$ gives

$$t(|h|+1) \le \tfrac{\delta}{2} < \delta, \qquad t(|h|+1) \le \tfrac{r-d(z,x_c)}{2} < r - d(z,x_c),$$

and $t|h| \le t(|h|+1)$ because $t>0$ and $|h| \le |h|+1$. Consequently:

- $z \pm th \in B(x_c,r)$. Indeed $|\pm th| = t|h|$, so by the triangle inequality $d(z\pm th, x_c) \le d(z\pm th, z) + d(z,x_c) = t|h| + d(z,x_c) < r$.
- (2) applies at $y := z + th$ and at $y := z - th$, since $|(z\pm th) - z| = t|h| \le t(|h|+1) < \delta$.

Step 4: midpoint convexity. The two points $z+th$ and $z-th$ lie in $B(x_c,r)$ by Step 3, and $\frac{1}{2}(z+th) + \frac{1}{2}(z-th) = z$. Convexity of g on $B(x_c,r)$, applied with the weights $\frac{1}{2},\frac{1}{2}$ (which are non-negative and sum to 1), therefore gives

$$g(z)\ \le\ \tfrac{1}{2}g(z+th) + \tfrac{1}{2}g(z-th). \tag{3}$$

Step 5: expansion, cancellation, and the ε-inequality. Apply (2) at $y = z+th$, where $y - z = th$, and at $y = z - th$, where $y - z = (-t)h$, and rewrite the linear and quadratic terms by Step 2:

$$\Big|g(z+th) - g(z) - t\,D_g(h) - \tfrac{1}{2}t^2 B_g(h,h)\Big| \le \varepsilon\, t^2|h|^2,$$

$$\Big|g(z-th) - g(z) - (-t)\,D_g(h) - \tfrac{1}{2}(-t)^2 B_g(h,h)\Big| \le \varepsilon\,(-t)^2|h|^2.$$

Each absolute-value bound gives an upper bound on the quantity inside it. Adding those two upper bounds, the first-order terms $\pm t\, D_g(h)$ cancel, and $(-t)^2 = t^2$, so

$$g(z+th) + g(z-th) - 2g(z) \;\le\; t^2 B_g(h,h) + 2\varepsilon t^2 |h|^2.$$

By (3) the left-hand side is non-negative, whence

$$0 \;\le\; t^2\Big(B_g(h,h) + 2\varepsilon|h|^2\Big).$$

Since $t > 0$ we have $t^2 > 0$, so the bracket is non-negative: were $B_g(h,h) + 2\varepsilon|h|^2 < 0$, multiplying this strict inequality by $t^2 > 0$ would make the right-hand side above strictly negative, contradicting the display. Hence, using (1) with $a = b = h$ and $\langle h, h\rangle = |h|^2$,

$$0 \;\le\; B(h,h) + \Lambda|h|^2 + 2\varepsilon|h|^2, \qquad \text{i.e.} \qquad -\Lambda|h|^2 - 2\varepsilon|h|^2 \;\le\; B(h,h). \tag{4}$$

This holds for every $\varepsilon > 0$.

Step 6: $\varepsilon \to 0^+$. Apply (4) with $\varepsilon := \frac{1}{n+1} > 0$ for each $n \in \mathbb{N}$:

$$-\Lambda|h|^2 - 2\cdot\tfrac{1}{n+1}\cdot|h|^2 \;\le\; B(h,h) \qquad (n \in \mathbb{N}).$$

As $n \to \infty$ we have $\frac{1}{n+1} \to 0$, so the left-hand side converges to $-\Lambda|h|^2$. A convergent sequence all of whose terms are $\le B(h,h)$ has limit $\le B(h,h)$, so $-\Lambda|h|^2 \le B(h,h)$. As $h \in \mathbb{R}^N$ was arbitrary, the lemma follows. □

2.22. `HgradTendstoOfTendsto`.

Lemma 2.22 (Continuity of the horizontal gradient along a convergent pair)**.** *Let $n, m \in \mathbb{N}$ and let $\sigma : \mathbb{R}^n \to (\mathbb{R}^n)^m$, written $x \mapsto (\sigma^1(x), \dots, \sigma^m(x))$, be a frame each of whose components $x \mapsto \sigma^i(x)$ is continuous. Let $x_k \to x_\infty$ in $\mathbb{R}^n$, and let $D_k \to D_\infty$ in the operator norm, where each D_k and D_∞ is a continuous linear functional on $\mathbb{R}^n$. Then*

$$\text{hgrad } \sigma\ x_k\ D_k \;\longrightarrow\; \text{hgrad } \sigma\ x_\infty\ D_\infty \qquad \text{in } \mathbb{R}^m,$$

with hgrad *as in* `hgrad`*.*

Proof. (The frame is taken bare here, with continuity of its components as the only hypothesis; in the application the σ^i are the horizontal fields of a Carnot setting and are continuous because they are smooth, but no group structure enters this argument.)

Step 0: the components of hgrad**.** By `hgrad`, for every $y \in \mathbb{R}^n$ and every continuous linear functional D on $\mathbb{R}^n$, the vector hgrad $\sigma\ y\ D \in \mathbb{R}^m$ is the one whose i-th component is $D(\sigma^i(y))$. So the assertion is that, in $\mathbb{R}^m$,

$$\big(D_k(\sigma^i(x_k))\big)_i \;\longrightarrow\; \big(D_\infty(\sigma^i(x_\infty))\big)_i. \tag{1}$$

Step 1: the jets converge uniformly. We have

$$\|D_k - D_\infty\| \;\longrightarrow\; 0. \tag{2}$$

Indeed $D_k \to D_\infty$, and the constant sequence D_∞ converges to D_∞, so $D_k - D_\infty \to D_\infty - D_\infty = 0$; the norm is continuous, so $\|D_k - D_\infty\| \to \|0\| = 0$.

Step 2: convergence of each component. Fix i. We claim

$$D_k(\sigma^i(x_k)) \;\longrightarrow\; D_\infty(\sigma^i(x_\infty)). \tag{3}$$

First, $\sigma^i(x_k) \to \sigma^i(x_\infty)$: the map $x \mapsto \sigma^i(x)$ is continuous by hypothesis, hence continuous at x_∞, and $x_k \to x_\infty$, so the composition converges. Consequently:

- $D_\infty(\sigma^i(x_k)) \to D_\infty(\sigma^i(x_\infty))$, because D_∞ is a continuous linear functional and is therefore continuous at $\sigma^i(x_\infty)$.
- $\|D_k - D_\infty\|\,\|\sigma^i(x_k)\| \to 0$: by (2) the first factor tends to 0, and the second converges to $\|\sigma^i(x_\infty)\|$ by continuity of the norm, so the product tends to $0 \cdot \|\sigma^i(x_\infty)\| = 0$.

Now compare the two sequences termwise. For each k, linearity gives $D_\infty(\sigma^i(x_k))-D_k(\sigma^i(x_k)) = (D_\infty - D_k)(\sigma^i(x_k))$, so by the defining bound of the operator norm and $\|-w\| = \|w\|$,

$$\Big|D_\infty(\sigma^i(x_k))-D_k(\sigma^i(x_k))\Big| = \Big|(D_\infty-D_k)(\sigma^i(x_k))\Big| \le \|D_\infty-D_k\|\,\|\sigma^i(x_k)\| = \|D_k-D_\infty\|\,\|\sigma^i(x_k)\|.$$

The left-hand side is the distance between the k-th terms of the two sequences; it is non-negative and, by the second bullet, dominated by a sequence tending to 0. Squeezing, that distance tends to 0. A sequence at vanishing distance from a convergent sequence has the same limit, so from the first bullet we get (3).

Step 3: from components to $\mathbb{R}^m$. Convergence in the product topology on functions $\{1,\dots,m\}\to\mathbb{R}$ is exactly convergence of each component, so by (3), applied to every i,

$$\big(i \mapsto D_k(\sigma^i(x_k))\big) \;\longrightarrow\; \big(i \mapsto D_\infty(\sigma^i(x_\infty))\big)$$

in that topology. The identification of such functions with $\mathbb{R}^m$ is a linear homeomorphism, hence continuous, and composing a convergent sequence with a continuous map preserves the limit. This gives (1), which by Step 0 is the assertion of the lemma. □

2.23. `ExpRelabelledMaxDifferentiability`.

Lemma 2.23 (Differentiability of u at a maximum point of the relabelled difference)**.** *Let $N\in\mathbb{N}$, let $\Omega\subseteq\mathbb{R}^N$, let $\lambda>0$ and let $u,v:\mathbb{R}^N\to\mathbb{R}$. Write $\psi(s):=s+\lambda e^s$ and assume*

(1) *$\psi\circ u$, i.e. $y\mapsto u(y)+\lambda e^{u(y)}$, is locally semi-convex on Ω (`LocallySemiConvexOn`);*
(2) *$-v$ is locally semi-convex on Ω;*
(3) *$x\in\Omega$ and x maximizes $y\mapsto \big(u(y)+\lambda e^{u(y)}\big)-v(y)$ over Ω.*

Then there is a continuous linear functional D on $\mathbb{R}^N$ such that u is Fréchet differentiable at x with derivative D.

Proof. **Step 1: the relabelling ψ.** The function $s\mapsto s$ has derivative 1 everywhere and $s\mapsto\lambda e^s$ has derivative λe^s (the derivative of the real exponential is itself, multiplied by the constant λ), so ψ is differentiable on $\mathbb{R}$ with

$$\psi'(s) = 1+\lambda e^s \qquad (s\in\mathbb{R}). \tag{1}$$

This is an elementary computation for an explicit function; no separate node is needed. Since $\lambda>0$ and $e^s>0$ we have $\psi'(s)\ge 1$ for every s. Put

$$t := 1+\lambda e^{u(x)},$$

so that $t\ge 1$, hence $t>0$, $t\neq 0$ and $t^{-1}>0$.

Step 2: ψ is expanding. For all $s,s'\in\mathbb{R}$,

$$|s-s'| \;\le\; |\psi(s)-\psi(s')|. \tag{2}$$

Indeed $\psi(s)-\psi(s') = (s-s')+\lambda\big(e^s-e^{s'}\big)$. If $s'\le s$ then $s-s'\ge 0$, and $e^{s'}\le e^s$ with $\lambda>0$ gives $\lambda(e^s-e^{s'})\ge 0$; so $\psi(s)-\psi(s')\ge 0$ and $|\psi(s)-\psi(s')| = (s-s')+\lambda(e^s-e^{s'}) \ge s-s' = |s-s'|$. If $s\le s'$ both summands are ≤ 0 and the same computation with reversed signs gives $|\psi(s)-\psi(s')| = -(s-s')-\lambda(e^s-e^{s'}) \ge -(s-s') = |s-s'|$.

Step 3: differentiability of $\psi\circ u$ at x. The first two hypotheses of the lemma say that $\psi\circ u$ and $-v$ are locally semi-convex on Ω, and the third says that $x\in\Omega$ lies in $\operatorname{argmax}_\Omega\big(\psi\circ u - v\big)$. These are exactly the hypotheses of part (2) of `SemiConvexDiffAtMax` for the pair $(\psi\circ u,\, v)$ on Ω, so that node supplies a continuous linear functional D_ℓ with

$$\psi\circ u \text{ Fréchet differentiable at } x \text{ with derivative } D_\ell \tag{3}$$

(the node also gives that v has the same derivative at x, which we do not use). Put $L := \|D_\ell\|+1$, so $L\ge 1>0$.

Step 4: $t^{-1}D_\ell$ is the derivative of u at x. We verify the Fréchet condition in its $o(\|y-x\|)$ form: for every $c>0$ there is $\delta>0$ with

$$\Big|u(y)-u(x)-\big(t^{-1}D_\ell\big)(y-x)\Big| \;\le\; c\|y-x\| \qquad \text{whenever } \|y-x\|<\delta. \tag{4}$$

Fix $c > 0$. From (3), written in the same o-form, choose

$$\text{(5a)} \qquad \delta_1 > 0 \text{ with } \quad \big|\psi(u(y)) - \psi(u(x)) - D_\ell(y-x)\big| \le 1 \cdot \|y-x\| \qquad \text{for } \|y-x\| < \delta_1,$$

$$\text{(5b)} \qquad \delta_2 > 0 \text{ with } \quad \big|\psi(u(y)) - \psi(u(x)) - D_\ell(y-x)\big| \le \tfrac{ct}{2}\|y-x\| \qquad \text{for } \|y-x\| < \delta_2,$$

the accuracies 1 and $\frac{ct}{2}$ being positive. From Step 1, ψ has derivative t at the point $u(x)$ by (1); writing that one-dimensional statement in o-form with the positive accuracy $\frac{ct}{2L}$, choose $\eta > 0$ with

$$\text{(6)} \qquad \Big|\psi(s) - \psi(u(x)) - \big(s-u(x)\big)t\Big| \le \frac{ct}{2L}\,\big|s-u(x)\big| \qquad \text{for } |s-u(x)| < \eta.$$

Put

$$\delta := \min\Big\{\delta_1,\ \delta_2,\ \frac{\eta}{L+1}\Big\} > 0$$

and let y satisfy $\|y-x\| < \delta$; then (5a), (5b) apply and $\|y-x\| < \eta/(L+1)$.

(4a) The crude Lipschitz bound. By (2) applied to $u(y)$ and $u(x)$, then the triangle inequality and (5a) together with $|D_\ell(y-x)| \le \|D_\ell\|\,\|y-x\|$,

$$|u(y)-u(x)| \le \big|\psi(u(y)) - \psi(u(x))\big| \le \big|\psi(u(y)) - \psi(u(x)) - D_\ell(y-x)\big| + \big|D_\ell(y-x)\big|$$

$$\text{(7)} \qquad \le \|y-x\| + \|D_\ell\|\,\|y-x\| = L\|y-x\|.$$

(4b) $u(y)$ lies in the η-ball about $u(x)$. Multiplying $\|y-x\| < \eta/(L+1)$ by $L > 0$ and using (7),

$$|u(y)-u(x)| \le L\|y-x\| \le L \cdot \frac{\eta}{L+1} < \eta,$$

the last inequality because $L\eta < (L+1)\eta$ (as $\eta > 0$) and $L+1 > 0$.

(4c) The relabelling estimate at $s := u(y)$. By (4b) we may use (6) at $s = u(y)$, and then bound $|u(y)-u(x)|$ by (7), the factor $\frac{ct}{2L}$ being non-negative:

$$\text{(8)} \quad \Big|\psi(u(y)) - \psi(u(x)) - \big(u(y)-u(x)\big)t\Big| \le \frac{ct}{2L}\big|u(y)-u(x)\big| \le \frac{ct}{2L} \cdot L\|y-x\| = \frac{ct}{2}\|y-x\|.$$

(4d) Combination. The algebraic identity

$$\big(u(y)-u(x)\big)t - D_\ell(y-x) = -\Big(\psi(u(y))-\psi(u(x))-\big(u(y)-u(x)\big)t\Big) + \Big(\psi(u(y))-\psi(u(x))-D_\ell(y-x)\Big)$$

holds because the two copies of $\psi(u(y))-\psi(u(x))$ cancel. By the triangle inequality, $|-w| = |w|$, and then (8) and (5b),

$$\text{(9)} \qquad \Big|\big(u(y)-u(x)\big)t - D_\ell(y-x)\Big| \le \frac{ct}{2}\|y-x\| + \frac{ct}{2}\|y-x\| = ct\,\|y-x\|.$$

(4e) Division by t. Since $t \neq 0$, and $\big(t^{-1}D_\ell\big)(y-x) = t^{-1}D_\ell(y-x)$ by homogeneity of D_ℓ,

$$u(y)-u(x)-\big(t^{-1}D_\ell\big)(y-x) = t^{-1}\Big(\big(u(y)-u(x)\big)t - D_\ell(y-x)\Big).$$

Taking absolute values, using $t^{-1} > 0$ and then (9),

$$\Big|u(y)-u(x)-\big(t^{-1}D_\ell\big)(y-x)\Big| = t^{-1}\Big|\big(u(y)-u(x)\big)t - D_\ell(y-x)\Big| \le t^{-1}\cdot ct\,\|y-x\| = c\|y-x\|,$$

which is (4) with this δ.

Since $c > 0$ was arbitrary, u is Fréchet differentiable at x with derivative $D := t^{-1}D_\ell$, which is the assertion. □

2.24. `ComparisonNondegenerateSemiconvex`.

Theorem 2.24 (Proposition 3.3, the non-degenerate case)**.** *Let $\mathbb{G}$ be a sub-elliptic setting (`CarnotSetting`) with frame $\sigma = [\sigma^1, \dots, \sigma^m]$, let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$ satisfy the structure conditions (E) and (S) (`StructureConditionsES`) together with the normalization (2.3) (`NormalizedAH`), and assume the cited results of the literature at $\mathbb{G}$ (`CitedResults`). Let $\Omega \subset \mathbb{R}^n$ be a bounded open set,* not necessarily connected. *Let $u, v : \mathbb{R}^n \to \mathbb{R}$ be continuous on $\bar\Omega$, with u locally semi-convex and v locally semi-concave in Ω (`LocallySemiConvexOn`), and assume*

(1) *$u \le v$ on $\partial\Omega$;*
(2) *$\mathcal{L}u \le 0 \le \mathcal{L}v$ in Ω in the viscosity sense (`IsViscSubsol`, `IsViscSupersol`);*
(3) *(non-degeneracy)* hgrad σ x $D \neq 0$ *(`hgrad`) whenever $x \in \operatorname{argmax}_\Omega(u - v)$ and D is a derivative of u at x.*

Then $u \le v$ in Ω.

Proof. This is Proposition 3.3. Throughout, balls, distances and $|\cdot|$ are Euclidean, and two matrix norms occur, on matrices of two different sizes. For an $m \times m$ horizontal matrix — $A(\xi)$, $\mathcal{X}\mathcal{X}^s$, N_k, $\mathcal{M}$ — we write $|X|$ for the Frobenius norm $\sqrt{\operatorname{Tr}(X^T X)}$, which is the norm the trace estimates of Step 4 use, through $|\operatorname{Tr}(XY)| \le |X|\,|Y|$. For an $n \times n$ Euclidean Hessian — D^2P, $D^2\tilde u$, $D^2 v_1$ — we write $\|X\|_{\mathrm{op}} := \sup_{|h|=1} |Xh|$ for the operator norm; that is the convention in which the paper states its bound on D^2P at (3.5), it is the one for which the constant 12μ below is exact, and it is the one the ordering $X \le \|X\|_{\mathrm{op}}\mathbb{I}$ of Step 4 needs. The two families live on different spaces, so naming each norm where it is used costs nothing and avoids a false constant: in the Frobenius norm the bound of (6) would read $4\sqrt{n+8}\,\mu|x-x_1|^2$, not $12\mu|x-x_1|^2$. We use repeatedly that $\bar\Omega$ is compact (Ω is bounded) and that $\partial\Omega = \bar\Omega \setminus \Omega$ (Ω is open). Connectedness of Ω is nowhere used; this matters because the reduced domains to which the proposition is later applied need not be connected.

Suppose, for contradiction, that $u(x^*) > v(x^*)$ for some $x^* \in \Omega$. Then

$$M_0 := \max_{\bar\Omega}(u - v) \;\ge\; u(x^*) - v(x^*) \;>\; 0,$$

the maximum being attained because $u - v$ is continuous on the compact set $\bar\Omega$. Put

$$E := \operatorname*{argmax}_{\Omega}(u - v) = \{x \in \Omega : \; (u - v)(x) = M_0\}.$$

Since $u - v \le 0 < M_0$ on $\partial\Omega$ by (1), every maximum point of $u - v$ over $\bar\Omega$ lies in Ω, so $E = \{x \in \bar\Omega : (u - v)(x) = M_0\}$ is non-empty; it is a closed subset of the compact set $\bar\Omega$, hence E is a non-empty compact subset of Ω.

Fix τ with $0 < \tau < M_0$ and put

$$v_1 := v + \tau.$$

Then: v_1 is continuous on $\bar\Omega$; $-v_1 = -v - \tau$ is locally semi-convex in Ω (adding a constant changes neither `ConvexOn` nor the constant Λ in `LocallySemiConvexOn`), i.e. v_1 is locally semi-concave; v_1 is a viscosity super-solution in Ω by `AddConstPreservesSubsol`; v_1 has the same derivatives as v, hence the same horizontal gradients; and

$$(0) \qquad M_1 := \max_{\bar\Omega}(u - v_1) = M_0 - \tau > 0, \qquad \operatorname*{argmax}_{\Omega}(u - v_1) = E, \qquad u - v_1 \le -\tau \text{ on } \partial\Omega.$$

This is the paper's "replacing v by $v + \tau$ …; this does not change E, $\mathcal{X}u$, $\mathcal{X}v$". We work with the pair (u, v_1) from now on and derive a contradiction; that contradiction refutes $M_0 > 0$ and so proves $u \le v$ in Ω.

Step 0: two facts used repeatedly.

(0a) Second-order jets are unique. If (D, B) and (D', B') are both second-order jets of w at x (`TwiceDiffAt`) then $D = D'$ and $B = B'$. Indeed, subtracting the two Peano estimates, for every $\varepsilon > 0$ there is $\delta > 0$ with

$$\Big|(D - D')(h) + \tfrac{1}{2}(B - B')(h, h)\Big| \le 2\varepsilon|h|^2 \qquad (|h| < \delta).$$

Fix $h \neq 0$ and apply this with h replaced by γh, $0 < \gamma < \delta/|h|$: by linearity and bilinearity, $\gamma(D-D')(h)+\frac{\gamma^2}{2}(B-B')(h,h)$ has absolute value $\leq 2\varepsilon\gamma^2|h|^2$. Dividing by γ and letting $\gamma \to 0^+$ gives $(D-D')(h) = 0$; then dividing the remaining inequality by $\gamma^2/2$ gives $|(B-B')(h,h)| \leq 4\varepsilon|h|^2$ for every $\varepsilon > 0$, i.e. $(B-B')(h,h) = 0$ for every h. Since $B-B'$ is symmetric, polarization $2(B-B')(h,k) = (B-B')(h+k,h+k)-(B-B')(h,h)-(B-B')(k,k)$ gives $B = B'$. Consequently "$\mathcal{X}w(x)$" and "$\mathcal{X}\mathcal{X}^s w(x)$" are unambiguous at a point of twice differentiability, and we use that notation below.

(0b) Adding a C^2 function preserves local semi-convexity. If w is locally semi-convex in Ω and $g \in C^2(\Omega)$, then $w+g$ is locally semi-convex in Ω. Indeed, given $x \in \Omega$ pick $\rho > 0$ and $\Lambda_1 \geq 0$ with $B(x,\rho) \subset \Omega$ and $w + \frac{\Lambda_1}{2}|\cdot|^2$ convex on $B(x,\rho)$; shrinking ρ we may assume $\bar{B}(x,\rho) \subset \Omega$, so $D^2 g$ is bounded on $B(x,\rho)$, say $D^2 g \geq -\Lambda_2 \mathbb{I}$ there with $\Lambda_2 \geq 0$. Then $g + \frac{\Lambda_2}{2}|\cdot|^2$ is a C^2 function with non-negative Hessian on the convex set $B(x,\rho)$, hence convex there, and the sum $(w+g) + \frac{\Lambda_1+\Lambda_2}{2}|\cdot|^2$ of two convex functions is convex on $B(x,\rho)$.

Step 1: relabelling.

(1a) $\theta := \min_E |\mathcal{X}u| > 0$. By (0), $E = \operatorname{argmax}_\Omega(u - v_1)$, so by the second clause of `SemiConvexDiffAtMax` (applied to the locally semi-convex u and the locally semi-concave v_1) both u and v_1 are differentiable at every point of E, with equal derivatives there. The Fréchet derivative of a function at a point is unique, by an elementary first-order argument that is *not* (0a): if D and D' both satisfy $u(x+h)-u(x)-D(h) = o(|h|)$ and $u(x+h)-u(x)-D'(h) = o(|h|)$, then $(D-D')(h) = o(|h|)$, and replacing h by γh and dividing by $\gamma > 0$ gives $(D-D')(h) = o(1)$ as $\gamma \to 0^+$ with the left-hand side independent of γ, so $(D-D')(h) = 0$ for every h. Hence $x \mapsto \mathcal{X}u(x)$ is a well-defined map on E. It is continuous on E: if $x_k \to x$ in E then all points lie in Ω and u is differentiable at each, so `SemiConvexGradPartialCont` gives $Du(x_k) \to Du(x)$ in the operator norm, and then $\mathcal{X}u(x_k) \to \mathcal{X}u(x)$ by `HgradTendstoOfTendsto`, applied with the frame σ — whose components are continuous, being smooth by `CarnotSetting` — the points $x_k \to x$ and the jets $Du(x_k) \to Du(x)$. That node is the two-term split of $D_k(\sigma^i(x_k)) - D_\infty(\sigma^i(x))$, in which the jets converge uniformly while the frame values stay bounded. Since E is non-empty and compact, the continuous function $|\mathcal{X}u|$ attains its minimum on E at some $\bar{x} \in E$, and hypothesis (3) at $\bar{x} \in \operatorname{argmax}_\Omega(u-v) = E$ gives $\mathcal{X}u(\bar{x}) \neq 0$. Hence $\theta := \min_E |\mathcal{X}u| = |\mathcal{X}u(\bar{x})| > 0$.

(1b) The relabelling and its elementary properties. Since u is continuous on the compact $\bar{\Omega}$ there is $M \geq 0$ with $|u| \leq M$ on $\bar{\Omega}$. For $\lambda > 0$ set

$$\psi_\lambda(s) := s + \lambda e^s, \qquad u_\lambda := \psi_\lambda \circ u.$$

By `ExpRelabelling`: $\psi_\lambda \in C^2(\mathbb{R})$, $\psi_\lambda' \geq 1$ and $\psi_\lambda'' \geq 0$ globally,

$$\psi_\lambda''(s) \;\geq\; \lambda' := \lambda e^{-M} > 0 \quad \text{for } |s| \leq M, \qquad \text{and} \qquad \|u_\lambda - u\|_{L^\infty(\bar{\Omega})} \;\leq\; \lambda e^M. \tag{1}$$

The first three properties are exactly the hypotheses of the first clause of `SemiConvexCompose`, so u_λ is locally semi-convex in Ω; and u_λ is continuous on $\bar{\Omega}$, being a composition of continuous maps. The last bound in (1) replaces the paper's $\|u_\lambda - u\|_{L^\infty} \leq 4\lambda\|u\|_{L^\infty}^2$; the reason for changing the relabelling, and the fact that the paper's own claim is only "$\psi_\lambda' \geq 1$ on $[u_0,\infty)$", are discussed in `ExpRelabelling`. Finally, ψ_λ is a C^2 diffeomorphism of $\mathbb{R}$ *onto all of* $\mathbb{R}$, so that $\psi_\lambda^{-1} \in C^2(\mathbb{R})$ with $(\psi_\lambda^{-1})' > 0$. Surjectivity has to be checked rather than assumed, because `PerturbationIdentity` requires its ψ to be of class C^2 on all of $\mathbb{R}$ and ψ_λ^{-1} will be fed to it in Step 3; a diffeomorphism onto the image would not suffice. It is short. By `ExpRelabelling`(2), $\psi_\lambda' \geq 1 > 0$ everywhere, so ψ_λ is continuous and strictly increasing, hence injective. For $s \geq 0$ the mean value theorem gives $\psi_\lambda(s)-\psi_\lambda(0) = \psi_\lambda'(\varsigma)\, s \geq s$ for some $\varsigma \in (0,s)$, so $\psi_\lambda(s) \geq \psi_\lambda(0)+s \to +\infty$ as $s \to +\infty$. For $s \leq 0$ we have $e^s \leq 1$, so $\psi_\lambda(s) = s + \lambda e^s \leq s + \lambda = \psi_\lambda(0) + s \to -\infty$ as $s \to -\infty$. A continuous real function that is unbounded above and unbounded below takes every real value, by the intermediate value theorem; so ψ_λ is a continuous strictly increasing bijection of $\mathbb{R}$ onto $\mathbb{R}$. Its inverse is therefore defined on all of $\mathbb{R}$, and since ψ_λ is C^2 with $\psi_\lambda' \geq 1 > 0$ nowhere vanishing, the inverse function theorem makes ψ_λ^{-1} of class C^2 on $\mathbb{R}$, with $(\psi_\lambda^{-1})'(\varsigma) = 1/\psi_\lambda'\big(\psi_\lambda^{-1}(\varsigma)\big) \in (0,1]$.

(1c) E_λ is a non-empty compact subset of Ω, for λ small. Assume

$$\lambda e^M \ \le\ \min\{\tau/2,\ M_1/2\}. \tag{2}$$

On $\partial\Omega$, by (0) and (1), $u_\lambda - v_1 \le (u - v_1) + \lambda e^M \le -\tau + \tau/2 = -\tau/2$. Also, picking any $\bar{x}_0 \in E$, $(u_\lambda - v_1)(\bar{x}_0) \ge (u - v_1)(\bar{x}_0) - \lambda e^M = M_1 - \lambda e^M \ge M_1/2 > 0$. Hence

$$M_\lambda := \max_{\bar{\Omega}}(u_\lambda - v_1)\ \ge\ M_1/2\ >\ 0\ >\ -\tau/2\ \ge\ \sup_{\partial\Omega}(u_\lambda - v_1),$$

the maximum again being attained by compactness and continuity. Exactly as for E, every maximum point lies in Ω, so

$$E_\lambda := \operatorname*{argmax}_{\Omega}(u_\lambda - v_1) = \{x \in \bar{\Omega} : (u_\lambda - v_1)(x) = M_\lambda\}$$

is a non-empty compact subset of Ω. Note also, from (1), that

$$|M_\lambda - M_1|\ \le\ \lambda e^M, \tag{3}$$

since $u_\lambda - v_1$ and $u - v_1$ differ by at most λe^M on $\bar{\Omega}$.

(1d) $\sup_{E_\lambda} \operatorname{dist}(\cdot, E) \to 0$ *as* $\lambda \to 0^+$. Let $\eta > 0$ and suppose the assertion fails for this η. Then there are $\lambda_k \to 0^+$ satisfying (2) and points $x_k \in E_{\lambda_k}$ with $\operatorname{dist}(x_k, E) > \eta$. The x_k lie in the bounded set Ω, so after passing to a subsequence $x_k \to \hat{x} \in \bar{\Omega}$. Now

$$\big|(u - v_1)(x_k) - M_1\big| \le \big|(u - v_1)(x_k) - (u_{\lambda_k} - v_1)(x_k)\big| + \big|M_{\lambda_k} - M_1\big| \le 2\lambda_k e^M \ \longrightarrow\ 0,$$

using (1), $(u_{\lambda_k} - v_1)(x_k) = M_{\lambda_k}$ and (3). Since $u - v_1$ is continuous on $\bar{\Omega}$, letting $k \to \infty$ gives $(u - v_1)(\hat{x}) = M_1 > 0$. But $u - v_1 \le -\tau < 0$ on $\partial\Omega$, so $\hat{x} \notin \partial\Omega$, i.e. $\hat{x} \in \Omega$ and $\hat{x} \in \operatorname{argmax}_\Omega(u - v_1) = E$. Then $\operatorname{dist}(x_k, E) \le |x_k - \hat{x}| \to 0$, contradicting $\operatorname{dist}(x_k, E) > \eta$. (This is the paper's "$E_\lambda \to E$ ... because $u_\lambda - v \to u - v$ uniformly", together with the remark of (2.8) that uniform limits of maximizers are maximizers.)

(1e) (3.4). The hypotheses of `RelabellingGradientBound` are now available: σ^i continuous; u locally semi-convex in Ω; $E \subset \Omega$ compact; u differentiable on E by (1a); $|\mathcal{X}u| \ge \theta > 0$ on E by (1a); $E_\lambda \subset \Omega$ by (1c); and (1d). Hence there is $\lambda_0 > 0$ such that

$$|\mathcal{X}u(x)|\ \ge\ \theta/2 \qquad \text{for all } x \in E_\lambda,\ 0 < \lambda < \lambda_0, \tag{4}$$

at every differentiability point x of u in E_λ — and every $x \in E_\lambda$ is one, by `ExpRelabelledMaxDifferentiabilit` applied with $N := n$, the parameter λ, the functions u and v_1, and the point x. Its hypotheses are exactly what is available: membership $x \in E_\lambda$ unfolds to $x \in \Omega$ together with $x \in \operatorname{argmax}_\Omega(u_\lambda - v_1)$, which is the node's IsMaxOn clause, and the two semi-convexity hypotheses are the local semi-convexity of u_λ from (1d) and the local semi-concavity of v_1. That node derives the differentiability of u at x from that of u_λ — which it obtains from `SemiConvexDiffAtMax`, second clause — by dividing the derivative by $\psi_\lambda'(u(x)) = 1 + \lambda e^{u(x)} \ge 1$, using that ψ_λ is expanding. In particular it does not invert ψ_λ: neither $\psi_\lambda^{-1} \in C^2$ nor a chain rule for the inverse is needed.

Fix λ with $0 < \lambda < \lambda_0$ and (2), and write

$$\tilde{u} := u_\lambda, \qquad \tilde{M} := M_\lambda = \max_{\Omega}(\tilde{u} - v_1) > 0, \qquad \lambda' = \lambda e^{-M} > 0.$$

Let $e_0 > 0$ be the ellipticity constant of (2.4), which exists by `EllipticityLowerBound` applied with (E) and (2.3), and set

$$\gamma := \lambda'\, e_0\, (\theta/4)^2\ >\ 0. \tag{5}$$

(The paper has $\gamma = 2\lambda e_0(\theta/4)^2$, with $2\lambda = \psi_\lambda''$ for its quadratic relabelling; ours has λ' in place of 2λ, by (1). The exponent $\theta/4$, not $\theta/2$, is the paper's: (3.4) gives $\theta/2$ on E_λ, and the points at which Corollary 3.2 is actually applied are the Jensen points z_k near a point of E_λ, where partial continuity only gives $\theta/4$; see Step 3.)

Step 2: penalization. Fix $x_1 \in E_\lambda$ and $\mu > 0$, and put

$$P(x) := \mu|x - x_1|^4, \qquad \tilde{v} := v_1 + P.$$

P is a polynomial, hence C^∞, and `QuarticPenaltyCalculus` collects everything the argument uses about it. Stated there for the explicit differentials

$$DP(x) = 4\mu|x-x_1|^2\langle x-x_1,\cdot\rangle, \qquad D^2P(x) = 4\mu|x-x_1|^2\,\mathbb{I} + 8\mu\,(x-x_1)\otimes(x-x_1),$$

that node gives: $D^2P(x)$ is symmetric; the two operator-norm bounds

$$\big\|DP(x)\big\|_{\mathrm{op}} \le 4\mu|x-x_1|^3, \qquad \big\|D^2P(x)\big\|_{\mathrm{op}} \le 12\mu|x-x_1|^2; \tag{6}$$

the fact that $\big(DP(x), D^2P(x)\big)$ is a second-order jet of P at every point — by part (1) of `SecondOrderJetCalculus`, the second-order remainder of a quartic being cubic; the values $P(x_1) = 0$ and $P(x) > 0$ for $x \neq x_1$; the parallelogram identity for $\|ta + (1-t)b\|^2$; and the convexity-defect bound

$$tP(x) + (1-t)P(y) - P\big(tx + (1-t)y\big) \;\le\; 6\mu R^2\, t(1-t)|x-y|^2 \qquad \text{when } |x-x_1|, |y-x_1| \le R,$$

which is the chord form that local semi-convexity of $-P$ runs on.

Since $|\nabla P(x)| = \big\|DP(x)\big\|_{\mathrm{op}}$, the first bound of (6) is the paper's $|\nabla P(x)| = 4\mu|x-x_1|^3$. The second bound is in fact an equality, and this is where the operator norm is the right convention. Writing $h := x - x_1$, differentiating $\nabla P(x) = 4\mu|h|^2 h$ gives

$$D^2P(x) = 4\mu\big(|h|^2\mathbb{I} + 2\,h\otimes h\big),$$

a symmetric matrix with eigenvalue $4\mu(|h|^2 + 2|h|^2) = 12\mu|h|^2$ on the line $\mathbb{R}h$ and eigenvalue $4\mu|h|^2$ on the orthogonal complement; a symmetric matrix has operator norm equal to the largest modulus of its eigenvalues, so $\|D^2P(x)\|_{\mathrm{op}} = 12\mu|h|^2$ (both sides vanishing when $h = 0$). The Frobenius norm of the same matrix is $\sqrt{(12\mu|h|^2)^2 + (n-1)(4\mu|h|^2)^2} = 4\mu|h|^2\sqrt{n+8}$, which exceeds $12\mu|h|^2$ for every $n \ge 2$; the paper's constant 12μ at (3.5) is therefore the operator-norm one, as declared in the preamble. By (0b) applied to $-v_1$ and $-P$, $-\tilde v$ is locally semi-convex in Ω, i.e. $\tilde v$ is locally semi-concave; and $\tilde v$ is continuous on $\bar\Omega$. Moreover x_1 is the *unique* maximum point of $\tilde u - \tilde v$ on $\bar\Omega$: for $x \in \bar\Omega$,

$$(\tilde u - \tilde v)(x) = (\tilde u - v_1)(x) - \mu|x-x_1|^4 \;\le\; \tilde M - \mu|x-x_1|^4 = (\tilde u - \tilde v)(x_1) - \mu|x-x_1|^4,$$

using $(\tilde u - v_1)(x) \le \tilde M = (\tilde u - v_1)(x_1)$ and $P(x_1) = 0$; the inequality is strict unless $x = x_1$. In particular, choosing $\nu > 0$ with $B(x_1,\nu) \subset \Omega$ (possible since $x_1 \in \Omega$ open),

$$(\tilde u - \tilde v)(y) < (\tilde u - \tilde v)(x_1) \qquad \text{for all } y \in B(x_1,\nu),\ y \neq x_1. \tag{7}$$

Step 3: Jensen's lemma. By (7), $\tilde u$ locally semi-convex, $\tilde v$ locally semi-concave, and $B(x_1,\nu) \subset \Omega$, the hypotheses of `JensenTwiceDiffPoint` hold for the pair $(\tilde u, \tilde v)$ at the strict maximum x_1, apart from its cited-results hypothesis, which is discharged as follows: that node asks for a Euclidean cited-results package (`EuclidCitedResults`), and by the definition in `CitedResults` a cited-results package at $\mathbb{G}$ — our hypothesis — *is* such a package together with the seven statements about $\mathbb{G}$, so it supplies one. No setting is involved: Aleksandrov's theorem and Jensen's lemma are statements about $\mathbb{R}^N$ alone. That node — which is Jensen's lemma together with Aleksandrov's theorem, applied to *both* functions, as the paper's Step 3 requires — yields sequences $p_k \to 0$ in $\mathbb{R}^n$ and $z_k \to x_1$ with, for every k: $z_k \in \Omega$; z_k is a local maximum point of $\tilde u - \tilde v + \langle p_k, \cdot\rangle$; and both $\tilde u$ and $\tilde v$ have second-order jets at z_k. Put

$$\delta_k := |p_k| + |z_k - x_1| \;\longrightarrow\; 0,$$

which plays the role of the paper's $1/k$. Fix $\rho > 0$ with $\bar B(x_1,\rho) \subset \Omega$ and discard finitely many k so that $z_k \in B(x_1,\rho)$ for all k; put $K := \bar B(x_1,\rho)$, a compact subset of Ω containing all the z_k and x_1.

(3a) Jets at z_k, and the first-order facts about u. Write (D_k, B_k) for the jet of $\tilde u$ at z_k and $(\tilde D_k, \tilde B_k)$ for that of $\tilde v$; by (0a) they are unique.

The facts about second-order jets used here and in Step 4 are collected in `SecondOrderJetCalculus`, stated there for jets at a point of $\mathbb{R}^N$ and applied here with $N := n$: a cubic remainder bound on the unit ball around the base point already produces a jet (part (1) there); jets add, negate and subtract, with their data adding, negating and subtracting (parts (2)–(4)); and a linear

functional $y \mapsto \langle p, y\rangle$ has at every point the jet $\big(\langle p,\cdot\rangle, 0\big)$, with vanishing second-order part (part (5)).

The cubic-bound form of part (1) is the one this proof needs, because every function to which it is applied below is a concrete polynomial. For the quartic P the second-order Taylor remainder at z is $O(|y-z|^3)$, since P is a polynomial of degree 4 whose second-order Taylor polynomial at z differs from it by terms of degree 3 and 4 in $y-z$; for the linear $\langle p_k,\cdot\rangle$ the remainder vanishes identically. (It is *not* exact for P: only the linear term has an exact second-order jet.)

By parts (1), (4) and (5) of `SecondOrderJetCalculus`, applied to the C^∞ functions P and $\langle p_k,\cdot\rangle$,

$$\big(D_k^v, B_k^v\big) := \big(\tilde D_k - DP(z_k),\ \tilde B_k - D^2P(z_k)\big)$$

is a second-order jet of $v_1 = \tilde v - P$ at z_k, and $\big(\tilde D_k - p_k, \tilde B_k\big)$ is one of $\tilde v - \langle p_k,\cdot\rangle$ at z_k.

The first-order facts about u at z_k. These are established now, *before* any appeal to `PerturbationIdentity`, because that node carries the hypothesis $\xi_0 \neq 0$ and the two instances of it that Step 3 needs will be discharged from these facts; taking them first is also the paper's own order — line 508 obtains $|\mathcal{X}\tilde u(z_k)| = \psi_\lambda'(u(z_k))|\mathcal{X}u(z_k)|$ by the plain chain rule and invokes Corollary 3.2 only afterwards, and line 489 makes the same inverse-composition move ("every $x \in E_\lambda$ is a differentiability point of u_λ, hence of u") at first order only.

First, $\tilde u$ is Fréchet differentiable at z_k with derivative D_k. Apply `TwiceDiffAt` to the jet (D_k, B_k) with accuracy $\varepsilon := 1$: there is $\delta > 0$ with $\big|\tilde u(z_k+h) - \tilde u(z_k) - D_k(h) - \frac12 B_k(h,h)\big| \le |h|^2$ for $|h| < \delta$. Since B_k is a continuous bilinear form, $\big|\frac12 B_k(h,h)\big| \le \frac12\|B_k\|\,|h|^2$, so $\tilde u(z_k+h) - \tilde u(z_k) - D_k(h) = O(|h|^2) = o(|h|)$, which is Fréchet differentiability at z_k with derivative D_k.

Second, $u = \psi_\lambda^{-1} \circ \tilde u$: by (1b) ψ_λ is a bijection of $\mathbb{R}$, and $\tilde u = u_\lambda = \psi_\lambda \circ u$ by the definition in (1b), so composing with ψ_λ^{-1} on the left gives the identity everywhere. By (1b) again ψ_λ^{-1} is differentiable at the point $\tilde u(z_k)$, with derivative $(\psi_\lambda^{-1})'(\tilde u(z_k)) = 1/\psi_\lambda'(u(z_k))$, using $\psi_\lambda^{-1}(\tilde u(z_k)) = u(z_k)$. The *first-order* chain rule — a differentiable real function of a Fréchet differentiable function — therefore gives that u is Fréchet differentiable at z_k with $Du(z_k) = \frac{1}{t_k} D_k$, i.e.

$$D_k = t_k\, Du(z_k) \quad\text{and}\quad \mathcal{X}\tilde u(z_k) = t_k\, \mathcal{X}u(z_k), \qquad t_k := \psi_\lambda'\big(u(z_k)\big) \ \ge\ 1, \tag{8}$$

where the second identity follows from the first because hgrad $\sigma\ z_k\ \cdot$ is linear in the 1-jet (its i-th component is $D \mapsto D(\sigma^i(z_k))$, `hgrad`), and $t_k \ge 1$ by `ExpRelabelling`(2). Note that no relabelling identity was used: (8) is the plain chain rule, and in particular $\mathcal{X}u(z_k)$ now denotes hgrad $\sigma\ z_k\ Du(z_k)$ for the Fréchet derivative just produced.

(3b) (3.5) and (3.6). z_k is a local maximum of $\tilde u - \big(\tilde v - \langle p_k,\cdot\rangle\big)$, and both functions have second-order jets at z_k. By `LocalMaxSecondOrder`,

$$D_k = \tilde D_k - p_k, \qquad \mathcal{X}\tilde u(z_k) = \mathrm{hgrad}\,\sigma\ z_k\ (\tilde D_k - p_k), \qquad \mathrm{hessS}\,\sigma\ z_k\ D_k\ B_k\ \le\ \mathrm{hessS}\,\sigma\ z_k\ (\tilde D_k - p_k)\ \tilde B_k \tag{9}$$

in the order (2.1) (`MatLe`). The first identity is the paper's $\nabla\tilde u(z_k) = \nabla v_1(z_k) + \nabla P(z_k) - p_k$ of (3.5), since $\tilde D_k = D_k^v + DP(z_k)$.

Next we convert (9) into the paper's (3.6). By `hgrad`, hgrad is linear in the 1-jet, so with

$$\xi_k := \mathcal{X}v_1(z_k) = \mathrm{hgrad}\ \sigma\ z_k\ D_k^v, \qquad \eta_k := \mathrm{hgrad}\ \sigma\ z_k\ \big(DP(z_k) - p_k\big)$$

we get from $\tilde D_k - p_k = D_k^v + \big(DP(z_k) - p_k\big)$ and (9) that

$$\mathcal{X}\tilde u(z_k) = \xi_k + \eta_k. \tag{10}$$

By `hhessS` and `Mfield`, $\mathrm{hessS}\ \sigma\ x\ D\ B = \big(B(\sigma^i(x), \sigma^j(x))\big)_{i,j} + \mathcal{M}(x, D)$ with $\mathcal{M}(x,\cdot)$ linear in the 1-jet, so hessS is additive in the pair (D, B), and (9) becomes

$$\mathcal{X}\mathcal{X}^s\tilde u(z_k)\ \le\ \mathcal{X}\mathcal{X}^s v_1(z_k) + N_k, \qquad N_k := \Big(D^2P(z_k)\big(\sigma^i(z_k), \sigma^j(z_k)\big)\Big)_{i,j} + \mathcal{M}\big(z_k,\ DP(z_k) - p_k\big), \tag{11}$$

which is (3.6), with $N_k = \sigma(z_k)^T D^2P(z_k)\sigma(z_k) + \mathcal{M}(z_k, \nabla P(z_k) - p_k)$. Since σ and $D\sigma$ are bounded on the compact K — say by c_1 — (6) gives

$$|\eta_k| \;\le\; c\,\delta_k, \qquad |N_k| \;\le\; c\,\delta_k \tag{12}$$

for a constant $c = c(\mathbb{G}, K, \mu)$ independent of k. Indeed $|DP(z_k) - p_k| \le 4\mu|z_k - x_1|^3 + |p_k| \le (4\mu\rho^2 + 1)\delta_k$ by (6) and $|z_k - x_1| \le \min\{\rho, \delta_k\}$, and $\|D^2P(z_k)\|_{\mathrm{op}} = 12\mu|z_k - x_1|^2 \le 12\mu\rho\,\delta_k$ by (6) and the same two bounds. For η_k: the i-th component of hgrad σ z_k $(DP(z_k) - p_k)$ is $(DP(z_k) - p_k)(\sigma^i(z_k))$, of modulus at most $|DP(z_k) - p_k|\,c_1$, so $|\eta_k| \le \sqrt{m}\,c_1(4\mu\rho^2 + 1)\delta_k$. For $N_k = \big(D^2P(z_k)(\sigma^i(z_k), \sigma^j(z_k))\big)_{i,j} + \mathcal{M}(z_k, DP(z_k) - p_k)$: the (i, j) entry of the first summand has modulus at most $\|D^2P(z_k)\|_{\mathrm{op}}c_1^2 \le 12\mu\rho c_1^2\delta_k$, so that summand has Frobenius norm at most $12m\mu\rho c_1^2\delta_k$ (an $m \times m$ matrix with entries at most a in modulus has Frobenius norm at most ma); and by `Mfield` each entry of $\mathcal{M}(z_k, DP(z_k) - p_k)$ has modulus at most $|DP(z_k) - p_k|\,c_1^2$, so that summand has Frobenius norm at most $m\,c_1^2(4\mu\rho^2+1)\delta_k$. Taking $c := \sqrt{m}\,c_1(4\mu\rho^2+1) + 12m\mu\rho c_1^2 + m\,c_1^2(4\mu\rho^2 + 1)$ gives (12).

(3c) The horizontal gradients are bounded below. We have $z_k \to x_1 \in E_\lambda$, u is differentiable at x_1 (by (1e)) and at each z_k (by (3a)), and all these points lie in Ω; so `SemiConvexGradPartialCont` gives $Du(z_k) \to Du(x_1)$ and hence $\mathcal{X}u(z_k) \to \mathcal{X}u(x_1)$, by `HgradTendstoOfTendsto` again — the same node used in (1a), which is why this step is no longer argued twice. By (4), $|\mathcal{X}u(x_1)| \ge \theta/2$, so for all large k

$$|\mathcal{X}u(z_k)| \;\ge\; \theta/4. \tag{13}$$

Discard finitely many k so that (13) holds for all k. By (8) and $t_k \ge 1$,

$$|\mathcal{X}\tilde{u}(z_k)| = t_k|\mathcal{X}u(z_k)| \;\ge\; \theta/4, \tag{14}$$

and then, since (10) gives $\xi_k = \mathcal{X}\tilde{u}(z_k) - \eta_k$, for k large enough that $c\,\delta_k \le \theta/8$ we get from (12) and (14)

$$|\xi_k| \;\ge\; |\mathcal{X}\tilde{u}(z_k)| - |\eta_k| \;\ge\; \theta/4 - c\delta_k \;\ge\; \theta/8. \tag{15}$$

Discard finitely many k again so that (15) holds for all k. In particular $\mathcal{X}u(z_k) \neq 0$ by (13) and $\mathcal{X}v_1(z_k) = \xi_k \neq 0$ by (15), and also $\mathcal{X}\tilde{u}(z_k) = \xi_k + \eta_k \neq 0$ by (14).

(3c′) The second-order jet of u at z_k, and the identification of (D_k, B_k). Only now, with $\mathcal{X}\tilde{u}(z_k) \neq 0$ and $\mathcal{X}u(z_k) \neq 0$ in hand from (3c), do we invoke `PerturbationIdentity`: that node carries the hypothesis $\xi_0 = \mathrm{hgrad}\ \sigma\ x_0\ D \neq 0$, and the two applications below need exactly those two facts to discharge it. Nothing in (3a), (3b) or (3c) used this node, so there is no circularity.

Backward: a second-order jet of u. This step is `ExpRelabellingJetDescent`, applied with $N := n$, the relabelling parameter λ, the function u, the point $z := z_k$, the constants $t := t_k$ and $c := c_k$ of (8), the jet $(D_j, B_j) := (D_k, B_k)$ of $\tilde{u} = \psi_\lambda \circ u$ at z_k from (3a), and the pair

$$\big(D_k^u, B_k^u\big) := \Big(\tfrac{1}{t_k}D_k,\ \tfrac{1}{t_k}\big(B_k - c_k\,D_k^u \otimes D_k^u\big)\Big),$$

which is how (D_k^u, B_k^u) is defined here. Its hypotheses are exactly $t_k = 1+\lambda e^{u(z_k)}$ and $c_k = \lambda e^{u(z_k)}$, the two displayed formulas for D_k^u and B_k^u, the symmetry of B_k^u — which follows from that of B_k, since B_k^u is built from B_k and the symmetric rank-one form $D_k^u \otimes D_k^u$ — and the jet property of (D_k, B_k) for $y \mapsto u(y) + \lambda e^{u(y)}$, which is $\tilde{u} = \psi_\lambda \circ u$ with ψ_λ unfolded. The node's conclusion is that $\big(D_k^u, B_k^u\big)$ is a second-order jet of u at z_k.

Note that this route uses neither ψ_λ^{-1} nor a separate first-order chain rule: the cited node derives the first-order expansion of u from the second-order estimate for $\tilde{u}$ itself, via the exact cubic Taylor remainder of the exponential. In particular it does not appeal to `PerturbationIdentity`, so the non-degeneracy facts of (3c) are not needed for the backward direction — they are needed only for the forward identification below. By (8), $D_k^u = D_k/t_k = Du(z_k)$, so the 1-jet of this pair is the Fréchet derivative of u at z_k and hgrad σ z_k $D_k^u = \mathcal{X}u(z_k)$, consistently with (3a).

Forward: the identification of (D_k, B_k). Apply part (1) of the same node again, now with $\psi := \psi_\lambda$ — in $C^2(\mathbb{R})$ by `ExpRelabelling`(1) — $w := u$, $x_0 := z_k$ and the jet (D_k^u, B_k^u) just

produced. Its hypotheses hold: $\xi_0 = \operatorname{hgrad}\sigma\ z_k\ D_k^u = \mathcal{X}u(z_k) \neq 0$ by (13), and $t = \psi_\lambda'(u(z_k)) = t_k \geq 1 > 0$. It gives that

$$\Big(t_k D_k^u,\ t_k B_k^u + \psi_\lambda''\big(u(z_k)\big)\, D_k^u \otimes D_k^u\Big)$$

is a second-order jet of $\psi_\lambda \circ u = \tilde{u}$ at z_k; since (D_k, B_k) is also one, (0a) gives that the two pairs coincide. This identification is what makes the conclusion of `StrictSubsolutionPerturbation` in (3e) — which is stated for the jet just displayed — a statement about $\mathcal{L}\tilde{u}(z_k) := \operatorname{Lop}\sigma\ A\ H\ z_k\ D_k\ B_k$, the quantity used in Step 4.

(3d) The viscosity inequalities hold classically at z_k. Ω is bounded and open, $u, v_1 \in C(\bar{\Omega})$, u has the second-order jet (D_k^u, B_k^u) at $z_k \in \Omega$ with $\mathcal{X}u(z_k) \neq 0$, and v_1 has the second-order jet (D_k^v, B_k^v) at z_k with $\mathcal{X}v_1(z_k) \neq 0$. Since u is a viscosity sub-solution and v_1 a viscosity super-solution in Ω, `ClassicalAtTwiceDiff` (part (1) for u, part (2) for v_1) gives

$$\mathcal{L}u(z_k)\ \leq\ 0\ \leq\ \mathcal{L}v_1(z_k), \tag{16}$$

where $\mathcal{L}u(z_k) := \operatorname{Lop}\sigma\ A\ H\ z_k\ D_k^u\ B_k^u$ and similarly for v_1 (`Lop`).

(3e) (3.7). Apply `StrictSubsolutionPerturbation` with $w := u$, $x_0 := z_k$, jet (D_k^u, B_k^u), $\psi := \psi_\lambda$, e_0 as in (5), θ replaced by $\theta/4$ and λ replaced by λ'. Its hypotheses hold: (2.3) for (A, H); $e_0 > 0$ with $e_0|\xi|^2 \leq \mathcal{E}(A, \xi)$; $\psi_\lambda \in C^2$; $\theta/4 > 0$ and $\theta/4 \leq |\mathcal{X}u(z_k)|$ by (13); $\lambda' > 0$; $\psi_\lambda'(u(z_k)) = t_k > 0$; $\lambda' \leq \psi_\lambda''(u(z_k))$ by (1), because $|u(z_k)| \leq M$; and $\mathcal{L}u(z_k) \leq 0$ by (16). Its conclusion is an inequality for Lop evaluated at the jet $\big(t_k D_k^u,\ t_k B_k^u + \psi_\lambda''(u(z_k)) D_k^u \otimes D_k^u\big)$ of $\psi_\lambda \circ u = \tilde{u}$ at z_k, which by (3c′) is (D_k, B_k); so the conclusion reads

$$\mathcal{L}\tilde{u}(z_k)\ \leq\ -\gamma\ <\ 0, \qquad \gamma = \lambda' e_0 (\theta/4)^2, \tag{17}$$

which is (3.7).

Step 4: conclusion.

(4a) $|\mathcal{X}\mathcal{X}^s\tilde{u}(z_k)| \leq C$ with C independent of k. By `LocallySemiConvexOn` applied at $x_1 \in \Omega$ to the locally semi-convex $\tilde{u}$ there are $\rho_u > 0$ and $\Lambda' \geq 0$ with $B(x_1, \rho_u) \subseteq \Omega$ and $\tilde{u} + \frac{\Lambda'}{2}|\cdot|^2$ convex on $B(x_1, \rho_u)$, and applied to the locally semi-convex $-v_1$ there are $\rho_v > 0$ and $\Lambda'' \geq 0$ with $-v_1 + \frac{\Lambda''}{2}|\cdot|^2$ convex on $B(x_1, \rho_v)$. Put $\rho_1 := \min\{\rho, \rho_u, \rho_v\} > 0$; a convex function on a ball is convex on every smaller concentric ball, so both functions are convex on $B(x_1, \rho_1)$. Since $z_k \to x_1$, discarding finitely many k we may assume $z_k \in B(x_1, \rho_1)$ for every k; note that ρ, K and the constants c_1, c of Step 3 are unchanged by this, since $\rho_1 \leq \rho$ and the bounds of (12) hold a fortiori.

The jet Hessian sign from convexity. This is `JetHessianLowerBoundOfConvex`, proved there over $\mathbb{R}^N$ and applied here with $N := n$: if $w + \frac{\Lambda}{2}|\cdot|^2$ is convex on a ball $B(x_c, r)$, if $z \in B(x_c, r)$ and if (D, B) is a second-order jet of w at z, then $B(h, h) \geq -\Lambda|h|^2$ for every $h \in \mathbb{R}^n$. Its proof there is the midpoint-convexity argument along $z \pm th$ combined with the Peano estimate of `TwiceDiffAt` at an accuracy $\varepsilon > 0$ and the limit $\varepsilon \to 0^+$, the jet of $y \mapsto \frac{\Lambda}{2}|y|^2$ being supplied by parts (1) and (2) of `SecondOrderJetCalculus`.

We use it twice, on the ball $B(x_1, \rho_1)$ of the previous paragraph, in which both $\tilde{u} + \frac{\Lambda'}{2}|\cdot|^2$ and $-v_1 + \frac{\Lambda''}{2}|\cdot|^2$ are convex and which contains every z_k. Applied with $w := \tilde{u}$ and $\Lambda := \Lambda'$, at $z := z_k$ with the jet (D_k, B_k) of (3a), it gives $B_k(h, h) \geq -\Lambda'|h|^2$ for every h, i.e. $D^2\tilde{u}(z_k) \geq -\Lambda' \mathbb{I}$ in the notation $D^2\tilde{u}(z_k) := B_k$. Applied with $w := -v_1$ and $\Lambda := \Lambda''$ — where the required jet of $-v_1$ at z_k is $\big(-D_k^v, -B_k^v\big)$, obtained from the jet $\big(D_k^v, B_k^v\big)$ of (3a) by part (3) of `SecondOrderJetCalculus` — it gives $-B_k^v(h, h) \geq -\Lambda''|h|^2$, i.e. $D^2 v_1(z_k) \leq \Lambda'' \mathbb{I}$. Both deductions are from a *pointwise jet* and not from a classical Hessian, which is the only form available here: pm-0003 makes local semi-convexity the notion in use, and `LocallySemiConvexOn` gives convexity of $w + \frac{\Lambda}{2}|\cdot|^2$ on a ball and nothing more.

The two-sided bound on B_k. Clause (2) of `LocalMaxSecondOrder`, in the application already made in (3b) — to the pair $\tilde{u}$ and $\tilde{v} - \langle p_k, \cdot \rangle$, which has a local maximum at z_k, with the jets (D_k, B_k) and $(\tilde{D}_k - p_k, \tilde{B}_k)$ — states

$$B_k(h, h)\ \leq\ \tilde{B}_k(h, h) \qquad (h \in \mathbb{R}^n),$$

and $\tilde{B}_k = B_k^v + D^2P(z_k)$ by the definition of B_k^v in (3a). (This, not the hhessS inequality of (9), is the Euclidean-Hessian ordering that is needed here.) Hence, for every h, using the two signs just proved and $\|D^2P(z_k)\|_{\mathrm{op}} = 12\mu|z_k - x_1|^2 \le 12\mu\rho^2$ from (6),

$$-\Lambda'|h|^2 \;\le\; B_k(h,h) \;\le\; B_k^v(h,h) + D^2P(z_k)(h,h) \;\le\; \big(\Lambda'' + 12\mu\rho^2\big)|h|^2.$$

So $|B_k(h,h)| \le \Lambda^*|h|^2$ with $\Lambda^* := \max\{\Lambda', \Lambda'' + 12\mu\rho^2\}$. A symmetric bilinear form whose quadratic form is bounded by Λ^* in modulus has operator norm at most Λ^* — for a symmetric matrix the operator norm is the largest modulus of a Rayleigh quotient — so $|B_k(h,h')| \le \Lambda^*|h|\,|h'|$ for all $h, h' \in \mathbb{R}^n$ (the letter k is reserved for the sequence index), and in particular

$$\big|B_k\big(\sigma^i(z_k), \sigma^j(z_k)\big)\big| \;\le\; \Lambda^* c_1^2 \qquad \text{for all } i, j. \tag{18b}$$

The bound on the correction matrix. By the fourth clause of `SemiConvexLocLipschitz`, applied to the locally semi-convex $\tilde{u}$ on the compact $K \subset \Omega$, there is $C_K \ge 0$ with $\|D_k\| \le C_K$ for every k (the hypothesis "D is a derivative of $\tilde{u}$ at a point of K" holds for D_k at z_k by (3a)). By `Mfield`, the (i,j) entry of $\mathcal{M}(z_k, D_k)$ is

$$\tfrac{1}{2}\Big(D_k\big(\mathrm{D}\sigma^j(z_k)\,\sigma^i(z_k)\big) + D_k\big(\mathrm{D}\sigma^i(z_k)\,\sigma^j(z_k)\big)\Big),$$

whose modulus is at most $\frac{1}{2}(\|D_k\|c_1 \cdot c_1 + \|D_k\|c_1 \cdot c_1) = \|D_k\|c_1^2 \le C_K c_1^2$, using $|\sigma^i| \le c_1$ and $\|\mathrm{D}\sigma^i\| \le c_1$ on K. An $m \times m$ matrix all of whose entries are at most a in modulus has Frobenius norm at most $\sqrt{m^2a^2} = ma$, so

$$\big|\mathcal{M}(z_k, D_k)\big| \;\le\; m\, C_K c_1^2. \tag{18c}$$

Conclusion. By `hhessS`, $\mathcal{X}\mathcal{X}^s\tilde{u}(z_k) = \big(B_k(\sigma^i(z_k), \sigma^j(z_k))\big)_{i,j} + \mathcal{M}(z_k, D_k)$. The first matrix has all entries at most $\Lambda^* c_1^2$ in modulus by (18b), hence Frobenius norm at most $m\Lambda^* c_1^2$; adding (18c) and using the triangle inequality for the Frobenius norm,

$$\big|\mathcal{X}\mathcal{X}^s\tilde{u}(z_k)\big| \;\le\; m\, c_1^2\big(\Lambda^* + C_K\big) \;=:\; C \tag{18}$$

with C depending on $\mathbb{G}$, K, μ, Λ', Λ'' and C_K, but not on k. This is the paper's "$|\mathcal{X}\mathcal{X}^s\tilde{u}(z_k)| \le C$ with C independent of k".

(4b) The compact set $\mathcal{Q}$ and the moduli of (3.2). By the fourth clause of `SemiConvexLocLipschitz`, applied to $\tilde{u}$ and to $-v_1$ on K, the derivatives D_k and D_k^v have norms bounded by a constant independent of k, so by `hgrad` and $|\sigma| \le c_1$ on K there is C' with $|\xi_k| \le C'$ and $|\xi_k + \eta_k| \le C'$ for every k. For the lower bounds: (15) gives $|\xi_k| \ge \theta/8$, and for $\xi_k + \eta_k$ we do *not* subtract $|\eta_k|$ from (15) — that would only give $\theta/4 - 2c\delta_k$, hence nothing better than 0 under $c\delta_k \le \theta/8$ — but use (10) and (14) directly: $|\xi_k + \eta_k| = |\mathcal{X}\tilde{u}(z_k)| \ge \theta/4 \ge \theta/8$. This is the paper's own direction at line 508, which bounds $|\xi_k|$ *from* $|\mathcal{X}\tilde{u}(z_k)|$ and not the reverse. Hence all the vectors ξ_k and $\xi_k + \eta_k$ lie in

$$\mathcal{Q} := \{\xi \in \mathbb{R}^m :\; \theta/8 \le |\xi| \le C'\},$$

which is a compact subset of $\mathbb{R}^m \setminus \{0\}$. By `StructureConditionsES`, A and H are continuous on $\mathbb{R}^m \setminus \{0\}$, hence uniformly continuous on $\mathcal{Q}$ and bounded there: there are non-decreasing $\omega_A, \omega_H : [0,\infty) \to [0,\infty)$ with $\omega_A(0^+) = \omega_H(0^+) = 0$ and

$$|A(\xi) - A(\xi')| \le \omega_A(|\xi - \xi'|), \quad |H(\xi) - H(\xi')| \le \omega_H(|\xi - \xi'|) \qquad (\xi, \xi' \in \mathcal{Q}), \tag{19}$$

and $\|A\|_{L^\infty(\mathcal{Q})} := \max_{\mathcal{Q}} |A| < \infty$. These are the moduli of (3.2); they are obtained here, inline, from uniform continuity on a compact subset of $\mathbb{R}^m \setminus \{0\}$, and no separate node is needed for them. Note that $\mathcal{Q}$ depends only on θ, C', hence not on k.

(4c) The trace estimate. Fix k. By `StructureConditionsES` and $\xi_k \ne 0$, the matrix $A(\xi_k)$ is symmetric and non-negative definite; the matrices $\mathcal{X}\mathcal{X}^s\tilde{u}(z_k)$ and $\mathcal{X}\mathcal{X}^s v_1(z_k) + N_k$ are symmetric (the forms B_k, B_k^v, $D^2P(z_k)$ are symmetric and $\mathcal{M}$ is symmetric in (i,j) by `Mfield`), and they are ordered by (11). Hence `TraceMonotone` applies and gives

$$0 \;\le\; \mathrm{Tr}\,\Big(A(\xi_k)\Big(\mathcal{X}\mathcal{X}^s v_1(z_k) + N_k - \mathcal{X}\mathcal{X}^s\tilde{u}(z_k)\Big)\Big). \tag{20}$$

We now expand the right-hand side exactly as the paper does, writing $M^u := \mathcal{X}\mathcal{X}^s\tilde{u}(z_k)$ and $M^v := \mathcal{X}\mathcal{X}^s v_1(z_k)$. By linearity of the trace,

$$\operatorname{Tr}\big(A(\xi_k)(M^v + N_k - M^u)\big) = \operatorname{Tr}\big(A(\xi_k)M^v\big) + \operatorname{Tr}\big(A(\xi_k)N_k\big) - \operatorname{Tr}\big(A(\xi_k)M^u\big). \tag{21}$$

By Lop and $\mathcal{X}v_1(z_k) = \xi_k$,

$$\mathcal{L}v_1(z_k) = -\operatorname{Tr}\big(A(\xi_k)M^v\big) + H(\xi_k), \qquad \text{i.e.} \qquad \operatorname{Tr}\big(A(\xi_k)M^v\big) = -\mathcal{L}v_1(z_k) + H(\xi_k). \tag{22}$$

By Lop and $\mathcal{X}\tilde{u}(z_k) = \xi_k + \eta_k$ from (10),

$$\mathcal{L}\tilde{u}(z_k) = -\operatorname{Tr}\big(A(\xi_k{+}\eta_k)M^u\big) + H(\xi_k{+}\eta_k), \qquad \text{i.e.} \qquad -\operatorname{Tr}\big(A(\xi_k{+}\eta_k)M^u\big) = \mathcal{L}\tilde{u}(z_k) - H(\xi_k{+}\eta_k), \tag{23}$$

and therefore, splitting $A(\xi_k) = A(\xi_k + \eta_k) - \big[A(\xi_k + \eta_k) - A(\xi_k)\big]$,

$$-\operatorname{Tr}\big(A(\xi_k)M^u\big) = -\operatorname{Tr}\big(A(\xi_k{+}\eta_k)M^u\big) + \operatorname{Tr}\big(\big[A(\xi_k{+}\eta_k) - A(\xi_k)\big]M^u\big) = \mathcal{L}\tilde{u}(z_k) - H(\xi_k{+}\eta_k) + \operatorname{Tr}\big(\big[A(\xi_k{+}\eta_k) - A($$

Substituting (22) and (24) into (21) and then into (20),

$$0 \;\le\; -\mathcal{L}v_1(z_k) + \mathcal{L}\tilde{u}(z_k) + \big[H(\xi_k) - H(\xi_k + \eta_k)\big] + \operatorname{Tr}\big(\big[A(\xi_k + \eta_k) - A(\xi_k)\big]M^u\big) + \operatorname{Tr}\big(A(\xi_k)N_k\big). \tag{25}$$

This is the paper's second and third displayed lines; the term $H(\xi_k)$ of (22) and the term $-H(\xi_k + \eta_k)$ of (24) do not cancel, and their difference is exactly the ω_H term below — that is the one place where the perturbation η_k of the horizontal gradient costs something.

Finally we bound the five terms of (25). First, $-\mathcal{L}v_1(z_k) \le 0$ by (16). Second, $\mathcal{L}\tilde{u}(z_k) \le -\gamma$ by (17). Third, $\xi_k, \xi_k + \eta_k \in \mathcal{Q}$ by (4b), so (19) and (12) give $|H(\xi_k) - H(\xi_k + \eta_k)| \le \omega_H(|\eta_k|) \le \omega_H(c\delta_k)$. Fourth, by the Cauchy–Schwarz inequality for the Frobenius inner product, $|\operatorname{Tr}(XY)| \le |X|\,|Y|$, so by (19) and (18)

$$\Big|\operatorname{Tr}\big(\big[A(\xi_k + \eta_k) - A(\xi_k)\big]M^u\big)\Big| \le \big|A(\xi_k + \eta_k) - A(\xi_k)\big|\,|M^u| \;\le\; C\,\omega_A(c\delta_k).$$

Fifth, similarly, $|\operatorname{Tr}(A(\xi_k)N_k)| \le |A(\xi_k)|\,|N_k| \le \|A\|_{L^\infty(\mathcal{Q})}\, c\,\delta_k$ by (19)'s boundedness statement and (12). Altogether (25) gives, for every large k,

$$0 \;\le\; -\gamma + \omega_H(c\delta_k) + C\,\omega_A(c\delta_k) + c\,\|A\|_{L^\infty(\mathcal{Q})}\,\delta_k. \tag{26}$$

All of γ, c, C, ω_A, ω_H, $\|A\|_{L^\infty(\mathcal{Q})}$ are independent of k, and $\delta_k \to 0$, so the right-hand side of (26) tends to $-\gamma < 0$ as $k \to \infty$. Choosing k so large that $\omega_H(c\delta_k) + C\omega_A(c\delta_k) + c\|A\|_{L^\infty(\mathcal{Q})}\delta_k < \gamma$ makes the right-hand side of (26) strictly negative, contradicting (26).

This contradiction refutes the assumption $M_0 > 0$. Hence $u(x) \le v(x)$ for every $x \in \Omega$. □

2.25. `LeftTranslateHgrad`.

Lemma 2.25 (First-order left-invariance of the horizontal gradient)**.** *Let $\mathbb{G}$ be a sub-elliptic setting (`CarnotSetting`) with group law $*$, homogeneous norm $\|\cdot\|$, first-layer projection π_1, horizontal exponential $\exp_{\mathbb{G}}$ and frame $\sigma = [\sigma^1, \dots, \sigma^m]$. Let $u : \mathbb{R}^n \to \mathbb{R}$, let $g, x \in \mathbb{R}^n$, and let D, D' be continuous linear functionals on $\mathbb{R}^n$ such that D is the Fréchet derivative of u at $g * x$ and D' is the Fréchet derivative of the left translate $y \mapsto u(g * y)$ at x. Then*

$$\operatorname{hgrad}\ \sigma\ x\ D' \;=\; \operatorname{hgrad}\ \sigma\ (g * x)\ D \qquad (\textit{hgrad}),$$

i.e. $\mathcal{X}(u \circ L_g)(x) = (\mathcal{X}u)(g * x)$, where $L_g(y) = g * y$.

Proof. This is the sentence "by left invariance $\mathcal{X}v_\varepsilon(x) = \mathcal{X}(v \circ L_{z_x})(x) = (\mathcal{X}v)(z_x * x)$" of the proof of Theorem 1.3(i), with the uniqueness step written out. It is *not* an instance of the left-invariance clause `leftJet` of `CarnotSetting`: that clause opens with the hypothesis that φ have a second-order jet at $g * x$, whereas here u is assumed only once differentiable there — which is the only regularity available at the use sites (the inf-convolution v_ε is differentiable at the maximal points but not twice, and a semi-convex u need not be C^2). No commutativity of $*$ is used either. Write $\langle \alpha, \beta \rangle := \sum_{i=1}^m \alpha_i \beta_i$ for $\alpha, \beta \in \mathbb{R}^m$.

Step 1: uniqueness of the Pansu horizontal differential. We first prove: *if $f : \mathbb{R}^n \to \mathbb{R}$, $y \in \mathbb{R}^n$ and $a, b \in \mathbb{R}^m$ are such that f is Pansu differentiable at y with horizontal differential a and also with horizontal differential b (`HasPansuGradAt`), then $a = b$.*

Suppose $a \neq b$ and put $\zeta := a - b$, so $\zeta \neq 0$ and $\|\zeta\| > 0$. Apply `HasPansuGradAt` to each of the two expansions with the accuracy $\varepsilon := \|\zeta\|/4 > 0$: there are $\delta_1 > 0$ and $\delta_2 > 0$ such that

$$\left|f(y * p) - f(y) - \langle a, \pi_1(p)\rangle\right| \leq \frac{\|\zeta\|}{4}\,\|p\| \qquad (\|p\| < \delta_1), \tag{1}$$

$$\left|f(y * p) - f(y) - \langle b, \pi_1(p)\rangle\right| \leq \frac{\|\zeta\|}{4}\,\|p\| \qquad (\|p\| < \delta_2). \tag{2}$$

Put $\delta := \min\{\delta_1, \delta_2\} > 0$, $t := \dfrac{\delta}{2\|\zeta\|} > 0$ and

$$p := \exp_{\mathbb{G}}(t\zeta).$$

By `CarnotSetting`, $\|\exp_{\mathbb{G}}(\eta)\| = |\eta|$ for every $\eta \in \mathbb{R}^m$ and $\pi_1 \circ \exp_{\mathbb{G}}$ is the identity of $\mathbb{R}^m$, so

$$\|p\| = \|t\zeta\| = t\,\|\zeta\| = \frac{\delta}{2} < \delta, \qquad \pi_1(p) = t\zeta. \tag{3}$$

These two identities are exact — this is why the horizontal one-parameter subgroups are the right test curves: on them the homogeneous norm and the first-layer projection are computed with no error term at all.

Both (1) and (2) therefore apply at this p, and subtracting the two expansions,

$$\left|\langle a, \pi_1(p)\rangle - \langle b, \pi_1(p)\rangle\right| \leq \left|f(y*p) - f(y) - \langle b, \pi_1(p)\rangle\right| + \left|f(y*p) - f(y) - \langle a, \pi_1(p)\rangle\right| \leq 2 \cdot \frac{\|\zeta\|}{4}\,\|p\| = \frac{\|\zeta\|}{2} \cdot t\|\zeta\|,$$

using (3). On the other hand, by (3) and $\zeta = a - b$,

$$\langle a, \pi_1(p)\rangle - \langle b, \pi_1(p)\rangle = \langle a - b, t\zeta\rangle = t\,\langle \zeta, \zeta\rangle = t\,\|\zeta\|^2.$$

Combining, $t\|\zeta\|^2 \leq \frac{1}{2}t\|\zeta\|^2$, i.e. $\frac{1}{2}t\|\zeta\|^2 \leq 0$, contradicting $t > 0$ and $\|\zeta\| > 0$. Hence $a = b$.

Step 2: the two Pansu expansions. By the Taylor formula (2.18), i.e. the clause `taylor2` of `CarnotSetting`, a function with a Fréchet derivative at a point is Pansu differentiable there, with horizontal differential the hgrad of that derivative. Applying it twice:

(A) $y \mapsto u(g * y)$ is Pansu differentiable at x with horizontal differential hgrad $\sigma\ x\ D'$;

(B) u is Pansu differentiable at $g * x$ with horizontal differential hgrad $\sigma\ (g * x)\ D$.

Step 3: (B) is an expansion of the same function at the same point. We claim that (B) says precisely that $y \mapsto u(g * y)$ is Pansu differentiable at x with horizontal differential hgrad $\sigma\ (g * x)\ D$. Let $\varepsilon > 0$ and let $\delta > 0$ be the radius furnished by `HasPansuGradAt` for (B) and this ε, so that

$$\left|u\big((g * x) * p\big) - u(g * x) - \big\langle \text{hgrad } \sigma\ (g * x)\ D,\ \pi_1(p)\big\rangle\right| \leq \varepsilon\|p\| \qquad (\|p\| < \delta). \tag{4}$$

By associativity of $*$ (`CarnotSetting`),

$$(g * x) * p = g * (x * p),$$

so writing $\bar{u}(y) := u(g * y)$ we have $u\big((g * x) * p\big) = \bar{u}(x * p)$ and $u(g * x) = \bar{u}(x)$. Substituting these two equalities into (4) turns it into

$$\left|\bar{u}(x * p) - \bar{u}(x) - \big\langle \text{hgrad } \sigma\ (g * x)\ D,\ \pi_1(p)\big\rangle\right| \leq \varepsilon\|p\| \qquad (\|p\| < \delta),$$

which is exactly the defining condition of `HasPansuGradAt` for $\bar{u}$ at x with horizontal differential hgrad $\sigma\ (g * x)\ D$, for the accuracy ε. As $\varepsilon > 0$ was arbitrary, the claim holds. (This is the only place the group structure enters, and it enters as one application of associativity: the increment p is applied on the *right*, which is what makes the left translation L_g transparent to the expansion.)

Step 4: conclusion. By (A) and Step 3, the single function $\bar{u} = u \circ L_g$ is Pansu differentiable at the single point x with horizontal differential hgrad $\sigma\ x\ D'$ and also with horizontal differential hgrad $\sigma\ (g * x)\ D$. By Step 1 these two vectors of $\mathbb{R}^m$ coincide:

$$\text{hgrad } \sigma\ x\ D' = \text{hgrad } \sigma\ (g * x)\ D.$$

□

2.26. `LeftTranslateSubsol`.

Lemma 2.26 (Lemma 2.9)**.** *Let $\mathbb{G}$ be a sub-elliptic setting (`CarnotSetting`) with group law $*$ and frame σ, let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$, let $\Omega \subseteq \mathbb{R}^n$, let $u : \mathbb{R}^n \to \mathbb{R}$ and let $h \in \mathbb{R}^n$. Write $L_h(x) = h * x$ and $u^h := u \circ L_h$. Then:*

1. *if u is a viscosity sub-solution of $\mathcal{L}\cdot = 0$ in Ω (`IsViscSubsol`), then u^h is a viscosity sub-solution of $\mathcal{L}\cdot = 0$ in $L_h^{-1}(\Omega) = h^{-1} * \Omega$;*
2. *if u is a viscosity super-solution of $\mathcal{L}\cdot = 0$ in Ω (`IsViscSupersol`), then u^h is a viscosity super-solution of $\mathcal{L}\cdot = 0$ in $L_h^{-1}(\Omega)$.*

Proof. This is Lemma 2.9. Write $\Omega' := L_h^{-1}(\Omega) = \{x : \ h * x \in \Omega\}$ and $\bar{u} := u^h = u \circ L_h$. We prove (1); (2) is the same argument with the two inequalities reversed, and is recorded separately because `IsViscSupersol` is a separate predicate.

Step 0: the two cancellations and the smoothness of $L_{\pm h}$. For every x,

$$(1)\quad (-h) * (h * x) = \big((-h) * h\big) * x = 0 * x = x, \qquad h * \big((-h) * x\big) = \big(h * (-h)\big) * x = 0 * x = x,$$

by associativity of $*$ and the identities $(-h)*h = 0$, $h*(-h) = 0$, $0*x = x$ of `CarnotSetting`. In particular L_h and L_{-h} are mutually inverse bijections of $\mathbb{R}^n$, and both are C^∞, being restrictions of the C^∞ group law to a fixed first argument. Moreover

$$(2)\qquad L_{-h}(\Omega) \subseteq \Omega' \qquad \text{and} \qquad L_h(\Omega') \subseteq \Omega,$$

the first because $h * \big((-h) * z\big) = z \in \Omega$ for $z \in \Omega$ by (1), the second by the definition of Ω'.

Step 1: upper semi-continuity of $\bar{u}$ on Ω'. Let $x \in \Omega'$ and let $y > \bar{u}(x) = u(h * x)$. Since $h * x \in \Omega$ and u is upper semi-continuous on Ω (first clause of `IsViscSubsol`), $u < y$ on a neighbourhood of $h*x$ in Ω. The map L_h is continuous and sends Ω' into Ω by (2), so it sends a neighbourhood of x in Ω' into any prescribed neighbourhood of $h*x$ in Ω; hence $\bar{u} = u \circ L_h < y$ on a neighbourhood of x in Ω'. Thus $\bar{u}$ is upper semi-continuous on Ω'.

Step 2: the transported test function. Let $\bar{x} \in \Omega'$, let $\bar{\varphi} : \mathbb{R}^n \to \mathbb{R}$ be C^2 on Ω' with $\bar{x} \in \operatorname{argmax}_{\Omega'}(\bar{u} - \bar{\varphi})$, and let (D, B) be a second-order jet of $\bar{\varphi}$ at $\bar{x}$ (`TwiceDiffAt`) with hgrad $\sigma\ \bar{x}\ D \neq 0$ (`hgrad`). Put

$$\varphi := \bar{\varphi} \circ L_{-h}, \qquad \text{i.e.} \qquad \varphi(z) = \bar{\varphi}\big((-h) * z\big),$$

which is the paper's $\bar{\varphi} \circ L_h^{-1}$ by (1). We check the three conditions that make φ admissible for u at the point $h * \bar{x} \in \Omega$.

(2a) φ is C^2 on Ω. L_{-h} is C^∞, hence C^2, and maps Ω into Ω' by (2); $\bar{\varphi}$ is C^2 on Ω'. The composition of a C^2 map into Ω' with a function C^2 on Ω' is C^2 on Ω.

(2b) $h * \bar{x} \in \operatorname{argmax}_\Omega(u - \varphi)$. First, by (1), $\varphi(h * \bar{x}) = \bar{\varphi}\big((-h) * (h * \bar{x})\big) = \bar{\varphi}(\bar{x})$, so $(u - \varphi)(h * \bar{x}) = u(h * \bar{x}) - \bar{\varphi}(\bar{x}) = (\bar{u} - \bar{\varphi})(\bar{x})$. Now let $z \in \Omega$ and put $w := (-h) * z$, which lies in Ω' by (2). Maximality of $\bar{u} - \bar{\varphi}$ over Ω' at $\bar{x}$ gives

$$u(h * w) - \bar{\varphi}(w) = (\bar{u} - \bar{\varphi})(w) \ \le \ (\bar{u} - \bar{\varphi})(\bar{x}),$$

and $h * w = h * \big((-h) * z\big) = z$ by (1) while $\bar{\varphi}(w) = \varphi(z)$ by the definition of φ. So $(u - \varphi)(z) \le (\bar{u} - \bar{\varphi})(\bar{x}) = (u - \varphi)(h * \bar{x})$. As $z \in \Omega$ was arbitrary, $h * \bar{x}$ maximizes $u - \varphi$ over Ω. This is the paper's "$u - \varphi = (u^h - \bar{\varphi}) \circ L_h^{-1}$ attains its maximum at $h * \bar{y}$".

(2c) the jet of φ at $h * \bar{x}$, with the same horizontal data. By (1), $(-h) * (h * \bar{x}) = \bar{x}$, so (D, B) is a second-order jet of $\bar{\varphi}$ at $(-h) * (h * \bar{x})$. Apply the left-invariance clause `leftJet` of `CarnotSetting` with the group element $-h$, the base point $h * \bar{x}$, the function $\bar{\varphi}$ and the jet (D, B): it provides a second-order jet (D', B') of $y \mapsto \bar{\varphi}\big((-h) * y\big) = \varphi(y)$ at $h * \bar{x}$ with

$$(3)\qquad \text{hgrad } \sigma\ (h * \bar{x})\ D' = \text{hgrad } \sigma\ \bar{x}\ D, \qquad \text{hhessS } \sigma\ (h * \bar{x})\ D'\ B' = \text{hhessS } \sigma\ \bar{x}\ D\ B$$

(`hgrad`, `hhessS`), after rewriting $(-h) * (h * \bar{x})$ as $\bar{x}$. This is exactly the paper's "$\mathcal{X}\varphi(h * \bar{y}) = \mathcal{X}\bar{\varphi}(\bar{y})$", together with the corresponding identity for the symmetrized horizontal second

derivative; it is the content of the left-invariance of the fields X_i, i.e. of $X_i(\varphi \circ L_h) = (X_i\varphi) \circ L_h$, read on 2-jets as pm-0002 prescribes.

In particular hgrad σ $(h * \bar{x})$ $D' =$ hgrad σ $\bar{x}$ $D \neq 0$.

Step 3: conclusion. By (2a), (2b), (2c) the function φ is admissible for `IsViscSubsol` at the point $h * \bar{x} \in \Omega$ with the jet (D', B'), so

$$\text{Lop } \sigma \ A \ H \ (h * \bar{x}) \ D' \ B' \ \leq \ 0 \qquad (\texttt{Lop}).$$

By `Lop`, Lop σ A H x D B depends on (x, D, B) only through hgrad σ x D and hhessS σ x D B, namely $\text{Lop} = -\operatorname{Tr}\big(A(\text{hgrad})\,\text{hhessS}\big) + H(\text{hgrad})$. Hence (3) gives

$$\text{Lop } \sigma \ A \ H \ (h * \bar{x}) \ D' \ B' = \text{Lop } \sigma \ A \ H \ \bar{x} \ D \ B,$$

and therefore Lop σ A H $\bar{x}$ D $B \leq 0$. Since $\bar{x}$, $\bar{\varphi}$ and (D, B) were arbitrary, and $\bar{u}$ is upper semi-continuous on Ω' by Step 1, $\bar{u} = u^h$ is a viscosity sub-solution of $\mathcal{L}\cdot = 0$ in Ω' in the sense of `IsViscSubsol`. This proves (1).

Step 4: the super-solution half. For (2), Step 0 and Steps (2a), (2c) do not involve u and are verbatim. Step 1 becomes lower semi-continuity of $\bar{u}$ on Ω', by the same pull-back of a neighbourhood through the continuous L_h. In (2b) the inequality reverses: with $w = (-h) * z \in \Omega'$, minimality of $\bar{u} - \bar{\varphi}$ over Ω' at $\bar{x}$ gives $(\bar{u} - \bar{\varphi})(\bar{x}) \leq (\bar{u} - \bar{\varphi})(w)$, i.e. $(u - \varphi)(h * \bar{x}) \leq (u - \varphi)(z)$, so $h * \bar{x} \in \operatorname{argmin}_\Omega(u - \varphi)$. Feeding φ and (D', B') to `IsViscSupersol` gives $0 \leq \text{Lop } \sigma \ A \ H \ (h * \bar{x}) \ D' \ B' = \text{Lop } \sigma \ A \ H \ \bar{x} \ D \ B$, which is (2). □

2.27. `EnvelopeIdentity`.

Lemma 2.27 (Lemma 3.6)**.** *Let $\mathbb{G}$ be a sub-elliptic setting (`CarnotSetting`) with group law $*$, homogeneous norm $\|\cdot\|$, first-layer projection π_1, horizontal exponential $\exp_{\mathbb{G}}$ and frame σ; let $\Omega \subseteq \mathbb{R}^n$, let $\delta > 0$ and let $u, v : \mathbb{R}^n \to \mathbb{R}$. Write Ω_δ for the inner domain (`TranslatedDomain`), $M_\delta(g)$ for the translated maximum value (`MaxTranslate`) and $u_g := u(\cdot * g)$. Assume*

(1) *for every g with $\|g\| < \delta$ there is $z \in \Omega_\delta$ such that $u_g - v$ is maximal over Ω_δ at z and $M_\delta(g) = u(z * g) - v(z)$.*

Let h satisfy $\|h\| < \delta$ and let $\xi \in \mathbb{R}^m$ be such that M_δ is Pansu differentiable at h with horizontal differential ξ (`HasPansuGradAt`). Let $x \in \Omega_\delta$ be a point at which $u_h - v$ is maximal over Ω_δ, and let D be the derivative of u at $x * h$. Then

$$\text{hgrad } \sigma \ (x * h) \ D \ = \ \xi \qquad (\textit{hgrad}).$$

Proof. This is Lemma 3.6. Hypothesis (1) is the attainment assertion of Lemma 3.4(c), restated here as a condition on the data. Write $\xi_0 :=$ hgrad σ $(x * h)$ D, which in the paper's notation is $\mathcal{X}u(x * h)$, and $\langle \alpha, \beta \rangle := \sum_{i=1}^m \alpha_i \beta_i$ for $\alpha, \beta \in \mathbb{R}^m$.

Step 0: $M_\delta(h) = u(x * h) - v(x)$. By hypothesis (1) at $g := h$ there is $z \in \Omega_\delta$ with $u_h - v$ maximal over Ω_δ at z and $M_\delta(h) = u(z * h) - v(z)$. Both x and z lie in Ω_δ and both maximize $u_h - v$ over Ω_δ, so $u(x * h) - v(x) = u(z * h) - v(z) = M_\delta(h)$.

Step 1: the curves g_t. Fix $\zeta \in \mathbb{R}^m$ and put

$$g_t := \exp_{\mathbb{G}}(t\zeta) \qquad (t \in \mathbb{R}),$$

which is the paper's $g_t = \exp\big(t \sum_i \zeta_i X_i\big)$. By `CarnotSetting`,

$$\pi_1(g_t) = t\zeta, \qquad \|g_t\| = |t|\,|\zeta|, \tag{1}$$

the first because $\pi_1 \circ \exp_{\mathbb{G}}$ is the identity of $\mathbb{R}^m$ and the second because $\|\exp_{\mathbb{G}}(\eta)\| = |\eta|$ for every $\eta \in \mathbb{R}^m$, applied with $\eta := t\zeta$. In particular $g_0 = \exp_{\mathbb{G}}(0)$ has $\|g_0\| = 0$, hence $g_0 = 0$ since $\|\cdot\|$ vanishes only at 0.

For $|t|$ small, $\|h * g_t\| < \delta$. The map $t \mapsto \|h * g_t\|$ is continuous: $\exp_{\mathbb{G}}$ is continuous, the group law is C^∞ hence continuous, and $\|\cdot\|$ is continuous (`CarnotSetting`). At $t = 0$ its value is $\|h * g_0\| = \|h * 0\| = \|h\| < \delta$. So there is $t_1 > 0$ with

$$\|h * g_t\| < \delta \qquad \text{for } |t| < t_1. \tag{2}$$

Note that no quasi-triangle inequality is used here; only continuity.

If $\zeta = 0$ then $\langle \xi - \xi_0, \zeta \rangle = 0$ trivially, so we may and do assume $\zeta \neq 0$, i.e. $|\zeta| > 0$.

Step 2: the lower bound from u. Let $\varepsilon > 0$. By `CarnotSetting` (the Taylor formula (2.18)), since D is the derivative of u at $x * h$, the function u is Pansu differentiable at $x * h$ with horizontal differential hgrad σ $(x * h)$ $D = \xi_0$. By `HasPansuGradAt` there is $\delta_u > 0$ with

$$\left| u\big((x * h) * g\big) - u(x * h) - \langle \xi_0, \pi_1(g) \rangle \right| \leq \varepsilon \|g\| \qquad \text{whenever } \|g\| < \delta_u. \tag{3}$$

Also, M_δ is Pansu differentiable at h with horizontal differential ξ, so by `HasPansuGradAt` there is $\delta_M > 0$ with

$$\left| M_\delta(h * g) - M_\delta(h) - \langle \xi, \pi_1(g) \rangle \right| \leq \varepsilon \|g\| \qquad \text{whenever } \|g\| < \delta_M. \tag{4}$$

Put

$$t_2 := \min\left\{ t_1,\ \frac{\delta_u}{2|\zeta|},\ \frac{\delta_M}{2|\zeta|} \right\} > 0.$$

For $0 < |t| < t_2$ we have, by (1), $\|g_t\| = |t||\zeta| < \delta_u$ and $\|g_t\| < \delta_M$, and $\|h * g_t\| < \delta$ by (2).

Fix such a t. Since $\|h * g_t\| < \delta$, hypothesis (1) applies at $g := h * g_t$ and gives $z_t \in \Omega_\delta$ with $u_{h * g_t} - v$ maximal over Ω_δ at z_t and $M_\delta(h * g_t) = u\big(z_t * (h * g_t)\big) - v(z_t)$. Applying that maximality at the point $x \in \Omega_\delta$,

$$M_\delta(h * g_t) \ \geq\ u\big(x * (h * g_t)\big) - v(x) = u\big((x * h) * g_t\big) - v(x), \tag{5}$$

the last equality by associativity of $*$. This is the paper's "since $x \in \Omega_\delta$, $M_\delta(h * g_t) \geq u(x * h * g_t) - v(x)$".

Combining (5), then (3) with $g := g_t$, then Step 0 and (1),

$$M_\delta(h * g_t) \ \geq\ u(x * h) + \langle \xi_0, t\zeta \rangle - \varepsilon|t||\zeta| - v(x) \ =\ M_\delta(h) + t\langle \xi_0, \zeta \rangle - \varepsilon|t||\zeta|. \tag{6}$$

Step 3: the upper bound from M_δ. By (4) with $g := g_t$ and (1),

$$M_\delta(h * g_t) \ \leq\ M_\delta(h) + t\langle \xi, \zeta \rangle + \varepsilon|t||\zeta|. \tag{7}$$

Step 4: comparing $t > 0$ and $t < 0$. From (6) and (7), for every t with $0 < |t| < t_2$,

$$t\langle \xi_0, \zeta \rangle - \varepsilon|t||\zeta| \ \leq\ t\langle \xi, \zeta \rangle + \varepsilon|t||\zeta|, \qquad \text{i.e.} \qquad t\,\langle \xi_0 - \xi, \zeta \rangle \ \leq\ 2\varepsilon|t||\zeta|. \tag{8}$$

Taking $t := t_2/2 > 0$ in (8) and dividing by $t > 0$ gives $\langle \xi_0 - \xi, \zeta \rangle \leq 2\varepsilon|\zeta|$. Taking $t := -t_2/2 < 0$ in (8) and dividing by $t < 0$ (which reverses the inequality) gives $\langle \xi_0 - \xi, \zeta \rangle \geq -2\varepsilon|\zeta|$. Hence

$$\big|\langle \xi_0 - \xi, \zeta \rangle\big| \ \leq\ 2\varepsilon|\zeta|.$$

Since $\varepsilon > 0$ was arbitrary and $|\zeta|$ does not depend on ε, we conclude $\langle \xi_0 - \xi, \zeta \rangle = 0$. This is the paper's "comparing for $t > 0$ and $t < 0$ yields $\langle \mathcal{X}u(x * h) - \mathcal{X}_h M_\delta(h), \zeta \rangle = 0$".

Step 5: conclusion. The vector $\zeta \in \mathbb{R}^m$ was arbitrary (the case $\zeta = 0$ being trivial, as noted in Step 1), so $\langle \xi_0 - \xi, \zeta \rangle = 0$ for every $\zeta \in \mathbb{R}^m$. Taking $\zeta := \xi_0 - \xi$ gives $|\xi_0 - \xi|^2 = 0$, i.e. $\xi_0 = \xi$, which is the assertion hgrad σ $(x * h)$ $D = \xi$. Since x was an arbitrary maximal point, all maximal points share this horizontal gradient, as the paper remarks. □

2.28. `InverseUniquenessAndDistanceSymmetry`.

Lemma 2.28 (Uniqueness of inverses, and the symmetry of d)**.** *Let $\mathbb{G}$ be a Carnot setting (`CarnotSetting`). Thus $(\mathbb{R}^n, *)$ is a group whose neutral element is 0 and whose inversion is $x \mapsto -x$ (the exponential coordinates of section Section 2.4, where $x^{-1} = -x$), and $\|\cdot\|$ is its homogeneous norm (2.15), which is even. Then:*

1. *(uniqueness of inverses) for all $a, b \in \mathbb{R}^n$, if $a * b = 0$ then $b = -a$;*
2. *for all $a, b \in \mathbb{R}^n$, $-(a * b) = (-b) * (-a)$, i.e. $(a * b)^{-1} = b^{-1} * a^{-1}$;*
3. *for all $a, b \in \mathbb{R}^n$, $\|(-b) * a\| = \|(-a) * b\|$; equivalently, the left-invariant distance $d(x, y) = \|y^{-1} * x\|$ of section Section 2.4 is symmetric, $d(a, b) = d(b, a)$.*

Proof. Only the following data of `CarnotSetting` are used: associativity $(x*y)*z = x*(y*z)$, the neutral-element laws $x*0 = x$ and $0*x = x$, the two inverse laws $x*(-x) = 0$ and $(-x)*x = 0$, and the evenness $\|-x\| = \|x\|$ of the homogeneous norm. No quasi-triangle inequality and no homogeneity property are needed.

Step 1: proof of (1). Let $a, b \in \mathbb{R}^n$ with $a*b = 0$. Then

$$b \;=\; 0*b \;=\; \big((-a)*a\big)*b \;=\; (-a)*(a*b) \;=\; (-a)*0 \;=\; -a,$$

where the first equality is the left neutral law $0*b = b$, the second is the left inverse law $(-a)*a = 0$, the third is associativity, the fourth is the hypothesis $a*b = 0$, and the last is the right neutral law $(-a)*0 = -a$. So a right inverse of a is necessarily the distinguished left inverse $-a$.

Step 2: proof of (2). Fix $a, b \in \mathbb{R}^n$. We show that $(-b)*(-a)$ is a right inverse of $a*b$:

$$(a*b)*\big((-b)*(-a)\big) = a*\Big(b*\big((-b)*(-a)\big)\Big) = a*\Big(\big(b*(-b)\big)*(-a)\Big) = a*\big(0*(-a)\big) = a*(-a) = 0,$$

the first two equalities being associativity, the third the right inverse law $b*(-b) = 0$, the fourth the left neutral law $0*(-a) = -a$, and the last the right inverse law $a*(-a) = 0$. Applying Step 1 with the pair $\big(a*b,\ (-b)*(-a)\big)$ in place of (a, b) gives $(-b)*(-a) = -(a*b)$, which is (2).

Step 3: proof of (3). Fix $a, b \in \mathbb{R}^n$ and apply (2) to the pair $(-b, a)$ in place of (a, b):

$$-\big((-b)*a\big) \;=\; (-a)*\big(-(-b)\big) \;=\; (-a)*b,$$

the second equality because $-(-b) = b$ in $\mathbb{R}^n$. Hence, by evenness of the homogeneous norm applied to the vector $(-b)*a$,

$$\|(-a)*b\| \;=\; \big\|-\big((-b)*a\big)\big\| \;=\; \|(-b)*a\|.$$

Since $x^{-1} = -x$, the left-invariant distance of section Section 2.4 is $d(x,y) = \|y^{-1}*x\| = \|(-y)*x\|$, so the displayed identity reads $d(b,a) = \|(-a)*b\| = \|(-b)*a\| = d(a,b)$, i.e. d is symmetric. □

2.29. `MaxValueLipschitz`.

Lemma 2.29 (Lemma 3.5, (3.12)). *Let $\mathbb{G}$ be a sub-elliptic setting (`CarnotSetting`) with group law $*$, homogeneous norm $\|\cdot\|$ and $d(x,y) := \|y^{-1}*x\|$, let $\Omega \subseteq \mathbb{R}^n$, let $\delta > 0$ and $\rho_1, \rho_2 \in \mathbb{R}$, and let $u, v : \mathbb{R}^n \to \mathbb{R}$. Write Ω_δ for the inner domain (`TranslatedDomain`), K_ρ for the compact core (`BoundaryCore`), $M_\delta(g)$ for the translated maximum value (`MaxTranslate`) and $u_g := u(\cdot * g)$. Assume*

1. *there is $L_u \geq 0$ with $|u(a) - u(b)| \leq L_u\, d(a,b)$ for all $a, b \in K_{\rho_2}$;*
2. *for every g with $\|g\| < \delta$ there is $x \in \Omega_\delta$ such that $u_g - v$ is maximal over Ω_δ at x and $M_\delta(g) = u(x*g) - v(x)$;*
3. *for every g with $\|g\| < \delta$, every $x \in \Omega_\delta$ at which $u_g - v$ is maximal over Ω_δ lies in K_{ρ_1};*
4. *$x * g \in K_{\rho_2}$ whenever $x \in K_{\rho_1}$ and $\|g\| < \delta$.*

Then

$$\big|M_\delta(h) - M_\delta(h')\big| \;\leq\; L_u\, d(h,h') \qquad \text{for all } h, h' \text{ with } \|h\| < \delta,\ \|h'\| < \delta.$$

Proof. This is Lemma 3.5. Hypotheses (2), (3), (4) are the conclusions of Lemma 3.4(c) and (a′), restated here as conditions on the data; hypothesis (1) is the paper's "L_u is the Lipschitz constant of u on $K_{\rho_0/4}$ with respect to d". The proof is the paper's, with the one exact identity $d(x*h, x*h') = d(h,h')$ written out.

Step 0: the group inverse reverses products, and d is symmetric. By `CarnotSetting`, if $a*b = 0$ then $b = 0*b = ((-a)*a)*b = (-a)*(a*b) = (-a)*0 = -a$. Since

$$(p*q)*\big((-q)*(-p)\big) = p*\Big(q*\big((-q)*(-p)\big)\Big) = p*\Big(\big(q*(-q)\big)*(-p)\Big) = p*\big(0*(-p)\big) = p*(-p) = 0,$$

it follows that

$$-(p*q) = (-q)*(-p) \qquad \text{for all } p, q. \tag{1}$$

Applying (1) with $p := -y$, $q := x$ and using $\|-z\| = \|z\|$ gives $(-x) * y = -\big((-y) * x\big)$ and hence

$$d(x,y) = d(y,x) \qquad \text{for all } x, y. \tag{2}$$

These are the two facts pm-0015 records as pre-solved and pm-0012 keeps inline.

Step 1: the exact translation identity. Let x, h, h' be arbitrary. By (1), $-(x * h') = (-h') * (-x)$, so by associativity and $(-x) * (x * h) = \big((-x) * x\big) * h = 0 * h = h$,

$$d(x*h,\, x*h') = \big\|-(x*h')*(x*h)\big\| = \Big\|\big((-h')*(-x)\big)*(x*h)\Big\| = \Big\|(-h')*\big((-x)*(x*h)\big)\Big\| = \big\|(-h')*h\big\| = d(h,h'). \tag{3}$$

This is exact: no quasi-triangle constant enters, exactly as the paper's $d(x*h, x*h') = \|h'^{-1}*h\| = d(h,h')$ asserts. (It is the second of the two inline derivations of pm-0012.)

Step 2: the one-sided estimate. Fix h, h' with $\|h\| < \delta$ and $\|h'\| < \delta$. By hypothesis (2) there are $x \in \Omega_\delta$ and $x' \in \Omega_\delta$ with

$$u_h - v \text{ maximal over } \Omega_\delta \text{ at } x, \qquad M_\delta(h) = u(x*h) - v(x), \tag{4}$$

$$u_{h'} - v \text{ maximal over } \Omega_\delta \text{ at } x', \qquad M_\delta(h') = u(x'*h') - v(x'). \tag{5}$$

Since $x \in \Omega_\delta$, maximality of $u_{h'} - v$ over Ω_δ at x', applied at the point x, gives

$$u(x*h') - v(x) \;\le\; u(x'*h') - v(x') \;=\; M_\delta(h'). \tag{6}$$

Therefore, by (4) and (6),

$$M_\delta(h) - M_\delta(h') \;\le\; \big(u(x*h) - v(x)\big) - \big(u(x*h') - v(x)\big) = u(x*h) - u(x*h'). \tag{7}$$

By (4) and hypothesis (3), $x \in K_{\rho_1}$; by hypothesis (4) applied with $g := h$ and with $g := h'$ (both admissible, $\|h\| < \delta$ and $\|h'\| < \delta$),

$$x*h \in K_{\rho_2} \qquad \text{and} \qquad x*h' \in K_{\rho_2}. \tag{8}$$

Hence hypothesis (1) applies to the pair $(x*h, x*h')$ and, with (3),

$$u(x*h) - u(x*h') \;\le\; \big|u(x*h) - u(x*h')\big| \;\le\; L_u\, d(x*h, x*h') \;=\; L_u\, d(h,h').$$

Combining with (7),

$$M_\delta(h) - M_\delta(h') \;\le\; L_u\, d(h,h'). \tag{9}$$

Step 3: conclusion. Inequality (9) holds for every admissible ordered pair, so exchanging the roles of h and h' gives $M_\delta(h') - M_\delta(h) \le L_u\, d(h', h) = L_u\, d(h,h')$ by (2). Together with (9),

$$\big|M_\delta(h) - M_\delta(h')\big| \;\le\; L_u\, d(h,h'),$$

which is (3.12). □

2.30. `NondegeneracyAtEveryScale`.

Lemma 2.30 (Lemma 3.7, non-degeneracy at every scale)**.** *Let $\mathbb{G}$ be a sub-elliptic setting (`CarnotSetting`) with group law $*$, homogeneous norm $\|\cdot\|$, frame σ and $d(x,y) := \|y^{-1} * x\|$, and assume the cited results of the literature at $\mathbb{G}$ (`CitedResults`). Let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$ satisfy the structure conditions (`StructureConditionsES`). Let $\Omega \subset \mathbb{R}^n$ be a bounded open set with $\mathbb{R}^n \setminus \Omega \neq \emptyset$, and let $u, v : \mathbb{R}^n \to \mathbb{R}$ be continuous on $\bar\Omega$ with u locally semi-convex in Ω (`LocallySemiConvexOn`), u a viscosity sub-solution (`IsViscSubsol`) and v a viscosity super-solution (`IsViscSupersol`) of $\mathcal{L}\cdot = 0$ in Ω. Let $\omega_u, \omega_v : \mathbb{R} \to \mathbb{R}$ be non-decreasing with*

$$|u(a) - u(b)| \le \omega_u\big(d(a,b)\big), \qquad |v(a) - v(b)| \le \omega_v\big(d(a,b)\big) \qquad (a, b \in \bar\Omega),$$

let $\tau > 0$ with $u - v \le -\tau$ on $\partial\Omega$, let $x_M \in \bar\Omega$ be a point at which $u - v$ attains its maximum over $\bar\Omega$ with $\tau \le M_0 := u(x_M) - v(x_M)$, and let $\rho_0 > 0$ satisfy $\omega_u(\rho_0) + \omega_v(\rho_0) \le \tau/4$. Write Ω_δ for the inner domain (`TranslatedDomain`) and $u_h := u(\cdot * h)$. Then there is $\delta_1 > 0$ such that for every $\delta \in (0, \delta_1]$ and every $\eta \in (0, \delta]$ there exists h with $\|h\| < \eta$ such that

$$\text{hgrad } \sigma\ (x*h)\ D \;\neq\; 0 \qquad (\textit{hgrad}) \tag{}$$

for every $x \in \Omega_\delta$ at which $u_h - v$ is maximal over Ω_δ and every derivative D of u at $x * h$.

Proof. This is Lemma 3.7. The hypotheses are the standing assumptions (3.10)–(3.11) of Section 3.1, except that semi-concavity of v is not assumed: it is not used anywhere below, so the lemma is stated without it. The hypothesis $\mathbb{R}^n \setminus \Omega \neq \emptyset$ makes explicit the paper's convention $\Omega \subset \mathbb{R}^n$; it is used only in Step 9, to know that a bounded non-empty open subset of $\mathbb{R}^n$ is not all of $\mathbb{R}^n$ and therefore has non-empty boundary. Write

$$E := \{y \in \Omega :\ u - v \text{ attains its maximum over } \bar{\Omega} \text{ at } y\} = \{y \in \Omega :\ (u-v)(y) = M_0\},$$

the second description because $u - v \le M_0$ on $\bar{\Omega}$, and $B_\rho(p) := \{z :\ d(z,p) < \rho\}$.

Step 0: the constants, and the facts of Lemma 3.4. All hypotheses of `BoundaryLocalization` are in force, so it provides $c' \ge 1$ such that, with

$$\rho_1 := \frac{\rho_0}{2c'}, \qquad \rho_2 := \frac{\rho_0}{4c'^2}, \qquad \delta_1 := \frac{\rho_0}{4c'^2}\ (>0),$$

the following hold: (a) $E \subseteq K_{\rho_0}$, and for $0 < \delta \le \delta_1$, $M_\delta(0) = M_0$ and $A_\delta(0) = E$, where K_ρ is the compact core (`BoundaryCore`), M_δ is the translated maximum value (`MaxTranslate`) and $A_\delta(g)$ denotes the set of points of Ω_δ at which $u_g - v$ is maximal over Ω_δ; (a′) $x * g \in K_{\rho_2}$ whenever $x \in K_{\rho_1}$, $0 < \delta \le \delta_1$ and $\|g\| < \delta$; (b) $u_g - v \le -\tau/2$ on $\Omega_\delta \setminus K_{\rho_1}$; and (c) for $0 < \delta \le \delta_1$ and $\|g\| < \delta$: $M_\delta(g) \ge M_0 - \tau/4$, there is $x \in \Omega_\delta$ with $u_g - v$ maximal over Ω_δ at x and $M_\delta(g) = u(x*g) - v(x)$, and every such x lies in K_{ρ_1} with $x * g \in K_{\rho_2}$. We take this δ_1 as the δ_1 of the statement. Note $\delta_1 \le \rho_0/4 < \rho_0$ and $\rho_2 = \delta_1 > 0$, both because $c' \ge 1$.

Fix δ with $0 < \delta \le \delta_1$ and η with $0 < \eta \le \delta$.

Step 1: u is d-Lipschitz on K_{ρ_2}. By items (4) and (2) of `InnerDomainProperties`, K_{ρ_2} is a compact subset of Ω (here $\rho_2 > 0$). By the third clause of `SemiConvexLocLipschitz`, applied to the locally semi-convex u on Ω and the compact $K_{\rho_2} \subseteq \Omega$, there is $C \ge 0$ with

$$|u(a) - u(b)| \le C\,|a - b| \qquad (a, b \in K_{\rho_2}). \tag{1}$$

The set $\bar{\Omega}$ is compact and non-empty (it contains x_M), so $\nu' := 1 + \sup\{|z| :\ z \in \bar{\Omega}\}$ is a finite real number with $\nu' > 0$ and $|z| \le \nu'$ for every $z \in \bar{\Omega}$; by the Hölder equivalence of `CarnotSetting`, applied at this ν', there is $c_0 > 0$ with $c_0^{-1}|a-b| \le d(a,b)$ whenever $|a| \le \nu'$ and $|b| \le \nu'$, i.e.

$$|a - b| \le c_0\, d(a,b) \qquad (a, b \in \bar{\Omega}). \tag{2}$$

Since $K_{\rho_2} \subseteq \Omega \subseteq \bar{\Omega}$, (1) and (2) give, with $L_u := C\,c_0 \ge 0$,

$$|u(a) - u(b)| \le L_u\, d(a,b) \qquad (a, b \in K_{\rho_2}). \tag{3}$$

This is the paper's L_u, "the Lipschitz constant of u on $K_{\rho_0/4}$ with respect to d", finite by Section 2.3(3).

Step 2: M_δ is d-Lipschitz on $B_\delta(0)$. We check the four hypotheses of `MaxValueLipschitz` for the data $(\mathbb{G}, \Omega, \delta, \rho_1, \rho_2, u, v, L_u)$: (1) is (3); (2) and (3) are the attainment and the inclusion $A_\delta(g) \subseteq K_{\rho_1}$ of Step 0(c); (4) is Step 0(a′). Hence

$$\big|M_\delta(g) - M_\delta(g')\big| \le L_u\, d(g,g') \qquad \text{whenever } \|g\| < \delta,\ \|g'\| < \delta. \tag{4}$$

Step 3: the ball $B_\eta(0)$. First, $-0 = 0$ and $0 * z = z$ (`CarnotSetting`), so

$$B_\eta(0) = \{z :\ \|(-0) * z\| < \eta\} = \{z :\ \|z\| < \eta\}.$$

It is open, because $\|\cdot\|$ is continuous (`CarnotSetting`), and it is connected by the ball-connectedness clause of `CarnotSetting` (with $\eta > 0$). It contains 0, since $\|0\| = 0 < \eta$. Finally $B_\eta(0) \subseteq B_\delta(0)$ because $\eta \le \delta$, so (4) gives in particular

$$\big|M_\delta(g) - M_\delta(g')\big| \le L_u\, d(g,g') \qquad (g, g' \in B_\eta(0)). \tag{5}$$

Step 4: the contradiction hypothesis. Suppose the conclusion fails for this δ and η. Then

(6)

for every $h \in B_\eta(0)$ there are $x_h \in A_\delta(h)$ and a derivative D_h of u at $x_h * h$ with $\operatorname{hgrad}\sigma\,(x_h * h)\,D_h = 0$.

Step 5: the horizontal differential of M_δ vanishes almost everywhere on $B_\eta(0)$. By Pansu's theorem, i.e. the clause `pansuDiff` of `CitedResults`, applied with the open set $B_\eta(0)$ (Step 3), the function M_δ and the constant L_u (whose hypothesis is (5)): for almost every $h \in B_\eta(0)$ there is $\xi_h \in \mathbb{R}^m$ such that M_δ is Pansu differentiable at h with horizontal differential ξ_h (`HasPansuGradAt`).

Let h be any such point of $B_\eta(0)$. Then $\|h\| < \eta \le \delta$, and by Step 0(c) the attainment hypothesis of `EnvelopeIdentity` holds for the data $(\mathbb{G}, \Omega, \delta, u, v)$. By (6) there are $x_h \in A_\delta(h)$ and a derivative D_h of u at $x_h * h$ with hgrad $\sigma\ (x_h * h)\ D_h = 0$. Applying `EnvelopeIdentity` with this h, $\xi := \xi_h$, $x := x_h$ and $D := D_h$ gives hgrad $\sigma\ (x_h * h)\ D_h = \xi_h$, whence $\xi_h = 0$. Therefore

(7) for almost every $h \in B_\eta(0)$, M_δ is Pansu differentiable at h with horizontal differential 0.

Step 6: $M_\delta \equiv M_0$ on $B_\eta(0)$. By the clause `pansuConst` of `CitedResults` — which is [DMV13, Prop. 4.8]: a d-Lipschitz function on a connected open set whose horizontal differential vanishes almost everywhere is constant — applied with the open connected set $B_\eta(0)$ (Step 3), the function M_δ, the constant L_u with (5), and (7), the function M_δ is constant on $B_\eta(0)$. Since $0 \in B_\eta(0)$ and $M_\delta(0) = M_0$ by Step 0(a),

$$M_\delta(g) = M_0 \qquad \text{for every } g \in B_\eta(0). \tag{8}$$

Step 7: $u \le u(y)$ on $B_\eta(y)$ for $y \in E$, and $B_\eta(y) \subseteq \Omega$. Fix $y \in E$; such a y exists, since $x_M \in E$ (if $x_M \in \partial\Omega$ then $M_0 \le -\tau < 0$, contradicting $M_0 \ge \tau > 0$, so $x_M \in \bar\Omega \setminus \partial\Omega = \Omega$, and $(u - v)(x_M) = M_0$).

By Step 0(a), $y \in K_{\rho_0}$; since $\delta < \rho_0$, item (5) of `InnerDomainProperties` gives $K_{\rho_0} \subseteq \Omega_\delta$, so $y \in \Omega_\delta$. Also, by item (7) of `InnerDomainProperties` applied at $y \in K_{\rho_0}$ (with $\rho_0 > 0$), $B_{\rho_0}(y) \subseteq \Omega$; and $\eta \le \delta < \rho_0$, so

$$B_\eta(y) \subseteq B_{\rho_0}(y) \subseteq \Omega. \tag{9}$$

Next, the set identity $B_\eta(y) = y * B_\eta(0)$. If $\|g\| < \eta$ then, by associativity and $0 * g = g$, $d(y * g, y) = \|(-y) * (y * g)\| = \|((-y) * y) * g\| = \|0 * g\| = \|g\| < \eta$ (this is (2.16), exact, kept inline by pm-0012), so $y * g \in B_\eta(y)$. Conversely if $z \in B_\eta(y)$, put $g := (-y) * z$; then $\|g\| = d(z, y) < \eta$ and $y * g = y * \big((-y) * z\big) = \big(y * (-y)\big) * z = 0 * z = z$. Hence

$$B_\eta(y) = \{y * g :\ g \in B_\eta(0)\}. \tag{10}$$

Now let $g \in B_\eta(0)$, so $\|g\| < \eta \le \delta$. By Step 0(c) there is $z_g \in \Omega_\delta$ with $u_g - v$ maximal over Ω_δ at z_g and $M_\delta(g) = u(z_g * g) - v(z_g)$; applying that maximality at the point $y \in \Omega_\delta$ and then (8),

$$u(y * g) - v(y) = (u_g - v)(y) \ \le\ M_\delta(g) = M_0 = u(y) - v(y),$$

the last equality because $y \in E$. Hence $u(y * g) \le u(y)$, and by (10)

$$u(z) \le u(y) \qquad \text{for every } z \in B_\eta(y). \tag{11}$$

This is the paper's "$u \le u(y)$ on $B_\eta(y) = y * B_\eta(0)$, a ball contained in Ω".

Step 8: the strong maximum and minimum principles give $B_\eta(y) \subseteq E$. The set $B_\eta(y) = \{z :\ \|(-y) * z\| < \eta\}$ is open (continuity of $*$ and $\|\cdot\|$, `CarnotSetting`) and connected (ball-connectedness clause of `CarnotSetting`, $\eta > 0$), and $y \in B_\eta(y)$ because $d(y, y) = \|0\| = 0 < \eta$.

$u \equiv u(y)$ on $B_\eta(y)$. Since $\bar\Omega$ is compact and u is continuous on it, u is bounded on $\bar\Omega$, hence bounded above on Ω; the same is then true of $\tilde u := u - u(y)$. By `AddConstPreservesSubsol` (with the constant $-u(y)$), $\tilde u$ is a viscosity sub-solution of $\mathcal{L}\cdot = 0$ in Ω. By part (1b) of `ViscTestLocalization` — whose hypotheses are that $\tilde u$ be bounded above on Ω and a viscosity sub-solution there — $\tilde u$ is a viscosity sub-solution in every open subset of Ω, in particular in $V := B_\eta(y)$, which is open and contained in Ω by (9). By (11), $\tilde u \le 0 = \tilde u(y)$ on V, so $\tilde u$ attains its maximum over V at $y \in V$, and that maximum value is $\tilde u(y) = 0 \ge 0$. The clause `strongMax` of `CitedResults` — the Bardi–Goffi strong maximum principle in the form Corollary 2.4 — applied

with the open connected set V, the sub-solution $\tilde{u}$, the point y and the structure conditions on (A, H), yields $\tilde{u}(z) = \tilde{u}(y) = 0$ for every $z \in V$, i.e.

$$u(z) = u(y) \qquad \text{for every } z \in B_\eta(y). \tag{12}$$

$v \equiv v(y)$ on $B_\eta(y)$. Let $z \in B_\eta(y)$. Then $z \in \Omega \subseteq \bar{\Omega}$ by (9), so $(u - v)(z) \leq M_0 = (u - v)(y)$; combined with $u(z) = u(y)$ from (12) this gives $v(z) \geq v(y)$. Put $\tilde{v} := v - v(y)$; then $\tilde{v} \geq 0 = \tilde{v}(y)$ on $B_\eta(y)$, so $\tilde{v}$ attains its minimum over $B_\eta(y)$ at y, with value $\tilde{v}(y) = 0 \leq 0$. As before, v is bounded below on Ω because $\bar{\Omega}$ is compact and v continuous on it, hence so is $\tilde{v}$; by `AddConstPreservesSubsol` $\tilde{v}$ is a viscosity super-solution in Ω, and by part (2b) of `ViscTestLocalization` it is one in the open subset $B_\eta(y) \subseteq \Omega$. The clause `strongMin` of `CitedResults`, applied with the open connected set $B_\eta(y)$, the super-solution $\tilde{v}$ and the point y, yields $\tilde{v}(z) = \tilde{v}(y) = 0$, i.e.

$$v(z) = v(y) \qquad \text{for every } z \in B_\eta(y). \tag{13}$$

Conclusion of Step 8. For $z \in B_\eta(y)$ we have $z \in \Omega$ by (9) and $(u - v)(z) = u(y) - v(y) = M_0$ by (12) and (13), so $z \in E$. Hence

$$B_\eta(y) \subseteq E \qquad \text{for every } y \in E, \tag{14}$$

and since $B_\eta(y)$ is open and contains y, the set E is open.

Step 9: E contains a connected component of Ω, and the final contradiction. E is non-empty ($x_M \in E$, Step 7) and open (Step 8). It is also relatively closed in Ω: $u - v$ is continuous on $\bar{\Omega} \supseteq \Omega$ and $E = \Omega \cap (u - v)^{-1}(\{M_0\})$, the preimage of a closed set under a function continuous on Ω.

Fix $y_0 \in E$ and let C be the connected component of Ω containing y_0. Since $\mathbb{R}^n$ is locally connected and Ω is open, C is open; it is connected and non-empty. Now $C \cap E$ is non-empty (y_0), open in C (as E is open), and closed in C (as E is relatively closed in Ω and $C \subseteq \Omega$). By connectedness of C, $C \cap E = C$, i.e.

$$C \subseteq E, \qquad \text{so} \qquad u - v \equiv M_0 > 0 \text{ on } C. \tag{15}$$

(We do not assume Ω connected; the paper is explicit that reduced domains, which need not be connected, will be used later.)

$\partial C \neq \emptyset$. Otherwise C would be both open and closed in $\mathbb{R}^n$; being non-empty and $\mathbb{R}^n$ connected, $C = \mathbb{R}^n$, hence $\Omega \supseteq C = \mathbb{R}^n$, contradicting $\mathbb{R}^n \setminus \Omega \neq \emptyset$.

$\partial C \subseteq \partial\Omega$. Let $z \in \partial C$. Since C is open, $\partial C = \bar{C} \setminus C$, so $z \in \bar{C} \subseteq \bar{\Omega}$ and $z \notin C$. If $z \in \Omega$, let C' be the connected component of Ω containing z; C' is open, so it is a neighbourhood of z, and $z \in \bar{C}$ forces $C' \cap C \neq \emptyset$; two components of Ω that meet are equal, so $C' = C$ and $z \in C$, a contradiction. Hence $z \notin \Omega$, and $z \in \bar{\Omega} \setminus \Omega = \partial\Omega$.

Now pick $z \in \partial C$. Then $z \in \bar{C}$, so there are points $z_k \in C$ with $z_k \to z$. By (15), $(u - v)(z_k) = M_0$ for every k; and $u - v$ is continuous on $\bar{\Omega}$ with $z_k \in C \subseteq \Omega \subseteq \bar{\Omega}$ and $z \in \bar{\Omega}$, so $(u - v)(z) = \lim_k (u - v)(z_k) = M_0 > 0$. But $z \in \partial C \subseteq \partial\Omega$, where $u - v \leq -\tau < 0$. This contradiction refutes (6).

Therefore the conclusion holds for this δ and η: there is h with $\|h\| < \eta$ such that hgrad σ $(x * h)$ $D \neq 0$ for every $x \in A_\delta(h)$ and every derivative D of u at $x * h$. Since $\delta \in (0, \delta_1]$ and $\eta \in (0, \delta]$ were arbitrary, the lemma is proved with the δ_1 of Step 0. □

2.31. `TranslatedPairDifferentiable`.

Lemma 2.31 (The sentence following (3.9)). *Let $\mathbb{G}$ be a sub-elliptic setting (`CarnotSetting`) with group law $*$ and homogeneous norm $\|\cdot\|$, let $\Omega \subset \mathbb{R}^n$ be a bounded open set, and let $u, v : \mathbb{R}^n \to \mathbb{R}$ with u locally semi-convex and v locally semi-concave in Ω, i.e. $-v$ locally semi-convex (`LocallySemiConvexOn`). Let $\delta > 0$ and let $h \in \mathbb{R}^n$ with $\|h\| < \delta$. Write Ω_δ for the inner domain (`TranslatedDomain`) and $u_h := u(\cdot * h)$. Then*

1. *$x * h \in \Omega$ for every $x \in \Omega_\delta$;*
2. *u_h is locally semi-convex on Ω_δ;*
3. *$-v$ is locally semi-convex on Ω_δ, i.e. v is locally semi-concave there;*

(4) *for every $x \in \Omega_\delta$ at which $u_h - v$ is maximal over Ω_δ, there is a continuous linear functional D which is the derivative of both u_h and v at x, and u is differentiable at $x * h$.*

Proof. This is the sentence the paper places immediately after (3.9): "By (2.16), $x * h \in \Omega$ whenever $x \in \Omega_\delta$ and $\|h\| < \delta$. ... The function u_h is semi-convex on compact subsets of Ω_δ (Section Section 2.3(4)), so at every point x of $\mathcal{A}_\delta(h)$ the functions u_h and v are differentiable, and u is differentiable at $x * h$."

Step 0: a restriction principle for local semi-convexity. We first record the following, used twice below. *If w is locally semi-convex on a set T and $S \subseteq T$ is open, then w is locally semi-convex on S.* Indeed, let $x \in S$. Then $x \in T$, so by `LocallySemiConvexOn` there are $\rho > 0$ and $\Lambda \geq 0$ with $B(x,\rho) \subseteq T$ and $y \mapsto w(y) + \frac{\Lambda}{2}|y|^2$ convex on the Euclidean ball $B(x,\rho)$. Since S is open there is $\rho' > 0$ with $B(x,\rho') \subseteq S$. Put $\rho'' := \min\{\rho, \rho'\} > 0$. Then $B(x,\rho'') \subseteq B(x,\rho') \subseteq S$, and $B(x,\rho'') \subseteq B(x,\rho)$ is a convex subset of $B(x,\rho)$, so the restriction of the convex function $y \mapsto w(y) + \frac{\Lambda}{2}|y|^2$ to $B(x,\rho'')$ is convex. Thus ρ'' and Λ witness the defining condition of `LocallySemiConvexOn` for w on S at x. As $x \in S$ was arbitrary, w is locally semi-convex on S.

Step 1: part (1). Let $x \in \Omega_\delta$. By associativity of $*$ and $0 * h = h$ (`CarnotSetting`),

$$d(x * h, x) = \big\|(-x) * (x * h)\big\| = \big\|\big((-x) * x\big) * h\big\| = \|0 * h\| = \|h\|,$$

which is the exact identity (2.16) (kept inline by pm-0012). Since $\|h\| < \delta$, the point $x * h$ lies in the d-ball of radius δ about x, and by item (6) of `InnerDomainProperties` that ball is contained in Ω. Hence $x * h \in \Omega$.

Step 2: part (2). Let $\Theta(y) := y * h$. By `CarnotSetting` the group law is C^∞, so Θ, a partial application of it, is C^∞ and in particular C^2. The second clause of `SemiConvexCompose`, applied with the set Ω, the function u (locally semi-convex on Ω by hypothesis) and this Θ, gives

$$u \circ \Theta = u_h \text{ is locally semi-convex on } \Theta^{-1}(\Omega) = \{y : \ y * h \in \Omega\}.$$

By Step 1, $\Omega_\delta \subseteq \Theta^{-1}(\Omega)$, and Ω_δ is open by item (3) of `InnerDomainProperties`. Step 0 with $T := \Theta^{-1}(\Omega)$ and $S := \Omega_\delta$ now gives that u_h is locally semi-convex on Ω_δ. This is the paper's appeal to Section 2.3(4); note that we obtain local semi-convexity on the open set Ω_δ itself, which is the form `SemiConvexDiffAtMax` consumes, rather than the paper's "on compact subsets".

Step 3: part (3). $-v$ is locally semi-convex on Ω by hypothesis, $\Omega_\delta \subseteq \Omega$ by item (1) of `InnerDomainProperties`, and Ω_δ is open by item (3) of the same node. Step 0 with $T := \Omega$ and $S := \Omega_\delta$ gives the claim.

Step 4: part (4), the common derivative at a maximal point. Let $x \in \Omega_\delta$ be a point at which $u_h - v$ is maximal over Ω_δ. Apply the second clause of `SemiConvexDiffAtMax` with the set Ω_δ, the locally semi-convex function u_h (Step 2) and the locally semi-concave function v (Step 3): since $x \in \Omega_\delta$ and $u_h - v$ attains its maximum over Ω_δ at x, there is a continuous linear functional D with

$$D \text{ is the derivative of } u_h \text{ at } x \quad \text{and} \quad D \text{ is the derivative of } v \text{ at } x. \tag{1}$$

Step 5: part (4), differentiability of u at $x * h$. Let $\Theta'(z) := z * (-h)$, which is C^∞ by `CarnotSetting` for the same reason as Θ. For every z, using associativity, $(-h) * h = 0$ and $z * 0 = z$,

$$u_h\big(\Theta'(z)\big) = u\Big(\big(z * (-h)\big) * h\Big) = u\Big(z * \big((-h) * h\big)\Big) = u(z * 0) = u(z),$$

so $u = u_h \circ \Theta'$ identically on $\mathbb{R}^n$. Moreover $\Theta'(x * h) = (x * h) * (-h) = x * \big(h * (-h)\big) = x * 0 = x$. Now Θ' is differentiable at $x * h$, with derivative the continuous linear map $L := D\Theta'(x * h)$, and u_h is differentiable at $\Theta'(x * h) = x$ with derivative D by (1). By the chain rule, $u = u_h \circ \Theta'$ is differentiable at $x * h$ with derivative $D \circ L$, a continuous linear functional. This proves the second half of part (4), and completes the proof. □

2.32. `AbelianNoStandingPair`.

Theorem 2.32 (Theorem 1.3(iii), the abelian case: no standing pair)**.** *Let $\mathbb{G}$ be a sub-elliptic setting (`CarnotSetting`) with group law $*$, homogeneous norm $\|\cdot\|$, frame σ and $d(x,y) := \|y^{-1} * x\|$, and assume the cited results of the literature at $\mathbb{G}$ (`CitedResults`). Assume $\mathbb{G}$ is* abelian*: $x * y = y * x$ for all $x, y \in \mathbb{R}^n$. Let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$ satisfy the structure conditions (`StructureConditionsES`) and the normalization (2.3) (`NormalizedAH`). Let $\Omega \subset \mathbb{R}^n$ be a bounded open set with $\mathbb{R}^n \setminus \Omega \neq \emptyset$ and let $u, v : \mathbb{R}^n \to \mathbb{R}$ be continuous on $\bar{\Omega}$, with u locally semi-convex and v locally semi-concave in Ω (`LocallySemiConvexOn`), u a viscosity sub-solution (`IsViscSubsol`) and v a viscosity super-solution (`IsViscSupersol`) of $\mathcal{L}\cdot = 0$ in Ω. Let $\omega_u, \omega_v : \mathbb{R} \to \mathbb{R}$ be non-decreasing with*

$$|u(a) - u(b)| \le \omega_u\big(d(a,b)\big), \qquad |v(a) - v(b)| \le \omega_v\big(d(a,b)\big) \qquad (a, b \in \bar{\Omega}),$$

let $\tau > 0$ satisfy $u - v \le -\tau$ on $\partial\Omega$, let $x_M \in \bar{\Omega}$ be a point at which $u - v$ attains its maximum over $\bar{\Omega}$ with $\tau \le M_0 := u(x_M) - v(x_M)$, and let $\rho_0 > 0$ satisfy $\omega_u(\rho_0) + \omega_v(\rho_0) \le \tau/4$. Then a contradiction follows: no such configuration exists.

Proof. This is the first paragraph of the proof of Theorem 1.3(iii), the case $r = 1$: "By the proof of Theorem Theorem 1.2, it suffices to show that under the standing assumptions (3.10)–(3.11) a semi-convex/semi-concave pair cannot exist. Given such a pair, apply Lemma Lemma 3.7 with $\eta = \delta \le \delta_1$: there is $h \in B_\delta(0)$ with $\mathcal{X}u(\cdot + h) \neq 0$ on $\mathcal{A}_\delta(h) = \operatorname{argmax}_{\Omega_\delta}(u_h - v)$. Since $u_h = u^h$ is a sub-solution in Ω_δ by Lemma Lemma 2.9, $u_h - v \le -\tau/2$ on $\partial\Omega_\delta$ by Lemma Lemma 3.4(b) and continuity, and $M_\delta(h) > 0$ is attained in Ω_δ, Proposition Proposition 3.3 applied to (u_h, v) on Ω_δ gives $u_h \le v$ on Ω_δ, a contradiction."

Write $u_h := u(\cdot * h)$, Ω_δ for the inner domain (`TranslatedDomain`), K_ρ for the compact core (`BoundaryCore`) and $M_\delta(h)$ for the translated maximum value (`MaxTranslate`). The hypothesis $x * y = y * x$ is used exactly twice, in Steps 3 and 6, and each use is identified there.

Step 1: the two thresholds and the scale δ. All hypotheses of `BoundaryLocalization` are in force for $(\mathbb{G}, \Omega, u, v, \omega_u, \omega_v, \tau, x_M, \rho_0)$, so it provides $c' \ge 1$ such that, with

$$\rho_1 := \frac{\rho_0}{2c'}, \qquad \rho_2 := \frac{\rho_0}{4c'^2}, \qquad \delta_1 := \frac{\rho_0}{4c'^2},$$

its clauses (a), (a′), (b), (c) hold for every $\delta \in (0, \delta_1]$ and every h with $\|h\| < \delta$. Note $\delta_1 > 0$, $\rho_1 > 0$ and

$$\rho_1 = \frac{\rho_0}{2c'} = 2c' \cdot \frac{\rho_0}{4c'^2} = 2c'\,\delta_1 \;\ge\; 2\delta_1, \tag{1}$$

using $c' \ge 1$.

All hypotheses of `NondegeneracyAtEveryScale` are also in force for the same data together with the cited results at $\mathbb{G}$ — that node is stated for the same $\mathbb{G}$, so our hypothesis is literally the one it asks for — the structure conditions, $\mathbb{R}^n \setminus \Omega \neq \emptyset$, the local semi-convexity of u and the two viscosity properties, so it provides $\delta_1' > 0$ with its conclusion for every $\delta \in (0, \delta_1']$ and every $\eta \in (0, \delta]$. Put

$$\delta := \min\{\delta_1, \delta_1'\} > 0,$$

so $0 < \delta \le \delta_1$ and $0 < \delta \le \delta_1'$. (We do not attempt to make the two nodes share a threshold; each states its conclusion for all smaller δ, so the minimum serves both.) By (1) and $\delta \le \delta_1$,

$$\delta < 2\delta \le 2\delta_1 \le \rho_1. \tag{2}$$

Step 2: the translation h. Apply `NondegeneracyAtEveryScale` with this δ and $\eta := \delta$ (admissible: $0 < \eta \le \delta$). It gives h with $\|h\| < \delta$ such that

(3)

for every $x \in \Omega_\delta$ at which $u_h - v$ is maximal over Ω_δ, and every derivative D of u at $x*h$: hgrad σ $(x*h)$ $D \neq$

This is the paper's "there is $h \in B_\delta(0)$ with $\mathcal{X}u(\cdot * h) \neq 0$ on $\mathcal{A}_\delta(h)$".

Since $\|h\| < \delta$ and $\delta > 0$, `TranslatedPairDifferentiable` applies to $(\mathbb{G}, \Omega, u, v, \delta, h)$; we use its four clauses below. Its clause 1 gives

$$x * h \in \Omega \qquad \text{for every } x \in \Omega_\delta. \tag{4}$$

Step 3: u_h is a viscosity sub-solution in Ω_δ. This is the first use of commutativity. By $x * y = y * x$,

$$u_h = \big(y \mapsto u(y * h)\big) = \big(y \mapsto u(h * y)\big) = u \circ L_h, \tag{5}$$

the two functions being equal at every point. By `LeftTranslateSubsol`, applied to u, h and the sub-solution hypothesis, $u \circ L_h$ is a viscosity sub-solution of $\mathcal{L}\cdot = 0$ in $L_h^{-1}(\Omega) = \{y : h * y \in \Omega\}$.

Next, $\Omega_\delta \subseteq L_h^{-1}(\Omega)$: for $x \in \Omega_\delta$ we have $x * h \in \Omega$ by (4), and $h * x = x * h$ by commutativity, so $h * x \in \Omega$.

Finally $u \circ L_h$ is bounded above on $L_h^{-1}(\Omega)$: if $h * y \in \Omega \subseteq \bar{\Omega}$ then $u(h * y) \le C_u$, where C_u is a bound for the continuous u on the compact set $\bar{\Omega}$ (Ω is bounded). Part (1b) of `ViscTestLocalization` — whose hypotheses are exactly that the function be bounded above on the domain and a viscosity sub-solution there — therefore gives that $u \circ L_h$ is a viscosity sub-solution in every open subset of $L_h^{-1}(\Omega)$. The set Ω_δ is open by item (3) of `InnerDomainProperties` and is such a subset, so $u \circ L_h$, i.e. u_h by (5), is a viscosity sub-solution of $\mathcal{L}\cdot = 0$ in Ω_δ. (This restriction is not free: pm-0009 records that our rendering of Definition 2.1 has the test condition global over the domain.)

Step 4: v is a viscosity super-solution in Ω_δ. By item (1) of `InnerDomainProperties`, $\Omega_\delta \subseteq \Omega$, and Ω_δ is open by item (3). The function v is bounded below on Ω, by $-C_v$ with C_v a bound for the continuous v on the compact $\bar{\Omega}$. Part (2b) of `ViscTestLocalization`, applied to v on Ω, therefore gives that v is a viscosity super-solution in Ω_δ.

Step 5: semi-convexity on Ω_δ. Clauses 2 and 3 of `TranslatedPairDifferentiable` give that u_h is locally semi-convex on Ω_δ and that $-v$ is locally semi-convex on Ω_δ, i.e. v is locally semi-concave there.

Step 6: the non-degeneracy hypothesis on Ω_δ. This is the second and essential use of commutativity. We claim

(6)
for every $x \in \Omega_\delta$ at which $u_h - v$ is maximal over Ω_δ, and every derivative D' of u_h at x : $\quad \mathrm{hgrad}\,\sigma\, x\, D' \neq 0.$

Let x and D' be as stated. Clause 4 of `TranslatedPairDifferentiable`, applied at this x, supplies both ends of what follows in one step: a derivative of u_h at x which is also a derivative of v at x, and — the part we use — a continuous linear functional D which is a derivative of u at $x * h$.

Now the two facts we hold are stated on opposite sides: D' is a derivative of the *right* translate $y \mapsto u(y * h)$ at x, while `LeftTranslateHgrad` consumes a derivative of the *left* translate $y \mapsto u(h * y)$ at x together with a derivative of u at $h * x$. Commutativity bridges this. By (5) the two translates are the same function, so D' is a derivative of $y \mapsto u(h * y)$ at x; and $h * x = x * h$, so D is a derivative of u at $h * x$. Therefore `LeftTranslateHgrad`, applied with the function u, the group element $g := h$ and the point x, gives

$$\mathrm{hgrad}\,\sigma\, x\, D' = \mathrm{hgrad}\,\sigma\, (h * x)\, D = \mathrm{hgrad}\,\sigma\, (x * h)\, D,$$

the last equality again by $h * x = x * h$. By (3), applied at this x and this D, the right-hand side is non-zero. This proves (6), which is precisely the non-degeneracy hypothesis of `ComparisonNondegenerateSemiconvex` for the pair (u_h, v) on Ω_δ.

(This is also where the abelian hypothesis is indispensable: without it the right translate u_h is neither a sub-solution nor expressible as a left translate, and `leftJet` of `CarnotSetting` cannot be used instead of `LeftTranslateHgrad` because it requires a second-order jet at $x * h$, whereas u is only semi-convex and so only once differentiable there.)

Step 7: $\overline{\Omega_\delta} * h \subseteq \bar{\Omega}$, and continuity of u_h up to the boundary of Ω_δ. Let $z \in \overline{\Omega_\delta}$. Pick $z_k \in \Omega_\delta$ with $z_k \to z$. By (4), $z_k * h \in \Omega \subseteq \bar{\Omega}$; the map $y \mapsto y * h$ is continuous (the group law is C^∞ by `CarnotSetting`), so $z_k * h \to z * h$, and $\bar{\Omega}$ is closed, whence

$$z * h \in \bar{\Omega} \qquad \text{for every } z \in \overline{\Omega_\delta}. \tag{7}$$

Consequently $u_h = u \circ (\cdot * h)$ is continuous on $\overline{\Omega_\delta}$: the map $y \mapsto y * h$ is continuous and sends $\overline{\Omega_\delta}$ into $\bar{\Omega}$ by (7), and u is continuous on $\bar{\Omega}$. Also $\overline{\Omega_\delta} \subseteq \bar{\Omega}$ because $\Omega_\delta \subseteq \Omega$, so v is continuous on $\overline{\Omega_\delta}$ as well.

Step 8: the boundary inequality $u_h \le v$ on $\partial\Omega_\delta$. This is the step the paper compresses into "by Lemma Lemma 3.4(b) and continuity"; clause (b) of `BoundaryLocalization` is an inequality on $\Omega_\delta \setminus K_{\rho_1}$, whereas we need one on $\partial\Omega_\delta$, and the two sets are disjoint.

Let $z \in \partial\Omega_\delta$. Since Ω_δ is open (item (3) of `InnerDomainProperties`), $\partial\Omega_\delta = \overline{\Omega_\delta} \setminus \Omega_\delta$, so $z \in \overline{\Omega_\delta}$ and $z \notin \Omega_\delta$.

(8a) $z \in \Omega$ and $z \notin K_{\rho_1}$. Fix $y \in \partial\Omega$. The function $w \mapsto d(w, y) = \|(-y) * w\|$ is continuous (`CarnotSetting`: $*$ is C^∞ and $\|\cdot\|$ is continuous), and $\delta < d(w, y)$ for every $w \in \Omega_\delta$ by `TranslatedDomain`; passing to the limit along $z_k \to z$ with $z_k \in \Omega_\delta$ gives $\delta \le d(z, y)$. As $y \in \partial\Omega$ was arbitrary,

$$\delta \le d(z, y) \qquad \text{for every } y \in \partial\Omega. \tag{8}$$

Also $z \in \overline{\Omega_\delta} \subseteq \bar{\Omega}$. If $z \notin \Omega$ then $z \in \bar{\Omega} \setminus \Omega = \partial\Omega$ (Ω is open), and (8) at $y := z$ would give $\delta \le d(z, z) = \|(-z) * z\| = \|0\| = 0$, contradicting $\delta > 0$. Hence $z \in \Omega$.

Now $z \notin \Omega_\delta$ and $z \in \Omega$, so by `TranslatedDomain` it is *not* the case that $\delta < d(z, y)$ for every $y \in \partial\Omega$: there is $y_0 \in \partial\Omega$ with $d(z, y_0) \le \delta$. By (2), $\delta < \rho_1$, so $d(z, y_0) < \rho_1$, and therefore $z \notin K_{\rho_1}$ by `BoundaryCore`.

(8b) a whole neighbourhood of z misses K_{ρ_1}. By item (4) of `InnerDomainProperties` the set K_{ρ_1} is compact (here $\rho_1 > 0$), hence closed, so its complement is open and contains z by (8a). Since $z \in \overline{\Omega_\delta}$, there are $z_k \in \Omega_\delta$ with $z_k \to z$, and for all large k we have $z_k \notin K_{\rho_1}$. Discard the finitely many smaller indices, so that

$$z_k \in \Omega_\delta \setminus K_{\rho_1} \qquad \text{for every } k. \tag{9}$$

(8c) passing to the limit. By clause (b) of `BoundaryLocalization`, applied at this δ and h (admissible: $0 < \delta \le \delta_1$ and $\|h\| < \delta$), (9) gives

$$u(z_k * h) - v(z_k) \le -\frac{\tau}{2} \qquad \text{for every } k.$$

By Step 7, u_h and v are continuous on $\overline{\Omega_\delta}$, and $z_k, z \in \overline{\Omega_\delta}$ with $z_k \to z$, so $u(z_k * h) - v(z_k) = (u_h - v)(z_k) \to (u_h - v)(z)$. Passing to the limit,

$$u_h(z) - v(z) \le -\frac{\tau}{2} < 0,$$

so $u_h(z) \le v(z)$. As $z \in \partial\Omega_\delta$ was arbitrary, $u_h \le v$ on $\partial\Omega_\delta$.

Step 9: Proposition 3.3 on Ω_δ. We now apply `ComparisonNondegenerateSemiconvex` with the following data. The cited results at $\mathbb{G}$ — that node is applied to the same $\mathbb{G}$, so our hypothesis is the one it asks for — and the structure conditions are hypotheses of the present statement, and so is the normalization (2.3) — which is why this node carries `NormalizedAH`, even though the comparison theorem one level up does not. The domain is Ω_δ: it is open by item (3) of `InnerDomainProperties` and bounded, being a subset of the bounded Ω by item (1); Proposition 3.3 does not assume it connected, which matters because Ω_δ need not be. The functions are u_h and v: continuous on $\overline{\Omega_\delta}$ by Step 7; u_h locally semi-convex and v locally semi-concave on Ω_δ by Step 5; $u_h \le v$ on $\partial\Omega_\delta$ by Step 8; a viscosity sub-solution and super-solution in Ω_δ by Steps 3 and 4; and the non-degeneracy (6) by Step 6. The conclusion is

$$u_h(x) \le v(x) \qquad \text{for every } x \in \Omega_\delta. \tag{10}$$

Step 10: the contradiction. By clause (c) of `BoundaryLocalization` at this δ and h there is $x_\star \in \Omega_\delta$ at which $u_h - v$ is maximal over Ω_δ with $M_\delta(h) = u(x_\star * h) - v(x_\star)$, and moreover $M_\delta(h) \ge M_0 - \tau/4$. Since $M_0 \ge \tau$,

$$u(x_\star * h) - v(x_\star) = M_\delta(h) \;\ge\; \tau - \frac{\tau}{4} = \frac{3\tau}{4} \;>\; 0.$$

But (10) at $x := x_\star \in \Omega_\delta$ gives $u_h(x_\star) = u(x_\star * h) \le v(x_\star)$, i.e. $u(x_\star * h) - v(x_\star) \le 0$. This contradiction completes the proof. □

2.33. `EllipPositiveAndQuadraticTwoHomogeneous.`

Lemma 2.33 (Positivity of $\mathcal{E}$ under (E), and 2-homogeneity of the quadratic form)**.** *Let $m \in \mathbb{N}$, let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$, and assume (A, H) satisfies the structure conditions (E) and (S) of section Section 1.2 (`StructureConditionsES`). Then, with $\mathcal{E}$ the ellipticity function (1.9) of `Ellip`:*

(1) *$\mathcal{E}_A(\eta) > 0$ for every $\eta \in \mathbb{R}^m$ with $\eta \neq 0$;*
(2) *for every matrix $M \in \mathbb{R}^{m\times m}$, every $s \in \mathbb{R}$ and every $\eta \in \mathbb{R}^m$,*

$$\sum_i \sum_j M_{i,j}(s\eta)_j(s\eta)_i \;=\; s^2 \sum_i \sum_j M_{i,j}\eta_j\eta_i.$$

(Part (2) uses no hypothesis on A, H or M: it is the 2-homogeneity of a quadratic form in its vector slot, the matrix being held fixed.)

Proof. **Part (1).** By `Ellip`, the ellipticity function (1.9) is by definition the double sum

$$\mathcal{E}_A(\eta) = \langle A(\eta)\eta, \eta\rangle = \sum_i \sum_j A(\eta)_{i,j}\eta_j\eta_i.$$

Condition (E) of `StructureConditionsES` asserts precisely that $\sum_i \sum_j A(\eta)_{i,j}\eta_j\eta_i > 0$ for every $\eta \neq 0$. The two expressions are the same, so (1) is condition (E) read through the definition of $\mathcal{E}$.

Part (2). Fix M, s and η. Scalar multiplication on $\mathbb{R}^m$ is coordinatewise, so $(s\eta)_k = s\,\eta_k$ for every index k. Hence, for each pair (i, j),

$$M_{i,j}(s\eta)_j(s\eta)_i = M_{i,j}\,(s\eta_j)(s\eta_i) = s^2\, M_{i,j}\eta_j\eta_i,$$

by commutativity and associativity of multiplication in $\mathbb{R}$. Both index sets are finite, so multiplication by the constant s^2 distributes over the sums: for each fixed i,

$$\sum_j M_{i,j}(s\eta)_j(s\eta)_i = \sum_j s^2 M_{i,j}\eta_j\eta_i = s^2 \sum_j M_{i,j}\eta_j\eta_i,$$

and summing these identities over i, again pulling the constant s^2 out of the finite outer sum,

$$\sum_i \sum_j M_{i,j}(s\eta)_j(s\eta)_i = \sum_i s^2 \sum_j M_{i,j}\eta_j\eta_i = s^2 \sum_i \sum_j M_{i,j}\eta_j\eta_i,$$

which is the asserted identity. □

2.34. `ScalingIdentityAllT.`

Theorem 2.34 (Remark 1.1(a): the scaling identity for every $t > 0$)**.** *Let $m \in \mathbb{N}$ and let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$ satisfy the structure conditions (E) and (S) (`StructureConditionsES`). Write $\mathcal{E}_A$ for the ellipticity function of A (`Ellip`). Then for every $t > 0$ and every $\xi \in \mathbb{R}^m \setminus \{0\}$,*

$$A(t\xi) = \frac{\mathcal{E}_A(t\xi)}{t^2\,\mathcal{E}_A(\xi)}\, A(\xi) \qquad \text{and} \qquad H(t\xi) = t\,\frac{\mathcal{E}_A(t\xi)}{t^2\,\mathcal{E}_A(\xi)}\, H(\xi).$$

Proof. Let $\kappa : [1,\infty)\times(\mathbb{R}^m\setminus\{0\}) \to (0,\infty)$ be a function for which clause (6) of `StructureConditionsES` — condition (S), (1.10) — holds, i.e.

$$\text{(S)} \qquad A(t\xi) = \kappa(t,\xi)\,A(\xi) \quad \text{and} \quad H(t\xi) = t\,\kappa(t,\xi)\,H(\xi) \qquad \text{for all } t \ge 1,\ \xi \neq 0.$$

Step 0: two preliminary remarks.

(0a) All denominators below are non-zero. If $t > 0$ and $\xi \neq 0$ then $t\xi \neq 0$, because $t\xi = 0$ with $t \neq 0$ forces $\xi = 0$. Hence condition (E) of `StructureConditionsES`, which asserts $\mathcal{E}_A(\eta) > 0$ for every $\eta \neq 0$, applies both at ξ and at $t\xi$:

$$\text{(0a)} \qquad \mathcal{E}_A(\xi) > 0, \qquad \mathcal{E}_A(t\xi) > 0.$$

In particular $t^2\mathcal{E}_A(\xi) > 0$, so the quotient in the statement is a well-defined positive real number.

(0b) The quadratic form is 2-homogeneous in its vector slot. Let $M \in \mathbb{R}^{m\times m}$, $\eta \in \mathbb{R}^m$ and $s \in \mathbb{R}$. Since $(s\eta)_k = s\,\eta_k$ for every k,

$$\sum_i \sum_j M_{i,j}\,(s\eta)_j\,(s\eta)_i = \sum_i \sum_j M_{i,j}\,s\,\eta_j\,s\,\eta_i = s^2 \sum_i \sum_j M_{i,j}\,\eta_j\,\eta_i. \tag{0b}$$

Applying this with $M = A(t\xi)$, $\eta = \xi$ and $s = t$, and recalling that $\mathcal{E}_A(t\xi)$ is by `Ellip` the quadratic form of the matrix $A(t\xi)$ evaluated at the vector $t\xi$,

$$\mathcal{E}_A(t\xi) = \sum_i \sum_j A_{i,j}(t\xi)\,(t\xi)_j\,(t\xi)_i = t^2 \sum_i \sum_j A_{i,j}(t\xi)\,\xi_j\,\xi_i. \tag{0b$'$}$$

Step 1: the case $t \geq 1$, and the formula for κ. Fix $t \geq 1$ and $\xi \neq 0$. Substituting $A(t\xi) = \kappa(t,\xi)A(\xi)$ from (S) into (0b$'$) and using that each entry satisfies $A_{i,j}(t\xi) = \kappa(t,\xi)A_{i,j}(\xi)$,

$$\mathcal{E}_A(t\xi) = t^2 \sum_i \sum_j \kappa(t,\xi)\,A_{i,j}(\xi)\,\xi_j\,\xi_i = t^2\,\kappa(t,\xi) \sum_i \sum_j A_{i,j}(\xi)\,\xi_j\,\xi_i = t^2\,\kappa(t,\xi)\,\mathcal{E}_A(\xi).$$

By (0a) we may divide by $t^2\mathcal{E}_A(\xi) > 0$:

$$\kappa(t,\xi) = \frac{\mathcal{E}_A(t\xi)}{t^2\,\mathcal{E}_A(\xi)}. \tag{1}$$

Substituting (1) back into the two identities (S) gives, for every $t \geq 1$ and $\xi \neq 0$,

$$A(t\xi) = \frac{\mathcal{E}_A(t\xi)}{t^2\mathcal{E}_A(\xi)}A(\xi), \qquad H(t\xi) = t\,\frac{\mathcal{E}_A(t\xi)}{t^2\mathcal{E}_A(\xi)}H(\xi), \tag{2}$$

which is the assertion of the theorem for $t \geq 1$. (For $t = 1$ formula (1) reads $\kappa(1,\xi) = \mathcal{E}_A(\xi)/\mathcal{E}_A(\xi) = 1$, so (2) is the trivial identity, as it must be.)

Step 2: the case $0 < t < 1$. Fix $0 < t < 1$ and $\xi \neq 0$, and set

$$\eta := t\xi, \qquad t' := \frac{1}{t}.$$

By Step 0(0a) applied to the pair (t,ξ) we have $\eta \neq 0$, and $t' > 1$ because $0 < t < 1$. So Step 1 is available for the pair (t',η). Note first that

$$t'\eta = \frac{1}{t}\,(t\xi) = \xi, \qquad (t')^2 = \frac{1}{t^2}, \qquad \mathcal{E}_A(t'\eta) = \mathcal{E}_A(\xi), \qquad \mathcal{E}_A(\eta) = \mathcal{E}_A(t\xi),$$

so that the scalar occurring in (2) for the pair (t',η) is

$$\frac{\mathcal{E}_A(t'\eta)}{(t')^2\,\mathcal{E}_A(\eta)} = \frac{\mathcal{E}_A(\xi)}{t^{-2}\,\mathcal{E}_A(t\xi)} = \frac{t^2\,\mathcal{E}_A(\xi)}{\mathcal{E}_A(t\xi)}. \tag{3}$$

By (0a) both $\mathcal{E}_A(\xi) > 0$ and $\mathcal{E}_A(t\xi) > 0$, so the scalar (3) is a positive real number; call it λ. Then:

The identity for A. The first identity of (2), applied to the pair (t',η) and read through $t'\eta = \xi$ and (3), says

$$A(\xi) = A(t'\eta) = \lambda\,A(\eta) = \lambda\,A(t\xi).$$

Multiplying both sides by $\lambda^{-1} = \mathcal{E}_A(t\xi)/\big(t^2\mathcal{E}_A(\xi)\big)$, which is legitimate because $\lambda > 0$, gives

$$A(t\xi) = \lambda^{-1}A(\xi) = \frac{\mathcal{E}_A(t\xi)}{t^2\,\mathcal{E}_A(\xi)}\,A(\xi),$$

as required.

The identity for H. The second identity of (2), applied to the pair (t',η) and read through $t'\eta = \xi$, $t' = 1/t$ and (3), says

$$H(\xi) = H(t'\eta) = t'\,\lambda\,H(\eta) = \frac{1}{t}\cdot\frac{t^2\mathcal{E}_A(\xi)}{\mathcal{E}_A(t\xi)}\cdot H(t\xi) = \frac{t\,\mathcal{E}_A(\xi)}{\mathcal{E}_A(t\xi)}\,H(t\xi).$$

The scalar $t\,\mathcal{E}_A(\xi)/\mathcal{E}_A(t\xi)$ is positive by (0a) and $t>0$; multiplying both sides by its reciprocal gives

$$H(t\xi)=\frac{\mathcal{E}_A(t\xi)}{t\,\mathcal{E}_A(\xi)}\,H(\xi)=t\cdot\frac{\mathcal{E}_A(t\xi)}{t^2\,\mathcal{E}_A(\xi)}\cdot H(\xi),$$

as required.

Step 3: conclusion. Every $t>0$ satisfies either $t\geq 1$, in which case Step 1 gives both identities, or $0<t<1$, in which case Step 2 does. Since $\xi\neq 0$ was arbitrary, the two identities hold for all $t>0$ and all $\xi\in\mathbb{R}^m\setminus\{0\}$. □

2.35. `NormalizationExists`.

Theorem 2.35 (Remark 1.1(a) and Section 2.1: the normalized factorization)**.** *Let $m\in\mathbb{N}$ and let $A:\mathbb{R}^m\to\mathbb{R}^{m\times m}$ and $H:\mathbb{R}^m\to\mathbb{R}$ satisfy the structure conditions (E) and (S) (`StructureConditionsES`). Then there exist $A_0:\mathbb{R}^m\to\mathbb{R}^{m\times m}$, $H_0:\mathbb{R}^m\to\mathbb{R}$ and $a:\mathbb{R}^m\to\mathbb{R}$ such that*

1. *$a(\xi)>0$ for every $\xi\in\mathbb{R}^m\setminus\{0\}$;*
2. *a is continuous on $\mathbb{R}^m\setminus\{0\}$;*
3. *$A(\xi)=a(\xi)\,A_0(\xi)$ for every $\xi\in\mathbb{R}^m\setminus\{0\}$;*
4. *$H(\xi)=a(\xi)\,H_0(\xi)$ for every $\xi\in\mathbb{R}^m\setminus\{0\}$;*
5. *(A_0,H_0) satisfies the structure conditions (E) and (S) (`StructureConditionsES`); and*
6. *(A_0,H_0) is normalized in the sense of (2.3) (`NormalizedAH`): $A_0(t\xi)=A_0(\xi)$ and $H_0(t\xi)=t\,H_0(\xi)$ for all $t>0$ and $\xi\in\mathbb{R}^m\setminus\{0\}$.*

Proof. Write $\mathcal{E}_A$ and $\mathcal{E}_{A_0}$ for the ellipticity functions of A and of A_0 (`Ellip`). Define the radial projection

$$\omega(\xi):=|\xi|^{-1}\xi\qquad(\xi\neq 0),$$

and put, for every $\xi\in\mathbb{R}^m$,

$$A_0(\xi):=A\big(\omega(\xi)\big),\qquad H_0(\xi):=|\xi|\,H\big(\omega(\xi)\big),\qquad a(\xi):=\frac{\mathcal{E}_A(\xi)}{|\xi|^2\,\mathcal{E}_A\big(\omega(\xi)\big)}.$$

(At $\xi=0$ these formulas are read with the convention $|0|^{-1}=0$, so that $\omega(0)=0$; the value of each of the three functions at the origin is irrelevant, because every one of the six assertions quantifies only over $\xi\neq 0$, including the clauses of `StructureConditionsES` and `NormalizedAH`.)

Throughout, fix $\xi\neq 0$ and abbreviate

$$t:=|\xi|>0,\qquad \omega:=\omega(\xi)=t^{-1}\xi,$$

so that $|\omega|=|\xi|^{-1}|\xi|=1$; in particular $\omega\neq 0$, and

$$(*)\qquad\qquad \xi=t\,\omega.$$

Step 1: assertion (1), $a>0$ away from the origin. Both ξ and ω are non-zero, so clause (5) of `StructureConditionsES` — condition (E), (1.9) — gives $\mathcal{E}_A(\xi)>0$ and $\mathcal{E}_A(\omega)>0$. Also $|\xi|^2>0$. Hence $a(\xi)=\mathcal{E}_A(\xi)/\big(|\xi|^2\mathcal{E}_A(\omega)\big)$ is a quotient of positive reals, so $a(\xi)>0$.

Step 2: assertions (3) and (4), the factorizations. Apply `ScalingIdentityAllT` to (A,H) at the pair (t,ω); this is legitimate because $t>0$ and $\omega\neq 0$. It gives

$$A(t\omega)=\frac{\mathcal{E}_A(t\omega)}{t^2\,\mathcal{E}_A(\omega)}\,A(\omega),\qquad H(t\omega)=t\,\frac{\mathcal{E}_A(t\omega)}{t^2\,\mathcal{E}_A(\omega)}\,H(\omega).$$

By $(*)$ we have $t\omega=\xi$ and $t=|\xi|$, so the first identity reads

$$A(\xi)=\frac{\mathcal{E}_A(\xi)}{|\xi|^2\,\mathcal{E}_A(\omega)}\,A(\omega)=a(\xi)\,A_0(\xi),$$

since $A_0(\xi)=A(\omega)$ by definition; this is (3). The second reads

$$H(\xi)=|\xi|\,\frac{\mathcal{E}_A(\xi)}{|\xi|^2\,\mathcal{E}_A(\omega)}\,H(\omega)=a(\xi)\cdot\Big(|\xi|\,H(\omega)\Big)=a(\xi)\,H_0(\xi),$$

since $H_0(\xi)=|\xi|H(\omega)$ by definition; this is (4).

Step 3: assertion (2), continuity of a on $\mathbb{R}^m \setminus \{0\}$. The map $\xi \mapsto |\xi|$ is continuous on $\mathbb{R}^m$ and does not vanish on $\mathbb{R}^m \setminus \{0\}$, so $\xi \mapsto |\xi|^{-1}$ is continuous there, and hence so is

$$\omega(\cdot) : \mathbb{R}^m \setminus \{0\} \to \mathbb{R}^m, \qquad \xi \mapsto |\xi|^{-1}\xi.$$

Moreover $|\omega(\xi)| = 1$ for $\xi \neq 0$, so ω maps $\mathbb{R}^m\setminus\{0\}$ into $\mathbb{R}^m\setminus\{0\}$. By clause (3) of `StructureConditionsES`, A is continuous on $\mathbb{R}^m \setminus \{0\}$; composing, $\xi \mapsto A(\omega(\xi))$ is continuous on $\mathbb{R}^m \setminus \{0\}$, and therefore so is each of its entries $\xi \mapsto A_{i,j}(\omega(\xi))$. Since

$$\mathcal{E}_A\big(\omega(\xi)\big) = \sum_i \sum_j A_{i,j}\big(\omega(\xi)\big)\, \omega(\xi)_j\, \omega(\xi)_i$$

is a finite sum of products of functions continuous on $\mathbb{R}^m \setminus \{0\}$ (`Ellip`), the map $\xi \mapsto \mathcal{E}_A(\omega(\xi))$ is continuous there; the same argument, with A in place of $A \circ \omega$ and ξ in place of $\omega(\xi)$, shows that $\xi \mapsto \mathcal{E}_A(\xi)$ is continuous on $\mathbb{R}^m \setminus \{0\}$. The denominator $\xi \mapsto |\xi|^2 \mathcal{E}_A(\omega(\xi))$ is then continuous on $\mathbb{R}^m \setminus \{0\}$ and, by Step 1, nowhere zero there. A quotient of continuous functions with non-vanishing denominator is continuous, so a is continuous on $\mathbb{R}^m \setminus \{0\}$.

Step 4: assertion (6), (A_0, H_0) is normalized. Let $s > 0$ and $\eta \neq 0$. Then $s\eta \neq 0$ and $|s\eta| = s|\eta|$, so

$$\omega(s\eta) = |s\eta|^{-1}(s\eta) = \big(s|\eta|\big)^{-1} s\,\eta = |\eta|^{-1}\eta = \omega(\eta).$$

Consequently

$$A_0(s\eta) = A\big(\omega(s\eta)\big) = A\big(\omega(\eta)\big) = A_0(\eta),$$
$$H_0(s\eta) = |s\eta|\, H\big(\omega(s\eta)\big) = s\,|\eta|\, H\big(\omega(\eta)\big) = s\, H_0(\eta),$$

which are exactly the two identities of `NormalizedAH` for (A_0, H_0).

Step 5: assertion (5), (A_0, H_0) satisfies the structure conditions. We verify clauses (1)–(6) of `StructureConditionsES` for (A_0, H_0), keeping the notation $t = |\xi|$, $\omega = \omega(\xi)$ for a fixed $\xi \neq 0$.

Clause (1), symmetry. $A_0(\xi) = A(\omega)$, and $A(\omega)$ is symmetric by clause (1) for A, applicable because $\omega \neq 0$.

Clause (2), positive semidefiniteness. For every $\eta \in \mathbb{R}^m$,

$$\sum_i \sum_j \big(A_0(\xi)\big)_{i,j} \eta_j \eta_i = \sum_i \sum_j A_{i,j}(\omega)\, \eta_j\, \eta_i \;\geq\; 0$$

by clause (2) for A at the point $\omega \neq 0$.

Clause (3), continuity of A_0. $A_0 = A \circ \omega$ is continuous on $\mathbb{R}^m \setminus \{0\}$, by the composition argument of Step 3.

Clause (4), continuity of H_0. $H_0(\xi) = |\xi|\, H(\omega(\xi))$ is a product of two functions continuous on $\mathbb{R}^m \setminus \{0\}$: $\xi \mapsto |\xi|$, and $H \circ \omega$, continuous there by clause (4) for H composed with the continuity of ω established in Step 3.

Clause (5), condition (E) for (A_0, H_0). Using $A_0(\xi) = A(\omega)$, then $(*)$ together with the 2-homogeneity of the quadratic form in its vector slot (for any matrix M, any η and any s, $\sum_i \sum_j M_{i,j}(s\eta)_j(s\eta)_i = s^2 \sum_i \sum_j M_{i,j}\eta_j\eta_i$, because $(s\eta)_k = s\eta_k$ contributes one factor s in each slot):

$$\mathcal{E}_{A_0}(\xi) = \sum_i \sum_j A_{i,j}(\omega)\, \xi_j\, \xi_i = t^2 \sum_i \sum_j A_{i,j}(\omega)\, \omega_j\, \omega_i = |\xi|^2\, \mathcal{E}_A(\omega) \;>\; 0,$$

because $|\xi|^2 > 0$ and $\mathcal{E}_A(\omega) > 0$ by clause (5) for A at $\omega \neq 0$.

Clause (6), condition (S) for (A_0, H_0). Take $\kappa_0(t, \eta) := 1$ for all t, η; then $\kappa_0 > 0$ everywhere. By Step 4, for every $t \geq 1$ (indeed every $t > 0$) and every $\eta \neq 0$,

$$A_0(t\eta) = A_0(\eta) = \kappa_0(t,\eta)\, A_0(\eta), \qquad H_0(t\eta) = t\, H_0(\eta) = t\, \kappa_0(t,\eta)\, H_0(\eta),$$

which is clause (6) for (A_0, H_0) with the constant factor $\kappa_0 \equiv 1$.

All six assertions are established with the above A_0, H_0 and a, which proves the theorem. (This is the decomposition of Remark 1.1(a): A_0 and H_0 are precisely the functions $A_0(\xi/|\xi|)$ and $|\xi|H_0(\xi/|\xi|)$ of that remark, written as functions on $\mathbb{R}^m \setminus \{0\}$ rather than on the unit sphere,

and the factor a is the positive continuous function of the remark. Step 5's clause (6) is the remark's observation that for this replacement (1.10) holds with $\kappa \equiv 1$.) □

2.36. `NormalizedSameViscositySolutions`.

Theorem 2.36 (Section 2.1: a positive factor does not change the viscosity solutions)**.** *Let $\mathbb{G}$ be a Carnot setting (`CarnotSetting`) with horizontal frame σ, and let $A, A_0 : \mathbb{R}^m \to \mathbb{R}^{m\times m}$, $H, H_0 : \mathbb{R}^m \to \mathbb{R}$ and $a : \mathbb{R}^m \to \mathbb{R}$ satisfy, for every $\xi \in \mathbb{R}^m \setminus \{0\}$,*

$$a(\xi) > 0, \qquad A(\xi) = a(\xi)\, A_0(\xi), \qquad H(\xi) = a(\xi)\, H_0(\xi).$$

Then for every $\Omega \subseteq \mathbb{R}^n$ and every $u : \mathbb{R}^n \to \mathbb{R}$,

u is a viscosity sub-solution of $\mathcal{L}u = 0$ in Ω $\iff$ u is a viscosity sub-solution of $\mathcal{L}_0 u = 0$ in Ω,

and the same equivalence holds for viscosity super-solutions. Here $\mathcal{L}$ and $\mathcal{L}_0$ denote the operators determined by (A, H) and by (A_0, H_0) in `IsViscSubsol` and `IsViscSupersol`.

Proof. **Step 1: the pointwise factorization of the operator.** Fix $x \in \mathbb{R}^n$, a 1-jet D and a 2-jet B on $\mathbb{R}^n$, and abbreviate

$$\xi := \mathrm{hgrad}\ \sigma\ x\ D \in \mathbb{R}^m \quad (\texttt{hgrad}), \qquad X := \mathrm{hhessS}\ \sigma\ x\ D\ B \in \mathbb{R}^{m\times m} \quad (\texttt{hhessS}).$$

Assume $\xi \neq 0$. By `Lop`,

$$\mathrm{Lop}\ \sigma\ A\ H\ x\ D\ B = -\mathrm{Tr}\big(A(\xi)\, X\big) + H(\xi), \qquad \mathrm{Lop}\ \sigma\ A_0\ H_0\ x\ D\ B = -\mathrm{Tr}\big(A_0(\xi)\, X\big) + H_0(\xi).$$

Since $\xi \neq 0$, the hypotheses give $A(\xi) = a(\xi)A_0(\xi)$ and $H(\xi) = a(\xi)H_0(\xi)$. Scalar multiplication passes through matrix multiplication, $\big(a(\xi)A_0(\xi)\big)X = a(\xi)\big(A_0(\xi)X\big)$, and the trace is homogeneous, $\mathrm{Tr}\big(a(\xi)M\big) = a(\xi)\mathrm{Tr}(M)$. Therefore

(1)

$$\mathrm{Lop}\,\sigma\ A\ H\ x\ D\ B = -a(\xi)\,\mathrm{Tr}\big(A_0(\xi)X\big) + a(\xi)\,H_0(\xi) = a(\xi)\Big[-\mathrm{Tr}\big(A_0(\xi)X\big) + H_0(\xi)\Big] = a(\xi)\ \mathrm{Lop}\,\sigma\ A_0\ H_0\ x\ D\ B.$$

This is the identity $\mathcal{L}\varphi(x) = a(\mathcal{X}\varphi(x))\,\mathcal{L}_0\varphi(x)$ of Section Section 2.1. Note where the hypothesis $\xi \neq 0$ was used: it is the only point at which the factorizations of A and H are available, since they are assumed only away from the origin.

Step 2: sub-solutions, the implication $\mathcal{L} \Rightarrow \mathcal{L}_0$. By `IsViscSubsol`, the statement that u is a viscosity sub-solution for (A, H) in Ω is the conjunction of

(a) u is upper semi-continuous on Ω; and
(b) for every $x \in \Omega$, every $\varphi \in C^2(\Omega)$ with $x \in \mathrm{argmax}_\Omega(u - \varphi)$ and every second-order jet (D, B) of φ at x with $\mathrm{hgrad}\ \sigma\ x\ D \neq 0$, one has $\mathrm{Lop}\ \sigma\ A\ H\ x\ D\ B \leq 0$,

and the corresponding statement for (A_0, H_0) is the conjunction of the same clause (a) with the clause (b_0) obtained from (b) by replacing (A, H) with (A_0, H_0). Clause (a) is literally the same in the two cases, since it does not mention A or H. Moreover the data over which (b) and (b_0) quantify are the same: the conditions $\varphi \in C^2(\Omega)$, $x \in \mathrm{argmax}_\Omega(u - \varphi)$, "$(D, B)$ is a second-order jet of φ at x" and $\mathrm{hgrad}\ \sigma\ x\ D \neq 0$ involve only σ, φ, u and Ω — by `hgrad`, the horizontal gradient of a jet depends on the frame and the jet alone, not on A or H.

So assume (b) and let x, φ, D, B be admissible data, with $\xi := \mathrm{hgrad}\ \sigma\ x\ D \neq 0$. By (b), $\mathrm{Lop}\ \sigma\ A\ H\ x\ D\ B \leq 0$, and by Step 1 this reads

$$a(\xi)\ \mathrm{Lop}\ \sigma\ A_0\ H_0\ x\ D\ B\ \leq\ 0 = a(\xi)\cdot 0.$$

Since $a(\xi) > 0$ by hypothesis, we may cancel the factor $a(\xi)$ from this inequality between products with a common positive left factor, obtaining $\mathrm{Lop}\ \sigma\ A_0\ H_0\ x\ D\ B \leq 0$. This is ($\mathrm{b}_0$).

Step 3: sub-solutions, the converse implication. Assume (b_0) and let x, φ, D, B be admissible data with $\xi \neq 0$. By (b_0), $\mathrm{Lop}\ \sigma\ A_0\ H_0\ x\ D\ B \leq 0$. Multiplying this inequality by the non-negative number $a(\xi)$ preserves it, so $a(\xi)\,\mathrm{Lop}\ \sigma\ A_0\ H_0\ x\ D\ B \leq a(\xi)\cdot 0 = 0$, and by Step 1 the left-hand side is $\mathrm{Lop}\ \sigma\ A\ H\ x\ D\ B$. This is (b).

Steps 2 and 3 together with the identity of clause (a) give the asserted equivalence for sub-solutions.

Step 4: super-solutions. The argument is the same with `IsViscSupersol` in place of `IsViscSubsol`. There clause (a) reads "u is lower semi-continuous on Ω", again not mentioning A or H; the admissible data are those with $x \in \operatorname{argmin}_\Omega(u-\varphi)$ in place of argmax, again not mentioning A or H; and the required conclusion is $0 \le \mathrm{Lop}\ \sigma\ A\ H\ x\ D\ B$, respectively $0 \le \mathrm{Lop}\ \sigma\ A_0\ H_0\ x\ D\ B$. Given the first, Step 1 turns it into $a(\xi)\cdot 0 = 0 \le a(\xi)\,\mathrm{Lop}\ \sigma\ A_0\ H_0\ x\ D\ B$, and cancelling the positive factor $a(\xi)$ gives the second; given the second, multiplying by the non-negative $a(\xi)$ and applying Step 1 gives the first.

Remark. The clause that makes the whole equivalence work is the restriction, built into `IsViscSubsol` and `IsViscSupersol` after Definition Definition 2.1, that test functions are only ever evaluated at points where their horizontal gradient does not vanish. That is exactly the hypothesis $\xi \neq 0$ under which Step 1 is available and under which $a(\xi)$ is known to be positive; without it the factor a could not be used at all, since nothing is assumed about A, H, A_0, H_0 or a at the origin. □

2.37. `ReducedDomainProperties`.

Theorem 2.37 (The reduced domains (3.15))**.** *Let $\mathbb{G}$ be a Carnot setting (`CarnotSetting`) with group law $*$ and homogeneous norm $\|\cdot\|$, put $q := 2\,r!$, let $\Omega \subseteq \mathbb{R}^n$ be open and bounded with $\mathbb{R}^n \setminus \Omega \neq \emptyset$, and let $R \ge 0$. Write $\Omega[\varepsilon]$ for the reduced domain of `ReducedDomain` and $d(x,y) := \|y^{-1} * x\|$ for the left-invariant distance. Then*

(1) *$\Omega[\varepsilon] \subseteq \Omega$ for every ε;*
(2) *$\Omega[\varepsilon]$ is bounded for every ε;*
(3) *$\varepsilon' \le \varepsilon$ implies $\Omega[\varepsilon] \subseteq \Omega[\varepsilon']$;*
(4) *$\Omega[\varepsilon]$ is open for every $\varepsilon > 0$;*
(5) *$\Omega[\varepsilon] \uparrow \Omega$ as $\varepsilon \to 0^+$, in the sense that for every $x \in \Omega$ there is $\varepsilon_0 > 0$ with $x \in \Omega[\varepsilon]$ for all $\varepsilon \in (0, \varepsilon_0)$;*
(6) *$\overline{\Omega[\varepsilon]} \subseteq \Omega$ for every $\varepsilon > 0$;*
(7) *$\sup_{z \in \partial\Omega[\varepsilon]} \operatorname{dist}(z, \partial\Omega) \to 0$ as $\varepsilon \to 0^+$, in the sense that for every $\eta > 0$ there is $\varepsilon_0 > 0$ such that for every $\varepsilon \in (0, \varepsilon_0)$ and every $z \in \partial\Omega[\varepsilon]$ there is $w \in \partial\Omega$ with $d(z,w) < \eta$.*

Proof. Throughout write

$$c := 1 + 4R, \qquad \kappa(y,x) := \left\|y * x^{-1}\right\|^q = \left\|y * (-x)\right\|^q, \qquad f(x) := \inf_{y \notin \Omega} \kappa(y,x),$$

so that by `ReducedDomain`

$$(*) \qquad \Omega[\varepsilon] = \{x \in \Omega :\ c\,\varepsilon < f(x)\}.$$

Step 0: preliminaries.

(0a) $c = 1 + 4R \ge 1 > 0$, since $R \ge 0$.

(0b) f is a genuine infimum. The index set $\mathbb{R}^n \setminus \Omega$ is non-empty by hypothesis, and $\kappa(y,x) = \|y * (-x)\|^q \ge 0$ for every y because $\|\cdot\| \ge 0$ (clause (1) of `CarnotNorm`) and q is a natural number. So the family $\{\kappa(y,x) : y \notin \Omega\}$ is non-empty and bounded below by 0, its infimum $f(x)$ is a real number with $f(x) \ge 0$, and for every $y \notin \Omega$ we have $f(x) \le \kappa(y,x)$ while for every $\theta > 0$ there is $y \notin \Omega$ with $\kappa(y,x) < f(x) + \theta$.

(0c) κ is jointly continuous. The group law is C^∞ (clause (2) of `CarnotGroup`), hence continuous; $x \mapsto -x$ is continuous; $\|\cdot\|$ is continuous (clause (1) of `CarnotNorm`); and $t \mapsto t^q$ is continuous. So $(x,y) \mapsto \kappa(y,x)$ is continuous on $\mathbb{R}^n \times \mathbb{R}^n$, and in particular $\kappa(y,\cdot)$ is continuous for each fixed y.

(0d) $\kappa(x,x) = 0$. $x * (-x) = 0$ (clause (1) of `CarnotGroup`) and $\|0\| = 0$ (clause (1) of `CarnotNorm`), so $\kappa(x,x) = 0^q = 0$.

(0e) $\|\cdot\|$ is bounded on Ω. Ω is bounded, so there is $\nu_0 \ge 0$ with $|a| \le \nu_0$ for every $a \in \Omega$. Apply clause (6) of `CarnotNorm` with $\nu := \nu_0 + 1$: there is $c_2 > 0$ with $\|(-b) * a\| \le c_2 |a - b|^{1/r}$ whenever $|a| \le \nu$ and $|b| \le \nu$. Taking $b := 0$ and using $-0 = 0$ and $0 * a = a$ (clause (1) of `CarnotGroup`) gives

$$(0e) \qquad \|a\| \le c_2 |a|^{1/r} \le c_2 (\nu_0 + 1)^{1/r} =: T \qquad \text{for every } a \in \Omega,$$

and $T \geq 0$.

Step 1: proof of (1). By $(*)$, every element of $\Omega[\varepsilon]$ lies in Ω.

Step 2: proof of (2). By Step 1, $\Omega[\varepsilon]$ is a subset of the bounded set Ω, hence bounded.

Step 3: proof of (3). Let $\varepsilon' \leq \varepsilon$ and $x \in \Omega[\varepsilon]$. Then $x \in \Omega$ and $c\varepsilon < f(x)$. Since $c > 0$ by (0a), $c\varepsilon' \leq c\varepsilon < f(x)$, so $x \in \Omega[\varepsilon']$ by $(*)$.

Step 4: f is continuous near every point. This is the step the paper does not supply, and the one on which (4) rests. *Claim:* for every $x_0 \in \mathbb{R}^n$ there are an open set $U \ni x_0$, a non-empty compact $K \subseteq \mathbb{R}^n \setminus \Omega$, and the identity

$$f(x) = \inf_{y \in K} \kappa(y, x) \qquad \text{for every } x \in U. \tag{4a}$$

Fix x_0 and put $U := B_1(x_0)$ (Euclidean), $U' := \overline{B_1(x_0)}$, a compact set. Fix $y_0 \notin \Omega$, which exists by hypothesis. By (0c) the function $\kappa(y_0, \cdot)$ is continuous, hence bounded on the compact U'; put

$$M := 1 + \max_{x \in U'} \kappa(y_0, x) \ \geq 1,$$

so that

$$\kappa(y_0, x) \leq M - 1 < M \qquad \text{for every } x \in U'. \tag{4b}$$

Put $S := M^{1/q}$, so that for $t \geq 0$ one has $t^q \leq M$ if and only if $t \leq S$; and put $T' := \max_{x \in U'} \|x\|$, finite because $\|\cdot\|$ is continuous and U' compact. Apply clause (4) of `CarnotNorm` with $\nu := S + T' + 1$: there is $c_1 > 0$ with

$$\|a * b\| \leq c_1\big(\|a\| + \|b\|\big) \qquad \text{whenever } \|a\| < \nu \text{ and } \|b\| < \nu. \tag{4c}$$

Set $C_1 := c_1(S + T')$ and

$$K := \big(\mathbb{R}^n \setminus \Omega\big) \cap \big\{y : \ \|y\| \leq C_1\big\}.$$

(4-i) K is compact. $\mathbb{R}^n \setminus \Omega$ is closed because Ω is open, and $\{y : \|y\| \leq C_1\}$ is closed because $\|\cdot\|$ is continuous; so K is closed. By clause (8) of `CarnotNorm` the sublevel set $\{y : \|y\| \leq C_1\}$ is bounded, so K is bounded. A closed bounded subset of the finite-dimensional space $\mathbb{R}^n$ is compact.

(4-ii) the key implication. Let $x \in U'$ and $y \in \mathbb{R}^n$ with $\kappa(y, x) \leq M$. Put $a := y * (-x)$. Then $\|a\|^q = \kappa(y, x) \leq M$, so $\|a\| \leq S < \nu$; and $\|x\| \leq T' < \nu$. Moreover $a * x = \big(y * (-x)\big) * x = y * \big((-x) * x\big) = y * 0 = y$ by associativity and clause (1) of `CarnotGroup`. Hence by (4c)

$$\|y\| = \|a * x\| \leq c_1\big(\|a\| + \|x\|\big) \leq c_1(S + T') = C_1. \tag{4d}$$

So: for $x \in U'$, every y with $\kappa(y, x) \leq M$ satisfies $\|y\| \leq C_1$.

(4-iii) $y_0 \in K$, so $K \neq \emptyset$. $y_0 \notin \Omega$, and by (4b) $\kappa(y_0, x_0) \leq M$ with $x_0 \in U'$, so (4d) gives $\|y_0\| \leq C_1$.

(4-iv) proof of (4a). Let $x \in U'$ and put $\alpha := \inf_{y \in K} \kappa(y, x)$, a real number since K is non-empty and $\kappa \geq 0$. As $K \subseteq \mathbb{R}^n \setminus \Omega$, every value $\kappa(y, x)$ with $y \in K$ occurs in the family defining $f(x)$, so $f(x) \leq \kappa(y, x)$ for all $y \in K$ and therefore $f(x) \leq \alpha$. Conversely, α is a lower bound for the whole family: let $y \notin \Omega$. If $y \in K$ then $\kappa(y, x) \geq \alpha$ by definition of α. If $y \notin K$ then, y being outside Ω, the failure must be $\|y\| > C_1$, so by the contrapositive of (4-ii) $\kappa(y, x) > M$; and $\alpha \leq \kappa(y_0, x) \leq M - 1 < M$ by (4-iii) and (4b), so again $\kappa(y, x) > \alpha$. Hence $\alpha \leq f(x)$, and (4a) holds on $U' \supseteq U$.

(4-v) continuity. By (0c) the map $(x, y) \mapsto \kappa(y, x)$ is jointly continuous and K is compact, so

$$F(x) := \inf_{y \in K} \kappa(y, x)$$

is a continuous function of x on all of $\mathbb{R}^n$: this is the standard fact that the infimum of a jointly continuous function over a compact parameter set depends continuously on the remaining variable. By (4a), $f = F$ on U, and U is an open neighbourhood of x_0. The claim is proved.

Step 5: proof of (4). Let $\varepsilon > 0$ and let $x_0 \in \Omega[\varepsilon]$; we exhibit a neighbourhood of x_0 inside $\Omega[\varepsilon]$. Take U, K, F as in Step 4 at x_0. By $(*)$, $x_0 \in \Omega$ and $c\varepsilon < f(x_0) = F(x_0)$. The set

$$V := \Omega \cap U \cap \{x : \ c\varepsilon < F(x)\}$$

is open (Ω and U are open and F is continuous by Step 4), and $x_0 \in V$. For $x \in V$ we have $x \in \Omega$ and, since $x \in U$, $f(x) = F(x) > c\varepsilon$; so $x \in \Omega[\varepsilon]$ by $(*)$. Hence $V \subseteq \Omega[\varepsilon]$ and x_0 is an interior point. As x_0 was arbitrary, $\Omega[\varepsilon]$ is open.

Step 6: proof of (5). Let $x \in \Omega$. Take K as in Step 4 at x; it is non-empty and compact and satisfies $f(x) = \inf_{y \in K} \kappa(y, x)$ by (4a). The function $\kappa(\cdot, x)$ is continuous by (0c), so on the non-empty compact K it attains its infimum: there is $y_* \in K$ with $f(x) = \kappa(y_*, x)$. Now $y_* \notin \Omega$ while $x \in \Omega$, so $y_* \neq x$; and $y_* * (-x) \neq 0$, since $y_* * (-x) = 0$ would give $y_* = \big(y_* * (-x)\big) * x = 0 * x = x$ by the computation in (4-ii) and clause (1) of `CarnotGroup`. Hence $\|y_* * (-x)\| \neq 0$ by clause (1) of `CarnotNorm`, so $\|y_* * (-x)\| > 0$ and

$$f(x) = \|y_* * (-x)\|^q > 0.$$

Put $\varepsilon_0 := f(x)/c > 0$ (using $c > 0$). For $0 < \varepsilon < \varepsilon_0$ we get $c\varepsilon < c\varepsilon_0 = f(x)$, so $x \in \Omega[\varepsilon]$ by $(*)$.

Step 7: proof of (6). Let $\varepsilon > 0$ and set

$$S_* := \bigcap_{y \notin \Omega} \{x : \ c\varepsilon \le \kappa(y, x)\}.$$

Each set in the intersection is closed by (0c), so S_* is closed. If $x \in \Omega[\varepsilon]$ then for every $y \notin \Omega$ we have $c\varepsilon < f(x) \le \kappa(y, x)$ by (0b), so $x \in S_*$; thus $\Omega[\varepsilon] \subseteq S_*$ and therefore $\overline{\Omega[\varepsilon]} \subseteq S_*$. Now suppose $x \in \overline{\Omega[\varepsilon]}$ but $x \notin \Omega$. Then $y := x$ is an admissible index, so membership of x in S_* gives $c\varepsilon \le \kappa(x, x) = 0$ by (0d), contradicting $c\varepsilon > 0$ (by (0a) and $\varepsilon > 0$). Hence $\overline{\Omega[\varepsilon]} \subseteq \Omega$.

Step 8: proof of (7). We first record two consequences of the group axioms of `CarnotGroup`, both standard in any group: the inverse of a product is $-(a * b) = (-b) * (-a)$, and consequently d is symmetric, since $(-a) * b = -\big((-b) * a\big)$ and $\| - g\| = \|g\|$ by clause (1) of `CarnotNorm`, so

$$d(a, b) = \|(-b) * a\| = \|(-a) * b\| = d(b, a). \tag{8a}$$

Let $\eta > 0$. Let $T \ge 0$ be as in (0e) and apply clause (5) of `CarnotNorm` with $\nu := T + 2$: there is $c_3 > 0$ with

$$\big\|(-a) * b * a\big\| \le c_3 \, \|b\|^{1/r} \qquad \text{whenever } \|a\| < \nu \text{ and } \|b\| < \nu. \tag{8b}$$

Since $(2c\varepsilon)^{1/q} \to 0$ and $c_3 (2c\varepsilon)^{1/(qr)} \to 0$ as $\varepsilon \to 0^+$, we may pick $\varepsilon_0 > 0$ so small that

$$(2c\varepsilon)^{1/q} < 1 \qquad \text{and} \qquad c_3 \, (2c\varepsilon)^{1/(qr)} < \eta \qquad \text{for every } \varepsilon \in (0, \varepsilon_0). \tag{8c}$$

Now fix $\varepsilon \in (0, \varepsilon_0)$ and $z \in \partial\Omega[\varepsilon]$.

(8-i) $z \in \Omega$ and $f(z) \le c\varepsilon$. By Step 5 the set $\Omega[\varepsilon]$ is open, so $\partial\Omega[\varepsilon] = \overline{\Omega[\varepsilon]} \setminus \Omega[\varepsilon]$; thus $z \in \overline{\Omega[\varepsilon]}$ and $z \notin \Omega[\varepsilon]$. Step 7 gives $z \in \Omega$. Since $z \in \Omega$ and $z \notin \Omega[\varepsilon]$, $(*)$ forces $f(z) \le c\varepsilon$. By (0e), $\|z\| \le T < \nu$.

(8-ii) a point outside Ω close to z for d. Apply (0b) with $\theta := c\varepsilon > 0$: there is $y \notin \Omega$ with $\kappa(y, z) < f(z) + c\varepsilon \le 2c\varepsilon$. Put $g := y * (-z)$ and $s := \|g\|$, so $s^q = \kappa(y, z) < 2c\varepsilon$ and hence, s being non-negative, $s < (2c\varepsilon)^{1/q} < 1 < \nu$ by (8c). Also $g * z = y$, by the computation in (4-ii). Therefore, using $-y = -(g * z) = (-z) * (-g)$,

$$d(z, y) = \big\|(-y) * z\big\| = \big\|\big((-z) * (-g)\big) * z\big\| \ \le \ c_3 \, \| - g\|^{1/r} = c_3 \, s^{1/r},$$

where (8b) was applied with $a := z$ and $b := -g$, legitimately because $\|z\| < \nu$ and $\|-g\| = s < \nu$. Combining with $s < (2c\varepsilon)^{1/q}$ and (8c),

$$d(z, y) \ \le \ c_3 \, s^{1/r} \ < \ c_3 \, (2c\varepsilon)^{1/(qr)} \ < \ \eta. \tag{8d}$$

(8-iii) the connected ball meets $\partial\Omega$. Let

$$B := \big\{w : \ \|(-z) * w\| < \eta\big\} = \{w : \ d(w, z) < \eta\}.$$

By clause (7) of `CarnotNorm`, B is non-empty and connected. It contains z, because $(-z)*z = 0$ and $\|0\| = 0 < \eta$; and it contains y, because $d(y,z) = d(z,y) < \eta$ by (8a) and (8d). So B meets Ω (at z, by (8-i)) and meets the complement of Ω (at y).

Suppose $B \cap \partial\Omega = \emptyset$. Since Ω is open, $\partial\Omega = \overline{\Omega} \setminus \Omega$, so for $w \in B$ either $w \in \Omega$ or $w \notin \overline{\Omega}$ — for if $w \notin \Omega$ and $w \in \overline{\Omega}$ then $w \in B \cap \partial\Omega$, excluded. Hence $B \subseteq \Omega \cup \big(\mathbb{R}^n \setminus \overline{\Omega}\big)$, a union of two disjoint open sets. The first meets B at z and the second meets B at y (indeed $y \in B$ and $y \notin \Omega$, so by the dichotomy $y \notin \overline{\Omega}$). This contradicts the connectedness of B. Therefore there is $w \in \partial\Omega$ with $\|(-z)*w\| < \eta$, i.e. $d(w,z) < \eta$, and by (8a) $d(z,w) = \|(-w)*z\| < \eta$, which is the required conclusion.

Remark. The hypothesis $\mathbb{R}^n \setminus \Omega \neq \emptyset$ is what makes the infimum of (3.15) an infimum over a non-empty index set, as (0b) records; it is automatic from boundedness of Ω whenever $n \geq 1$, which is the only case the paper considers, and without it the reduced domain would be governed by the empty-infimum convention rather than by (3.15). Clause (4) is the assertion the paper makes without proof at the line introducing (3.15). It cannot be obtained from upper semicontinuity of f: f is an infimum of continuous functions, hence upper semicontinuous, and $\Omega[\varepsilon]$ is a *super*level set of f, which an upper semicontinuous function does not make open. Nor does the quasi-triangle inequality (4) of `CarnotNorm` suffice, since from $y * x^{-1} = (y * x'^{-1}) * (x' * x^{-1})$ it yields only $f(x) \leq c^q\big(f(x')^{1/q} + \|x' * x^{-1}\|\big)^q$ and hence $\liminf_{x' \to x} f(x') \geq f(x)/c^q$, which is weaker than lower semicontinuity whenever $c > 1$. Step 4 instead proves that f is genuinely continuous, by showing that locally in x the infimum over the closed unbounded set $\mathbb{R}^n \setminus \Omega$ coincides with the infimum over a fixed compact subset of it; properness of the homogeneous norm, clause (8) of `CarnotNorm`, is exactly what makes that subset bounded. □

2.38. `RegularizedPairStandingAssumptions.`

Theorem 2.38 (The standing assumptions for the regularized pair)**.** *Let $\mathbb{G}$ be a Carnot setting (`CarnotSetting`) with group law $*$ and homogeneous norm $\|\cdot\|$, write $d(x,y) := \|y^{-1} * x\|$, and let the cited results at $\mathbb{G}$ hold (`CitedResults`). Let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$ satisfy the structure conditions (`StructureConditionsES`). Let $\Omega \subseteq \mathbb{R}^n$ be open and bounded with $\mathbb{R}^n \setminus \Omega \neq \emptyset$; let $u, v : \mathbb{R}^n \to \mathbb{R}$ be continuous on $\overline{\Omega}$ with u a viscosity sub-solution (`IsViscSubsol`) and v a viscosity super-solution (`IsViscSupersol`) of $\mathcal{L}\cdot = 0$ in Ω, and $u \leq v$ on $\partial\Omega$; let $R \geq 0$ satisfy $|u| \leq R/2$ and $|v| \leq R/2$ on $\overline{\Omega}$; and let $x_M \in \overline{\Omega}$ be a point at which $u - v$ attains its maximum over $\overline{\Omega}$, with*

$$M_0 := u(x_M) - v(x_M) > 0.$$

Put $\tau := M_0/2$ and

$$E := \big\{y \in \Omega : \ u - v \text{ attains its maximum over } \overline{\Omega} \text{ at } y\big\} \quad \big(= \operatorname{argmax}_{\Omega}(u - v)\big).$$

Write $\Omega[\varepsilon]$ for the reduced domain (`ReducedDomain`), $u^\varepsilon := \mathrm{SupConv}\ \mathbb{G}\ \Omega\ u\ \varepsilon$ (`SupConv`) and $\tilde{v}_\varepsilon := \mathrm{InfConv}\ \mathbb{G}\ \Omega\ v\ \varepsilon + \tau$ (`InfConv`). Then:

1. *E is non-empty and compact;*
2. *there is $\varepsilon_1 > 0$ such that for every $\varepsilon \in (0, \varepsilon_1)$:*
   - (a) *$E \subseteq \Omega[\varepsilon]$;*
   - (b) *u^ε and $\tilde{v}_\varepsilon$ are continuous on $\overline{\Omega[\varepsilon]}$;*
   - (c) *u^ε is locally semi-convex and $\tilde{v}_\varepsilon$ is locally semi-concave on $\Omega[\varepsilon]$, i.e. $-\tilde{v}_\varepsilon$ is locally semi-convex there (`LocallySemiConvexOn`);*
   - (d) *u^ε is a viscosity sub-solution and $\tilde{v}_\varepsilon$ a viscosity super-solution of $\mathcal{L}\cdot = 0$ in $\Omega[\varepsilon]$;*
   - (e) *$u^\varepsilon - \tilde{v}_\varepsilon \leq -\tau/2$ on $\partial\Omega[\varepsilon]$;*
   - (f) *$u^\varepsilon - \tilde{v}_\varepsilon$ attains its maximum over $\overline{\Omega[\varepsilon]}$ at some point of $\Omega[\varepsilon]$, and the maximum value is $\geq \tau$;*
   - (g) *every point of $\overline{\Omega[\varepsilon]}$ at which $u^\varepsilon - \tilde{v}_\varepsilon$ attains its maximum over $\overline{\Omega[\varepsilon]}$ lies in $\Omega[\varepsilon]$;*
3. *$\sup_{z \in E_\varepsilon} \operatorname{dist}(z, E) \to 0$ as $\varepsilon \to 0^+$: for every $\eta > 0$ there is $\varepsilon_2 > 0$ such that for every $\varepsilon \in (0, \varepsilon_2)$ and every $z \in \Omega[\varepsilon]$ at which $u^\varepsilon - \tilde{v}_\varepsilon$ attains its maximum over $\overline{\Omega[\varepsilon]}$ there is $p \in E$ with $d(z,p) < \eta$.*

Proof. Note first that $\tau = M_0/2 > 0$ and $M_0 = 2\tau$. Since Ω is open, $\partial\Omega = \overline{\Omega} \setminus \Omega$ and hence $\overline{\Omega} = \Omega \cup \partial\Omega$; likewise $\overline{\Omega[\varepsilon]} = \Omega[\varepsilon] \cup \partial\Omega[\varepsilon]$ for every $\varepsilon > 0$, because $\Omega[\varepsilon]$ is open by clause (4) of `ReducedDomainProperties`. Also $\overline{\Omega}$ is compact: it is closed, and it is bounded because Ω is. Throughout, "$\mathcal{L}w \le 0$ in V" means that w is a viscosity sub-solution in V in the sense of `IsViscSubsol`, and dually.

Step 1: $x_M \in \Omega$, and E is non-empty. If $x_M \in \partial\Omega$ then $u(x_M) \le v(x_M)$ by hypothesis, i.e. $M_0 \le 0$, contradicting $M_0 > 0$. Since $x_M \in \overline{\Omega} = \Omega \cup \partial\Omega$, we get $x_M \in \Omega$. As $u - v$ attains its maximum over $\overline{\Omega}$ at x_M by hypothesis, $x_M \in E$.

Step 2: E is compact. We claim

$$E = \{y \in \overline{\Omega} : \ M_0 \le u(y) - v(y)\}. \tag{2a}$$

If $y \in E$ then $y \in \Omega \subseteq \overline{\Omega}$ and, $u-v$ attaining its maximum over $\overline{\Omega}$ at y and x_M lying in $\overline{\Omega}$, we get $M_0 = u(x_M) - v(x_M) \le u(y) - v(y)$. Conversely let $y \in \overline{\Omega}$ with $M_0 \le u(y) - v(y)$. Since $u - v$ attains its maximum over $\overline{\Omega}$ at x_M and $y \in \overline{\Omega}$, also $u(y) - v(y) \le M_0$; hence $u(y) - v(y) = M_0$, and therefore $u(w) - v(w) \le M_0 = u(y) - v(y)$ for every $w \in \overline{\Omega}$, i.e. $u - v$ attains its maximum over $\overline{\Omega}$ at y. Finally $y \notin \partial\Omega$: otherwise $u(y) - v(y) \le 0 < M_0$. So $y \in \Omega$ and $y \in E$. This proves (2a).

In particular $u - v = M_0$ on E: for $y \in E$ the inclusion just proved gives $M_0 \le u(y) - v(y)$, while maximality at x_M gives $u(y) - v(y) \le M_0$.

By (2a), E is the intersection of $\overline{\Omega}$ with the preimage of the closed set $[M_0, \infty)$ under $u - v$, which is continuous on $\overline{\Omega}$; so E is closed in the subspace $\overline{\Omega}$. A closed subset of the compact set $\overline{\Omega}$ is compact, so E is compact.

Step 3: uniform continuity of u and v on $\overline{\Omega}$ with respect to d. We claim: for every $\theta > 0$ there is $\rho > 0$ such that

$$a, b \in \overline{\Omega},\ d(a,b) < \rho \implies |u(a) - u(b)| + |v(a) - v(b)| < \theta. \tag{3a}$$

Indeed, u and v are continuous on the compact set $\overline{\Omega}$, hence uniformly continuous there for the Euclidean metric: there is $\sigma > 0$ with $|u(a) - u(b)| + |v(a) - v(b)| < \theta$ whenever $a, b \in \overline{\Omega}$ and $|a - b| < \sigma$. Since Ω is bounded, choose $\nu > 0$ with $|a| \le \nu$ for all $a \in \overline{\Omega}$, and let $c_2 > 0$ be the constant of clause (6) of `CarnotNorm` for this ν, so that $c_2^{-1}|a - b| \le \|(-b) * a\| = d(a,b)$ for $a, b \in \overline{\Omega}$. Then $|a - b| \le c_2\, d(a,b)$, and $\rho := \sigma/c_2 > 0$ does the job.

Step 4: the quantitative boundary gap for $u - \tilde{v}$. Apply Step 3 with $\theta := \tau/8 > 0$ to get $\rho_0 > 0$ such that

$$a, b \in \overline{\Omega},\ d(a,b) < \rho_0 \implies |u(a) - u(b)| + |v(a) - v(b)| < \tau/8. \tag{4a}$$

This is the paper's choice of ρ_0 with $\omega_u(\rho_0) + \omega_v(\rho_0) \le \tau/8$.

Step 5: three smallness requirements on ε. We now fix $\varepsilon_1 > 0$ so that each of (5a), (5b), (5c) below holds for $0 < \varepsilon < \varepsilon_1$.

(5a) $E \subseteq \Omega[\varepsilon]$. For $k \in \mathbb{N}$, $k \ge 1$, the set $\Omega[1/k]$ is open by clause (4) of `ReducedDomainProperties`, and $\Omega[1/k] \subseteq \Omega[1/(k+1)]$ by clause (3) of the same node, since $1/(k+1) \le 1/k$. Moreover $\bigcup_{k \ge 1} \Omega[1/k] \supseteq E$: for $x \in E \subseteq \Omega$ clause (5) of `ReducedDomainProperties` gives $\varepsilon_0 > 0$ with $x \in \Omega[\varepsilon]$ for all $\varepsilon \in (0, \varepsilon_0)$, and any k with $1/k < \varepsilon_0$ then has $x \in \Omega[1/k]$. So $\{\Omega[1/k]\}_{k \ge 1}$ is an open cover of the compact set E (Step 2); a finite subcover exists, and since the family is increasing in k a single member contains E: there is $k_0 \ge 1$ with $E \subseteq \Omega[1/k_0]$. For $0 < \varepsilon < 1/k_0$, clause (3) gives $\Omega[1/k_0] \subseteq \Omega[\varepsilon]$ and hence $E \subseteq \Omega[\varepsilon]$.

(5b) the regularizations are $\tau/8$-close to u and v. By the third clause of WANG2 in `CitedResults`, applied to u and to v on the bounded open Ω (both are continuous on $\overline{\Omega}$), with $\eta := \tau/8$, there is $\varepsilon' > 0$ such that for $0 < \varepsilon < \varepsilon'$ and every $x \in \overline{\Omega}$

$$\big|u^\varepsilon(x) - u(x)\big| \le \tau/8 \qquad \text{and} \qquad \big|\mathrm{InfConv}\ \mathbb{G}\ \Omega\ v\ \varepsilon\ (x) - v(x)\big| \le \tau/8. \tag{5b}$$

(5c) the boundary of $\Omega[\varepsilon]$ is ρ_0-close to $\partial\Omega$. By clause (7) of `ReducedDomainProperties` with $\eta := \rho_0$ there is $\varepsilon'' > 0$ such that for $0 < \varepsilon < \varepsilon''$ and every $z \in \partial\Omega[\varepsilon]$ there is $w \in \partial\Omega$ with $d(z, w) < \rho_0$.

Put $\varepsilon_1 := \min\{1/k_0, \varepsilon', \varepsilon''\} > 0$ and fix $\varepsilon \in (0, \varepsilon_1)$ for the rest of Steps 6–10. Write $\Omega_\varepsilon := \Omega[\varepsilon]$, which is open, bounded, contained in Ω, with $\overline{\Omega_\varepsilon} \subseteq \Omega$ (clauses (1)–(4) and (6) of `ReducedDomainProperties`); and $\Omega_\varepsilon \neq \emptyset$ because $E \subseteq \Omega_\varepsilon$ and $E \neq \emptyset$. Hence $\overline{\Omega_\varepsilon}$ is a non-empty compact subset of Ω.

Step 6: 2(a), 2(c) and 2(d). Clause 2(a) is (5a).

By the first clause of WANG1 in `CitedResults`, u^ε is locally semi-convex on Ω and $-\mathrm{InfConv}\ \mathbb{G}\ \Omega\ v\ \varepsilon$ is locally semi-convex on Ω. Two elementary remarks turn this into 2(c).

(6a) local semi-convexity restricts to open subsets. If f is locally semi-convex on Ω and $V \subseteq \Omega$ is open, then f is locally semi-convex on V: given $x \in V$, `LocallySemiConvexOn` supplies $\rho > 0$, $\Lambda \geq 0$ with $B_\rho(x) \subseteq \Omega$ and $f + \frac{\Lambda}{2}|\cdot|^2$ convex on $B_\rho(x)$; choosing $\sigma > 0$ with $B_\sigma(x) \subseteq V$ and replacing ρ by $\min\{\rho, \sigma\}$ gives the requirement at x for V, since a convex function on $B_\rho(x)$ is convex on the convex subset $B_{\min\{\rho,\sigma\}}(x)$.

(6b) local semi-convexity is invariant under adding a constant. If $f + \frac{\Lambda}{2}|\cdot|^2$ is convex on a set, so is $(f + c) + \frac{\Lambda}{2}|\cdot|^2$ for any constant c, since adding a constant to a convex function preserves convexity. Hence $-\tilde{v}_\varepsilon = -\mathrm{InfConv}\ \mathbb{G}\ \Omega\ v\ \varepsilon - \tau$ is locally semi-convex wherever $-\mathrm{InfConv}\ \mathbb{G}\ \Omega\ v\ \varepsilon$ is.

Applying (6a) with $V := \Omega_\varepsilon$ (open, contained in Ω) to u^ε and, after (6b), to $-\tilde{v}_\varepsilon$, gives 2(c).

For 2(d): by WANG3 in `CitedResults`, applied with the given A, H and structure conditions, with Ω bounded open, u, v continuous on $\overline{\Omega}$, the bound $|u|, |v| \leq R/2$ on $\overline{\Omega}$, and the viscosity hypotheses on u and v, we get that u^ε is a viscosity sub-solution and $\mathrm{InfConv}\ \mathbb{G}\ \Omega\ v\ \varepsilon$ a viscosity super-solution of $\mathcal{L}\cdot = 0$ in $\Omega[\varepsilon] = \Omega_\varepsilon$. The super-solution property transfers to $\tilde{v}_\varepsilon = \mathrm{InfConv}\ \mathbb{G}\ \Omega\ v\ \varepsilon + \tau$ by `AddConstPreservesSubsol`, whose super-solution half says exactly that adding a constant to a viscosity super-solution of $\mathcal{L}\cdot = 0$ in a set again gives one. This is 2(d).

Step 7: 2(b). By 2(c) and part (1) of `SemiConvexLocLipschitz`, a locally semi-convex function on a set is continuous on it; applied on Ω (where WANG1 gives the semi-convexity) this makes u^ε and $-\mathrm{InfConv}\ \mathbb{G}\ \Omega\ v\ \varepsilon$, hence also $\mathrm{InfConv}\ \mathbb{G}\ \Omega\ v\ \varepsilon$ and $\tilde{v}_\varepsilon$, continuous on Ω. Since $\overline{\Omega_\varepsilon} \subseteq \Omega$ by clause (6) of `ReducedDomainProperties`, both are continuous on $\overline{\Omega_\varepsilon}$. This is 2(b); in particular $u^\varepsilon - \tilde{v}_\varepsilon$ is continuous on the non-empty compact set $\overline{\Omega_\varepsilon}$.

Step 8: 2(e). Let $z \in \partial\Omega_\varepsilon$. By clause (6) of `ReducedDomainProperties`, $z \in \overline{\Omega_\varepsilon} \subseteq \Omega \subseteq \overline{\Omega}$. By (5c) there is $w \in \partial\Omega \subseteq \overline{\Omega}$ with $d(z, w) < \rho_0$, so by (4a)

$$|u(z) - u(w)| + |v(z) - v(w)| < \tau/8.$$

Since $w \in \partial\Omega$, the boundary hypothesis gives $u(w) - v(w) \leq 0$. Hence

$$u(z) - v(z) = \big(u(w) - v(w)\big) + \big(u(z) - u(w)\big) - \big(v(z) - v(w)\big) \ \leq\ 0 + \tau/8 = \tau/8,$$

so, as $\tilde{v} = v + \tau$ and $M_0 = 2\tau$,

$$u(z) - \big(v(z) + \tau\big) \ \leq\ \tau/8 - \tau = -\tfrac{7}{8}\tau. \tag{8a}$$

Now by (5b), $u^\varepsilon(z) \leq u(z) + \tau/8$ and $\mathrm{InfConv}\ \mathbb{G}\ \Omega\ v\ \varepsilon(z) \geq v(z) - \tau/8$, so $\tilde{v}_\varepsilon(z) \geq v(z) + \tau - \tau/8$. Subtracting and using (8a),

$$u^\varepsilon(z) - \tilde{v}_\varepsilon(z) \ \leq\ \Big(u(z) + \frac{\tau}{8}\Big) - \Big(v(z) + \tau - \frac{\tau}{8}\Big) \ =\ \big(u(z) - v(z) - \tau\big) + \frac{\tau}{4} \ \leq\ -\frac{7}{8}\tau + \frac{\tau}{4} = -\frac{5}{8}\tau \ \leq\ -\frac{\tau}{2},$$

the last inequality because $\tau > 0$. This is 2(e).

Step 9: 2(f) and 2(g). By the first clause of WANG2 in `CitedResults`, $u \leq u^\varepsilon$ and $\mathrm{InfConv}\ \mathbb{G}\ \Omega\ v\ \varepsilon \leq v$ on $\overline{\Omega}$; hence $\tilde{v}_\varepsilon \leq v + \tau = \tilde{v}$ on $\overline{\Omega}$ and so

$$u^\varepsilon - \tilde{v}_\varepsilon \ \geq\ u - \tilde{v} \qquad \text{on } \overline{\Omega}. \tag{9a}$$

Pick $x \in E$, which is non-empty by Step 1 and contained in Ω_ε by (5a); then $x \in \overline{\Omega}$ and $u(x) - v(x) = M_0 = 2\tau$ by the last remark of Step 2, so by (9a)

$$u^\varepsilon(x) - \tilde{v}_\varepsilon(x) \ \geq\ u(x) - v(x) - \tau = 2\tau - \tau = \tau. \tag{9b}$$

By Step 7 the function $u^\varepsilon - \tilde v_\varepsilon$ is continuous on the non-empty compact set $\overline{\Omega_\varepsilon}$, so it attains its maximum there at some $x_e \in \overline{\Omega_\varepsilon}$, and by (9b) and $x \in \Omega_\varepsilon$ the maximum value satisfies

$$u^\varepsilon(x_e) - \tilde v_\varepsilon(x_e) \;\geq\; u^\varepsilon(x) - \tilde v_\varepsilon(x) \;\geq\; \tau. \tag{9c}$$

Now let $z \in \overline{\Omega_\varepsilon}$ be *any* maximum point of $u^\varepsilon - \tilde v_\varepsilon$ over $\overline{\Omega_\varepsilon}$. Then $u^\varepsilon(z) - \tilde v_\varepsilon(z) \geq \tau > 0$ by the same comparison as in (9c). If $z \in \partial\Omega_\varepsilon$ then 2(e) gives $u^\varepsilon(z) - \tilde v_\varepsilon(z) \leq -\tau/2 < 0$, a contradiction. Since $\overline{\Omega_\varepsilon} = \Omega_\varepsilon \cup \partial\Omega_\varepsilon$, we conclude $z \in \Omega_\varepsilon$. This is 2(g), and applying it to x_e gives $x_e \in \Omega_\varepsilon$, which together with (9c) is 2(f).

Step 10: proof of 3. Let $\eta > 0$. Put

$$K_\eta := \big\{ y \in \overline{\Omega} :\; d(y,p) \geq \eta \text{ for every } p \in E \big\}.$$

If $K_\eta = \emptyset$ then every $y \in \overline{\Omega}$ admits $p \in E$ with $d(y,p) < \eta$, so $\varepsilon_2 := \varepsilon_1$ works and there is nothing more to prove (every $z \in \Omega_\varepsilon \subseteq \overline{\Omega}$ has such a p). So assume $K_\eta \neq \emptyset$.

(10a) K_η is compact and disjoint from E. For each fixed p, the map $y \mapsto d(y,p) = \|(-p)*y\|$ is continuous, since the group law is continuous (clause (2) of `CarnotGroup`) and $\|\cdot\|$ is continuous (clause (1) of `CarnotNorm`); so $\{y : d(y,p) \geq \eta\}$ is closed, and K_η is the intersection of $\overline{\Omega}$ with an intersection of closed sets, hence closed, hence compact as a closed subset of the compact $\overline{\Omega}$. It is disjoint from E: for $p \in E$ we have $d(p,p) = \|(-p) * p\| = \|0\| = 0 < \eta$, so $p \notin K_\eta$.

(10b) $u - \tilde v < \tau$ on K_η, with a gap. Let $y \in K_\eta$. Then $y \in \overline{\Omega}$ and $y \notin E$, so by (2a) $u(y) - v(y) < M_0 = 2\tau$, whence $u(y) - \tilde v(y) = u(y) - v(y) - \tau < \tau$. The function $u - \tilde v = u - v - \tau$ is continuous on $\overline{\Omega}$, hence on the non-empty compact K_η, so it attains its maximum there:

$$\beta := \max_{K_\eta} \big(u - \tilde v\big) < \tau.$$

Put $\theta := (\tau - \beta)/3 > 0$.

(10c) conclusion. By the third clause of WANG2 in `CitedResults`, applied with $\eta := \theta$, there is $\varepsilon''' > 0$ such that for $0 < \varepsilon < \varepsilon'''$ and every $x \in \overline{\Omega}$ both $|u^\varepsilon(x) - u(x)| \leq \theta$ and $|\mathrm{InfConv}\ \mathbb{G}\ \Omega\ v\ \varepsilon(x) - v(x)| \leq \theta$. Put $\varepsilon_2 := \min\{\varepsilon_1, \varepsilon'''\} > 0$ and let $\varepsilon \in (0, \varepsilon_2)$. For $y \in K_\eta$,

$$u^\varepsilon(y) - \tilde v_\varepsilon(y) \;\leq\; \big(u(y)+\theta\big) - \big(v(y) - \theta + \tau\big) \;=\; \big(u(y) - \tilde v(y)\big) + 2\theta \;\leq\; \beta + 2\theta = \frac{\beta + 2\tau}{3} \;<\; \tau, \tag{10d}$$

the final strict inequality because $\beta < \tau$.

Now let $z \in \Omega_\varepsilon$ be a maximum point of $u^\varepsilon - \tilde v_\varepsilon$ over $\overline{\Omega_\varepsilon}$. Since $\varepsilon < \varepsilon_1$, Step 9 applies and gives $u^\varepsilon(z) - \tilde v_\varepsilon(z) \geq \tau$. By (10d) this forces $z \notin K_\eta$. But $z \in \Omega_\varepsilon \subseteq \Omega \subseteq \overline{\Omega}$, so the failure of membership in K_η must be the second condition: there is $p \in E$ with $d(z,p) < \eta$. This is 3.

Remark. This node is the common first page of three proofs in the paper — of Theorem 1.2, of Theorem 1.3(i) and of Theorem 1.3(iii) — all of which begin by assuming $M_0 > 0$, setting $\tau := M_0/2$ and $\tilde v := v + \tau$, fixing ρ_0 with $\omega_u(\rho_0) + \omega_v(\rho_0) \leq \tau/8$, producing ε_1, and checking that the pair $(u^\varepsilon, \tilde v_\varepsilon)$ satisfies on $\Omega[\varepsilon]$ the standing assumptions (3.10) with $\tau/2$ in place of τ. Proving it once is the content of the node; no other single node states it, and each of the three proofs consumes all of clauses 1–3. Since $\tilde v_\varepsilon = v_\varepsilon + \tau$ and argmax is unchanged by adding a constant to the subtrahend's companion, the set of maximum points of $u^\varepsilon - \tilde v_\varepsilon$ over $\overline{\Omega[\varepsilon]}$ appearing in clauses 2(f), 2(g) and 3 is literally the set E_ε of (1.15), namely the set of points of $\Omega[\varepsilon]$ at which $u^\varepsilon - v_\varepsilon$ attains its maximum over $\overline{\Omega[\varepsilon]}$; clauses 2(f) and 2(g) say that it is a non-empty subset of $\Omega[\varepsilon]$, and it is compact, being closed in the compact $\overline{\Omega[\varepsilon]}$ by continuity (Step 7). Clause 3 is the convergence $\sup_{E_\varepsilon} \mathrm{dist}(\cdot, E) \to 0$ that Theorem 1.3(i) uses to place the displaced points $z_x * x$ inside the neighbourhood on which v is C^1. □

2.39. `ComparisonAbelian`.

Theorem 2.39 (Comparison in an abelian setting, under (E) and (S) alone)**.** *Let $\mathbb{G}$ be a sub-elliptic setting (`CarnotSetting`) with group law $*$, homogeneous norm $\|\cdot\|$, step r, frame σ, $q := 2\,r!$ and $d(x,y) := \|y^{-1} * x\|$, and assume the cited results of the literature at $\mathbb{G}$ (`CitedResults`). Assume $\mathbb{G}$ is* abelian*: $x * y = y * x$ for all $x, y \in \mathbb{R}^n$. Let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$ be continuous on $\mathbb{R}^m \setminus \{0\}$, with $A(\xi)$ symmetric and positive semidefinite for $\xi \neq 0$, and satisfy the*

conditions of strict ellipticity in the gradient direction (E) (1.9) and of homogeneity along rays (S) (1.10) — that is, let (A, H) satisfy the structure conditions of `StructureConditionsES`*. No normalization of (A, H) is assumed. Let $\Omega \subset \mathbb{R}^n$ be a bounded open set with $\mathbb{R}^n \setminus \Omega \neq \emptyset$, let $u, v : \mathbb{R}^n \to \mathbb{R}$ be continuous on $\bar{\Omega}$, and assume*

(1) *u is a viscosity sub-solution (`IsViscSubsol`) and v a viscosity super-solution (`IsViscSupersol`) of $\mathcal{L}\cdot = 0$ in Ω;*
(2) *$u \leq v$ on $\partial\Omega$.*

Let $R \geq 0$ satisfy $|u| \leq R/2$ and $|v| \leq R/2$ on $\bar{\Omega}$. Then $u \leq v$ in Ω.

Proof. This is the conclusion the paper reaches in the first paragraph of Theorem 1.3(iii): "By the proof of Theorem Theorem 1.2, it suffices to show that under the standing assumptions (3.10)–(3.11) a semi-convex/semi-concave pair cannot exist." The comparison machinery is not re-derived: the regularized pair on the reduced domain is such a pair, and `AbelianNoStandingPair` forbids it.

Two points of the formulation first, as in Theorem 1.2.

The parameter R. The paper defines $R := 2\max\{\|u\|_{L^\infty}, \|v\|_{L^\infty}\}$ just before (3.15), whereas `ReducedDomain` carries R as a parameter. We therefore quantify over every admissible R, i.e. every $R \geq 0$ with $|u|, |v| \leq R/2$ on $\bar{\Omega}$. This contains the paper's statement: the paper's R is admissible, since $\|u\|_{L^\infty} \leq \max\{\|u\|_{L^\infty}, \|v\|_{L^\infty}\} = R/2$ and likewise for v, and it is the smallest admissible one. It is also not stronger than the paper's statement in any way the paper does not justify: R enters only through the definition of $\Omega[\varepsilon]$, where a larger R makes $\Omega[\varepsilon]$ smaller, and the role of $|u| \leq R/2$ is exactly the one the paper gives it in the attainment step of Lemma 3.8(4), namely to make $\|z_x\|^q \leq 4\varepsilon\|u\|_{L^\infty} \leq 2R\varepsilon < (1 + 4R)\varepsilon$.

The hypothesis $\mathbb{R}^n \setminus \Omega \neq \emptyset$. This is the paper's own convention made explicit: (3.15) takes an infimum over $\mathbb{R}^n \setminus \Omega$, and the paper's $\Omega \subset \mathbb{R}^n$ is a proper subset. It is automatic as soon as $n \geq 1$, since no bounded subset of $\mathbb{R}^n$ is then all of $\mathbb{R}^n$.

Why the normalization is not in the statement. The printed theorem is for general A, H satisfying (E) and (S); (2.3) is an internal step, carried out once in Step 1 below. It appears as a hypothesis of `AbelianNoStandingPair` only because `ComparisonNondegenerateSemiconvex`, which that node uses, is stated for a normalized pair; this is the level at which it is discharged.

Suppose the conclusion fails, so $u(x_0) > v(x_0)$ for some $x_0 \in \Omega$; we derive a contradiction.

Step 0: $M_0 > 0$. Ω is bounded, so $\bar{\Omega}$ is compact, and it is non-empty ($x_0 \in \Omega \subseteq \bar{\Omega}$); $u - v$ is continuous on $\bar{\Omega}$, so it attains its maximum there at some $x_M \in \bar{\Omega}$ and

$$M_0 := u(x_M) - v(x_M) \;\geq\; u(x_0) - v(x_0) \;>\; 0.$$

Put $\tau := M_0/2 > 0$.

Step 1: normalize once. The only hypothesis of `NormalizationExists` is that (A, H) satisfy the structure conditions, which we have. It provides A_0, H_0 and a with $a(\xi) > 0$, $A(\xi) = a(\xi)A_0(\xi)$ and $H(\xi) = a(\xi)H_0(\xi)$ for $\xi \neq 0$, with (A_0, H_0) satisfying both the structure conditions (`StructureConditionsES`) and the normalization (2.3) (`NormalizedAH`). Write $\mathcal{L}_0$ for the operator built from (A_0, H_0).

The hypotheses of `NormalizedSameViscositySolutions` are exactly those three identities, and its open set and function are arbitrary parameters; applying it with Ω and u, and with Ω and v, gives that u is a viscosity sub-solution and v a viscosity super-solution of $\mathcal{L}_0\cdot = 0$ in Ω.

Step 2: the regularized pair. All hypotheses of `RegularizedPairStandingAssumptions` hold for ($\mathbb{G}$, cited results at $\mathbb{G}, A_0, H_0, \Omega, u, v, R, x_M$): the structure conditions on (A_0, H_0) by Step 1; Ω open, bounded, with non-empty complement; u, v continuous on $\bar{\Omega}$ and a viscosity sub-/super-solution pair for $\mathcal{L}_0$ in Ω by Step 1; $u \leq v$ on $\partial\Omega$; $R \geq 0$ with $|u|, |v| \leq R/2$ on $\bar{\Omega}$; x_M a maximum point of $u - v$ over $\bar{\Omega}$; and $M_0 > 0$. Taking $\tau = M_0/2$ and $E = \operatorname{argmax}_{\bar{\Omega}}(u - v)$ and writing

$$u^\varepsilon := \text{SupConv } \mathbb{G}\ \Omega\ u\ \varepsilon \ (\texttt{SupConv}), \qquad \tilde{v}_\varepsilon := \big(y \mapsto \text{InfConv } \mathbb{G}\ \Omega\ v\ \varepsilon\ (y) + \tau\big) \ (\texttt{InfConv}),$$

and $\Omega[\varepsilon]$ for the reduced domain (`ReducedDomain`) with this R, it yields an $\varepsilon_1 > 0$ such that for every $\varepsilon \in (0, \varepsilon_1)$:

(b) u^ε and $\tilde{v}_\varepsilon$ are continuous on $\overline{\Omega[\varepsilon]}$;
(c) u^ε is locally semi-convex and $\tilde{v}_\varepsilon$ locally semi-concave on $\Omega[\varepsilon]$, i.e. $-\tilde{v}_\varepsilon$ is locally semi-convex there (`LocallySemiConvexOn`);
(d) u^ε is a viscosity sub-solution and $\tilde{v}_\varepsilon$ a viscosity super-solution of $\mathcal{L}_0 \cdot = 0$ in $\Omega[\varepsilon]$;
(e) $u^\varepsilon - \tilde{v}_\varepsilon \le -\tau/2$ on $\partial\Omega[\varepsilon]$;
(f) there is $x_e \in \Omega[\varepsilon]$ at which $u^\varepsilon - \tilde{v}_\varepsilon$ attains its maximum over $\overline{\Omega[\varepsilon]}$, with $\tau \le u^\varepsilon(x_e) - \tilde{v}_\varepsilon(x_e)$.

(It also yields items 1, 2(a), 2(g) and 3, which we do not need here.) Fix

$$\varepsilon := \frac{\varepsilon_1}{2} \in (0, \varepsilon_1),$$

and write $\Omega' := \Omega[\varepsilon]$, $U := u^\varepsilon$, $V := \tilde{v}_\varepsilon$ and $\tau' := \tau/2 > 0$.

Step 3: the geometry of Ω'. By items (1), (2) and (4) of `ReducedDomainProperties` — applicable since Ω is open and bounded with non-empty complement and $R \ge 0$ — the set Ω' satisfies $\Omega' \subseteq \Omega$, is bounded, and is open (here $\varepsilon > 0$). Hence $\overline{\Omega'}$ is compact. Moreover $\mathbb{R}^n \setminus \Omega' \neq \emptyset$: taking $p \notin \Omega$ from our hypothesis, $p \notin \Omega'$ because $\Omega' \subseteq \Omega$.

Step 4: the (3.11) package, built as a step function. We must supply `AbelianNoStandingPair` with non-decreasing d-moduli ω_U, ω_V for U and V on $\overline{\Omega'}$ and a radius $\rho_0 > 0$ with $\omega_U(\rho_0) + \omega_V(\rho_0) \le \tau'/4$. The paper asserts at (3.11) that such moduli exist "because $\bar{\Omega}$ is compact and $|x-y| \le c\, d(x,y)$" and does not construct them; a supremum $\omega(s) = \sup\{|w(a)-w(b)| : d(a,b) \le s\}$ would need a separate boundedness argument, and none is needed, because a *two-valued step function* already does everything the two consumers ask of ω: it is non-decreasing, it dominates the oscillation at every scale, and it is as small as we like at one prescribed radius. We construct it.

The construction. Let $w : \mathbb{R}^n \to \mathbb{R}$ be continuous on the compact set $K := \overline{\Omega'}$, let M be such that $|w| \le M$ on K (such an M exists: w is continuous on the compact K), and let $\eta > 0$. Then there are $\rho > 0$ and a non-decreasing $\omega : \mathbb{R} \to \mathbb{R}$ with

$$|w(a) - w(b)| \le \omega\big(d(a,b)\big) \quad (a, b \in K), \qquad \omega(\rho) \le \eta. \tag{1}$$

Indeed: w is uniformly continuous on the compact K, so there is $\varrho > 0$ with $|w(a) - w(b)| \le \eta$ whenever $a, b \in K$ and $|a-b| < \varrho$. The set K is bounded, so $\nu := 1 + \sup\{|a| : a \in K\}$ is a finite real number with $\nu > 0$ and $|a| \le \nu$ for every $a \in K$ (the supremum of the empty set being 0, which only makes ν larger); by the Hölder equivalence clause of `CarnotSetting` at this ν there is $c > 0$ with $c^{-1}|a-b| \le d(a,b)$ whenever $|a| \le \nu$ and $|b| \le \nu$, i.e.

$$|a - b| \le c\, d(a,b) \qquad (a, b \in K). \tag{2}$$

Put

$$\rho := \frac{\varrho}{2c} > 0, \qquad \omega(s) := \begin{cases} \eta, & s \le \rho, \\ \eta + 2|M| + 1, & s > \rho. \end{cases}$$

ω is non-decreasing, because $2|M| + 1 > 0$ and ω takes the smaller value exactly on the lower half-line. If $a, b \in K$ with $d(a,b) \le \rho$ then by (2) $|a-b| \le c\rho = \varrho/2 < \varrho$, so $|w(a) - w(b)| \le \eta = \omega(d(a,b))$. If $d(a,b) > \rho$ then $\omega(d(a,b)) = \eta + 2|M| + 1$, while $|w(a) - w(b)| \le |w(a)| + |w(b)| \le 2M \le 2|M| < \eta + 2|M| + 1$. Finally $\omega(\rho) = \eta$. This proves (1).

The instantiation. Apply the construction twice on $K = \overline{\Omega'}$, with $\eta := \tau/16 > 0$: to $w := U$ (continuous on $\overline{\Omega'}$ by (b)), giving $\rho_U > 0$ and ω_U; and to $w := V$ (continuous on $\overline{\Omega'}$ by (b)), giving $\rho_V > 0$ and ω_V. Put $\rho_0 := \min\{\rho_U, \rho_V\} > 0$. Since $\rho_0 \le \rho_U$ and ω_U is non-decreasing, $\omega_U(\rho_0) \le \omega_U(\rho_U) \le \tau/16$, and likewise $\omega_V(\rho_0) \le \tau/16$. Hence

$$\omega_U(\rho_0) + \omega_V(\rho_0) \;\le\; \frac{\tau}{8} \;=\; \frac{\tau/2}{4} \;=\; \frac{\tau'}{4}, \tag{3}$$

which is (3.11) for the standing constant $\tau' = \tau/2$.

Step 5: the standing assumptions for (U,V) on Ω', and the contradiction. We now check every hypothesis of `AbelianNoStandingPair` for the data $(\mathbb{G}, \text{cited results at } \mathbb{G}, A_0, H_0, \Omega', U, V, \omega_U, \omega_V, \tau', x_e, \rho$

- The cited results at $\mathbb{G}$: our hypothesis, and $\mathbb{G}$ is the setting that node is applied to. Commutativity $x*y = y*x$: our hypothesis.
- (A_0, H_0) satisfies the structure conditions and (2.3): Step 1. This is where the normalization is spent.
- Ω' is open and bounded with $\mathbb{R}^n \setminus \Omega' \neq \emptyset$: Step 3.
- U, V are continuous on $\overline{\Omega'}$: (b).
- U is locally semi-convex and V locally semi-concave in Ω': (c).
- U is a viscosity sub-solution and V a viscosity super-solution of $\mathcal{L}_0 \cdot = 0$ in Ω': (d).
- ω_U, ω_V are non-decreasing and are d-moduli for U, V on $\overline{\Omega'}$: Step 4.
- $\tau' > 0$ and $U - V \leq -\tau'$ on $\partial\Omega'$: $\tau' = \tau/2 > 0$, and (e) is exactly $U - V \leq -\tau/2 = -\tau'$ on $\partial\Omega'$.
- $x_e \in \overline{\Omega'}$ is a point at which $U - V$ attains its maximum over $\overline{\Omega'}$, and $\tau' \leq U(x_e) - V(x_e)$: by (f), $x_e \in \Omega' \subseteq \overline{\Omega'}$ is such a maximum point with $\tau \leq U(x_e) - V(x_e)$, and $\tau' = \tau/2 \leq \tau$ since $\tau > 0$.
- $\rho_0 > 0$ and $\omega_U(\rho_0) + \omega_V(\rho_0) \leq \tau'/4$: Step 4, (3).

`AbelianNoStandingPair` therefore applies and yields a contradiction. Since the only assumption made beyond the hypotheses of the present statement was $u(x_0) > v(x_0)$ for some $x_0 \in \Omega$, that assumption is false: $u(x) \leq v(x)$ for every $x \in \Omega$, i.e. $u \leq v$ in Ω. □

2.40. `SupConvAttainment`.

Lemma 2.40 (Part (4) of Lemma 3.8, on $\Omega[\varepsilon]$)**.** *Let $\mathbb{G}$ be a Carnot setting (`CarnotSetting`) with group law $*$ and homogeneous norm $\|\cdot\|$, put $q := 2\,r!$ and $d(x,y) := \|y^{-1} * x\|$. Let $\Omega \subseteq \mathbb{R}^n$ be open and bounded, let $u, v : \mathbb{R}^n \to \mathbb{R}$ be continuous on $\overline{\Omega}$, let $M_u, M_v, R \in \mathbb{R}$ satisfy $R \geq 0$, $2M_u \leq R$, $2M_v \leq R$, $|u| \leq M_u$ and $|v| \leq M_v$ on $\overline{\Omega}$, and let $\omega_u, \omega_v : \mathbb{R} \to \mathbb{R}$ be moduli of continuity of u and v on $\overline{\Omega}$ with respect to d, i.e.*

$$|u(a) - u(b)| \leq \omega_u\big(d(a,b)\big), \qquad |v(a) - v(b)| \leq \omega_v\big(d(a,b)\big) \qquad \text{for all } a, b \in \overline{\Omega}.$$

Let $\varepsilon > 0$ and write $u^\varepsilon := \mathrm{SupConv}\ \mathbb{G}\ \Omega\ u\ \varepsilon$ (`SupConv`), $v_\varepsilon := \mathrm{InfConv}\ \mathbb{G}\ \Omega\ v\ \varepsilon$ (`InfConv`) and $\Omega[\varepsilon]$ for the reduced domain (`ReducedDomain`). Then

(1) $|u^\varepsilon(x) - u^\varepsilon(x')| \leq \omega_u\big(d(x,x')\big)$ *for all $x, x' \in \Omega[\varepsilon]$;*

(2) *for every $x \in \Omega[\varepsilon]$ there is z with $z * x \in \overline{\Omega}$ such that the supremum defining $u^\varepsilon(x)$ is attained at z, i.e. $u^\varepsilon(x) = u(z*x) - \|z\|^q/(2\varepsilon)$, and*

$$\|z\|^q \leq 4\varepsilon M_u, \qquad \text{and} \qquad z * x' \in \Omega \ \text{ for every } x' \in \Omega[\varepsilon];$$

(3) $|v_\varepsilon(x) - v_\varepsilon(x')| \leq \omega_v\big(d(x,x')\big)$ *for all $x, x' \in \Omega[\varepsilon]$;*

(4) *for every $x \in \Omega[\varepsilon]$ there is z with $z * x \in \overline{\Omega}$ such that the infimum defining $v_\varepsilon(x)$ is attained at z, i.e. $v_\varepsilon(x) = v(z*x) + \|z\|^q/(2\varepsilon)$, and $\|z\|^q \leq 4\varepsilon M_v$ and $z * x' \in \Omega$ for every $x' \in \Omega[\varepsilon]$.*

Proof. Write $c := 1 + 4R$ and, as in `ReducedDomain`,

$$(*) \qquad \Omega[\varepsilon] = \Big\{ x \in \Omega : \ c\varepsilon < \inf_{y \notin \Omega} \|y * (-x)\|^q \Big\}.$$

For $x \in \mathbb{R}^n$ set

$$Z_x := \{ z \in \mathbb{R}^n : \ z * x \in \overline{\Omega} \}, \qquad \Phi_x(z) := u(z*x) - \frac{\|z\|^q}{2\varepsilon}, \qquad \Psi_x(z) := v(z*x) + \frac{\|z\|^q}{2\varepsilon},$$

so that by `SupConv` and `InfConv`

$$(**) \qquad u^\varepsilon(x) = \sup_{z \in Z_x} \Phi_x(z), \qquad v_\varepsilon(x) = \inf_{z \in Z_x} \Psi_x(z).$$

Step 0: group identities used repeatedly. By associativity and clause (1) of `CarnotGroup`, for all a, x

(0a) $\big(a * (-x)\big) * x = a * \big((-x) * x\big) = a * 0 = a, \qquad \big(a * x\big) * (-x) = a * \big(x * (-x)\big) = a * 0 = a.$

Also d is left-invariant: for all a, x, x',

(0b) $d(a*x,\, a*x') = \big\|(a*x')^{-1} * (a*x)\big\| = \big\|(-x') * \big((-a)*a\big) * x\big\| = \big\|(-x') * x\big\| = d(x, x'),$

using $-(a*x') = (-x') * (-a)$, which is the inverse-of-a-product identity valid in any group.

Step 1: Z_x is compact, and non-empty for $x \in \Omega$. The map $\Theta : y \mapsto y*(-x)$ is continuous, because the group law is C^∞ (clause (2) of `CarnotGroup`) hence continuous. By (0a) the maps $z \mapsto z*x$ and $y \mapsto y*(-x)$ are mutually inverse bijections of $\mathbb{R}^n$, so

$$Z_x = \Theta\big(\overline{\Omega}\big).$$

Since Ω is bounded, $\overline{\Omega}$ is closed and bounded, hence compact; so Z_x is compact as the image of a compact set under a continuous map. If $x \in \Omega$ then $0 \in Z_x$, because $0*x = x \in \Omega \subseteq \overline{\Omega}$; so $Z_x \neq \emptyset$.

Step 2: the extrema in $(**)$ are attained for $x \in \Omega$. Let $x \in \Omega$. On Z_x the map $z \mapsto z*x$ is continuous with values in $\overline{\Omega}$, and u is continuous on $\overline{\Omega}$, so $z \mapsto u(z*x)$ is continuous on Z_x; and $z \mapsto \|z\|^q/(2\varepsilon)$ is continuous because $\|\cdot\|$ is continuous (clause (1) of `CarnotNorm`) and $\varepsilon \neq 0$. Hence Φ_x is continuous on the non-empty compact set Z_x (Step 1) and therefore attains its supremum: there is $z \in Z_x$ with

$$u^\varepsilon(x) = \sup_{Z_x} \Phi_x = \Phi_x(z) = u(z*x) - \frac{\|z\|^q}{2\varepsilon}, \tag{2a}$$

and in particular $u^\varepsilon(x)$ is a real number and $u^\varepsilon(x) \geq \Phi_x(z')$ for every $z' \in Z_x$. The same argument applied to Ψ_x gives z with $v_\varepsilon(x) = \Psi_x(z)$, and $v_\varepsilon(x) \leq \Psi_x(z')$ for every $z' \in Z_x$.

Step 3: proof of (2). Let $x \in \Omega[\varepsilon]$; by $(*)$, $x \in \Omega$, so Step 2 gives $z \in Z_x$ with (2a). We verify the two extra properties.

(3a) the bound $\|z\|^q \leq 4\varepsilon M_u$. Since $0 \in Z_x$ (Step 1) and $\|0\| = 0$ (clause (1) of `CarnotNorm`), comparing with $z' = 0$ in Step 2 gives

$$u(z*x) - \frac{\|z\|^q}{2\varepsilon} = u^\varepsilon(x) \;\geq\; \Phi_x(0) = u(0*x) - 0 = u(x).$$

Hence $\|z\|^q \leq 2\varepsilon\big(u(z*x) - u(x)\big)$. Both $z*x$ and x lie in $\overline{\Omega}$ (the former because $z \in Z_x$, the latter because $x \in \Omega$), so $u(z*x) \leq M_u$ and $-u(x) \leq M_u$ by the bound $|u| \leq M_u$ on $\overline{\Omega}$. Therefore

$$\|z\|^q \;\leq\; 2\varepsilon\big(M_u + M_u\big) = 4\varepsilon M_u. \tag{3a}$$

(3b) $z*x' \in \Omega$ for every $x' \in \Omega[\varepsilon]$. First, the chain of the paper: by (3a), $2M_u \leq R$ and $\varepsilon > 0$,

$$\|z\|^q \;\leq\; 4\varepsilon M_u = 2\varepsilon\,(2M_u) \;\leq\; 2\varepsilon R \;<\; (1 + 4R)\,\varepsilon = c\,\varepsilon, \tag{3b}$$

the last inequality because $2R < 1 + 4R$, i.e. $0 < 1 + 2R$, which holds since $R \geq 0$.

Now let $x' \in \Omega[\varepsilon]$ and suppose $z * x' \notin \Omega$. Then $y := z * x'$ is an admissible index in the infimum of $(*)$ at the point x', so

$$\inf_{y' \notin \Omega} \big\|y' * (-x')\big\|^q \;\leq\; \big\|(z*x') * (-x')\big\|^q = \|z\|^q,$$

using (0a) for the last equality. Combining with (3b) this gives $\inf_{y' \notin \Omega} \|y' * (-x')\|^q < c\varepsilon$, contradicting $c\varepsilon < \inf_{y' \notin \Omega} \|y' * (-x')\|^q$, which is what $x' \in \Omega[\varepsilon]$ asserts by $(*)$. Hence $z * x' \in \Omega$.

This proves (2), with $z * x \in \overline{\Omega}$ holding because $z \in Z_x$.

Step 4: proof of (1). Let $x, x' \in \Omega[\varepsilon]$. Let z be as in Step 3 for the point x, so (2a), (3a) and (3b) hold and, by Step 3(3b) applied with the point x',

$$z * x' \in \Omega \subseteq \overline{\Omega}, \qquad \text{i.e.} \qquad z \in Z_{x'}.$$

Therefore, by the second assertion of Step 2 at the point x' (legitimate since $x' \in \Omega$),

$$u^\varepsilon(x') \;\geq\; \Phi_{x'}(z) = u(z*x') - \frac{\|z\|^q}{2\varepsilon}.$$

Subtracting this from (2a), the two penalty terms $\|z\|^q/(2\varepsilon)$ cancel:

$$u^\varepsilon(x) - u^\varepsilon(x') \;\le\; \Big(u(z*x) - \frac{\|z\|^q}{2\varepsilon}\Big) - \Big(u(z*x') - \frac{\|z\|^q}{2\varepsilon}\Big) \;=\; u(z*x) - u(z*x').$$

Both $z*x$ and $z*x'$ lie in $\overline{\Omega}$, so the modulus hypothesis and (0b) give

$$u(z*x) - u(z*x') \;\le\; \big|u(z*x) - u(z*x')\big| \;\le\; \omega_u\big(d(z*x, z*x')\big) = \omega_u\big(d(x,x')\big),$$

whence $u^\varepsilon(x) - u^\varepsilon(x') \le \omega_u(d(x,x'))$. Exchanging the roles of x and x' gives $u^\varepsilon(x') - u^\varepsilon(x) \le \omega_u(d(x',x))$, and $d(x',x) = d(x,x')$ because $(-x)*x' = -\big((-x')*x\big)$ and $\|-g\| = \|g\|$ by clause (1) of `CarnotNorm`. The two inequalities together give $|u^\varepsilon(x) - u^\varepsilon(x')| \le \omega_u(d(x,x'))$, which is (1).

Step 5: proof of (4) and (3). The arguments are those of Steps 3 and 4 with the inequalities reversed. Let $x \in \Omega[\varepsilon]$ and let $z \in Z_x$ be the minimizer of Ψ_x from Step 2, so $v_\varepsilon(x) = v(z*x) + \|z\|^q/(2\varepsilon)$. Comparing with $z' = 0$,

$$v(z*x) + \frac{\|z\|^q}{2\varepsilon} = v_\varepsilon(x) \;\le\; \Psi_x(0) = v(x),$$

so $\|z\|^q \le 2\varepsilon\big(v(x) - v(z*x)\big) \le 2\varepsilon(M_v + M_v) = 4\varepsilon M_v$, using $|v| \le M_v$ on $\overline{\Omega}$ at the two points x and $z*x$ of $\overline{\Omega}$. Then $\|z\|^q \le 4\varepsilon M_v = 2\varepsilon(2M_v) \le 2\varepsilon R < c\varepsilon$ exactly as in (3b), and the same contradiction argument with $y := z*x'$ gives $z*x' \in \Omega$ for every $x' \in \Omega[\varepsilon]$. This is (4).

For (3), let $x, x' \in \Omega[\varepsilon]$ and let z be the minimizer at x. By (4), $z*x' \in \Omega \subseteq \overline{\Omega}$, so $z \in Z_{x'}$ and, by the minimality at x',

$$v_\varepsilon(x') \;\le\; \Psi_{x'}(z) = v(z*x') + \frac{\|z\|^q}{2\varepsilon}.$$

Subtracting the identity $v_\varepsilon(x) = v(z*x) + \|z\|^q/(2\varepsilon)$, the penalty terms cancel and

$$v_\varepsilon(x') - v_\varepsilon(x) \;\le\; v(z*x') - v(z*x) \;\le\; \omega_v\big(d(z*x', z*x)\big) = \omega_v\big(d(x',x)\big) = \omega_v\big(d(x,x')\big),$$

using (0b) and the symmetry of d. Exchanging x and x' gives the other inequality, and (3) follows.

Remark. The statement is part (4) of Lemma 3.8 as printed: both halves on $\Omega[\varepsilon]$, for both convolutions. The constant M_u plays the role of $\|u\|_{L^\infty}$, and the sharpest instance is $M_u = \|u\|_{L^\infty(\overline{\Omega})}$, $M_v = \|v\|_{L^\infty(\overline{\Omega})}$ and $R = 2\max\{M_u, M_v\}$, which is the paper's choice and satisfies $2M_u \le R$, $2M_v \le R$; the quantified form contains it. The reason the printed version suffices — and no restatement on $\Omega[\varepsilon/2]$ is needed — is that both halves are used only with both points in $\Omega[\varepsilon]$, which is exactly what (1)–(4) provide; the continuity of u^ε on $\overline{\Omega[\varepsilon]}$ that the comparison proofs also need is a separate matter, coming from local semi-convexity of u^ε on Ω together with $\overline{\Omega[\varepsilon]} \subseteq \Omega$, and not from the modulus half of this lemma. Note finally that no monotonicity or continuity of ω_u is used: the proof only ever evaluates ω_u at the single value $d(x,x')$. □

2.41. `UncondCaseC1`.

Theorem 2.41 (Theorem 1.3(i)). *Let $\mathbb{G}$ be a sub-elliptic setting (`CarnotSetting`) with group law $*$, homogeneous norm $\|\cdot\|$, step r, frame σ, $q := 2\,r!$ and $d(x,y) := \|y^{-1}*x\|$, and assume the cited results of the literature at $\mathbb{G}$ (`CitedResults`). Let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$ satisfy the structure conditions (`StructureConditionsES`). Let $\Omega \subset \mathbb{R}^n$ be a bounded open set with $\mathbb{R}^n \setminus \Omega \neq \emptyset$, let $u, v : \mathbb{R}^n \to \mathbb{R}$ be continuous on $\bar{\Omega}$ with u a viscosity sub-solution (`IsViscSubsol`) and v a viscosity super-solution (`IsViscSupersol`) of $\mathcal{L}\cdot = 0$ in Ω and $u \le v$ on $\partial\Omega$, and let $R \ge 0$ satisfy $|u| \le R/2$ and $|v| \le R/2$ on $\bar{\Omega}$. Let $N \subseteq \mathbb{R}^n$ be open with*

$$E := \big\{y \in \Omega :\; u - v \text{ attains its maximum over } \bar{\Omega} \text{ at } y\big\} \;\subseteq\; N.$$

Then:

1. *if v is C^1 on N and* hgrad σ z $D \neq 0$ *(`hgrad`) whenever $z \in N$ and D is the derivative of v at z, then $u \le v$ in Ω;*
2. *if u is C^1 on N and* hgrad σ z $D \neq 0$ *whenever $z \in N$ and D is the derivative of u at z, then $u \le v$ in Ω.*

Proof. This is Theorem 1.3(i), both orientations ("the case of u is symmetric, with u^ε in place of v_ε"). The comparison machinery is not re-derived here: `ComparisonUnderNondegeneracy` is Theorem 1.2, and the entire work of this node is to verify its non-degeneracy hypothesis for a suitable sequence $\varepsilon_k \to 0^+$. We prove (1) in Steps 0–8 and (2) in Step 9.

Step 0: reduction to building the non-degeneracy hypothesis. Assume the hypotheses of (1) and suppose the conclusion fails, so $u(x_0) > v(x_0)$ for some $x_0 \in \Omega$. We shall produce $\varepsilon_k > 0$ with $\varepsilon_k \to 0$ such that the non-degeneracy hypothesis of `ComparisonUnderNondegeneracy` holds for the family (ε_k); that node then gives $u \le v$ in Ω, contradicting $u(x_0) > v(x_0)$ and proving (1). This is the paper's "Theorem Theorem 1.2 gives $u \le v$, against $M_0 > 0$".

Since Ω is bounded, $\bar\Omega$ is compact, and it is non-empty ($x_0 \in \Omega \subseteq \bar\Omega$); $u - v$ is continuous on it, so $u - v$ attains its maximum over $\bar\Omega$ at some $x_M \in \bar\Omega$ and

$$M_0 := u(x_M) - v(x_M) \;\ge\; u(x_0) - v(x_0) \;>\; 0.$$

Put $\tau := M_0/2 > 0$ and write, for $\varepsilon > 0$,

$$u^\varepsilon := \mathrm{SupConv}\ \mathbb{G}\ \Omega\ u\ \varepsilon\ (\texttt{SupConv}), \qquad v_\varepsilon := \mathrm{InfConv}\ \mathbb{G}\ \Omega\ v\ \varepsilon\ (\texttt{InfConv}), \qquad \tilde v_\varepsilon := v_\varepsilon + \tau,$$

and $\Omega[\varepsilon]$ for the reduced domain (`ReducedDomain`) with this R.

All the hypotheses of `RegularizedPairStandingAssumptions` hold for ($\mathbb{G}$, cited results at $\mathbb{G}, A, H, \Omega, u, v, R$, Ω is open, bounded and has non-empty complement; u, v are continuous on $\bar\Omega$ and are a viscosity sub-/super-solution pair in Ω; $u \le v$ on $\partial\Omega$; $R \ge 0$ with $|u|, |v| \le R/2$ on $\bar\Omega$; $x_M \in \bar\Omega$ is a maximum point of $u - v$ over $\bar\Omega$; $M_0 > 0$; and $\tau = M_0/2$, $E = \mathrm{argmax}_\Omega(u - v)$ are as the node prescribes. It therefore yields

- (item 1) E is non-empty and compact;
- (item 2) an $\varepsilon_1 > 0$ such that for every $\varepsilon \in (0, \varepsilon_1)$: (b) u^ε and $\tilde v_\varepsilon$ are continuous on $\overline{\Omega[\varepsilon]}$; (c) u^ε is locally semi-convex and $\tilde v_\varepsilon$ locally semi-concave on $\Omega[\varepsilon]$ (`LocallySemiConvexOn`); and the remaining clauses (a), (d)–(g), which we do not need;
- (item 3) for every $\eta > 0$ an $\varepsilon_2 > 0$ such that for every $\varepsilon \in (0, \varepsilon_2)$ and every $z \in \Omega[\varepsilon]$ at which $u^\varepsilon - \tilde v_\varepsilon$ attains its maximum over $\overline{\Omega[\varepsilon]}$ there is $p \in E$ with $d(z, p) < \eta$.

Note once and for all that $\tilde v_\varepsilon = v_\varepsilon + \tau$ differs from v_ε by a constant, so for every set S and every x,

$$u^\varepsilon - \tilde v_\varepsilon \text{ is maximal over } S \text{ at } x \iff u^\varepsilon - v_\varepsilon \text{ is maximal over } S \text{ at } x, \tag{1}$$

since subtracting τ from both sides of an inequality between reals changes nothing; and a derivative of $\tilde v_\varepsilon$ at a point is the same continuous linear functional as a derivative of v_ε there.

Step 1: a uniform Euclidean thickness for N around E. E is compact by item 1 and $E \subseteq N$ with N open. Hence there is $\rho_N > 0$ such that

$$\text{for every } p \in E \text{ and every } w \text{ with } |w - p| < \rho_N, \quad w \in N. \tag{2}$$

Indeed, for each $p \in E$ pick $r_p > 0$ with the Euclidean ball $B(p, 2r_p) \subseteq N$; the balls $B(p, r_p)$ cover the compact E, so finitely many $B(p_1, r_{p_1}), \dots, B(p_k, r_{p_k})$ do; put $\rho_N := \min_j r_{p_j} > 0$. If $p \in E$ then $p \in B(p_j, r_{p_j})$ for some j, and $|w - p| < \rho_N \le r_{p_j}$ gives $|w - p_j| < 2r_{p_j}$, so $w \in N$. (The covering is a finite non-empty family because $E \neq \emptyset$ by item 1, so the minimum defining ρ_N is taken over a non-empty finite set and is positive.)

Step 2: converting d-closeness into Euclidean closeness. The homogeneous norm $\|\cdot\|$ is continuous and $\bar\Omega$ is compact, so

$$\nu_0 := 1 + \max\{|w| :\ w \in \bar\Omega\} \quad \text{and} \quad \mu_0 := 1 + \max\{\|w\| :\ w \in \bar\Omega\}$$

are finite and positive. By the Hölder equivalence clause of `CarnotSetting`, applied at ν_0, there is $c_0 > 0$ with $c_0^{-1}|a - b| \le d(a, b)$ whenever $|a| \le \nu_0$ and $|b| \le \nu_0$; since $|w| \le \nu_0$ for $w \in \bar\Omega$, this gives

$$|a - b| \;\le\; c_0\, d(a, b) \qquad (a, b \in \bar\Omega). \tag{3}$$

Step 3: the translation z_x of Lemma 3.8(4). Fix $\varepsilon > 0$. We apply `SupConvAttainment` with $M_u := R/2$, $M_v := R/2$ and the *constant* moduli

$$\omega_u := \big(s \mapsto R\big), \qquad \omega_v := \big(s \mapsto R\big).$$

Its hypotheses hold: Ω is open and bounded; u, v are continuous on $\bar\Omega$; $R \geq 0$, $2M_u = R \leq R$ and $2M_v = R \leq R$; $|u| \leq M_u$ and $|v| \leq M_v$ on $\bar\Omega$ by hypothesis; and the modulus conditions hold because for $a, b \in \bar\Omega$

$$|u(a) - u(b)| \leq |u(a)| + |u(b)| \leq \tfrac{R}{2} + \tfrac{R}{2} = R = \omega_u\big(d(a,b)\big),$$

and likewise for v. (No d-modulus of continuity has to be constructed: only the attainment clause is used below, and the constant moduli discharge the node's remaining hypotheses.) Its fourth clause gives, for every $x \in \Omega[\varepsilon]$, a point $z_x \in \mathbb{R}^n$ with

(4)

$$z_x * x \in \bar\Omega, \qquad v_\varepsilon(x) = v(z_x * x) + \frac{\|z_x\|^q}{2\varepsilon}, \qquad \|z_x\|^q \leq 4\varepsilon \cdot \frac{R}{2} = 2R\varepsilon, \qquad z_x * x' \in \Omega \ \ \forall x' \in \Omega[\varepsilon].$$

Step 4: the paper's display, both halves. Put $K := \dfrac{\|z_x\|^q}{2\varepsilon}$ and

$$\varphi(x') := v(z_x * x') + K.$$

The equality $v_\varepsilon(x) = \varphi(x)$ is the second item of (4). For the inequality, let $x' \in \Omega[\varepsilon]$. By the last item of (4), $z_x * x' \in \Omega \subseteq \bar\Omega$, so z_x belongs to the index set of the infimum defining $v_\varepsilon(x') = \mathrm{InfConv}\ \mathbb{G}\ \Omega\ v\ \varepsilon\ x'$ (`InfConv`), namely $\{p : \ p * x' \in \bar\Omega\}$. That infimum is over a non-empty family bounded below: each member is $v(p * x') + \|p\|^q/(2\varepsilon) \geq -R/2 + 0 = -R/2$, using $|v| \leq R/2$ on $\bar\Omega$ and $\|p\| \geq 0$, $\varepsilon > 0$. An infimum over a non-empty family bounded below is $\leq$ each member, so

$$v_\varepsilon(x') \ \leq\ v(z_x * x') + \frac{\|z_x\|^q}{2\varepsilon} = \varphi(x') \qquad (x' \in \Omega[\varepsilon]), \qquad \text{with equality at } x' = x. \tag{5}$$

This is exactly the paper's display; its inequality half is not a separate lemma but the defining property of an infimum, the only side condition being the lower bound just established.

Step 5: $z_x * x$ lies in N for small ε. First, $\|z_x\| \leq (2R\varepsilon)^{1/q}$ by (4) and $q \geq 2$ (as $q = 2\,r!$ and $r \geq 1$), and $\|z_x\| \geq 0$; so for $\varepsilon \leq 1$ we have $\|z_x\| \leq (2R+1)^{1/q}$. Put $\mu := \mu_0 + (2R+1)^{1/q} + 1 > 0$. Then, for $\varepsilon \leq 1$, $x \in \Omega[\varepsilon] \subseteq \Omega \subseteq \bar\Omega$ gives $\|x\| < \mu$ and also $\|z_x\| < \mu$. By the conjugation clause `hnorm_conj_le` of `CarnotSetting` (the second inequality of (2.17)), applied at this μ, there is $c_1 > 0$ with

$$\Big\|\big((-a) * b\big) * a\Big\| \leq c_1 \|b\|^{1/r} \qquad (\|a\| < \mu,\ \|b\| < \mu).$$

Taking $a := x$, $b := z_x$ and using associativity of $*$ in the form $(-x) * (z_x * x) = \big((-x) * z_x\big) * x$,

$$d(z_x * x,\ x) = \big\|(-x) * (z_x * x)\big\| \leq c_1 \|z_x\|^{1/r} \leq c_1 (2R\varepsilon)^{1/(qr)}. \tag{6}$$

The right-hand side tends to 0 as $\varepsilon \to 0^+$ and does not depend on x, so there is $\varepsilon_3 \in (0, 1]$ such that

$$d(z_x * x,\ x) < \frac{\rho_N}{2c_0} \qquad \text{for all } \varepsilon \in (0, \varepsilon_3) \text{ and all } x \in \Omega[\varepsilon]. \tag{7}$$

Both $z_x * x$ and x lie in $\bar\Omega$ (by (4) and $\Omega[\varepsilon] \subseteq \Omega$), so (3) and (7) give

$$|z_x * x - x| \ \leq\ c_0\, d(z_x * x, x) \ <\ \frac{\rho_N}{2}. \tag{8}$$

Now let $\varepsilon_2 > 0$ be the radius furnished by item 3 of Step 0 for $\eta := \dfrac{\rho_N}{2c_0}$, and set

$$\varepsilon^* := \min\{\varepsilon_1, \varepsilon_2, \varepsilon_3\} > 0.$$

Let $\varepsilon \in (0, \varepsilon^*)$ and let $x \in \Omega[\varepsilon]$ be a point at which $u^\varepsilon - \tilde v_\varepsilon$ is maximal over $\overline{\Omega[\varepsilon]}$. By item 3 there is $p \in E$ with $d(x, p) < \rho_N/(2c_0)$, and $x, p \in \bar\Omega$, so (3) gives $|x - p| < \rho_N/2$. Combined with (8),

$$|z_x * x - p| \leq |z_x * x - x| + |x - p| < \frac{\rho_N}{2} + \frac{\rho_N}{2} = \rho_N,$$

and therefore, by (2),

$$z_x * x \in N. \tag{9}$$

Step 6: the two derivatives at x. Keep $\varepsilon \in (0, \varepsilon^*)$ and x as at the end of Step 5.

(6a) u^ε and v_ε are differentiable at x, with the same derivative. By item 2(c), u^ε is locally semi-convex and $\tilde{v}_\varepsilon$ locally semi-concave on $\Omega[\varepsilon]$. Since $x \in \Omega[\varepsilon] \subseteq \overline{\Omega[\varepsilon]}$ and $u^\varepsilon - \tilde{v}_\varepsilon$ is maximal over $\overline{\Omega[\varepsilon]}$ at x, it is in particular maximal over the subset $\Omega[\varepsilon]$ at x. The second clause of `SemiConvexDiffAtMax`, applied with the set $\Omega[\varepsilon]$, the locally semi-convex u^ε and the locally semi-concave $\tilde{v}_\varepsilon$, therefore gives a continuous linear functional D_0 which is the derivative of u^ε at x *and* the derivative of $\tilde{v}_\varepsilon$ at x. As $\tilde{v}_\varepsilon = v_\varepsilon + \tau$ with τ constant, D_0 is also the derivative of v_ε at x. This is the paper's "v_ε is differentiable at x (Section Section 2.3(1))" together with "$\mathcal{X}u^\varepsilon = \mathcal{X}v_\varepsilon$ on E_ε".

(6b) φ is differentiable at x. By (9), $z_x * x \in N$; N is open and v is C^1 on N, so v is differentiable at $z_x * x$; let D_v be its derivative there. The map $y \mapsto z_x * y$ is C^∞ (a partial application of the C^∞ group law, `CarnotSetting`), so by the chain rule $y \mapsto v(z_x * y)$ is differentiable at x; let D' be its derivative. Adding the constant K does not change the derivative, so D' is the derivative of φ at x.

(6c) the derivatives coincide. By (5), $v_\varepsilon - \varphi \le 0$ on $\Omega[\varepsilon]$ with equality at x. The set $\Omega[\varepsilon]$ is open (item (4) of `ReducedDomainProperties`, using $\varepsilon > 0$, Ω open and bounded with non-empty complement, $R \ge 0$) and contains x, so $v_\varepsilon - \varphi$ has a local maximum at x. Both v_ε and φ are differentiable at x, by (6a) and (6b), so $v_\varepsilon - \varphi$ is differentiable at x with derivative $D_0 - D'$; a differentiable function with a local maximum at an interior point has vanishing derivative there, so

$$D_0 = D'. \tag{10}$$

This is the paper's "the right-hand side is then a C^1 function of x' near x touching the differentiable function v_ε from above at x, so their gradients coincide at x". Only differentiability of φ *at* x is used, which is why (6b) needs v differentiable only at the single point $z_x * x$.

Step 7: transporting the non-vanishing. We have: D_v is the derivative of v at $z_x * x$, and D' is the derivative of $y \mapsto v(z_x * y)$ at x. `LeftTranslateHgrad`, applied with the function v, the group element $g := z_x$ and the point x, gives

$$\text{hgrad } \sigma\ x\ D' \ =\ \text{hgrad } \sigma\ (z_x * x)\ D_v. \tag{11}$$

(This is the paper's "by left invariance $\mathcal{X}v_\varepsilon(x) = \mathcal{X}(v \circ L_{z_x})(x) = (\mathcal{X}v)(z_x * x)$". The second-order left-invariance clause `leftJet` of `CarnotSetting` is unusable here: it requires a second-order jet of the function at $z_x * x$, whereas v is assumed only C^1 and v_ε is only once differentiable at x. That is precisely what `LeftTranslateHgrad` supplies.)

By (9) and the hypothesis of (1), hgrad σ $(z_x * x)$ $D_v \neq 0$; so by (11) and (10),

$$\text{hgrad } \sigma\ x\ D_0 = \text{hgrad } \sigma\ x\ D' \neq 0. \tag{12}$$

Step 8: the sequence, and conclusion of (1). Define

$$\varepsilon_k := \frac{\varepsilon^*}{k+2} \qquad (k \in \mathbb{N}),$$

so $0 < \varepsilon_k < \varepsilon^*$ for every k and $\varepsilon_k \to 0$.

We check the non-degeneracy hypothesis of `ComparisonUnderNondegeneracy` for this family. Let $k \in \mathbb{N}$, let $x \in \Omega[\varepsilon_k]$ be such that $u^{\varepsilon_k} - v_{\varepsilon_k}$ is maximal over $\overline{\Omega[\varepsilon_k]}$ at x, and let D be the derivative of u^{ε_k} at x. By (1) of Step 0, $u^{\varepsilon_k} - \tilde{v}_{\varepsilon_k}$ is also maximal over $\overline{\Omega[\varepsilon_k]}$ at x, so Steps 3–7 apply with $\varepsilon := \varepsilon_k$ and this x, giving D_0 as in (6a) with (12). Since the Fréchet derivative of u^{ε_k} at x is unique and both D and D_0 are derivatives of u^{ε_k} at x, we get $D = D_0$ and hence hgrad σ x $D \neq 0$ by (12). That is the required hypothesis.

All the other hypotheses of `ComparisonUnderNondegeneracy` are our own: the cited results, the structure conditions on (A, H), Ω bounded open with non-empty complement, u, v continuous on $\bar{\Omega}$, the viscosity inequalities, $u \le v$ on $\partial\Omega$, and $R \ge 0$ with $|u|, |v| \le R/2$ on $\bar{\Omega}$. It therefore yields $u \le v$ in Ω, contradicting $u(x_0) > v(x_0)$. Hence (1) holds.

Step 9: part (2), the symmetric orientation. Assume now the hypotheses of (2) and suppose again $u(x_0) > v(x_0)$ for some $x_0 \in \Omega$. Steps 0, 1, 2 and the choice of $\varepsilon_1, \varepsilon_2$ are unchanged. In Step 3 we use instead the *second* clause of `SupConvAttainment` (the one for SupConv), with the same $M_u = M_v = R/2$ and the same constant moduli: for every $x \in \Omega[\varepsilon]$ there is z_x with

$$z_x * x \in \bar{\Omega}, \quad u^\varepsilon(x) = u(z_x * x) - \frac{\|z_x\|^q}{2\varepsilon}, \quad \|z_x\|^q \le 2R\varepsilon, \quad z_x * x' \in \Omega \ \ \forall x' \in \Omega[\varepsilon].$$

In Step 4, put $\psi(x') := u(z_x * x') - \|z_x\|^q/(2\varepsilon)$; for $x' \in \Omega[\varepsilon]$ the point z_x belongs to the index set of the supremum defining $u^\varepsilon(x')$ (`SupConv`), and that supremum is over a non-empty family bounded above (each member is $u(p * x') - \|p\|^q/(2\varepsilon) \le R/2$, by $|u| \le R/2$ on $\bar{\Omega}$ and $\|p\| \ge 0$, $\varepsilon > 0$), so a supremum is $\ge$ each member:

$$u^\varepsilon(x') \ \ge \ \psi(x') \qquad (x' \in \Omega[\varepsilon]), \qquad \text{with equality at } x' = x.$$

Steps 5 and (6a) are verbatim; in (6b), u is C^1 on N and $z_x * x \in N$, so u is differentiable at $z_x * x$ with derivative D_u, and $y \mapsto u(z_x * y)$ is differentiable at x with some derivative D', which is also the derivative of ψ at x. In (6c) the roles reverse: $u^\varepsilon - \psi \ge 0$ on the open $\Omega[\varepsilon]$ with equality at x, so $u^\varepsilon - \psi$ has a local *minimum* at x; it is differentiable there with derivative $D_0 - D'$ (with D_0 the derivative of u^ε at x from (6a)), and a differentiable function with a local minimum at an interior point has vanishing derivative, so again $D_0 = D'$. Step 7 is `LeftTranslateHgrad` applied to the function u with $g := z_x$, giving hgrad σ x D' = hgrad σ $(z_x * x)$ $D_u \ne 0$ by the hypothesis of (2) and (9). Step 8 is unchanged — indeed slightly shorter, since D_0 is already a derivative of u^ε at x and uniqueness of the Fréchet derivative gives $D = D_0$ directly. So hgrad σ x $D \ne 0$, and `ComparisonUnderNondegeneracy` gives $u \le v$ in Ω, contradicting $u(x_0) > v(x_0)$. Hence (2) holds. □

2.42. `UncondCaseConstant`.

Theorem 2.42 (Theorem 1.3(ii))**.** *Let $\mathbb{G}$ be a sub-elliptic setting (`CarnotSetting`) with frame σ, and assume the cited results of the literature at $\mathbb{G}$ (`CitedResults`). Let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$ satisfy the structure conditions (`StructureConditionsES`). Let $\Omega \subset \mathbb{R}^n$ be a bounded open set with $\mathbb{R}^n \setminus \Omega \ne \emptyset$, and let $u, v : \mathbb{R}^n \to \mathbb{R}$ be continuous on $\bar{\Omega}$ with u a viscosity sub-solution (`IsViscSubsol`) and v a viscosity super-solution (`IsViscSupersol`) of $\mathcal{L}\cdot = 0$ in Ω, and $u \le v$ on $\partial\Omega$. Then*

(1) *if v is constant on $\bar{\Omega}$, then $u \le v$ in Ω;*
(2) *if u is constant on $\bar{\Omega}$, then $u \le v$ in Ω.*

Proof. This is Theorem 1.3(ii), both orientations as the paper states them ("The case u constant uses the strong minimum principle for v"). Throughout, $\bar{\Omega}$ is compact because Ω is bounded, $u - v$ is continuous on $\bar{\Omega}$, and $\partial\Omega = \bar{\Omega} \setminus \Omega$ because Ω is open.

Step 0: the topological core. We first prove the following, which is the paper's last clause and is common to both cases. *Let $y \in \Omega$, let $C :=$ the connected component of Ω containing y, and let $M > 0$ be such that $u - v \equiv M$ on C. Then a contradiction follows.*

Since $\mathbb{R}^n$ is locally connected and Ω is open, C is open; $C \subseteq \Omega$, and $y \in C$.

$\partial C \ne \emptyset$. If $\partial C = \emptyset$ then C is both closed and open in $\mathbb{R}^n$; $\mathbb{R}^n$ is connected and $C \ne \emptyset$ (it contains y), so $C = \mathbb{R}^n$, whence $\Omega \supseteq C = \mathbb{R}^n$. That contradicts $\mathbb{R}^n \setminus \Omega \ne \emptyset$. (This is the only use of that hypothesis; it makes explicit the paper's convention $\Omega \subset \mathbb{R}^n$, and it is automatic for a bounded Ω once $n \ge 1$.)

Fix $z \in \partial C$. Since C is open, $\partial C = \bar{C} \setminus C$, so $z \in \bar{C}$ and $z \notin C$.

$\partial C \subseteq \partial\Omega$. Suppose $z \in \Omega$, and let C' be the connected component of Ω containing z. Then C' is open and $z \in C'$, so, z being in $\bar{C}$, the open set C' meets C: there is $w \in C' \cap C$. Two

components of Ω containing a common point coincide, so $C' = C$ (both equal the component of Ω containing w), and therefore $z \in C$, contradicting $z \notin C$. Hence $z \notin \Omega$; and $z \in \bar{C} \subseteq \bar{\Omega}$, so $z \in \bar{\Omega} \setminus \Omega = \partial\Omega$.

The contradiction. Since $z \in \bar{C}$, there is a sequence (z_k) in C with $z_k \to z$. For every k, $(u - v)(z_k) = M$ by hypothesis. All z_k lie in $C \subseteq \Omega \subseteq \bar{\Omega}$ and $z \in \partial\Omega \subseteq \bar{\Omega}$, and $u - v$ is continuous on $\bar{\Omega}$, so $(u - v)(z_k) \to (u - v)(z)$; the left-hand side is the constant sequence M, so $(u - v)(z) = M > 0$. But $z \in \partial\Omega$, where $u \leq v$, i.e. $(u - v)(z) \leq 0$. This is the required contradiction.

Step 1: part (1), v constant. Let $v \equiv c_v$ on $\bar{\Omega}$ and suppose the conclusion fails: $u(x_0) > v(x_0)$ for some $x_0 \in \Omega$.

Since $\bar{\Omega}$ is compact and non-empty ($x_0 \in \Omega \subseteq \bar{\Omega}$) and $u - v$ is continuous on it, $u - v$ attains its maximum over $\bar{\Omega}$ at some $x_M \in \bar{\Omega}$, and

$$M_0 := u(x_M) - v(x_M) \;\geq\; u(x_0) - v(x_0) \;>\; 0.$$

If $x_M \notin \Omega$ then $x_M \in \bar{\Omega} \setminus \Omega = \partial\Omega$ and $M_0 = (u - v)(x_M) \leq 0$, a contradiction; so $x_M \in \Omega$.

u attains its maximum over $\bar{\Omega}$ at x_M. For $z \in \bar{\Omega}$ we have $(u - v)(z) \leq (u - v)(x_M)$, and $v(z) = c_v = v(x_M)$, so $u(z) \leq u(x_M)$. This is the paper's "$\operatorname{argmax}_\Omega(u - v) = \operatorname{argmax}_\Omega u$ … u attains an interior maximum".

Let C be the connected component of Ω containing x_M; it is open (local connectedness of $\mathbb{R}^n$, Ω open), connected, contains x_M and is contained in Ω.

The shifted sub-solution. Put $\tilde{u} := u - u(x_M)$, which is the paper's "adding a constant we may take it non-negative". By `AddConstPreservesSubsol`, applied with the constant $-u(x_M)$, $\tilde{u}$ is a viscosity sub-solution of $\mathcal{L}\cdot = 0$ in Ω. Moreover $\tilde{u} \leq 0$ on Ω by the previous paragraph, so $\tilde{u}$ is bounded above on Ω (by 0). Part (1b) of `ViscTestLocalization` — whose hypotheses are exactly that $\tilde{u}$ be bounded above on Ω and a viscosity sub-solution there — gives that $\tilde{u}$ is a viscosity sub-solution in every open subset of Ω, in particular in C. This step is not free: pm-0009 records that our rendering of Definition 2.1 has the test condition $x \in \operatorname{argmax}_\Omega(w - \varphi)$ global over the domain, so restriction to a smaller open set needs `ViscTestLocalization`.

The strong maximum principle. For $z \in C \subseteq \Omega \subseteq \bar{\Omega}$ we have $u(z) \leq u(x_M)$, i.e. $\tilde{u}(z) \leq 0 = \tilde{u}(x_M)$; so $\tilde{u}$ attains its maximum over C at $x_M \in C$, and that maximum value is $\tilde{u}(x_M) = 0 \geq 0$. The clause `strongMax` of `CitedResults` — the Bardi–Goffi strong maximum principle in the form Corollary 2.4 — applied with the structure conditions on (A, H), the open connected set C, the sub-solution $\tilde{u}$ and the point x_M, gives $\tilde{u}(z) = \tilde{u}(x_M) = 0$ for every $z \in C$, i.e.

$$u(z) = u(x_M) \qquad (z \in C).$$

Conclusion. For $z \in C$ we therefore have, using $v(z) = c_v = v(x_M)$,

$$u(z) - v(z) = u(x_M) - v(x_M) = M_0 > 0,$$

so $u - v \equiv M_0$ on C with $M_0 > 0$. Step 0, applied with $y := x_M$ and $M := M_0$, gives a contradiction. Hence $u \leq v$ in Ω, which is (1).

Step 2: part (2), u constant. Let $u \equiv c_u$ on $\bar{\Omega}$ and suppose $u(x_0) > v(x_0)$ for some $x_0 \in \Omega$. Exactly as in Step 1, $u - v$ attains its maximum over the compact non-empty $\bar{\Omega}$ at some x_M, $M_0 := (u - v)(x_M) > 0$, and $x_M \in \Omega$ because $u - v \leq 0$ on $\partial\Omega$.

v attains its minimum over $\bar{\Omega}$ at x_M. For $z \in \bar{\Omega}$, $(u - v)(z) \leq (u - v)(x_M)$ and $u(z) = c_u = u(x_M)$, so $v(x_M) \leq v(z)$.

Let C be the connected component of Ω containing x_M, open and connected as before. Put $\tilde{v} := v - v(x_M)$. By `AddConstPreservesSubsol` (super-solution half, with the constant $-v(x_M)$), $\tilde{v}$ is a viscosity super-solution in Ω; and $\tilde{v} \geq 0$ on Ω by the previous paragraph, so it is bounded below on Ω (by 0). Part (2b) of `ViscTestLocalization` then gives that $\tilde{v}$ is a viscosity super-solution in every open subset of Ω, in particular in C.

For $z \in C$ we have $\tilde{v}(x_M) = 0 \leq \tilde{v}(z)$, so $\tilde{v}$ attains its minimum over C at $x_M \in C$, with value $\tilde{v}(x_M) = 0 \leq 0$. The clause `strongMin` of `CitedResults` — the super-solution side of Corollary 2.4 — applied with the structure conditions, the open connected C, the super-solution $\tilde{v}$ and the point x_M, gives $\tilde{v}(z) = \tilde{v}(x_M) = 0$ for every $z \in C$, i.e. $v(z) = v(x_M)$ on C.

Hence for $z \in C$, using $u(z) = c_u = u(x_M)$,

$$u(z) - v(z) = u(x_M) - v(x_M) = M_0 > 0,$$

and Step 0 with $y := x_M$, $M := M_0$ gives a contradiction. Hence $u \le v$ in Ω, which is (2). □

2.43. `EuclideanCarnotSettingExists`.

Theorem 2.43 (The Euclidean additive group on $\mathbb{R}^N$ is a Carnot setting)**.** *For every $N \in \mathbb{N}$ there exists a Carnot setting $\mathbb{G}$ (`CarnotSetting`) whose data (n, m, r), group law $*$, first-layer projection π_1, horizontal exponential $\exp_{\mathrm{h}}$, homogeneous norm $\|\cdot\|$ and horizontal frame $\sigma = [\sigma^1, \dots, \sigma^m]$ satisfy*

(1) *$n = N$, $m = N$ and $r = 1$;*
(2) *$x * y = x + y$ for all $x, y \in \mathbb{R}^n$;*
(3) *$\|x\| = |x|$ for all $x \in \mathbb{R}^n$, where $|\cdot|$ is the Euclidean norm;*
(4) *$\exp_{\mathrm{h}}(\pi_1(x)) = x$ for all $x \in \mathbb{R}^n$;*
(5) *$\sigma^i(x) = \exp_{\mathrm{h}}(e_i)$ for all $x \in \mathbb{R}^n$ and all $i \in \{1, \dots, m\}$, where $e_1, \dots, e_m$ is the standard basis of $\mathbb{R}^m$.*

Proof. Fix $N \in \mathbb{N}$ and write $e_1, \dots, e_N$ for the standard basis of $\mathbb{R}^N$. We exhibit the data and verify, one at a time, every condition of `CarnotGroup`, `CarnotNorm` and `CarnotSetting`, and then conditions (1)–(5).

The data. Put

$$n := N, \qquad m := N, \qquad r := 1,$$

$$x * y := x + y, \qquad \pi_1 := \mathrm{id}_{\mathbb{R}^N}, \qquad \exp_{\mathrm{h}} := \mathrm{id}_{\mathbb{R}^N}, \qquad \|x\| := |x|, \qquad \sigma^i(x) := e_i \ \ (i = 1, \dots, N).$$

Here $\mathrm{id}_{\mathbb{R}^N}$ is the identity, which is a continuous linear map $\mathbb{R}^N \to \mathbb{R}^N$, as required of π_1; and σ is the constant frame, so the associated first-order operators are $X_i = e_i \cdot \nabla = \partial_{x_i}$, which is the frame the paper attaches to the Euclidean group (Section Section 1.2: "the case $m = n$ corresponds to $r = 1$ and $\mathbb{G}$ being the Euclidean additive group, with $X_i = \partial_{x_i}$"). Since $r = 1$ we have $q := 2\,r! = 2$, and the left-invariant distance of `CarnotNorm` is

$$d(x, y) = \|(-y) * x\| = |(-y) + x| = |x - y|,$$

the Euclidean distance.

Step 1: the conditions of `CarnotGroup`.

(i) $r \ge 1$. Immediate from $r = 1$.

(ii) $(\mathbb{R}^N, *)$ *is a group with neutral element* 0 *and inversion* $x \mapsto -x$. The five required identities are $(x + y) + z = x + (y + z)$, $x + 0 = x$, $0 + x = x$, $x + (-x) = 0$ and $(-x) + x = 0$; these are the axioms of the additive group $\mathbb{R}^N$.

(iii) The group law is C^∞. The map $\mathbb{R}^N \times \mathbb{R}^N \to \mathbb{R}^N$, $(x, y) \mapsto x + y$, is the sum of the two coordinate projections. Each projection is continuous and linear, hence C^∞, and a sum of C^∞ maps is C^∞.

(iv) $\pi_1(x * y) = \pi_1(x) + \pi_1(y)$. Both sides equal $x + y$ because π_1 is the identity.

(v) $\exp_{\mathrm{h}}$ *is continuous and* $\pi_1(\exp_{\mathrm{h}}(\zeta)) = \zeta$. The identity is continuous, and $\pi_1 \circ \exp_{\mathrm{h}} = \mathrm{id}$.

(vi) $r = 1$ *implies* $*$ *commutative.* Addition on $\mathbb{R}^N$ is commutative.

Step 2: the conditions of `CarnotNorm`.

(i) $\|x\| \ge 0$; $\|x\| = 0$ *iff* $x = 0$; $\|-x\| = \|x\|$; $\|\cdot\|$ *continuous.* These are the defining properties of the Euclidean norm $|\cdot|$ together with its continuity.

(ii) $\|\exp_{\mathrm{h}}(\zeta)\| = |\zeta|$. Since $\exp_{\mathrm{h}}$ is the identity and $\|\cdot\| = |\cdot|$, both sides are $|\zeta|$.

(iii) $x \mapsto \|x\|^q$ *is C^∞.* Here $q = 2$, and $|x|^2 = \sum_{i=1}^N x_i^2$ is a polynomial in the coordinates of x, hence C^∞.

(iv) First inequality of (2.17). Given $\nu > 0$, take $c := 1 > 0$. For all x, y (in particular whenever $\|x\| < \nu$ and $\|y\| < \nu$) the triangle inequality for the Euclidean norm gives $\|x * y\| = |x + y| \le |x| + |y| = c\,(\|x\| + \|y\|)$.

(v) Second inequality of (2.17). Given $\nu > 0$, take $c := 1 > 0$. Since addition is commutative, $x^{-1} * y * x = ((-x) + y) + x = y + ((-x) + x) = y$, so $\|x^{-1} * y * x\| = |y|$. As $r = 1$ we have $1/r = 1$, and $c\,\|y\|^{1/r} = |y|^1 = |y|$. Hence the required inequality holds with equality.

(vi) Hölder equivalence on bounded sets. Given $\nu > 0$, take $c := 1 > 0$. For all x, y we computed $d(x, y) = \|(-y) * x\| = |x - y|$, and with $1/r = 1$ the two required inequalities read $c^{-1}|x - y| = |x - y| \le |x - y|$ and $|x - y| \le c\,|x - y|^1 = |x - y|$; both hold with equality, in particular whenever $|x| \le \nu$ and $|y| \le \nu$.

(vii) Balls are non-empty and connected. Fix $x \in \mathbb{R}^N$ and $\rho > 0$. By the computation of d,

$$B_\rho(x) = \{y : \ \|(-x) * y\| < \rho\} = \{y : \ |y - x| < \rho\},$$

the Euclidean open ball of radius ρ about x. It contains x, since $|x - x| = 0 < \rho$, so it is non-empty; and it is convex, hence connected.

(viii) Properness. Let $C \in \mathbb{R}$. Then $\{x : \|x\| \le C\} = \{x : |x| \le C\}$ is the closed Euclidean ball of radius C about the origin (empty if $C < 0$), which is bounded.

Step 3: two computations for the constant frame. Both are used repeatedly below.

(3a) The horizontal gradient reads off the frame directions. By `hgrad`, for every x, every 1-jet D and every i,

$$\big(\text{hgrad } \sigma\ x\ D\big)_i = D\big(\sigma^i(x)\big) = D(e_i).$$

In particular this value does not depend on x.

(3b) The correction matrix vanishes, so the symmetrized horizontal second derivative is the matrix of the 2-jet in the frame directions. By `Mfield`,

$$\mathcal{M}(x, D)_{i,j} = \tfrac{1}{2}\Big(D\big(D\sigma^j(x)\,\sigma^i(x)\big) + D\big(D\sigma^i(x)\,\sigma^j(x)\big)\Big).$$

Each σ^k is a constant map, so its Fréchet derivative $D\sigma^k(x)$ is the zero linear map; hence $D\sigma^j(x)\sigma^i(x) = 0$ and $D\sigma^i(x)\sigma^j(x) = 0$, and since D is linear, $D(0) = 0$. Therefore $\mathcal{M}(x, D) = 0$ for every x and D, and by `hhessS`, for every x, every 1-jet D and every 2-jet B,

$$\big(\text{hhessS } \sigma\ x\ D\ B\big)_{i,j} = B\big(\sigma^i(x), \sigma^j(x)\big) + \mathcal{M}(x, D)_{i,j} = B(e_i, e_j),$$

again independently of x.

Step 4: the conditions of `CarnotSetting`.

(i) Each σ^i is C^∞. Each σ^i is constant, hence C^∞.

(ii) $\sigma^1(x), \dots, \sigma^N(x)$ are linearly independent. They are $e_1, \dots, e_N$, the standard basis of $\mathbb{R}^N$, which is linearly independent.

(iii) Hörmander's rank condition (1.6). Fix $x \in \mathbb{R}^N$; we must show that the span of $\{V(x) : V \in \text{LieHull } \sigma\}$ is all of $\mathbb{R}^N$. By `LieHull`, LieHull σ is the intersection of all families of vector fields that contain each σ^i and are closed under the Lie bracket. Every such family contains the vector field $y \mapsto \sigma^i(y) = e_i$, by the first clause of admissibility; hence that vector field belongs to the intersection, i.e. $(y \mapsto e_i) \in \text{LieHull } \sigma$ for each i. Evaluating at x gives

$$\{e_1, \dots, e_N\} \ \subseteq \ \{V(x) : V \in \text{LieHull } \sigma\},$$

and therefore $\mathbb{R}^N = \text{span}\{e_1, \dots, e_N\} \subseteq \text{span}\{V(x) : V \in \text{LieHull } \sigma\} \subseteq \mathbb{R}^N$, so the two spans coincide with $\mathbb{R}^N$. Note that only the inclusion $\sigma^i \in \text{LieHull } \sigma$ is used; the hull is never computed. (It is in fact exhausted at once, since the bracket $[V, W](x) = DW(x)V(x) - DV(x)W(x)$ of two constant fields is 0, but this is not needed.)

(iv) The Taylor formula (2.18). Let $u : \mathbb{R}^N \to \mathbb{R}$ be differentiable at x with differential D. We must show that u is Pansu differentiable at x with horizontal differential hgrad $\sigma\ x\ D$ in the sense of `HasPansuGradAt`: for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$\Big|\, u(x * g) - u(x) - \big\langle \text{hgrad } \sigma\ x\ D,\ \pi_1(g) \big\rangle \,\Big| \ \le \ \varepsilon\,\|g\| \qquad \text{whenever } \|g\| < \delta.$$

Here $x * g = x + g$, $\pi_1(g) = g$ and $\|g\| = |g|$. The pairing simplifies: writing $g = \sum_{i=1}^N g_i\, e_i$ and using (3a) and the linearity of D,

$$\langle \operatorname{hgrad} \sigma\, x\, D,\ \pi_1(g)\rangle = \sum_{i=1}^N \big(\operatorname{hgrad} \sigma\, x\, D\big)_i\, g_i = \sum_{i=1}^N D(e_i)\, g_i = \sum_{i=1}^N D\big(g_i\, e_i\big) = D\Big(\sum_{i=1}^N g_i\, e_i\Big) = D(g).$$

So the assertion to be proved is: for every $\varepsilon > 0$ there is $\delta > 0$ with $|u(x+g)-u(x)-D(g)| \le \varepsilon|g|$ whenever $|g| < \delta$. This is exactly differentiability of u at x with differential D: by definition of the Fréchet derivative, $u(y) - u(x) - D(y-x) = o(|y-x|)$ as $y \to x$, so given $\varepsilon > 0$ there is $\delta > 0$ with $|u(y) - u(x) - D(y-x)| \le \varepsilon|y-x|$ for $|y-x| < \delta$; substituting $y = x+g$ and using $(x+g) - x = g$ gives the claim with the same δ.

(v) Left-invariance of the horizontal derivatives. Let $h, x \in \mathbb{R}^N$, let $\varphi : \mathbb{R}^N \to \mathbb{R}$ and let (D, B) be a second-order jet of φ at $h*x = h+x$ in the sense of `TwiceDiffAt`. We claim that $(D', B') := (D, B)$ has the three required properties for $\varphi \circ L_h$, $L_h(y) = h + y$. (This is the expected transport $D' = D \circ DL_h(x)$, $B'(u,v) = B(DL_h(x)u, DL_h(x)v) + D(D^2L_h(x)(u,v))$ specialized to $DL_h(x) = \mathrm{id}$ and $D^2L_h(x) = 0$, which is what makes the Euclidean case immediate.)

(D, B) is a second-order jet of $\varphi \circ L_h$ at x. The symmetry of B is part of the hypothesis. For the expansion, let $\varepsilon > 0$ and take $\delta > 0$ from the hypothesis at $h + x$, so that

$$\Big|\varphi(y') - \varphi(h+x) - D\big(y' - (h+x)\big) - \tfrac{1}{2}B\big(y' - (h+x), y' - (h+x)\big)\Big| \le \varepsilon\, \big|y' - (h+x)\big|^2$$

for every y' with $|y' - (h+x)| < \delta$. Now let $y \in \mathbb{R}^N$ with $|y - x| < \delta$ and put $y' := h + y$. Then $y' - (h+x) = (h+y) - (h+x) = y - x$, so $|y' - (h+x)| = |y - x| < \delta$, the displayed inequality applies, and after substituting $y' - (h+x) = y - x$ it reads

$$\Big|(\varphi \circ L_h)(y) - (\varphi \circ L_h)(x) - D(y-x) - \tfrac{1}{2}B(y-x, y-x)\Big| \le \varepsilon\, |y-x|^2,$$

because $(\varphi \circ L_h)(y) = \varphi(h+y)$ and $(\varphi \circ L_h)(x) = \varphi(h+x)$. This is the required expansion, with the same δ.

Equality of the horizontal gradients. By (3a), the i-th component of both hgrad $\sigma\ x\ D$ and hgrad $\sigma\ (h+x)\ D$ is $D(e_i)$; hence the two vectors are equal.

Equality of the symmetrized horizontal second derivatives. By (3b), the (i,j) entry of both hhessS $\sigma\ x\ D\ B$ and hhessS $\sigma\ (h+x)\ D\ B$ is $B(e_i, e_j)$; hence the two matrices are equal.

(vi) The form of the X_i on functions of the first layer. Let $\psi : \mathbb{R}^m \to \mathbb{R}$ and let (E, C) be a second-order jet of ψ at $\pi_1(x)$ in the sense of `TwiceDiffAt`. Since $m = n = N$ and π_1 is the identity we have $\pi_1(x) = x$ and $\psi \circ \pi_1 = \psi$. Take $(D, B) := (E, C)$. Then (D, B) is a second-order jet of $\psi \circ \pi_1 = \psi$ at $x = \pi_1(x)$, which is the hypothesis itself. By (3a) the i-th component of hgrad $\sigma\ x\ E$ is $E(e_i)$, so hgrad $\sigma\ x\ E = (E(e_i))_{i=1}^m$ as required; and by (3b) the (i,j) entry of hhessS $\sigma\ x\ E\ C$ is $C(e_i, e_j)$, so hhessS $\sigma\ x\ E\ C = (C(e_i, e_j))_{i,j=1}^m$ as required.

Step 5: the five listed properties. By construction $n = N$, $m = N$ and $r = 1$, which is (1); $x * y = x + y$, which is (2); and $\|x\| = |x|$, which is (3). For (4), $\exp_{\mathrm{h}}$ and π_1 are both the identity, so $\exp_{\mathrm{h}}(\pi_1(x)) = x$. For (5), $\sigma^i(x) = e_i$ and $\exp_{\mathrm{h}}(e_i) = e_i$, so $\sigma^i(x) = \exp_{\mathrm{h}}(e_i)$.

This exhibits the required Carnot setting and completes the proof.

Remark on the role of this node. Item 3 of the paper's note for the formalization permits the structural facts of Section Section 2.4 to be built into the definition of the setting only "provided every Carnot group in the coordinates of Section Section 2.4 satisfies that definition". The theorem above discharges that proviso at the one instance that can be checked outright, and it is the instance Theorem Theorem 1.3(iii) needs: the comparison principle for $r = 1$ is applied in $\mathbb{R}^m$, on the abelianization, so a Carnot setting on $\mathbb{R}^m$ with $x * y = x + y$ and $X_i = \partial_{x_i}$ must exist for that step to be available at all. Properties (4) and (5) record how that setting is identified with the Euclidean one: by (4) and the field $\pi_1(\exp_{\mathrm{h}}(\zeta)) = \zeta$ of `CarnotGroup`, the continuous linear map π_1 is a bijection with two-sided inverse $\exp_{\mathrm{h}}$, so $\exp_{\mathrm{h}}$ is linear as well; by (3) and the field $\|\exp_{\mathrm{h}}(\zeta)\| = |\zeta|$ of `CarnotNorm` it is an isometry; and by (5) the frame is the image $\exp_{\mathrm{h}}(e_1), \dots, \exp_{\mathrm{h}}(e_m)$ of the standard basis, so by (3a) and (3b) the horizontal gradient and the symmetrized horizontal second derivative of a jet are its Euclidean gradient and Hessian

read in that orthonormal basis. For $N = 0$ the setting is the trivial group, which is a Carnot group of step 1 with no generators; the case used in the paper is $N \geq 1$. □

2.44. `EuclidFirstLayerComparison`.

Theorem 2.44 (The case $r = 1$ applied in $\mathbb{R}^m$)**.** *Let $m \in \mathbb{N}$, write $\mathbb{G}_E$ for the Euclidean setting on $\mathbb{R}^m$ (`EuclidSetting` at $N := m$), and assume the cited results of the literature* at *$\mathbb{G}_E$ (`CitedResults`). Let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$ satisfy the structure conditions (`StructureConditionsES`). Let $\Omega' \subseteq \mathbb{R}^m$ be open and bounded with $\mathbb{R}^m \setminus \Omega' \neq \emptyset$, and let $F, Gf : \mathbb{R}^m \to \mathbb{R}$ be continuous on $\overline{\Omega'}$ with $F \leq Gf$ on $\partial\Omega'$. Let $R \geq 0$ satisfy $|F| \leq R/2$ and $|Gf| \leq R/2$ on $\overline{\Omega'}$. Write e_i for the i-th standard basis vector of $\mathbb{R}^m$ and, for a continuous linear functional E on $\mathbb{R}^m$ and a continuous bilinear form C on $\mathbb{R}^m$,*

$$\xi(E) := \big(E(e_1), \dots, E(e_m)\big) \in \mathbb{R}^m, \qquad \mathcal{C}(C) := \big(C(e_i, e_j)\big)_{i,j} \in \mathbb{R}^{m\times m}.$$

Assume

(1) *for every $\zeta \in \Omega'$, every $\psi : \mathbb{R}^m \to \mathbb{R}$ that is C^2 on Ω' with $\zeta \in \operatorname{argmax}_{\Omega'}(F - \psi)$ and every second-order jet (E, C) of ψ at ζ (`TwiceDiffAt`) with $\xi(E) \neq 0$: $-\operatorname{Tr}\big(A(\xi(E))\,\mathcal{C}(C)\big) + H(\xi(E)) \leq 0$;*

(2) *for every $\zeta \in \Omega'$, every ψ that is C^2 on Ω' with $\zeta \in \operatorname{argmin}_{\Omega'}(Gf - \psi)$ and every second-order jet (E, C) of ψ at ζ with $\xi(E) \neq 0$: $0 \leq -\operatorname{Tr}\big(A(\xi(E))\,\mathcal{C}(C)\big) + H(\xi(E))$.*

Then $F \leq Gf$ in Ω'.

Proof. This is the paper's final sentence of Theorem 1.3(iii), "By the case $r = 1$ applied in $\mathbb{R}^m$, $F \leq Gf$ in Ω'". The case $r = 1$ of the paper is our `ComparisonAbelian`, which is stated for a sub-elliptic setting; the work of this node is to translate the two inline viscosity conditions into the notion of viscosity sub- and super-solution of the Euclidean setting $\mathbb{G}_E$ on $\mathbb{R}^m$. The translation is not a cast: pm-0010 forbids assuming that the setting's first-layer projection is the identity, and the proof below never does — every passage between $\mathbb{R}^{n_E}$ and $\mathbb{R}^m$ goes through $\exp_E$ and π_1^E.

Note also where the cited results are assumed. `ComparisonAbelian` is applied below *at* $\mathbb{G}_E$, so what Step 7 needs is a cited-results package at $\mathbb{G}_E$, and that is exactly the hypothesis this statement carries. A package at some other setting would not serve, and a package quantified over every sub-elliptic setting is not available: the three Carnot-side items of `CitedResults` are theorems about genuine Carnot groups. It is because $\mathbb{G}_E$ *is* a Carnot group — $(\mathbb{R}^m, +)$ with the Euclidean norm — that assuming them there is a satisfiable hypothesis.

Step 1: the Euclidean setting. The setting $\mathbb{G}_E$ named in the statement is a sub-elliptic setting (`CarnotSetting`), and by `EuclidSetting` at $N := m$ it satisfies
(1)

$$n_E = m, \quad m_E = m, \quad r_E = 1, \quad x *_E y = x + y, \quad \|x\|_E = |x|, \quad \exp_E(\pi_1^E(x)) = x, \quad \sigma_E(x)_i = \exp_E(e_i)$$

for all x, y and all i, where we write $n_E, m_E, r_E, *_E, \|\cdot\|_E, \pi_1^E, \exp_E, \sigma_E$ for the data of $\mathbb{G}_E$. These seven identities are all that is used about $\mathbb{G}_E$ below; in particular π_1^E is never assumed to be the identity.

Because $m_E = m$, the spaces $\mathbb{R}^{m_E}$ and $\mathbb{R}^m$ are the same space and the index sets $\operatorname{Fin} m_E$ and $\operatorname{Fin} m$ are the same, with the standard basis vectors e_i corresponding to themselves. We use that identification silently from now on, so that A and H are regarded as defined on $\mathbb{R}^{m_E}$ and the structure conditions on (A, H) read as `StructureConditionsES` m_E A H; in Lean this is a single transport along the equality $m_E = m$ of natural numbers, involving no choice and no `subst`. We do *not* identify $\mathbb{R}^{n_E}$ with $\mathbb{R}^m$; all movement between $\mathbb{R}^{n_E}$ and $\mathbb{R}^m$ goes through $\exp_E$ and π_1^E, as the next step makes precise.

Step 2: $\exp_E$ and π_1^E are mutually inverse linear isometries. By `CarnotSetting`, $\pi_1^E : \mathbb{R}^{n_E} \to \mathbb{R}^m$ is a continuous linear map with $\pi_1^E(\exp_E(\zeta)) = \zeta$ for every $\zeta \in \mathbb{R}^m$, and $\exp_E$ is continuous. By (1), $\exp_E(\pi_1^E(x)) = x$ for every $x \in \mathbb{R}^{n_E}$. So $\exp_E$ and π_1^E are mutually inverse bijections; since π_1^E is linear and bijective, its inverse $\exp_E$ is linear as well, and both are

continuous, hence homeomorphisms. Moreover, by the field $\|\exp_{\mathbb{G}}(\zeta)\| = |\zeta|$ of `CarnotSetting` together with $\|x\|_E = |x|$ from (1),

$$|\exp_E(\zeta)| = \|\exp_E(\zeta)\|_E = |\zeta| \qquad (\zeta \in \mathbb{R}^m), \tag{2}$$

so $\exp_E$ is a linear isometry and so is π_1^E.

Step 3: the transported domain and functions. Put

$$\Omega'' := \exp_E(\Omega') \subseteq \mathbb{R}^{n_E}, \qquad U := F \circ \pi_1^E, \qquad V := Gf \circ \pi_1^E.$$

Because $\exp_E$ and π_1^E are mutually inverse,

$$\Omega'' = (\pi_1^E)^{-1}(\Omega'), \tag{3}$$

for $z = \exp_E(\zeta)$ with $\zeta \in \Omega'$ has $\pi_1^E(z) = \zeta \in \Omega'$, and conversely $\pi_1^E(z) \in \Omega'$ gives $z = \exp_E(\pi_1^E(z)) \in \exp_E(\Omega')$. Consequently:

- Ω'' is open, being the preimage of the open Ω' under the continuous π_1^E, by (3);
- Ω'' is bounded, by (2) and the boundedness of Ω';
- $\mathbb{R}^{n_E} \setminus \Omega'' \neq \emptyset$: pick $\zeta_0 \notin \Omega'$; then $\pi_1^E(\exp_E(\zeta_0)) = \zeta_0 \notin \Omega'$, so $\exp_E(\zeta_0) \notin \Omega''$ by (3);
- since $\exp_E$ is a homeomorphism with inverse π_1^E, $\overline{\Omega''} = \exp_E(\overline{\Omega'})$ and $\partial\Omega'' = \exp_E(\partial\Omega')$; in particular $\pi_1^E(\overline{\Omega''}) = \overline{\Omega'}$ and $\pi_1^E(\partial\Omega'') = \partial\Omega'$.

From the last item: U and V are continuous on $\overline{\Omega''}$, because π_1^E is continuous and maps $\overline{\Omega''}$ into $\overline{\Omega'}$ where F and Gf are continuous. Also, for $z \in \partial\Omega''$ we have $\pi_1^E(z) \in \partial\Omega'$, so $U(z) = F(\pi_1^E(z)) \leq Gf(\pi_1^E(z)) = V(z)$; and for $z \in \overline{\Omega''}$ we have $\pi_1^E(z) \in \overline{\Omega'}$, so $|U(z)| \leq R/2$ and $|V(z)| \leq R/2$.

Finally $\mathbb{G}_E$ is abelian: $r_E = 1$, and by `CarnotSetting` a setting of step 1 has a commutative group law, so $x *_E y = y *_E x$ for all x, y. (This is also visible directly from $x *_E y = x + y$ in (1).)

Step 4: the two jet identities. Let $z \in \mathbb{R}^{n_E}$, let D be a continuous linear functional on $\mathbb{R}^{n_E}$ and let B be a continuous symmetric bilinear form on $\mathbb{R}^{n_E}$. Define

$$E := D \circ \exp_E, \qquad C(a,b) := B\big(\exp_E(a), \exp_E(b)\big) \qquad (a, b \in \mathbb{R}^m),$$

which are a continuous linear functional and a continuous symmetric bilinear form on $\mathbb{R}^m$, since $\exp_E$ is continuous and linear by Step 2 and B is symmetric. Then

$$\text{hgrad } \sigma_E\ z\ D = \xi(E) \qquad \text{and} \qquad \text{hhessS } \sigma_E\ z\ D\ B = \mathcal{C}(C). \tag{4}$$

Indeed, by `hgrad` the i-th component of hgrad $\sigma_E\ z\ D$ is $D(\sigma_E(z)_i)$, which by (1) equals $D(\exp_E(e_i)) = E(e_i)$; that is the i-th component of $\xi(E)$. For the second identity, by `hhessS`,

$$\text{hhessS } \sigma_E\ z\ D\ B = \Big(B\big(\sigma_E(z)_i, \sigma_E(z)_j\big)\Big)_{i,j} + \mathcal{M}(z, D),$$

where $\mathcal{M}$ is the correction matrix of `Mfield`. By (1) the frame $x \mapsto \sigma_E(x)$ is *constant* in x — each $\sigma_E(x)_i$ equals $\exp_E(e_i)$, independently of x — so every $\mathrm{D}\sigma_E^j$ occurring in `Mfield` is the derivative of a constant map and vanishes; hence $\mathcal{M}(z, D) = 0$. And $B(\sigma_E(z)_i, \sigma_E(z)_j) = B(\exp_E(e_i), \exp_E(e_j)) = C(e_i, e_j)$ by (1) and the definition of C, so the matrix is $\mathcal{C}(C)$. This proves (4).

Step 5: U is a viscosity sub-solution in Ω''. Upper semi-continuity of U on Ω'' follows from its continuity on $\overline{\Omega''} \supseteq \Omega''$ (Step 3). For the test-function clause of `IsViscSubsol`, let $z \in \Omega''$, let $\varphi : \mathbb{R}^{n_E} \to \mathbb{R}$ be C^2 on Ω'' with $z \in \operatorname{argmax}_{\Omega''}(U - \varphi)$, let (D, B) be a second-order jet of φ at z (`TwiceDiffAt`) and assume hgrad $\sigma_E\ z\ D \neq 0$. Put

$$\zeta := \pi_1^E(z) \in \Omega' \quad \text{(by (3))}, \qquad \psi := \varphi \circ \exp_E,$$

and let E, C be as in Step 4 for this (D, B). Note $\exp_E(\zeta) = z$ by Step 2.

(5a) ψ is C^2 on Ω'. $\exp_E$ is linear and continuous, hence C^∞, and it maps Ω' into Ω'' by the definition of Ω''; and φ is C^2 on Ω''.

(5b) $\zeta \in \operatorname{argmax}_{\Omega'}(F-\psi)$. First $\psi(\zeta)=\varphi(\exp_E(\zeta))=\varphi(z)$ and $F(\zeta)=F(\pi_1^E(z))=U(z)$, so $(F-\psi)(\zeta)=(U-\varphi)(z)$. Next, for $w\in\Omega'$ we have $\exp_E(w)\in\Omega''$ and $\pi_1^E(\exp_E(w))=w$, so

$$(F-\psi)(w)=F(w)-\varphi(\exp_E(w))=U\big(\exp_E(w)\big)-\varphi\big(\exp_E(w)\big)=(U-\varphi)\big(\exp_E(w)\big)\ \le\ (U-\varphi)(z)=(F-\psi)(\zeta)$$

by maximality of $U-\varphi$ over Ω'' at z. As $w\in\Omega'$ was arbitrary, the claim holds.

(5c) (E,C) *is a second-order jet of* ψ *at* ζ. Symmetry of C was noted in Step 4. For the Peano estimate, let $\varepsilon>0$ and let $\delta>0$ be given by `TwiceDiffAt` for φ, (D,B) and ε, so that

$$\Big|\varphi(y)-\varphi(z)-D(y-z)-\tfrac12 B(y-z,y-z)\Big|\le\varepsilon|y-z|^2\qquad(|y-z|<\delta).$$

Let $w\in\mathbb{R}^m$ with $|w-\zeta|<\delta$ and put $y:=\exp_E(w)$. Since $\exp_E$ is linear and $\exp_E(\zeta)=z$ we get $y-z=\exp_E(w-\zeta)$, and $|y-z|=|w-\zeta|<\delta$ by (2). Therefore

$$\Big|\psi(w)-\psi(\zeta)-E(w-\zeta)-\tfrac12 C(w-\zeta,w-\zeta)\Big|=\Big|\varphi(y)-\varphi(z)-D(y-z)-\tfrac12 B(y-z,y-z)\Big|\le\varepsilon|y-z|^2=\varepsilon|w-\zeta|^2,$$

using $\psi(w)=\varphi(y)$, $\psi(\zeta)=\varphi(z)$, $E(w-\zeta)=D(\exp_E(w-\zeta))=D(y-z)$ and $C(w-\zeta,w-\zeta)=B(y-z,y-z)$. As $\varepsilon>0$ was arbitrary, (E,C) is a second-order jet of ψ at ζ.

(5d) conclusion. By (4), $\xi(E)=\operatorname{hgrad}\sigma_E\ z\ D\neq 0$. Hypothesis (1) of the statement, applied at $\zeta\in\Omega'$ with ψ (which is C^2 on Ω' by (5a)), with the maximality (5b) and the jet (5c), gives

$$-\operatorname{Tr}\big(A(\xi(E))\,\mathcal{C}(C)\big)+H(\xi(E))\ \le\ 0.$$

By `Lop`, $\operatorname{Lop}\ \sigma_E\ A\ H\ z\ D\ B=-\operatorname{Tr}\big(A(\operatorname{hgrad}\sigma_E\ z\ D)\ \operatorname{hhessS}\ \sigma_E\ z\ D\ B\big)+H(\operatorname{hgrad}\sigma_E\ z\ D)$, which by (4) equals $-\operatorname{Tr}(A(\xi(E))\mathcal{C}(C))+H(\xi(E))$. Hence $\operatorname{Lop}\ \sigma_E\ A\ H\ z\ D\ B\le 0$, which is what `IsViscSubsol` requires. So U is a viscosity sub-solution of $\mathcal{L}_E\cdot=0$ in Ω'', where $\mathcal{L}_E$ is the operator of $\mathbb{G}_E$ built from (A,H).

Step 6: V is a viscosity super-solution in Ω''. Lower semi-continuity of V on Ω'' follows from its continuity on $\overline{\Omega''}$. Steps (5a), (5c) and Step 4 do not involve F or the direction of the inequality and are verbatim with V, Gf in place of U, F. In (5b) the inequality reverses: for $w\in\Omega'$, minimality of $V-\varphi$ over Ω'' at z gives $(Gf-\psi)(\zeta)=(V-\varphi)(z)\le(V-\varphi)(\exp_E(w))=(Gf-\psi)(w)$, so $\zeta\in\operatorname{argmin}_{\Omega'}(Gf-\psi)$. Hypothesis (2) of the statement then gives $0\le-\operatorname{Tr}(A(\xi(E))\mathcal{C}(C))+H(\xi(E))=\operatorname{Lop}\ \sigma_E\ A\ H\ z\ D\ B$, which is what `IsViscSupersol` requires.

Step 7: the abelian comparison, and descent. We apply `ComparisonAbelian` with: the setting $\mathbb{G}_E$; the cited results at $\mathbb{G}_E$, which is the hypothesis of the present statement; its commutativity from Step 3; the pair (A,H), which satisfies the structure conditions by Step 1; the set Ω'', open and bounded with non-empty complement by Step 3; the functions U and V, continuous on $\overline{\Omega''}$ by Step 3; their viscosity properties from Steps 5 and 6; $U\le V$ on $\partial\Omega''$ from Step 3; and $R\ge 0$ with $|U|,|V|\le R/2$ on $\overline{\Omega''}$ from Step 3. Its conclusion is

$$U(z)\le V(z)\qquad\text{for every } z\in\Omega''.$$

Finally let $\zeta\in\Omega'$. Then $\exp_E(\zeta)\in\Omega''$, and $U(\exp_E(\zeta))=F(\pi_1^E(\exp_E(\zeta)))=F(\zeta)$ and similarly $V(\exp_E(\zeta))=Gf(\zeta)$, by $\pi_1^E\circ\exp_E=\mathrm{id}$. Hence $F(\zeta)\le Gf(\zeta)$. As $\zeta\in\Omega'$ was arbitrary, $F\le Gf$ in Ω'. □

2.45. `FirstLayerEuclideanSubsol`.

Theorem 2.45 (The first-layer transfer of Theorem 1.3(iii))**.** *Let $\mathbb{G}$ be a sub-elliptic setting (`CarnotSetting`) with frame σ and first-layer projection $\pi_1:\mathbb{R}^n\to\mathbb{R}^m$, and let $A:\mathbb{R}^m\to\mathbb{R}^{m\times m}$ and $H:\mathbb{R}^m\to\mathbb{R}$. Let $\Omega\subset\mathbb{R}^n$ be a bounded open set and let $F,Gf:\mathbb{R}^m\to\mathbb{R}$ be continuous on $\pi_1(\bar\Omega)$. Assume*

1. *$F\circ\pi_1$ is a viscosity sub-solution (`IsViscSubsol`) and $Gf\circ\pi_1$ a viscosity super-solution (`IsViscSupersol`) of $\mathcal{L}\cdot=0$ in Ω;*
2. *$F(\pi_1(x))\le Gf(\pi_1(x))$ for every $x\in\partial\Omega$.*

Put $\Omega':=\pi_1(\Omega)\subseteq\mathbb{R}^m$. Then

1. *Ω' is open;*
2. *Ω' is bounded;*

(3) $\overline{\Omega'} \subseteq \pi_1(\bar{\Omega})$;
(4) $\partial\Omega' \subseteq \pi_1(\partial\Omega)$;
(5) $F \le Gf$ *on* $\partial\Omega'$;
(6) F *and* Gf *are continuous on* $\overline{\Omega'}$;
(7) *for every* $\zeta \in \Omega'$, *every* $\psi : \mathbb{R}^m \to \mathbb{R}$ *that is* C^2 *on* Ω' *with* $\zeta \in \operatorname{argmax}_{\Omega'}(F - \psi)$, *and every second-order jet* (E, C) *of* ψ *at* ζ *(`TwiceDiffAt`), writing*

$$\xi := \big(E(e_1), \dots, E(e_m)\big) \in \mathbb{R}^m, \qquad \mathcal{C} := \big(C(e_i, e_j)\big)_{i,j} \in \mathbb{R}^{m\times m}$$

with e_i *the* i*-th standard basis vector of* $\mathbb{R}^m$, *one has*

$$\xi \neq 0 \implies -\operatorname{Tr}\big(A(\xi)\,\mathcal{C}\big) + H(\xi) \le 0;$$

(8) *dually, for every* $\zeta \in \Omega'$, *every* ψ *that is* C^2 *on* Ω' *with* $\zeta \in \operatorname{argmin}_{\Omega'}(Gf - \psi)$, *and every second-order jet* (E, C) *of* ψ *at* ζ, *with* ξ *and* $\mathcal{C}$ *as above,*

$$\xi \neq 0 \implies 0 \le -\operatorname{Tr}\big(A(\xi)\,\mathcal{C}\big) + H(\xi).$$

Proof. This is the content of the second paragraph of the proof of Theorem 1.3(iii), up to but not including the appeal to the case $r = 1$: "The set $\Omega' := \pi_1(\Omega) \subset \mathbb{R}^m$ is open and bounded, and $\partial\Omega' \subset \pi_1(\partial\Omega)$ ... Hence $F \le Gf$ on $\partial\Omega'$. ... thus $\mathcal{L}\varphi = (\mathcal{L}^{\mathrm{eucl}}\psi) \circ \pi_1$ ... F is a viscosity sub-solution of $\mathcal{L}^{\mathrm{eucl}}F = 0$ in Ω', and likewise Gf is a super-solution."

The quantities ξ and $\mathcal{C}$ of (7) are exactly the shapes that the clause `pi1Jet` of `CarnotSetting` produces for hgrad (`hgrad`) and hhessS (`hhessS`); that is deliberate, and it is what makes the Euclidean condition of (7) literally the inequality $\mathcal{L}\varphi(x) \le 0$ read through those two identities.

Step 1: π_1 is a surjective continuous linear map; (1) and (2). By `CarnotSetting`, π_1 is a continuous linear map $\mathbb{R}^n \to \mathbb{R}^m$ and $\pi_1(\exp_{\mathbb{G}}(\zeta)) = \zeta$ for every $\zeta \in \mathbb{R}^m$. The latter exhibits $\exp_{\mathbb{G}}$ as a right inverse, so π_1 is surjective. A surjective continuous linear map between Banach spaces is open, so π_1 maps open sets to open sets; as Ω is open, $\Omega' = \pi_1(\Omega)$ is open. This is (1).

For (2): π_1 is a continuous linear map, so $|\pi_1(x)| \le \|\pi_1\|\,|x|$ for every x. If $|x| \le \kappa$ for all $x \in \Omega$ (such a κ exists, Ω being bounded) then $|\pi_1(x)| \le \|\pi_1\|\kappa$ for all $x \in \Omega$, so Ω' is bounded.

Step 2: (3) and (6). $\bar{\Omega}$ is compact (Ω is bounded) and π_1 is continuous, so $\pi_1(\bar{\Omega})$ is compact, hence closed. It contains $\pi_1(\Omega) = \Omega'$, so it contains $\overline{\Omega'}$; this is (3). Then (6) follows from (3) and the hypothesis that F and Gf are continuous on $\pi_1(\bar{\Omega})$: a function continuous on a set is continuous on every subset of it.

Step 3: (4), the limit-point argument. We transcribe the paper's own argument. Let $\zeta \in \partial\Omega'$. Since Ω' is open by (1), $\partial\Omega' = \overline{\Omega'} \setminus \Omega'$, so $\zeta \in \overline{\Omega'}$ and $\zeta \notin \Omega'$. By $\zeta \in \overline{\Omega'}$ there are $x_k \in \Omega$ with $\pi_1(x_k) \to \zeta$. The points x_k lie in the bounded set Ω, so (x_k) has a convergent subsequence $x_{k_j} \to \bar{x}$, and $\bar{x} \in \bar{\Omega}$ because $x_{k_j} \in \Omega \subseteq \bar{\Omega}$ and $\bar{\Omega}$ is closed. By continuity of π_1,

$$\pi_1(\bar{x}) = \lim_j \pi_1(x_{k_j}) = \zeta.$$

Now $\bar{x} \notin \Omega$: otherwise $\zeta = \pi_1(\bar{x}) \in \pi_1(\Omega) = \Omega'$, contradicting $\zeta \notin \Omega'$. Hence $\bar{x} \in \bar{\Omega} \setminus \Omega = \partial\Omega$ (Ω is open), and $\zeta = \pi_1(\bar{x}) \in \pi_1(\partial\Omega)$. This is (4).

Step 4: (5). Let $\zeta \in \partial\Omega'$. By (4) there is $x \in \partial\Omega$ with $\pi_1(x) = \zeta$. Hypothesis (2) of the statement gives $F(\pi_1(x)) \le Gf(\pi_1(x))$, i.e. $F(\zeta) \le Gf(\zeta)$.

Step 5: (7). Let $\zeta \in \Omega'$, let ψ be C^2 on Ω' with $\zeta \in \operatorname{argmax}_{\Omega'}(F - \psi)$, let (E, C) be a second-order jet of ψ at ζ and let $\xi, \mathcal{C}$ be as in the statement; assume $\xi \neq 0$. Write $u := F \circ \pi_1$ and $\varphi := \psi \circ \pi_1$.

Since $\zeta \in \Omega' = \pi_1(\Omega)$ there is $x \in \Omega$ with $\pi_1(x) = \zeta$. Fix such an x.

(5a) φ is C^2 on Ω. π_1 is a continuous linear map, hence C^∞, and it maps Ω into Ω' by the definition of Ω'; and ψ is C^2 on Ω'. The composition of a C^2 map into Ω' with a function that is C^2 on Ω' is C^2 on Ω.

(5b) $x \in \operatorname{argmax}_\Omega(u - \varphi)$. Let $y \in \Omega$. Then $\pi_1(y) \in \Omega'$, so by $\zeta \in \operatorname{argmax}_{\Omega'}(F - \psi)$,

$$(u - \varphi)(y) = F(\pi_1(y)) - \psi(\pi_1(y)) \le F(\zeta) - \psi(\zeta) = F(\pi_1(x)) - \psi(\pi_1(x)) = (u - \varphi)(x).$$

As $y \in \Omega$ was arbitrary, $u - \varphi$ attains its maximum over *all* of Ω at x.

This is the paper's clause "φ touches u from above at every $x \in \Omega$ with $\pi_1(x) = \zeta$", and it is worth pausing on it: pm-0009 records that our rendering of Definition 2.1 keeps the test condition $x \in \operatorname{argmax}_\Omega(w - \varphi)$ *global* over the domain, which elsewhere in this development forces an appeal to the localization lemma for test functions before a sub-solution may be used on a smaller set. Here the global form is exactly what the paper supplies for free, because ψ is composed with π_1 and the maximality over Ω' pulls back to maximality over all of Ω. No localization step is needed in this proof.

(5c) the jet of φ at x. We have $\pi_1(x) = \zeta$, so (E, C) is a second-order jet of ψ at $\pi_1(x)$. The clause `pi1Jet` of `CarnotSetting`, applied with ψ, x, E and C, therefore provides a second-order jet (D, B) of $y \mapsto \psi(\pi_1(y)) = \varphi(y)$ at x with

$$(*) \qquad \text{hgrad } \sigma\ x\ D = \big(E(e_1), \dots, E(e_m)\big) = \xi, \qquad \text{hhessS } \sigma\ x\ D\ B = \big(C(e_i, e_j)\big)_{i,j} = \mathcal{C}.$$

(This is the jet form of the paper's coordinate computation "$X_i\varphi = (\partial_i\psi) \circ \pi_1$ and $X_jX_i\varphi = (\partial_j\partial_i\psi) \circ \pi_1$"; the field `pi1Jet` is exactly that statement, carried in the definition of the setting as pm-0005 permits.)

In particular hgrad $\sigma\ x\ D = \xi \neq 0$.

(5d) conclusion. By (5a), (5b), (5c) and $\xi \neq 0$, the function φ is admissible for `IsViscSubsol` at the point $x \in \Omega$ with the jet (D, B), so

$$\text{Lop } \sigma\ A\ H\ x\ D\ B\ \le\ 0 \qquad (\texttt{Lop}).$$

By `Lop`, Lop $\sigma\ A\ H\ x\ D\ B = -\operatorname{Tr}\big(A(\text{hgrad } \sigma\ x\ D)\ \text{hhessS } \sigma\ x\ D\ B\big) + H(\text{hgrad } \sigma\ x\ D)$, so by $(*)$ this is exactly $-\operatorname{Tr}(A(\xi)\mathcal{C}) + H(\xi)$. Hence $-\operatorname{Tr}(A(\xi)\mathcal{C}) + H(\xi) \le 0$, which is (7).

Step 6: (8), the dual. Let $\zeta \in \Omega'$, let ψ be C^2 on Ω' with $\zeta \in \operatorname{argmin}_{\Omega'}(Gf - \psi)$, let (E, C) be a second-order jet of ψ at ζ, let $\xi, \mathcal{C}$ be as above and assume $\xi \neq 0$. Write $v := Gf \circ \pi_1$ and $\varphi := \psi \circ \pi_1$, and pick $x \in \Omega$ with $\pi_1(x) = \zeta$.

Steps (5a) and (5c) do not involve F or the direction of the inequality and are verbatim. In (5b) the inequality reverses: for $y \in \Omega$, $\pi_1(y) \in \Omega'$ and $\zeta \in \operatorname{argmin}_{\Omega'}(Gf - \psi)$ give

$$(v - \varphi)(x) = Gf(\zeta) - \psi(\zeta)\ \le\ Gf(\pi_1(y)) - \psi(\pi_1(y)) = (v - \varphi)(y),$$

so $x \in \operatorname{argmin}_\Omega(v - \varphi)$. Feeding φ and the jet (D, B) of (5c) to `IsViscSupersol` at x, with hgrad $\sigma\ x\ D = \xi \neq 0$, gives $0 \le$ Lop $\sigma\ A\ H\ x\ D\ B$, which by `Lop` and $(*)$ is $0 \le -\operatorname{Tr}(A(\xi)\mathcal{C}) + H(\xi)$. This is (8). □

2.46. `UncondCaseFirstLayer`.

Theorem 2.46 (Theorem 1.3(iii))**.** *Let $\mathbb{G}$ be a sub-elliptic setting (`CarnotSetting`) with step r, frame σ and first-layer projection $\pi_1 : \mathbb{R}^n \to \mathbb{R}^m$. Assume the cited results of the literature* at $\mathbb{G}$ *(`CitedResults`), and also the cited results of the literature* at the Euclidean setting $\mathbb{G}_E$ on $\mathbb{R}^m$ *(`EuclidSetting` at $N := m$, and `CitedResults` at that setting). Let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$ satisfy the structure conditions (`StructureConditionsES`). Let $\Omega \subset \mathbb{R}^n$ be a bounded open set with $\mathbb{R}^n \setminus \Omega \neq \emptyset$, let $u, v : \mathbb{R}^n \to \mathbb{R}$ be continuous on $\bar\Omega$ with u a viscosity sub-solution (`IsViscSubsol`) and v a viscosity super-solution (`IsViscSupersol`) of $\mathcal{L}\cdot = 0$ in Ω and $u \le v$ on $\partial\Omega$, and let $R \ge 0$ satisfy $|u| \le R/2$ and $|v| \le R/2$ on $\bar\Omega$. Then:*

1. *if $r = 1$, then $u \le v$ in Ω;*
2. *if there are $F, Gf : \mathbb{R}^m \to \mathbb{R}$, continuous on $\pi_1(\bar\Omega)$, with $u = F \circ \pi_1$ and $v = Gf \circ \pi_1$, then $u \le v$ in Ω.*

Proof. This is Theorem 1.3(iii), both disjuncts as the paper prints them ("$r = 1$, i.e. $\mathbb{G}$ is the Euclidean group; more generally, $u = F \circ \pi_1$ and $v = Gf \circ \pi_1$ for continuous F, Gf").

The two cited-results hypotheses are used in disjoint places, and each part uses exactly one of them. Part (1) applies `ComparisonAbelian` to $\mathbb{G}$ itself and so needs the package at $\mathbb{G}$. Part (2) descends to $\mathbb{R}^m$ and applies `EuclidFirstLayerComparison`, which works inside the Euclidean setting $\mathbb{G}_E$ and so needs the package at $\mathbb{G}_E$; the package at $\mathbb{G}$ would not serve there, because the three Carnot-side items of `CitedResults` are asserted only at the setting carried as that node's parameter.

Part (1): $r = 1$**.** By `CarnotSetting`, a setting of step 1 has a commutative group law: the field `abelian_of_step_one` gives $x * y = y * x$ for all x, y when $r = 1$. All the remaining hypotheses of `ComparisonAbelian` are hypotheses of the present statement — the cited results at $\mathbb{G}$, which is the setting that node is applied to here; the structure conditions on (A, H); Ω bounded, open, with non-empty complement; u, v continuous on $\bar{\Omega}$; the two viscosity properties; $u \le v$ on $\partial\Omega$; and $R \ge 0$ with $|u|, |v| \le R/2$ on $\bar{\Omega}$. Applying `ComparisonAbelian` to $\mathbb{G}$ itself therefore gives $u \le v$ in Ω. This is the paper's "Suppose first $r = 1$, so that $\mathbb{G} = (\mathbb{R}^n, +)$".

Part (2): the first-layer case. Let F, Gf be as in (2).

(2a) The degenerate corner $m = 0$. Suppose $m = 0$. Then the frame is the empty family of vector fields, so the two conditions defining an admissible family in `LieHull` — containing each σ^i, and closure under the Lie bracket — are both vacuous for the empty family; hence $\emptyset$ is admissible and, LieHull σ being the intersection of all admissible families, LieHull $\sigma = \emptyset$. Hörmander's rank condition (1.6), carried as a field of `CarnotSetting`, says that for every x the span of $\{V(x) : V \in \text{LieHull } \sigma\}$ is all of $\mathbb{R}^n$; with LieHull $\sigma = \emptyset$ that set of values is empty and its span is $\{0\}$, so $\mathbb{R}^n = \{0\}$ and therefore $n = 0$ (for $n \ge 1$ the vector e_1 is a non-zero element of $\mathbb{R}^n$). So $\mathbb{R}^n$ has exactly one point, 0; by $\mathbb{R}^n \setminus \Omega \ne \emptyset$ that point is not in Ω, so $\Omega = \emptyset$ and the conclusion "$u \le v$ in Ω" holds vacuously. We may therefore assume $m \ge 1$ from now on.

(2b) The descent to $\mathbb{R}^m$. Put $\Omega' := \pi_1(\Omega)$. The hypotheses of `FirstLayerEuclideanSubsol` hold for $(\mathbb{G}, A, H, \Omega, F, Gf)$: Ω is bounded and open; F and Gf are continuous on $\pi_1(\bar{\Omega})$; $F \circ \pi_1 = u$ is a viscosity sub-solution and $Gf \circ \pi_1 = v$ a viscosity super-solution of $\mathcal{L}\cdot = 0$ in Ω; and for $x \in \partial\Omega$, $F(\pi_1(x)) = u(x) \le v(x) = Gf(\pi_1(x))$. It therefore gives

(i) Ω' is open;
(ii) Ω' is bounded;
(iii) $\overline{\Omega'} \subseteq \pi_1(\bar{\Omega})$;
(v) $F \le Gf$ on $\partial\Omega'$;
(vi) F and Gf are continuous on $\overline{\Omega'}$;
(vii) the Euclidean sub-solution condition for F in Ω': for every $\zeta \in \Omega'$, every ψ that is C^2 on Ω' with $\zeta \in \operatorname{argmax}_{\Omega'}(F - \psi)$ and every second-order jet (E, C) of ψ at ζ (`TwiceDiffAt`) with $\xi(E) := (E(e_1), \dots, E(e_m)) \ne 0$, one has $-\operatorname{Tr}\big(A(\xi(E))\,\mathcal{C}(C)\big) + H(\xi(E)) \le 0$, where $\mathcal{C}(C) := (C(e_i, e_j))_{i,j}$;
(viii) the dual Euclidean super-solution condition for Gf in Ω'.

(It also gives $\partial\Omega' \subseteq \pi_1(\partial\Omega)$, which we do not need separately, (v) being the form in which it is used.)

(2c) The complement of Ω' is non-empty. Since $m \ge 1$ by (2a), the space $\mathbb{R}^m$ is unbounded: the points $t\,e_1$, $t \in \mathbb{R}$, have $|t\,e_1| = |t|$, unbounded in t. By (ii) the set Ω' is bounded, so $\Omega' \ne \mathbb{R}^m$, i.e. there is $\zeta_0 \in \mathbb{R}^m$ with $\zeta_0 \notin \Omega'$.

(2d) The bound on F and Gf. Let $\zeta \in \overline{\Omega'}$. By (iii) there is $x_0 \in \bar{\Omega}$ with $\pi_1(x_0) = \zeta$. Then $F(\zeta) = F(\pi_1(x_0)) = u(x_0)$ and $Gf(\zeta) = Gf(\pi_1(x_0)) = v(x_0)$, so $|F(\zeta)| \le R/2$ and $|Gf(\zeta)| \le R/2$ by the hypothesis $|u|, |v| \le R/2$ on $\bar{\Omega}$.

(2e) The Euclidean comparison. All hypotheses of `EuclidFirstLayerComparison` are now available with the dimension m, the cited-results package at the Euclidean setting $\mathbb{G}_E$ on $\mathbb{R}^m$ — which is the second cited-results hypothesis of the present statement, and is what that node requires, since it applies `ComparisonAbelian` inside $\mathbb{G}_E$ and not inside $\mathbb{G}$ — the pair (A, H) (which satisfies the structure conditions by hypothesis — note that `StructureConditionsES` m A H is literally the hypothesis we carry, so no transport is involved here), the set Ω' (open by (i), bounded by (ii), with non-empty complement by (2c)), the functions F, Gf (continuous on $\overline{\Omega'}$ by (vi), with $F \le Gf$ on $\partial\Omega'$ by (v)), the constant $R \ge 0$ with the bound (2d), and the two inline viscosity conditions (vii) and (viii). It gives

$$F(\zeta) \le Gf(\zeta) \qquad \text{for every } \zeta \in \Omega'.$$

This is the paper's "By the case $r = 1$ applied in $\mathbb{R}^m$, $F \le Gf$ in Ω'".

(2f) Descent back to Ω. Let $x \in \Omega$. Then $\pi_1(x) \in \pi_1(\Omega) = \Omega'$, so (2e) gives $F(\pi_1(x)) \le Gf(\pi_1(x))$, i.e. $u(x) \le v(x)$. As $x \in \Omega$ was arbitrary, $u \le v$ in Ω, which is the paper's "i.e. $u \le v$ in Ω". □

3. Main results

3.1. `ComparisonUnderNondegeneracy` (paper, Theorem 1.2).

Theorem 3.1 (Theorem 1.2)**.** *Let $\mathbb{G}$ be a sub-elliptic setting (`CarnotSetting`) with group law $*$, homogeneous norm $\|\cdot\|$, step r, frame $\sigma = [\sigma^1, \dots, \sigma^m]$ and $q := 2\,r!$, and assume the cited results of the literature at $\mathbb{G}$ (`CitedResults`). Let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$ be continuous on $\mathbb{R}^m \setminus \{0\}$, with $A(\xi)$ symmetric and positive semidefinite for $\xi \neq 0$, and satisfy the conditions of strict ellipticity in the gradient direction (E) (1.9) and of homogeneity along rays (S) (1.10) — that is, let (A, H) satisfy the structure conditions of `StructureConditionsES`. No normalization of (A, H) is assumed. Let $\Omega \subset \mathbb{R}^n$ be a bounded open set with $\mathbb{R}^n \setminus \Omega \neq \emptyset$, let $u, v : \mathbb{R}^n \to \mathbb{R}$ be continuous on $\bar\Omega$, and assume*

1. *u is a viscosity sub-solution (`IsViscSubsol`) and v a viscosity super-solution (`IsViscSupersol`) of $\mathcal{L}\cdot = 0$ in Ω;*
2. *$u \le v$ on $\partial\Omega$.*

Let $R \ge 0$ satisfy $|u| \le R/2$ and $|v| \le R/2$ on $\bar\Omega$, and with this R write $\Omega[\varepsilon]$ for the reduced domain (3.15) (`ReducedDomain`),

$$u^\varepsilon := \mathrm{SupConv}\ \mathbb{G}\ \Omega\ u\ \varepsilon \quad (\texttt{SupConv}), \qquad v_\varepsilon := \mathrm{InfConv}\ \mathbb{G}\ \Omega\ v\ \varepsilon \quad (\texttt{InfConv}),$$

and, as in (1.15),

$$E_\varepsilon := \Big\{x \in \Omega[\varepsilon] :\ (u^\varepsilon - v_\varepsilon)(y) \le (u^\varepsilon - v_\varepsilon)(x) \text{ for every } y \in \overline{\Omega[\varepsilon]}\Big\} = \Big\{x \in \Omega[\varepsilon] :\ (u^\varepsilon - v_\varepsilon)(x) = \max_{\overline{\Omega[\varepsilon]}}(u^\varepsilon - v_\varepsilon)\Big\}.$$

Assume finally that there is a sequence $(\varepsilon_k)_{k\in\mathbb{N}}$ of positive reals with $\varepsilon_k \to 0$ such that

3. *for every k, every $x \in E_{\varepsilon_k}$ and every continuous linear functional D that is the derivative of u^{ε_k} at x,*

$$\mathcal{X}u^{\varepsilon_k}(x) = \mathrm{hgrad}\ \sigma\ x\ D\ \neq\ 0 \qquad (\texttt{hgrad}).$$

Then $u \le v$ in Ω.

Proof. This is Theorem 1.2. Two points of the formulation deserve a word before the argument, because each is a place where a reader could suspect a departure from the paper and there is none.

The parameter R. The paper *defines* $R := 2\max\{\|u\|_{L^\infty}, \|v\|_{L^\infty}\}$ just before (3.15), whereas `ReducedDomain` carries R as a parameter. We therefore quantify over every admissible R, i.e. every $R \ge 0$ with $|u|, |v| \le R/2$ on $\bar\Omega$, and assume (3) for that R's family E_ε. This contains the paper's statement: the paper's R is admissible, since $\|u\|_{L^\infty} \le \max\{\|u\|_{L^\infty}, \|v\|_{L^\infty}\} = R/2$ and likewise for v, and it is the smallest admissible one. It is also not stronger than the paper's statement in any way the paper does not justify: R enters the argument only through the definition of $\Omega[\varepsilon]$, where a larger R makes $\Omega[\varepsilon]$ smaller, and the role of the bound $|u| \le R/2$ is exactly the one the paper gives it in the attainment step of Lemma 3.8(4), namely to make $\|z_x\|^q \le 4\varepsilon\|u\|_{L^\infty} \le 2R\varepsilon < (1 + 4R)\varepsilon$. Every step below uses R only through that bound and through the properties of $\Omega[\varepsilon]$, and all of them are available for an arbitrary admissible R.

The hypothesis $\mathbb{R}^n \setminus \Omega \neq \emptyset$. This is the paper's own convention, made explicit: (3.15) reads $\Omega[\varepsilon] = \{x \in \Omega :\ \inf_{y\in\mathbb{R}^n\setminus\Omega} \|y * x^{-1}\|^q > (1 + 4R)\varepsilon\}$, an infimum over $\mathbb{R}^n \setminus \Omega$, and the paper's $\Omega \subset \mathbb{R}^n$ is a proper subset. The assumption is automatic as soon as $n \ge 1$, since no bounded subset of $\mathbb{R}^n$ is then all of $\mathbb{R}^n$; it excludes only the degenerate setting $n = 0$, in which $\partial\Omega = \emptyset$ and the hypothesis $u \le v$ on $\partial\Omega$ carries no information at all.

Suppose the conclusion fails, so that $u(x^*) > v(x^*)$ for some $x^* \in \Omega$. We derive a contradiction.

Step 0: $M_0 > 0$ and a maximum point. Ω is bounded, so $\bar\Omega$ is a closed bounded subset of $\mathbb{R}^n$, hence compact; $u - v$ is continuous on $\bar\Omega$, so it attains its maximum there, say at $x_M \in \bar\Omega$.

Since $x^* \in \Omega \subset \bar{\Omega}$,

$$M_0 := \max_{\bar{\Omega}}(u-v) = u(x_M) - v(x_M) \;\geq\; u(x^*) - v(x^*) \;>\; 0.$$

Put $\tau := M_0/2 > 0$, as in the paper's proof.

Step 1: normalize once. The only hypothesis of `NormalizationExists` is that (A, H) satisfy the structure conditions (`StructureConditionsES`), which we have. It therefore provides $A_0 : \mathbb{R}^m \to \mathbb{R}^{m\times m}$, $H_0 : \mathbb{R}^m \to \mathbb{R}$ and $a : \mathbb{R}^m \to \mathbb{R}$ with

$$\text{(i) } a(\xi) > 0 \ (\xi \neq 0), \qquad \text{(ii) } A(\xi) = a(\xi)\, A_0(\xi) \ (\xi \neq 0), \qquad \text{(iii) } H(\xi) = a(\xi)\, H_0(\xi) \ (\xi \neq 0),$$

and with (iv) (A_0, H_0) satisfying the structure conditions (`StructureConditionsES`) and (v) (A_0, H_0) satisfying the normalization (2.3) (`NormalizedAH`). (It also gives continuity of a away from the origin, which we do not use.) Write $\mathcal{L}_0$ for the operator built from (A_0, H_0).

The hypotheses of `NormalizedSameViscositySolutions` are exactly (i), (ii), (iii), and its open set and function are arbitrary parameters. Applying it with Ω and u, and then with Ω and v, gives

$$\mathcal{L}u \leq 0 \text{ in } \Omega \iff \mathcal{L}_0 u \leq 0 \text{ in } \Omega, \qquad \mathcal{L}v \geq 0 \text{ in } \Omega \iff \mathcal{L}_0 v \geq 0 \text{ in } \Omega,$$

both in the viscosity sense. By hypothesis (1), u is therefore a viscosity sub-solution and v a viscosity super-solution of $\mathcal{L}_0 \cdot = 0$ in Ω. This is the standing normalization of Section 3 ("with A and H as in (1.9)–(1.10), normalized as in (2.3)"), carried out once and for all.

The pivot costs nothing in hypothesis (3), and it is worth saying why explicitly. By `SupConv` and `InfConv`, u^ε and v_ε are determined by $\mathbb{G}$, Ω, the function and ε alone; by `ReducedDomain`, $\Omega[\varepsilon]$ is determined by $\mathbb{G}$, Ω, R and ε alone. Neither A nor H occurs in any of them. Hence E_ε is literally the same set for (A, H) and for (A_0, H_0), and (3) is the same statement in both cases: there is nothing to transport.

Step 2: the standing assumptions for the regularized pair. Apply `RegularizedPairStandingAssumptio` with: $\mathbb{G}$; the cited results at $\mathbb{G}$, which that node asks for at the same setting; the pair (A_0, H_0), whose structure conditions are Step 1(iv); the set Ω, which is open, bounded and satisfies $\mathbb{R}^n \setminus \Omega \neq \emptyset$; the functions u and v, continuous on $\bar{\Omega}$, with u a viscosity sub-solution and v a viscosity super-solution of $\mathcal{L}_0 \cdot = 0$ in Ω by Step 1 and $u \leq v$ on $\partial\Omega$ by hypothesis (2); the constant $R \geq 0$ with $|u|, |v| \leq R/2$ on $\bar{\Omega}$; and the point $x_M \in \bar{\Omega}$ of Step 0, at which $u - v$ attains its maximum over $\bar{\Omega}$ and where $u(x_M) - v(x_M) = M_0 > 0$. Taking $\tau = M_0/2$ (as Step 0) and $E = \{y \in \Omega : \ u - v \text{ is maximal over } \bar{\Omega} \text{ at } y\}$, and writing

$$\tilde{v}_\varepsilon := \big(y \mapsto v_\varepsilon(y) + \tau\big),$$

it yields an $\varepsilon_1 > 0$ such that for every ε with $0 < \varepsilon < \varepsilon_1$:

(a) $E \subseteq \Omega[\varepsilon]$;
(b) u^ε and $\tilde{v}_\varepsilon$ are continuous on $\overline{\Omega[\varepsilon]}$;
(c) u^ε is locally semi-convex and $\tilde{v}_\varepsilon$ is locally semi-concave on $\Omega[\varepsilon]$, i.e. $-\tilde{v}_\varepsilon$ is locally semi-convex there (`LocallySemiConvexOn`);
(d) u^ε is a viscosity sub-solution and $\tilde{v}_\varepsilon$ a viscosity super-solution of $\mathcal{L}_0 \cdot = 0$ in $\Omega[\varepsilon]$;
(e) $u^\varepsilon - \tilde{v}_\varepsilon \leq -\tau/2$ on $\partial\Omega[\varepsilon]$;
(f) there is $x_e \in \Omega[\varepsilon]$ at which $u^\varepsilon - \tilde{v}_\varepsilon$ attains its maximum over $\overline{\Omega[\varepsilon]}$, with $\tau \leq u^\varepsilon(x_e) - \tilde{v}_\varepsilon(x_e)$;
(g) every point of $\overline{\Omega[\varepsilon]}$ at which $u^\varepsilon - \tilde{v}_\varepsilon$ attains its maximum over $\overline{\Omega[\varepsilon]}$ lies in $\Omega[\varepsilon]$.

(The node also asserts that E is non-empty and compact and that $\sup_{E_\varepsilon} \operatorname{dist}(\cdot, E) \to 0$; we need neither here. Clause (g) is likewise not needed below, but it is what makes E_ε an interior set, as (1.15) requires.)

We record the observation the paper makes at the start of its proof. Since $\tilde{v}_\varepsilon = v_\varepsilon + \tau$ with τ a constant, for every set $S \subseteq \mathbb{R}^n$ and every x,

$$(1) \qquad \big[(u^\varepsilon - \tilde{v}_\varepsilon)(y) \leq (u^\varepsilon - \tilde{v}_\varepsilon)(x) \ \forall y \in S\big] \iff \big[(u^\varepsilon - v_\varepsilon)(y) \leq (u^\varepsilon - v_\varepsilon)(x) \ \forall y \in S\big],$$

because $(u^\varepsilon - \tilde{v}_\varepsilon)(y) \leq (u^\varepsilon - \tilde{v}_\varepsilon)(x)$ reads $(u^\varepsilon - v_\varepsilon)(y) - \tau \leq (u^\varepsilon - v_\varepsilon)(x) - \tau$, and subtracting the same constant from both sides of an inequality between reals changes nothing. In particular, E_ε is also the set of points of $\Omega[\varepsilon]$ at which $u^\varepsilon - \tilde{v}_\varepsilon$ attains its maximum over $\overline{\Omega[\varepsilon]}$.

Step 3: choose ε from the sequence, and the geometry of $\Omega[\varepsilon]$. Since $\varepsilon_k \to 0$ and $\varepsilon_1 > 0$, we have $\varepsilon_k < \varepsilon_1$ for all large k; in particular there is an index k with $\varepsilon_k < \varepsilon_1$, and $\varepsilon_k > 0$ by hypothesis, so $0 < \varepsilon_k < \varepsilon_1$. Fix such a k and put $\varepsilon := \varepsilon_k$ and $\Omega' := \Omega[\varepsilon]$, so that (a)–(g) of Step 2 hold for this ε. (The sequence enters only here: it is used solely to produce one sufficiently small ε, and the hypothesis (3) is used only at this single index, exactly as in the paper's proof.)

By `ReducedDomainProperties`, applied with $\mathbb{G}$, the open bounded set Ω with $\mathbb{R}^n \setminus \Omega \neq \emptyset$, and $R \geq 0$: its second clause gives that $\Omega' = \Omega[\varepsilon]$ is bounded, and its fourth clause, with $\varepsilon > 0$, gives that Ω' is open.

Step 4: the non-degeneracy hypothesis transfers to Ω'. We claim

(2)

for every $x \in \Omega'$ at which $u^\varepsilon - \tilde{v}_\varepsilon$ is maximal over Ω', and every derivative D of u^ε at x : $\quad \operatorname{hgrad} \sigma\, x\, D \neq 0$.

So let $x \in \Omega'$ satisfy $(u^\varepsilon - \tilde{v}_\varepsilon)(y) \leq (u^\varepsilon - \tilde{v}_\varepsilon)(x)$ for every $y \in \Omega'$, and let D be a derivative of u^ε at x.

First we upgrade maximality from Ω' to $\overline{\Omega'}$. Let $z \in \overline{\Omega'}$. Because $\overline{\Omega'}$ is the closure of Ω' in the metric space $\mathbb{R}^n$, there is a sequence $(z_j)_{j\in\mathbb{N}}$ in Ω' with $z_j \to z$. For every j we have $z_j \in \Omega'$, hence

$$(u^\varepsilon - \tilde{v}_\varepsilon)(z_j) \;\leq\; (u^\varepsilon - \tilde{v}_\varepsilon)(x).$$

By (b), u^ε and $\tilde{v}_\varepsilon$ are continuous on $\overline{\Omega'}$, hence so is their difference; since $z_j \in \Omega' \subseteq \overline{\Omega'}$ for every j, $z \in \overline{\Omega'}$ and $z_j \to z$, continuity on $\overline{\Omega'}$ gives $(u^\varepsilon - \tilde{v}_\varepsilon)(z_j) \to (u^\varepsilon - \tilde{v}_\varepsilon)(z)$. Passing to the limit in the displayed inequality,

$$(u^\varepsilon - \tilde{v}_\varepsilon)(z) \;\leq\; (u^\varepsilon - \tilde{v}_\varepsilon)(x).$$

As $z \in \overline{\Omega'}$ was arbitrary, $u^\varepsilon - \tilde{v}_\varepsilon$ attains its maximum over $\overline{\Omega'} = \overline{\Omega[\varepsilon]}$ at x.

By (1) with $S = \overline{\Omega[\varepsilon]}$, the same holds for $u^\varepsilon - v_\varepsilon$: $(u^\varepsilon - v_\varepsilon)(y) \leq (u^\varepsilon - v_\varepsilon)(x)$ for every $y \in \overline{\Omega[\varepsilon]}$. Together with $x \in \Omega[\varepsilon]$ this is precisely the statement $x \in E_\varepsilon = E_{\varepsilon_k}$ in the sense of (1.15). Hypothesis (3) at the index k therefore applies to x and to the derivative D of $u^{\varepsilon_k} = u^\varepsilon$ at x, and gives $\operatorname{hgrad} \sigma\, x\, D \neq 0$. This proves (2).

Step 5: Proposition 3.3 on Ω'. We now apply `ComparisonNondegenerateSemiconvex` with the following data, checking its hypotheses one by one.

- The cited results at $\mathbb{G}$: our hypothesis (`CitedResults`).
- The sub-elliptic setting $\mathbb{G}$.
- The pair (A_0, H_0): it satisfies the structure conditions by Step 1(iv) and the normalization (2.3) by Step 1(v). This is where the normalization of Step 1 is spent; Proposition 3.3 is stated for a normalized pair, and the printed Theorem 1.2 is not.
- The domain Ω': bounded and open by Step 3. It is not assumed connected, and Proposition 3.3 does not assume it — which is why the reduced domains may be used here.
- The functions u^ε and $\tilde{v}_\varepsilon$: continuous on $\overline{\Omega'}$ by (b).
- u^ε locally semi-convex on Ω' and $-\tilde{v}_\varepsilon$ locally semi-convex on Ω': clause (c).
- $u^\varepsilon \leq \tilde{v}_\varepsilon$ on $\partial\Omega'$: let $z \in \partial\Omega'$. By (e), $u^\varepsilon(z) - \tilde{v}_\varepsilon(z) \leq -\tau/2$, and $\tau > 0$, so $u^\varepsilon(z) - \tilde{v}_\varepsilon(z) \leq -\tau/2 < 0$, whence $u^\varepsilon(z) \leq \tilde{v}_\varepsilon(z)$.
- The viscosity inequalities $\mathcal{L}_0 u^\varepsilon \leq 0 \leq \mathcal{L}_0 \tilde{v}_\varepsilon$ in Ω': clause (d).
- The non-degeneracy on $\operatorname{argmax}_{\Omega'}(u^\varepsilon - \tilde{v}_\varepsilon)$: this is exactly (2) of Step 4.

Its conclusion is

$$u^\varepsilon(x) \;\leq\; \tilde{v}_\varepsilon(x) \qquad \text{for every } x \in \Omega'. \tag{3}$$

Step 6: the contradiction. By (f) there is $x_e \in \Omega'$ with $\tau \leq u^\varepsilon(x_e) - \tilde{v}_\varepsilon(x_e)$. By (3) applied at $x = x_e \in \Omega'$, $u^\varepsilon(x_e) - \tilde{v}_\varepsilon(x_e) \leq 0$. Hence $\tau \leq 0$, contradicting $\tau = M_0/2 > 0$ from Step 0.

This contradiction refutes the assumption that $u(x^*) > v(x^*)$ for some $x^* \in \Omega$. Therefore $u(x) \leq v(x)$ for every $x \in \Omega$, i.e. $u \leq v$ in Ω.

Remark on the role of the sequence. The paper's hypothesis is the existence of a sequence $\varepsilon_k \to 0$ along which the non-degeneracy holds, and the proof uses it, as above, only to select one index k with $\varepsilon_k < \varepsilon_1$. The sequence is the natural form of the hypothesis because ε_1 is produced by

the proof and not known in advance: a hypothesis "for some ε small enough" would not be checkable, whereas "along some sequence tending to 0" is, and the two are equivalent once ε_1 is in hand. We have kept the printed form. □

3.2. `ComparisonUnconditional` (paper, Theorem 1.3).

Theorem 3.2 (Theorem 1.3). *Let $\mathbb{G}$ be a sub-elliptic setting (`CarnotSetting`) with group law $*$, homogeneous norm $\|\cdot\|$, step r, frame $\sigma = [\sigma^1, \dots, \sigma^m]$ and first-layer projection $\pi_1 : \mathbb{R}^n \to \mathbb{R}^m$. Assume the cited results of the literature* at $\mathbb{G}$ *(`CitedResults`), and also the cited results of the literature* at the Euclidean setting $\mathbb{G}_E$ on $\mathbb{R}^m$ *(`EuclidSetting` at $N := m$, and `CitedResults` at that setting). Let $A : \mathbb{R}^m \to \mathbb{R}^{m\times m}$ and $H : \mathbb{R}^m \to \mathbb{R}$ be continuous on $\mathbb{R}^m \setminus \{0\}$, with $A(\xi)$ symmetric and positive semidefinite for $\xi \neq 0$, and satisfy the conditions of strict ellipticity in the gradient direction (E) (1.9) and of homogeneity along rays (S) (1.10) — that is, let (A, H) satisfy the structure conditions of `StructureConditionsES`. No normalization of (A, H) is assumed. Let $\Omega \subset \mathbb{R}^n$ be a bounded open set with $\mathbb{R}^n \setminus \Omega \neq \emptyset$, let $u, v : \mathbb{R}^n \to \mathbb{R}$ be continuous on $\bar\Omega$, and assume*

(1) *u is a viscosity sub-solution (`IsViscSubsol`) and v a viscosity super-solution (`IsViscSupersol`) of $\mathcal{L}\cdot = 0$ in Ω;*
(2) *$u \le v$ on $\partial\Omega$.*

Let $R \ge 0$ satisfy $|u| \le R/2$ and $|v| \le R/2$ on $\bar\Omega$. Write

$$E := \operatorname*{argmax}_{\Omega}(u - v) = \big\{y \in \Omega :\ u - v \text{ attains its maximum over } \bar\Omega \text{ at } y\big\}.$$

Assume finally that at least one of the following three cases holds.

(i) *There is an open set $N \subseteq \mathbb{R}^n$ with $E \subseteq N$ such that either v is C^1 on N and* hgrad $\sigma\ z\ D \neq 0$ *(`hgrad`) whenever $z \in N$ and D is the derivative of v at z, or u is C^1 on N and* hgrad $\sigma\ z\ D \neq 0$ *whenever $z \in N$ and D is the derivative of u at z.*
(ii) *u is constant on $\bar\Omega$, or v is constant on $\bar\Omega$.*
(iii) *$r = 1$; or there are $F, Gf : \mathbb{R}^m \to \mathbb{R}$, continuous on $\pi_1(\bar\Omega)$, with $u = F \circ \pi_1$ and $v = Gf \circ \pi_1$.*

Then $u \le v$ in Ω.

Proof. This is Theorem 1.3. The paper states it as "Let Ω, $\mathcal{L}$, u, v be as in Theorem Theorem 1.2, without the hypothesis on E_{ε_k}", a citation of another theorem's hypotheses; FILESPEC and the substantiveness rubric require those hypotheses to be written out, and the statement above does so. Two of them deserve the same justification they were given on Theorem 1.2.

The parameter R. The paper defines $R := 2\max\{\|u\|_{L^\infty}, \|v\|_{L^\infty}\}$ just before (3.15), whereas `ReducedDomain` carries R as a parameter. We therefore quantify over every admissible R, i.e. every $R \ge 0$ with $|u|, |v| \le R/2$ on $\bar\Omega$. This contains the paper's statement: the paper's R is admissible, since $\|u\|_{L^\infty} \le \max\{\|u\|_{L^\infty}, \|v\|_{L^\infty}\} = R/2$ and likewise for v, and it is the smallest admissible one. It is also not stronger than the paper's statement in any way the paper does not justify: R enters the argument only through the definition of $\Omega[\varepsilon]$, where a larger R makes $\Omega[\varepsilon]$ smaller, and the role of the bound $|u| \le R/2$ is exactly the one the paper gives it in the attainment step of Lemma 3.8(4), namely to make $\|z_x\|^q \le 4\varepsilon\|u\|_{L^\infty} \le 2R\varepsilon < (1 + 4R)\varepsilon$.

The hypothesis $\mathbb{R}^n \setminus \Omega \neq \emptyset$. This is the paper's own convention, made explicit: (3.15) reads $\Omega[\varepsilon] = \{x \in \Omega :\ \inf_{y\in\mathbb{R}^n\setminus\Omega} \|y * x^{-1}\|^q > (1 + 4R)\varepsilon\}$, an infimum over $\mathbb{R}^n \setminus \Omega$, and the paper's $\Omega \subset \mathbb{R}^n$ is a proper subset. It is automatic as soon as $n \ge 1$, since no bounded subset of $\mathbb{R}^n$ is then all of $\mathbb{R}^n$; it excludes only the degenerate setting $n = 0$, in which $\partial\Omega = \emptyset$ and hypothesis (2) carries no information at all.

"Constant" in case (ii). The paper writes "u or v is constant" without naming the set. We read it on $\bar\Omega$. That costs nothing and loses nothing: if v is constant on Ω then, v being continuous on $\bar\Omega$ and Ω dense in $\bar\Omega$, v is constant on $\bar\Omega$ as well (and if $\Omega = \emptyset$ then $\bar\Omega = \emptyset$ and the condition holds vacuously), so the paper's reading implies ours and the theorem below is at least as strong as the printed one.

The structure conditions are kept in the form (E) and (S), with no normalization: the printed theorem is for general $A, H \in C(\mathbb{R}^m \setminus \{0\})$ satisfying (1.9) and (1.10), and (2.3) is an internal step of the proofs, discharged inside the nodes cited below. And the cited results of the literature are carried as an explicit hypothesis, as pm-0001 and item 2 of the paper's "Note for the formalization" require, so that the formalized theorem reads "cited results ⇒ Theorem Theorem 1.3".

Why there are two cited-results hypotheses, and not one. Three of the five cited results — the Bardi–Goffi principles, Pansu's theorem with [DMV13, Prop. 4.8], and parts (1)–(3) of Lemma 3.8 — are theorems about Carnot groups, and `CitedResults` carries the setting at which they are asserted as a parameter. Asserting them instead at every instance of `CarnotSetting` would over-reach: that structure has no dilations and no stratification, which is precisely what their proofs use, so for an arbitrary instance they are not known and may fail; a hypothesis package that may be unsatisfiable would make this theorem vacuous. Item 3 of the paper's note for the formalization licenses building the structural facts of Section 2.4 into the setting only "provided every Carnot group in the coordinates of Section Section 2.4 satisfies that definition", which guarantees that every Carnot group is an instance and says nothing in the converse direction. Case (iii) applies the cited results a second time, inside the Euclidean setting on $\mathbb{R}^m$ to which it descends, and a package at $\mathbb{G}$ does not supply one at that setting; hence the second hypothesis. It is satisfiable for the same reason the first is: $\mathbb{G}_E$ is the Carnot group $(\mathbb{R}^m, +)$ with the Euclidean norm, where Pansu's theorem is Rademacher's, Bardi–Goffi is the classical strong maximum principle, and [CYW07, Prop. 3.3] is the classical sup-convolution lemma. Neither package is proved here; both are assumed, as item 2 of the note directs.

The proof is the three-way case split the paper itself makes ("Only case (i) is obtained through the hypothesis of Theorem Theorem 1.2; case (ii) follows from the strong maximum principle alone, and case (iii) from the Euclidean translation argument"). Each case is a separate node, and the three arguments are genuinely different; the split is the one contemplated by the note on cases in the substantiveness rubric.

Case (i). Let N be as in (i): open, with $E \subseteq N$. The hypotheses of `UncondCaseC1` are exactly the ambient hypotheses of the present statement together with this N: the cited results at $\mathbb{G}$; the structure conditions on (A, H); Ω bounded, open, with non-empty complement; u, v continuous on $\bar{\Omega}$; the two viscosity properties; $u \le v$ on $\partial\Omega$; $R \ge 0$ with $|u|, |v| \le R/2$ on $\bar{\Omega}$; N open; and $E \subseteq N$. That node has two conclusions, one for each orientation of the C^1 hypothesis:

- if v is C^1 on N and hgrad σ z $D \neq 0$ for every $z \in N$ and every derivative D of v at z, then $u \le v$ in Ω;
- if u is C^1 on N and hgrad σ z $D \neq 0$ for every $z \in N$ and every derivative D of u at z, then $u \le v$ in Ω.

Case (i) supplies one of the two antecedents, so one of the two conclusions applies and gives $u \le v$ in Ω.

Case (ii). The hypotheses of `UncondCaseConstant` are the ambient hypotheses of the present statement, minus R and the bound on $|u|, |v|$, which that node does not use: the cited results at $\mathbb{G}$; the structure conditions on (A, H); Ω bounded, open, with non-empty complement; u, v continuous on $\bar{\Omega}$; the two viscosity properties; and $u \le v$ on $\partial\Omega$. Its two conclusions are

- if v is constant on $\bar{\Omega}$ then $u \le v$ in Ω;
- if u is constant on $\bar{\Omega}$ then $u \le v$ in Ω.

Case (ii) supplies one of the two antecedents, so one of the two conclusions applies.

Case (iii). The hypotheses of `UncondCaseFirstLayer` are again exactly the ambient hypotheses of the present statement, and this is the one case that consumes both cited-results hypotheses: that node asks for a package at $\mathbb{G}$ and a package at the Euclidean setting $\mathbb{G}_E$ on $\mathbb{R}^m$ (`EuclidSetting` at $N := m$), and the present statement carries exactly those two. Its two conclusions are

- if $r = 1$ then $u \le v$ in Ω;

- if there are F, Gf continuous on $\pi_1(\bar{\Omega})$ with $u = F \circ \pi_1$ and $v = Gf \circ \pi_1$, then $u \leq v$ in Ω.

Case (iii) supplies one of the two antecedents, so one of the two conclusions applies.

In each of the three cases we obtain $u(x) \leq v(x)$ for every $x \in \Omega$, which is the assertion. $\square$